I.
On the Stability of Equilibrium.

By G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, vol. 32, pp. 85–88.

On the Stability of Equilibrium.

[Extract from a paper read in the Royal Academy of Sciences on 22 January 1846.][1]

When attractive or repulsive forces, depending only on distance, act on a system of material points, either directed toward fixed centers, or such that, when they occur mutually between two masses, action and reaction are equal; and when, moreover, the constraint equations connecting the coordinates of the points with one another do not contain the time, then the so-called integral of vis viva always holds. In its full generality it was first set out by D. Bernoulli, and is contained in the equation

$$\Sigma mv^2 = f(x, y, z, x', \ldots) + C.$$

The summation sign extends over all masses of the system; each is denoted by m, its velocity by v, and C denotes the arbitrary constant. The function of the coordinates depends only on the nature of the forces and can always be expressed by a definite number of independent variables $\lambda, \mu, \nu, \ldots$, whereby the equation becomes

$$\Sigma mv^2 = \varphi(\lambda, \mu, \nu, \ldots) + C.$$

The function φ is intimately related to the equilibrium positions of the system, since the condition that equilibrium should take place for a position corresponding to specified values of $\lambda, \mu, \nu, \ldots$ coincides with the vanishing of the total differential of φ for these values; thus, in general, for every equilibrium position the function will have a maximum or a minimum. If a maximum actually occurs, then the equilibrium has the character of stability; that is, if the points are displaced only slightly from such a position and receive small initial velocities, the system will never, in the course of time, move away from that position beyond certain narrow bounds.

The theorem just mentioned is unquestionably among the most important in all mechanics; in particular, the theory of small oscillations, which contains so many interesting physical applications, rests upon it. One must indeed wonder that its truth has not until now been demonstrated with the necessary rigor. Suppose that the equilibrium position, or the maximum of the function φ, corresponds to the values $\lambda = 0$, $\mu = 0$, …; this can always be done without loss of generality. The proof given by Lagrange[2] consists in expanding the function in powers of $\lambda, \mu, \nu, \ldots$; since this expansion begins with terms of the second order, he restricts it to those terms, and then from the known criterion for a maximum, according to which the second-order terms can be represented as a sum of negative squares, derives bounds for $\lambda, \mu, \nu, \ldots$ which they cannot exceed. This line of reasoning, which is also not seldom applied to other questions of stability, especially in physical astronomy, is plainly without demonstrative force. It can rightly be doubted whether quantities for which, under the supposition that they always remain small — for only in this supposition lies the permission to neglect the higher terms —, one finds small

[1] In the 1846 volume of the Academy reports the extract is introduced on p. 34 with the words: Mr. Lejeune Dirichlet read *on the conditions for the stability of equilibrium.* For the printing in the Journal, Dirichlet changed the title, as well as the text in a few places, and newly added the entire final paragraph. K.

[2] Mécanique analytique, première partie, section III.

bounds, will after an arbitrary time actually be enclosed within these, or even within narrow bounds at all.

So far as I know, the proof just mentioned has hitherto been reproduced without essential alteration by all writers who have dealt with this subject; and what Poisson[3] in particular added to it, in order to take account of the terms of higher order, rests on the quite untenable assumption that each term of the second order exceeds the aggregate of all higher terms. But even if Lagrange's considerations were completed for the case to which they apply, and in which the maximum is revealed in the terms of second order, the theorem in question would still not thereby be proved in its full extent. As is well known, the existence of a maximum is compatible with the vanishing of the second-order terms; only the lowest order for which not all terms vanish must always be even, and the aggregate of all terms of that order must always remain negative. The complete criteria for this latter condition have not yet been established, even when the fourth order is concerned (Théorie des fonctions, seconde partie, no. 57);[4] they would therefore have to be sought first, which would necessarily introduce great complication into the proof of the mechanical theorem.

Fortunately, the principle of the stability of equilibrium can be established independently of these criteria, and by very simple considerations directly connected with the concept of a maximum.

Keeping the above supposition that the equilibrium position corresponds to vanishing values of $\lambda, \mu, \nu, \ldots$, we make the second supposition that the value of $\varphi(0, 0, 0, \ldots)$ is likewise zero; this is allowed because of the arbitrary constant. If one now determines the constant from the given initial state, for which $v, \lambda, \mu, \nu, \ldots$ are to be denoted by $v_0, \lambda_0, \mu_0, \nu_0, \ldots$, one obtains

$$\Sigma m v^2 = \varphi(\lambda, \mu, \nu, \ldots) - \varphi(\lambda_0, \mu_0, \nu_0, \ldots) + \Sigma m v_0^2.$$

By the assumption that $\varphi(\lambda, \mu, \nu, \ldots)$ has, for $\lambda = 0$, $\mu = 0$, $\nu = 0$, $\ldots$, a maximum whose value itself vanishes, one can always specify such small positive quantities $l, m, n, \ldots$ that for every system $\lambda, \mu, \nu, \ldots$, if the numerical values of these variables are respectively no greater than $l, m, n, \ldots$, $\varphi(\lambda, \mu, \nu, \ldots)$ is always negative, except in the single case where $\lambda, \mu, \nu, \ldots$ vanish simultaneously. This case is excluded if we consider only such systems in which at least one of the variables $\lambda, \mu, \nu, \ldots$, apart from sign, is equal to its bound $l, m, n, \ldots$. If, among all these negative values of the function, $-p$ is the smallest apart from sign, it is easy to show that, if $\lambda_0, \mu_0, \nu_0, \ldots$ are taken numerically smaller than $l, m, n, \ldots$ and at the same time the condition

$$-\varphi(\lambda_0, \mu_0, \nu_0, \ldots) + \Sigma m v_0^2 < p$$

is satisfied, then each of the variables $\lambda, \mu, \nu, \ldots$ will remain, in the course of the motion, numerically below its bound $l, m, n, \ldots$. Were the contrary to occur, then, since the initial values $\lambda_0, \mu_0, \nu_0, \ldots$ satisfy the condition just stated by hypothesis, and since the variables $\lambda, \mu, \nu, \ldots$ are continuous,[5] there would have to be a time at which equality first occurs between one or more of the numerical values of $\lambda, \mu, \nu, \ldots$ and the corresponding bound $l, m, n, \ldots$, without any of the remaining variables having yet exceeded its bound. For this instant the negative value of $\varphi(\lambda, \mu, \nu, \ldots)$
would not be numerically smaller than p, and therefore the right-hand side of our equation, in view of the preceding inequality relating to the initial state, would be negative; this contradicts the nature of the left-hand side. At the same time it is clear that the velocities v will also always remain enclosed within bounds, since one always has

$$\Sigma m v^2 \leq \Sigma m v_0^2 - \varphi(\lambda_0, \mu_0, \nu_0, \ldots),$$

[3] Traité de Mécanique, tome second, page 492.

[4] This citation is absent in the Academy report. K.

[5] The words "and since the variables $\lambda, \mu, \nu, \ldots$ are continuous" are absent in the Academy report. K.

and also that the bounds for each v, as well as for each of the variables $\lambda, \mu, \nu, \ldots$ determining the position of the system, can be made arbitrarily narrow, since $l, m, n, \ldots$ are capable of every degree of smallness.[6]

Finally, we wish to point out a remarkable error, found in various writers, concerning the subject just discussed.[7] It is asserted that if the system passes through several equilibrium positions in the course of its motion, then these are alternately positions of stable and unstable equilibrium, that is, positions to which a maximum or a minimum of the function $\varphi(\lambda, \mu, \nu, \ldots)$ corresponds. This assertion is based, first, on the fact that a maximum or minimum does not lose this character when the variables $\lambda, \mu, \nu, \ldots$ formerly regarded as independent obtain the special values for which such a maximum or minimum occurs by becoming functions of a new quantity t; and second, on the fact that for a function of a single variable t, maxima and minima follow one another alternately. Both statements are correct, but they do not justify the conclusion drawn from them. Rather, it would be necessary for the first statement also to remain valid conversely: that is, that the function $\varphi(\lambda, \mu, \nu, \ldots)$ depending only on t, if it presents a maximum or minimum for a particular t, should retain the same property when the corresponding values of $\lambda, \mu, \nu, \ldots$ are regarded as special values of the quantities $\lambda, \mu, \nu, \ldots$ considered as independent variables. This need not be the case, even if only one quantity, for example λ, occurs. Thus the function λ^2 has only a minimum and no maximum at all, whereas the function $\sin^2 t$, into which the former passes under the assumption $\lambda = \sin t$, has not only infinitely many minima, which are equal to one another and to the former one, but also infinitely many maxima. The error into which one has fallen in this respect must be all the more surprising since the incorrectness of the assertion made is already recognizable in the ordinary motion of the pendulum.

[6] Here the extract in the Academy report closes; cf. p. 37 in the 1846 volume. K.

[7] Traité de Mécanique par Poisson, tome second, page 491.

II.

On a General Means of Verifying the Expression for the Potential Relative to an Arbitrary Mass, Homogeneous or Heterogeneous.

By M. G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, Bd. 32, p. 80–84.

On a General Means of Verifying the Expression for the Potential Relative to an Arbitrary Mass, Homogeneous or Heterogeneous.

Given an arbitrary mass, bounded in every direction, forming a continuous whole or composed of several separated parts, and having at each of its points a finite density, if one forms the sum of all the elements of this mass, each divided by its distance from an arbitrary point m, one obtains what geometers have agreed to call the potential of the mass with respect to that point. It is known that the triple integral thus defined, which is always a determinate function of the rectangular coordinates x, y, z of the point m, has a great number of important properties. I shall recall here only those on which the observation that is the object of this note bears.

1) The potential v and its first-order differential coefficients

$$\frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial z},$$

which express the components of the attraction exerted by the mass at the point m, are finite and continuous functions of x, y, z throughout all of space.

2) There always exist determinate bounds which the products

$$xv, \quad yv, \quad zv, \quad x^2\frac{\partial v}{\partial x}, \quad y^2\frac{\partial v}{\partial y}, \quad z^2\frac{\partial v}{\partial z}$$

cannot exceed at any point of space.

3) If one leaves aside the particular points around which the density varies abruptly, each of the three derivatives

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will always have a finite and unique value, and these simultaneous values will be such that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = -4\pi\varrho, \tag{a}$$

where ϱ denotes the density at the point (x, y, z), to be considered zero when this point lies outside the mass. For the points just excepted, which will always be situated on certain surfaces, among which are also included those which bound the mass, or at least the parts of the latter where the density is not zero, the functions

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will generally each have two distinct values, but always finite ones; for the first, for example, these will be the two limits of the quotient

$$\frac{\Delta \dfrac{\partial v}{\partial x}}{\Delta x},$$

corresponding to an infinitely small positive or negative value of Δx. From the combination of these double values there will in general result eight different values for the trinomial (a); it is not necessary to consider them. It suffices for our object to know that they are all finite.

Here is now the very simple observation to which the properties just stated give rise, and which, so far as I know, had not yet been made. These properties completely characterize the potential v, and there exists no other function v' possessing all these properties together.

To prove this, let us suppose that the contrary takes place, and see what will result for the difference $v' - v = u$. From the nature of the properties in question, it is evident that the new function u will still satisfy conditions (1) and (2), while equation (a) will be replaced by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

The points mentioned above must, however, be excepted; we know only that finite values of the trinomial correspond to such points. These finite values - unknown for the moment, but in fact all equal to zero, since we are going to see that $u = 0$ - occurring only along certain

surfaces, will have no influence on the result of a triple integration, and we shall have rigorously

$$\iiint u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx\, dy\, dz = 0,$$

the limits being arbitrary. Let us extend each of the three integrations from $-a$ to $+a$, and consider in particular the integral of the first term. If we begin with the operation relative to x, integration by parts gives

$$\int_{-a}^{+a} u \frac{\partial^2 u}{\partial x^2}\, dx = \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} - \int_{-a}^{+a} \left(\frac{\partial u}{\partial x} \right)^2 dx,$$

the indices appended to the first term indicating that one must take the difference of the two values corresponding to $+a$ and to $-a$. It is indeed to this difference alone that the terms produced outside the sign by the integration by parts reduce, $u \frac{\partial u}{\partial x}$ being a continuous function of x. Proceeding in an analogous manner on the other two integrals, one obtains

$$\begin{aligned} &\iiint \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 \right) dx\, dy\, dz \\ &\quad = \iint \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} dy\, dz + \iint \left(u \frac{\partial u}{\partial y} \right)_{-a}^{+a} dx\, dz + \iint \left(u \frac{\partial u}{\partial z} \right)_{-a}^{+a} dx\, dy. \end{aligned}$$

By virtue of conditions (2), the function of y and z that stands under the sign in the first double integral will, throughout the extent of that integral, be numerically less than $\frac{k}{a^3}$, where k does not depend on a; the integral is therefore less than $\frac{4k}{a}$, and vanishes for $a = \infty$. The same thing holding for the other two, it follows that the triple integral, taken between infinite limits, is likewise zero. But the function subjected to the triple integration, being continuous and never capable of becoming negative, must be identically zero; consequently u has a constant value, which can only be zero, since u must vanish at infinity. This was to be shown.

The result just obtained supplies a general means of verifying the expression for the potential when this potential is given throughout all of space. It will suffice, for this purpose, to make sure that the given expression satisfies all the conditions enumerated above.

The application of this procedure to the known formulae relating to a homogeneous ellipsoid perhaps offers the most expeditious means of establishing the theorems concerning that celebrated question, provided, however, that one seeks only a simple verification and does not have in view a method that naturally leads to the supposedly unknown results at the same time as it proves them.

The equation of the ellipsoid being

$$\frac{x^2}{\alpha^2}+\frac{y^2}{\beta^2}+\frac{z^2}{\gamma^2}=1,$$

the trinomial forming its left-hand side will be less than or greater than unity according as the point (x, y, z) is inside or outside the ellipsoid. In the second case, the equation

$$\frac{x^2}{\alpha^2+\sigma}+\frac{y^2}{\beta^2+\sigma}+\frac{z^2}{\gamma^2+\sigma}=1$$

has a unique positive root σ, which varies continuously with the variables x, y and z. This is seen by observing that the derivatives

$$\frac{\partial\sigma}{\partial x},\quad \frac{\partial\sigma}{\partial y},\quad \frac{\partial\sigma}{\partial z},$$

given by the equations

$$l\frac{\partial\sigma}{\partial x}=\frac{2x}{\alpha^2+\sigma},\quad l\frac{\partial\sigma}{\partial y}=\frac{2y}{\beta^2+\sigma},\quad l\frac{\partial\sigma}{\partial z}=\frac{2z}{\gamma^2+\sigma},\tag{b}$$

where

$$l=\frac{x^2}{(\alpha^2+\sigma)^2}+\frac{y^2}{(\beta^2+\sigma)^2}+\frac{z^2}{(\gamma^2+\sigma)^2},$$

always have finite values. This being set, if for brevity we put

$$D=\sqrt{\left(1+\frac{s}{\alpha^2}\right)\left(1+\frac{s}{\beta^2}\right)\left(1+\frac{s}{\gamma^2}\right)},$$

it remains to prove that one has

$$v=\pi\int_0^\infty\left(1-\frac{x^2}{\alpha^2+s}-\frac{y^2}{\beta^2+s}-\frac{z^2}{\gamma^2+s}\right)\frac{ds}{D},$$

or

$$v=\pi\int_\sigma^\infty\left(1-\frac{x^2}{\alpha^2+s}-\frac{y^2}{\beta^2+s}-\frac{z^2}{\gamma^2+s}\right)\frac{ds}{D},$$

according as the point (x, y, z) is located inside or outside the mass, whose density is assumed equal to unity. Differentiation gives

$$\frac{\partial v}{\partial x}=-2\pi x\int_0^\infty\frac{ds}{(\alpha^2+s)D},$$

or

$$\frac{\partial v}{\partial x}=-2\pi x\int_\sigma^\infty\frac{ds}{(\alpha^2+s)D},$$

the term arising from the variability of the limit σ in the second case vanishing by virtue of the equation of which σ is a root. It is manifest that the first expressions for v and $\frac{\partial v}{\partial x}$, and the analogous expressions for $\frac{\partial v}{\partial y}$, $\frac{\partial v}{\partial z}$, vary continuously with the coordinates x, y, z, and that the same is true of those corresponding to an exterior point; one also sees that continuity is not broken at the surface where $\sigma = 0$, which makes the corresponding expressions coincide.

Conditions (2) plainly hold for an interior point and, in this case, are already included in conditions (1). To verify them outside the ellipsoid, let λ be the smallest of the semiaxes α, β, γ. Observing that the factor $\frac{ds}{D}$ in the second expression for v is less than unity, and attending only to numerical values, we evidently have

$$v < \frac{2\pi\alpha\beta\gamma}{(\lambda^2+\sigma)^{1/2}}, \quad \frac{1}{x}\cdot\frac{\partial v}{\partial x} < \frac{4\pi\alpha\beta\gamma}{3(\lambda^2+\sigma)^{3/2}}.$$

As, on the other hand, the equation in σ gives, apart from the sign, $x < (\alpha^2+\sigma)^{1/2}$, one concludes

$$xv < 2\pi\alpha\beta\gamma\left(\frac{\alpha^2+\sigma}{\lambda^2+\sigma}\right)^{1/2}, \quad x^2\frac{\partial v}{\partial x} < \frac{4}{3}\pi\alpha\beta\gamma\left(\frac{\alpha^2+\sigma}{\lambda^2+\sigma}\right)^{3/2},$$

inequalities whose right-hand sides remain finite as σ increases from zero to infinity.

Let us pass to the derivatives of the second order. One finds

$$\frac{\partial^2 v}{\partial x^2} = -2\pi\int_0^\infty \frac{ds}{(\alpha^2+s)D}$$

or

$$\frac{\partial^2 v}{\partial x^2} = \frac{2\pi x}{(\alpha^2+\sigma)\Delta}\cdot\frac{\partial\sigma}{\partial x} - 2\pi\int_\sigma^\infty \frac{ds}{(\alpha^2+s)D},$$

where Δ denotes the value of D corresponding to $s = \sigma$. These expressions are still finite and determinate, but they no longer coincide at the surface. If we now add to each of these equations the two analogous equations, and observe that one has, indefinitely,

$$\int\left(\frac{1}{\alpha^2+s}+\frac{1}{\beta^2+s}+\frac{1}{\gamma^2+s}\right)\frac{ds}{D} = -\frac{2}{D} + \text{const.},$$

there will come, according to the two cases,

$$\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2} = -4\pi,$$

or

$$\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2} = \left(\frac{2x}{\alpha^2+\sigma}\cdot\frac{\partial\sigma}{\partial x}+\frac{2y}{\beta^2+\sigma}\cdot\frac{\partial\sigma}{\partial y}+\frac{2z}{\gamma^2+\sigma}\cdot\frac{\partial\sigma}{\partial z}-4\right)\frac{\pi}{\Delta}.$$

The first of these two equations already agrees with condition (a), and to make sure that the same is true of the second it suffices to take the sum of the three equations (b), each multiplied by its right-hand side, and then to suppress the common factor l, which is evidently different from zero. One thus recognizes that the right-hand side of the last equation vanishes, as it must for an exterior point.

We have supposed up to now that the given mass has three dimensions. When this mass has only two and is distributed over one or over several surfaces, the method of verification just set out will have to undergo certain modifications, which nevertheless follow too simply from the known theorems relating to that case for it to be necessary to enter into further details on this point.[1]

[1]Cf. the immediately following paper. K.

III.

On the Characteristic Properties of the Potential of a Mass Distributed over One or More Finite Surfaces.

By G. Lejeune Dirichlet.

Bericht über die Verhandlungen der Königl. Preuss. Akademie der Wissenschaften. Jahrg. 1846, S. 211–212.

On the Characteristic Properties of the Potential of a Mass Distributed over One or More Finite Surfaces.

As was already observed in an earlier paper,[1] the method developed there for the case of a mass extended in three dimensions can also be applied to the question just mentioned, and it follows that, for a mass distributed over surfaces, the characteristic properties of the potential, that is, those which completely determine its expression, are the following:

1) The potential v is, throughout all of space, a finite and continuous function of the rectangular coordinates x, y, z.

2) The three partial differential quotients of the potential taken with respect to the coordinates are likewise continuous everywhere outside the surfaces. For a point lying on them, however, there is an interruption of continuity consisting in this: if the potential at such a point and at the points lying on the normal on both sides at distance ε are denoted by v, v', v'', then the quotient

$$\frac{v' + v'' - 2v}{\varepsilon}$$

has, for an infinitely small positive ε, the expression $-4\pi\varrho$ as its limit, where ϱ denotes the density at the point of the surface.

3) Everywhere outside the surfaces the equation

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

holds.

4) At infinite distance from the surfaces, the same conditions are fulfilled as for a mass extended in three dimensions.

Apart from the utility that may be afforded by the proof that the properties just listed completely characterize the potential, in order to verify an expression for the potential known from wherever it may have been obtained, this proof also presents another and more essential interest. It follows from it, namely, that the important theorem stated by Gauss, according to which a mass can always be distributed over given surfaces in such a way that the potential corresponding to this distribution receives, at every point of the surfaces, an arbitrarily prescribed value varying according to continuity, is wholly identical with the assertion that in a homogeneous mass filling all of space, provided that initially only temperatures vanishing at infinite distance

[1] *Crelle's Journal*, Band 32, p. 80. Editorial note in the Werke: P. 9 of the second volume of this edition of G. Lejeune Dirichlet's works. The observation is found at the end of the paper on p. 16. K.

occur, a fixed temperature will always be established everywhere after an infinite time under the influence of constant heat sources distributed over surfaces. The identity of the two theorems, and the new relation between two branches of mathematical physics already exhibiting such great kinship that is thereby established, is immediately clear from what has been said above, since the known conditions determining the permanent state of heat coincide exactly with those by which the potential corresponding to the required mass distribution is characterized.

IV.
On the Reduction of Positive Quadratic Forms with Three Indeterminate Integers.

By G. Lejeune Dirichlet.

Report on the proceedings of the Royal Prussian Academy of Sciences, year 1848, pp. 285–288.

On
the reduction of positive quadratic forms
with three indeterminate integers.

It is well known that Lagrange was the first to show that every binary quadratic form can be reduced, that is, transformed into another one whose coefficients satisfy certain inequality conditions; he also showed at the same time that in every class of positive forms there always exists only a single such form, so that in this case the various reduced forms corresponding to a given determinant can serve as the representatives of the various classes. After the ternary forms had later been considered in the *Disquisitiones arithmeticae* from a general point of view, it became necessary for the further development of this theory to extend Lagrange's investigation to the positive forms of this kind, that is, to find such inequality conditions among the coefficients that, in every class, they are satisfied by one and only one form. This extension, which involved great difficulties, was accomplished by Seeber in a work devoted especially to the positive ternary forms, of which it forms the principal content. The great complication of Seeber's method made it desirable to find a simpler path leading to the same results; this path, however, could be found only after many fruitless attempts.

For an easier description of this new procedure, which is of surprising simplicity, it will be expedient to clothe the matter in a geometric form,

using here the interesting geometric construction by which Gauss, in a short essay in which he discusses Seeber's work,[1] represents the principal properties of positive binary and ternary forms. According to this geometric representation, every positive ternary form corresponds to an infinite system of points arranged parallelepipedally, and the various subformations of the form are nothing other than changed arrangements of the same system according to another elementary parallelepiped. Expressed in this language, Seeber's results consist essentially in the following.

1. Every system of parallelepipedally arranged points can always be divided in such a way that, in the corresponding elementary parallelepiped, neither the sides of the faces are greater than their diagonals, nor the edges of the parallelepiped greater than the diagonals of the parallelepiped.

2. For a given system, such an arrangement can in general be carried out in only one way.

One is easily convinced of the correctness of the first of these propositions by the following very simple consideration.

Let (0) be any point of the system. The remaining points of the system evidently always occur in pairs at equal distance and in opposite directions from (0). Let (1) be one of the points

[1] *Crelle's Journal*, vol. 20, p. 312. Editorial note in the *Werke*: Gauss' Werke, vol. II, p. 188. K.

in the pair for which the distance from (0) is smaller than for every other pair. If the same least distance occurs for several pairs, choose (1) arbitrarily in any one of them. Now place through the line (01) and any point of the system outside that line an infinite plane; all points falling in this plane then form a parallelogrammatic system. In one of the two nearest parallel lines of this system take the point lying nearest to (0), or arbitrarily one of them if the same shortest distance occurs for two points of this line. Among all planes that can be placed through (01) in the indicated way, this shortest distance will be smaller in one plane, or in some of them, than in all the others. Fix this plane, or one of them if the same shortest distance should be common to several, and the point in it which we shall denote by (2). Then one evidently has:

$$(02) \geq (01),$$

and just as easily one sees that, in the parallelogram determined by the sides (01) and (02), the two diagonals are not smaller than these sides. Once the plane (012) has been fixed in this way, observe that the entire system of points lies in the plane (012) and in other planes parallel to it and mutually equidistant. Now take, in one of the two planes adjacent to the plane (012), the point nearest to (0), or one of the nearest points if the same shortest distance should occur for several. If this point is called (3), then one evidently has:

$$(03) \geq (02),$$

and the fundamental parallelepiped determined by the edges (01), (02), and (03) will have the required properties.

After what was remarked above about the parallelogram with sides (01) and (02), it remains for us to show that, both in the parallelograms (013) and (023) and in the parallelepiped, the sides and edges are not greater than the diagonals. In view of the two inequalities

$$(03) \geq (02) \geq (01)$$

this evidently amounts to proving that (03) is no greater than any one of the four diagonals of the named parallelograms, nor greater than any one of the four diagonals of the parallelepiped. But these eight diagonals evidently coincide, in length, with the eight straight lines that can be drawn from (0) to the eight corners of the parallelograms that lie around (3) in the plane through (3) parallel to (012). That none of these eight joining lines is smaller than (03) follows immediately from the condition by which the point (3) was chosen.

If, in the construction just indicated, (1), (2), and (3) are completely determined points (apart from the possibility, always present, that for each of them one can also take the point that lies, with respect to (0), at equal distance and in the opposite direction), then the edges (01), (02), and (03) of the fundamental parallelepiped obtained are unequal, and the diagonals are actually greater than the sides and edges by which, in general, they are not to be exceeded. In this case one then easily proves

that this system admits only this single elementary parallelepiped with the required properties.

The matter takes a somewhat different form when, in our construction, a choice can be made among different non-opposite positions for the points (1), (2), and (3). In such singular cases there may exist, for the system, several non-congruent fundamental parallelepipeds possessing the required properties. All these parallelepipeds, however, can be completely enumerated without difficulty, and one can always distinguish one of them from the others by certain subsidiary conditions, so that, with the addition of these secondary conditions, the theorem that every class contains only one reduced form does not lose its validity. Something similar is already well known in the theory of positive binary forms, where in two special cases one must prescribe a definite sign for the middle coefficient if one wishes to retain only one reduced form in each class.

V.

On the Reduction of Positive Quadratic Forms with Three Undetermined Integral Numbers.

By G. Lejeune Dirichlet.

Crelle, *Journal für die reine und angewandte Mathematik*, vol. 40, pp. 209–227.

On

the Reduction of Positive Quadratic Forms

with Three Undetermined Integral Numbers.

[Presented in the session of the physical-mathematical class of the Academy on 31 July 1848[1].]

It is well known that Lagrange first showed that every binary quadratic form can be reduced, that is, transformed into an equivalent form whose coefficients satisfy certain inequalities. At the same time he showed that in each class of positive forms there is always only one such form, so that, in this case, the several reduced forms belonging to a given determinant may serve as representatives of the several classes. After the ternary forms had later been treated from a general point of view in the *Disquisitiones arithmeticae*, the further development of this theory required the extension to positive ternary forms of the investigation carried out by Lagrange for the positive binary forms: one had to find inequalities among the coefficients such that in each class they are satisfied by one and only one form.

This extension, beset with great difficulties, was carried out by Seeber in a work devoted specifically to positive ternary forms. It forms the principal content of that work, which Gauss characterized as follows in a highly interesting review[2]:

> Full justice must be done to the spirit of thoroughness with which these matters are treated. If we must regret that this thoroughness is accompanied by great length, perhaps discouraging to some–the solution of the problem taking 41 pages and the proof of the theorem 91 pages–this is not to be taken as a reproach. When a difficult problem or theorem is to be solved or proved, the first step, and one to be acknowledged with due thanks, is that a solution or proof be found at all. The question whether this could have been done in an easier and simpler manner remains idle until the possibility has also been decided by the deed. We therefore consider it premature to dwell here on this question.

The great complication of Seeber's method long ago tempted me to try to found the theory of reduced ternary forms in a simpler way. In now communicating the result of these efforts, I believe that, in the interest of brevity and, if I may so speak, of transparency of exposition, I must retain the geometrical form in which I have conducted the investigation. It is based on the remarkable relations which hold between quadratic forms with two or three elements and certain spatial configurations. I begin by carrying out the indications already given by Gauss, in the review just mentioned, concerning these relations.

[1]An extract from this memoir had already been given in the Monthly Report of the Academy. There the principle of this new treatment of the reduction of positive ternary forms–the consideration of successive minima–was indicated, and the proof of the first of Seeber's two results was carried out completely according to this principle.

[2]Crelle's Journal, vol. 20, p. 312.

§. 1.

The ternary form

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy = \varphi, \tag{1}$$

in which we regard x, y, z as first, second, and third element, is called positive when φ never becomes negative for real values of these elements. In such a form the coefficients

$$a, \quad b, \quad c$$

are always positive, while the combinations of coefficients

$$a'^2 - bc, \quad b'^2 - ac, \quad c'^2 - ab, \quad aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -D, \tag{2}$$

of which the last, $-D$, is called the determinant of the form, are negative[3]. By virtue of these conditions there are always three acute or obtuse angles λ, μ, ν, completely determined by the equations

$$\cos\lambda = \frac{a'}{\sqrt{bc}}, \qquad \cos\mu = \frac{b'}{\sqrt{ac}}, \qquad \cos\nu = \frac{c'}{\sqrt{ab}},$$

from which a trihedral angle can be formed, since the required condition

$$\cos^2\lambda + \cos^2\mu + \cos^2\nu - 2\cos\lambda\cos\mu\cos\nu < 1$$

coincides with $D > 0$.

Since, however, two mutually symmetric trihedral angles can be formed with the same angles λ, μ, ν, we agree always to choose that one of the two in which the edges, as they lie opposite these angles in order, follow one another from right to left with respect to a straight line drawn from the vertex 0 into the interior of the angle and thought of as going upward. If we now regard the three edges as the positive axes of a coordinate system, the whole infinite space can be referred to our form by taking the products $x\sqrt{a}$, $y\sqrt{b}$, $z\sqrt{c}$ as the coordinates of an arbitrary point. Then φ expresses the square of the distance of this point from the vertex, or, more generally, the square of the distance between two points whose like-named coordinates have those products as differences.

Now form, with three new undetermined elements x', y', z', the linear expressions

$$x = \alpha x' + \beta y' + \gamma z', \quad y = \alpha' x' + \beta' y' + \gamma' z', \quad z = \alpha'' x' + \beta'' y' + \gamma'' z', \tag{3}$$

where the only restriction is that the determinant formed from the nine coefficients

$$\alpha\beta'\gamma'' + \beta\gamma'\alpha'' + \gamma\alpha'\beta'' - \gamma\beta'\alpha'' - \alpha\gamma'\beta'' - \beta\alpha'\gamma'' = E \tag{4}$$

be not zero. Then φ goes over into a new form φ', and the corresponding quantities for that form will be denoted by accented letters[4].

If one again lets an infinite space correspond to the new form, the two infinite spaces are thereby put in pointwise correspondence: two points correspond when, in the expressions for their coordinates

$$x\sqrt{a},\ y\sqrt{b},\ z\sqrt{c}; \qquad x'\sqrt{a'},\ y'\sqrt{b'},\ z'\sqrt{c'},$$

the elements x, y, z and x', y', z' are connected by equations (3). If these expressions are the coordinate differences for two pairs of corresponding points, then the same relations among $x, y, \ldots$ hold as before; hence, by what has just been said and because $\varphi = \varphi'$, the distance

[3] *Disquisitiones arithmeticae*, art. 271.

[4] This is the meaning determined here for the accented letters in this paragraph; it is therefore essentially different from the meaning of a', b', c' in the first display of §. 1. K.

between any two points of the one space is equal to the distance between the corresponding two points of the other. The two pointwise corresponding spaces are therefore either congruent or symmetric: when the origins 0 and $0'$ are made to coincide, either every point falls on its corresponding point or on the opposite point of the latter, where, for brevity, two points of the same space which lie at equal distance and in opposite directions from the origin are called opposite points.

To decide which of these two cases occurs, one draws from the vertex in one space to three arbitrary points and examines whether the straight lines drawn in the other space from its vertex to the corresponding points have the same or the opposite order. For example, in the second space take the lines to the points with coordinates

$$\sqrt{a'}, 0, 0; \qquad 0, \sqrt{b'}, 0; \qquad 0, 0, \sqrt{c'},$$

lying on the positive axes. By the convention above these follow one another from right to left. For the corresponding points in the first space the coordinates are

$$\alpha\sqrt{a'}, \alpha'\sqrt{a'}, \alpha''\sqrt{a'}; \quad \beta\sqrt{b'}, \beta'\sqrt{b'}, \beta''\sqrt{b'}; \quad \gamma\sqrt{c'}, \gamma'\sqrt{c'}, \gamma''\sqrt{c'}.$$

Their order is the same or the opposite according as the determinant E is positive or negative. Thus the transformation is a motion when $E > 0$ and a motion followed by reflection when $E < 0$.

If the elements x, y, z had arbitrary values up to now, let them henceforth be restricted to integers. In place of the whole space we then have an infinite system of parallelepipedally arranged points, that is, a system formed by the intersections of three families of parallel equidistant planes. Suppose now that the substitution coefficients $\alpha, \beta, \gamma, \ldots$ are also integers and that E has the value ± 1. Then to each integral combination x, y, z there corresponds an integral combination x', y', z', and conversely. By what has just been shown, the parallelepipedal systems so put in correspondence can be brought into such a position that one coincides either with the other or with the opposite points of the other. Since, however, the opposite points of such a system again form the same system, the two cases are not distinct. This is also clear from the fact that φ' is unchanged when one changes the signs of $\alpha, \beta, \gamma, \ldots$, which changes E into $-E$.

Thus the two systems are always congruent. Systems corresponding to two equivalent ternary forms φ and φ' are therefore the same spatial configuration in two different arrangements. Conversely, any two different parallelepipedal arrangements of the same system correspond to equivalent forms. Indeed, choose any point of the system as common origin. Between the coordinates relative to the two systems of axes, and hence also between the proportional elements x, y, z, x', y', z', there are linear equations without constant term, that is, equations of the form (3). Since, by assumption, integral values of x, y, z are accompanied by integral values of x', y', z' and conversely, the coefficients $\alpha, \beta, \gamma, \ldots$ are also integers and $E = \pm 1$. On the other hand, for corresponding integral values of the elements one has $\varphi = \varphi'$, and the equality therefore holds identically.

Analogous relations hold between a positive binary form

$$lx^2 + 2mxy + ny^2$$

and a system of parallelogrammatically arranged points. Here take two axes inclined to one another by the angle ϑ determined by

$$m = \sqrt{ln}\cos\vartheta,$$

where the distinction between the axes is made uniformly, for instance by choosing the second to the left of the first after one side of the plane has been designated as the upper side. If $x\sqrt{l}$, $y\sqrt{n}$

are then regarded as coordinates, one obtains a system of points completely determined by the quadratic form, which may be regarded as the intersections of two families of equidistant parallel lines. If between two forms there is the so-called proper equivalence, so that $\alpha\delta - \beta\gamma$ in the substitutions $x = \alpha x' + \beta y'$, $y = \gamma x' + \delta y'$ equals positive unity, then the corresponding systems may be brought to coincide by a motion in the plane. In the other case, where $\alpha\delta - \beta\gamma = -1$, generally speaking one of the systems must be turned over for this purpose.

§. 2.

Having established the connection between quadratic forms and certain geometrical configurations, we must develop some further properties of these configurations. For brevity, a system of parallelogrammatically or parallelepipedally arranged points will be called a *system of second* or *third order*, while an infinite row of equidistant points in a straight line will be called a *system of first order*.

The common character of all three kinds of systems is plainly this: if such a system is brought by a motion without rotation, which we shall call a displacement, into a new position such that one of its points passes into the place originally occupied by another, then the same happens for all points; the system in its new position coincides completely with the system in its original position. It is easy to prove that this displacement property completely characterizes all three kinds of systems: a system with this property is respectively a system of first, second, or third order according as it lies in a straight line and contains two points, lies in a plane and contains three non-collinear points, or contains at least four points not lying in one plane.

For example, suppose all points of a system lie on the same straight line, and let a and a' be two neighbouring points of it. By a displacement carrying a to a', the point a' goes to a point a'' which is as far from a' as a' is from a. Thus a'' also belongs to the system, and the system has no point between a' and a'', since such a point would before the motion have lain between a and a'. Continuing this consideration indefinitely in both directions proves the assertion.

Now suppose a planar system has the displacement property, and let a and a' be two neighbouring points such that no point of the system lies between them on the line aa'. Since the displacement from a to a' carries the infinite line aa' into itself, all points of the system on this line form a system of first order $\ldots a'' a a' a'' \ldots$. Since, by hypothesis, the system has at least one further point outside this line, let b be one of the points nearest to the line. If now a displacement carries a to b, the first-order system passes into the new position $\ldots b'' b b' b'' \ldots$ and in that position belongs to the original system. It is also clear that neither between the points $\ldots b'', b, b', b'', \ldots$ nor between the lines $\ldots bbb' \ldots$, $\ldots aaa' \ldots$ can any point of the system lie. Continuing in this way, one sees that the whole system can be arranged parallelogrammatically, and that $aa'bb'$ may be chosen as one of its fundamental parallelograms. This construction also gives all parallelogrammatic arrangements of which the system is capable: a' may be chosen arbitrarily subject only to the necessary restriction that no point lie between a and a', and then b may be chosen arbitrarily in the nearest parallel line.

Finally, suppose a system with the displacement property contains at least four points not lying in the same plane. Through any three non-collinear points of it draw a plane. Since every displacement parallel to this plane moves the plane into itself, the points lying in it form, by the preceding result, a system of second order. After this planar system has been divided parallelogrammatically in any way, take one of the remaining points of the spatial system nearest to the plane and displace the system so that an arbitrary point in the plane goes to the chosen point outside it. Repeated application of this motion and of its inverse evidently gives a parallelepipedal arrangement of the given system. At the same time the construction plainly has the required generality, since the first plane, the arrangement of the system of second order in it, and finally the choice of the point in the nearest plane may all be made arbitrarily.

At the end of this paragraph we also show that however one divides the same system of second or third order, the parallelogram or parallelepiped underlying the division always has the same content. This is the geometrical meaning of the theorem that equivalent forms have equal determinants. Namely, in the plane of a system of second order think of a closed curve, for instance a circle. Let z denote the area enclosed by it and s the number of points inside it; it is immaterial whether points on the circumference are counted or not. The quotient z/s, as the radius increases, clearly has the content of a fundamental parallelogram as its limit. Since s and z are independent of the kind of arrangement of the system, the theorem follows for systems of second order. The assertion for spatial systems follows in exactly the same way.

§. 3.

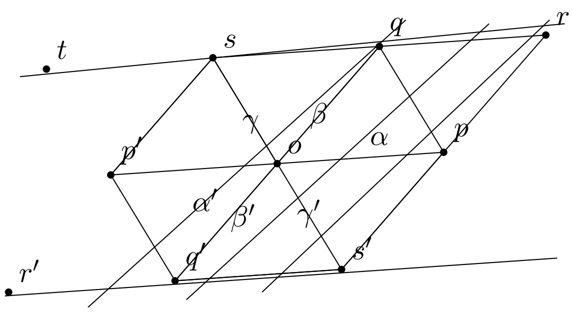
We now show that every system of second order can be divided according to a fundamental parallelogram whose sides are not greater than its diagonals.

I. Let o be any point of the system. The remaining points occur in pairs equidistant from o and in opposite directions. Let p be one of the pair for which the distance from o is smaller than for every other pair; if this shortest distance occurs for more than one pair, choose one of them arbitrarily.

The given system consists of infinitely many mutually congruent and equidistant systems of first order, one of which contains o and p. In one of the two adjacent systems take the point q nearest to o, or, if the same shortest distance occurs for two points, either one. The parallelogram $poqr$ so obtained has the required properties, since by construction

$$op \leq oq, \qquad oq \leq or, \qquad oq \leq os = pq.$$

A fundamental parallelogram satisfying these conditions will henceforth be called *reduced*.



II. We now establish the relation between such a parallelogram and the planar system to which it belongs. If $poqr$ is reduced, then without loss of generality the angle poq may be assumed not obtuse; if it were obtuse, the adjacent parallelogram belonging to the same arrangement would have an acute angle at o. Similarly we may assume $op \leq oq$. Then $or > oq$, and it remains only to consider the condition $pq \geq oq$. Put $op = \sqrt{l}$, $oq = \sqrt{n}$, so that $l \leq n$. Then the relation to the whole system is this: the minimum distance of any point of the system from o is $\sqrt{l}$, and after a point at this distance has been chosen, the second minimum, in all remaining directions outside the line through o and that first point, is $\sqrt{n}$.

This statement is general. The further assertion that the first minimum is attained only at p (opposite points being counted only once), and the second only at q, has the following exceptions:

1°. If $op < oq$ and $oq = pq = os$, the first minimum occurs only at p, the second at q and s.

2°. If $op = oq$ and $oq < pq = os$, the minima are equal, and p and q may be interchanged.

3°. If $op = oq = pq = os$, one of p, q, s may be chosen as first point, and then either of the remaining two as second.

To prove this it is enough, since opposite points are always equally distant from o, to show that q is nearer to o first than all other points on the line sqr, except s, whose distance from o equals $os = pq \geq oq$, and secondly than all points on the following parallel lines.

Since $pq \geq op$, $pq \geq oq$, and the angle poq is not obtuse, the triangle opq, and hence also the congruent triangle oqs, has no obtuse angle. Thus the perpendicular from o to qs falls between s and q (inclusive), proving the first point. Write

$$\cos poq = \frac{m}{\sqrt{ln}},$$

with $m \geq 0$. Then

$$pq^2 = l - 2m + n \geq oq^2 = n,$$

and hence

$$2m \leq l, \qquad 2m \leq n, \qquad 4m^2 \leq ln.$$

If the square of the height of the parallelogram (with $op = \sqrt{l}$ as base) is k, then the square Δ of its area is

$$\Delta = lk = ln - m^2 \geq \frac{3}{4} ln,$$

so that

$$\sqrt{k} \geq \frac{1}{2}\sqrt{3n}.$$

Thus already the second parallel line is at least $\sqrt{3n} = oq\sqrt{3}$ away, proving the second point.

III. Since the successive minima $\sqrt{l}$, $\sqrt{n}$ are determined by the system itself and independently of any particular arrangement, while by what has just been shown they agree in magnitude with the sides of a reduced parallelogram, it follows that if the system admits several such arrangements, the sides of the reduced parallelograms always keep the values $\sqrt{l}$ and $\sqrt{n}$.

All possible fundamental parallelograms are therefore obtained by drawing from o lines to all nearest points (again excluding opposite points), and then taking, in the corresponding nearest parallel line, the nearest point or the two nearest points. By II, these nearest points are nearer to o than all points of the following parallel lines; hence one may omit the separate requirement that the second points be in the first parallel line. Thus all possible arrangements of the system arise by connecting o successively with all pairs of points for which the successive minima occur. Taking account of II, there is in general, and in the second singular case mentioned there, only one arrangement; in the first and third exceptional cases there are respectively two and three.

In the present notation those singular cases correspond to

$$2m = l < n, \qquad 2m < l = n, \qquad 2m = l = n.$$

§. 4.

We have so far considered only properties of the geometrical configurations which may be regarded as constructive representations of known results from the theory of forms. We now solve a different problem: given a system of second order and a fixed point o of it, determine the part of the plane in which every point is nearer to o than to any other point of the system. Since the condition that a point is not farther from o than from another point v is that it lie on the same side as o of the perpendicular erected at the midpoint of ov, we must combine o with all other points and construct the convex polygon formed by the corresponding perpendiculars. Of these infinitely many perpendiculars only finitely many matter; the others do not meet the polygon determined by these. We keep the previous assumptions, so that in the reduced parallelogram $(poqr)$ one has $op \leq oq$, the angle poq is not obtuse, and the angles opq, oqp are acute.

Under these assumptions, one easily proves that only the six vertices p, q, s, p', q', s' of the four parallelograms meeting at o need be considered, and that even the perpendiculars corresponding to s and s' merely touch the desired figure in the special case where poq is a right angle; in that case the same is true of those corresponding to r and r'.

Draw the lines $pq, os, p'q', os'$. One obtains the congruent triangles

$$poq, \quad qos, \quad sop', \quad p'oq, \quad q'os, \quad s'op.$$

If only p, q, s, p', q', s' are considered, one erects perpendiculars at the midpoints of the lines from o to those points, that is, one performs the same construction as if one wanted the circumcentres of the triangles just listed. Since no angle in these triangles is obtuse, two consecutive perpendiculars do not meet outside the corresponding triangle. One obtains the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, with centre o and equal opposite angles and sides, as the region in which every point is nearer to o than to any one of the points p, q, s, p', q', s'. One readily verifies that, except for r and r', the perpendiculars corresponding to the remaining points do not meet this hexagon. By symmetry it suffices to prove this for points in and above the line pop'. For points on the line this is clear; for the points above it, it follows because their distance from o is greater than the diameter of the circle circumscribed about the hexagon. If ϱ denotes the square of its radius, then

$$4\varrho\Delta = ln(l - 2m + n),$$

and from $2m \leq l$, $2m \leq n$, $\Delta \geq \frac{3}{4}ln$ follows

$$4\varrho \leq \frac{4}{3}(l - 2m + n) \leq \frac{8}{3}n.$$

Since the square of the distance from o of points of the second and following parallel lines is at least $3n$, it remains only to consider the points in $tsqr$ other than s, q, r. None of these is nearer to o than t, for which the square of the distance is $4l - 4m + n$. That this is greater than 4ϱ is seen at once after multiplication by Δ and use of $2m \leq l \leq n$[5]. For the point r one likewise sees that the square of its distance from o is $l + 2m + n > 4\varrho$, except in the single case $m = 0$, where the corresponding perpendicular only touches.

It is thus proved that a point lies inside the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, and only then, precisely when it is nearer to o than to any other point of the system. On each side, the distance from o becomes equal to the distance from a second point–for example, for $\alpha\beta$ this point is q–and each vertex of the figure is equally distant from o and from two further points of the system. This last statement changes only in the special case where poq is a right angle; then β and γ, and likewise β' and γ', coincide, and the hexagon becomes a rectangle whose vertices are equidistant from o and from three further points of the system.

It is clear that the same hexagon is obtained whichever reduced parallelogram is used in the singular cases where more than one exists. Likewise, the hexagons or quadrilaterals corresponding to all points of the system are congruent and cover the whole plane.

We also note that the expression

$$\varrho = \frac{ln(l - 2m + n)}{4(ln - m^2)}$$

decreases when, with l and n held constant, m increases from zero to its limiting value $\frac{1}{2}l$. Hence

$$\varrho \leq \frac{1}{4}(l + n) \leq \frac{1}{2}n. \tag{5}$$

[5] The difference $4l - 4m + n - 4\varrho$, multiplied by Δ, can be written as $4m^3 + n(l^2 - m^2) + ln(l - 2m) + l(ln - 4m^2)$, in which the first two terms are positive and the last two non-negative. K.

There is also the inequality

$$2\Delta(n-\varrho) \geq ln^2, \tag{6}$$

whose truth follows immediately after multiplying by 2, bringing all terms to one side, and substituting $\Delta = ln - m^2$, $4\Delta\varrho = ln(l-2m+n)$, whereby it becomes

$$ln(l-2m) + 2mn(n-l) \geq 0.$$

§. 5.

We now come to the proper subject. We must show that every system of third order can be arranged according to a parallelepiped whose faces are reduced parallelograms, and whose edges, four by four equal, do not exceed its diagonals.

Having fixed an arbitrary point (0) of the system, choose in the pair of opposite points for which the distance from (0) is a minimum, or if this minimum occurs for several pairs, arbitrarily in one of these pairs, a point (1).

Among all points outside the line (01) choose again one of the two nearest, call it (2), with arbitrary choice if the same shortest distance occurs for several pairs. Since in the whole system, apart from points on (01), no point is nearer to (0) than (2), the same holds in the plane (102); for the system contained in that plane, (102) is a reduced parallelogram (§. 3, III.). Now take in one of the two nearest parallel planes the point nearest to (0), or, if the minimum occurs more than once, one of the nearest, and join (0) to the chosen point (3). The parallelepiped with edges (01), (02), (03) will then satisfy the requirement. First, by construction,

$$(01) \leq (02) \leq (03).$$

For the base faces of the parallelepiped–that is, the opposite faces containing the two edges that do not exceed the third in size; the other four faces will be called side faces–we already know that they are reduced. By the double inequality just noted we have only to show that the four diagonals of the side faces and the four body diagonals are not smaller than (03). These eight diagonals agree in length with the eight lines from (0) to the eight points lying around (3) in the plane of the upper base face, the eight vertices of the four parallelograms meeting at (3). None of these connecting lines is smaller than (03), by the condition under which (3) was chosen.

Having seen that a system of third order can always be divided according to a reduced parallelepiped, we now establish the relation between such a parallelepiped and the system, and especially compare the distances of the points of the system from (0). Put

$$(01) = \sqrt{a}, \qquad (02) = \sqrt{b}, \qquad (03) = \sqrt{c},$$

and keep throughout the assumption $a \leq b \leq c$.

1°. In the plane of the base face, the relations discussed above (§. 3, II.) hold; hence the successive minima of distance have magnitudes $\sqrt{a}$, $\sqrt{b}$, with the possible freedom in the singular cases described there.

2°. Consider now the points outside the plane of the lower base face, first those in the plane of the upper base face. Since, by the assumption that the parallelepiped is reduced, the line (03) is not greater than any line drawn from (0) to the eight points lying around (3), the foot of the perpendicular from (0) to the plane of the upper base face is not farther from (3) than from any of those eight points. Thus this foot does not fall outside the hexagon or quadrilateral belonging to (3) that was constructed in the preceding paragraph. Among the eight points, exceptionally, if the foot falls on a side, one point, or if the foot coincides with a vertex, two points (three when the polygon becomes a rectangle), may be as near to the foot as (3) is; all other points of the plane are farther away from it. It follows that the shortest distance from (0) to a point of the upper base face, equal to $\sqrt{c}$, occurs generally only for (3), but exceptionally may occur for one, two, or even three further points.

3°. For the following parallel planes we need a bound for the square h of the perpendicular just mentioned. Since its foot does not fall outside the hexagon belonging to (3), if ϱ denotes the square of the radius of the circumscribed circle, then

$$h \geq c - \varrho.$$

But by §. 4 also $\varrho \leq \frac{1}{2}b \leq \frac{1}{2}c$, hence $h \geq \frac{1}{2}c$. Thus already the second parallel plane is at least $\sqrt{2c}$ distant, so above the upper base face there are only points whose distance from (0) is greater than $\sqrt{c}$.

Summing up, the minimum distance in the whole system has value $\sqrt{a}$; after a point at this distance has been chosen, the minimum in the remaining directions is $\sqrt{b}$; and after the second point has also been fixed, the least distance from (0) to points outside the plane determined by (0) and the two first points is $\sqrt{c}$. The successive minima $\sqrt{a}$, $\sqrt{b}$, $\sqrt{c}$ are therefore completely determined in magnitude, though not always in position. The possible exceptions are exactly those arising from the singular cases in the base plane, together with the exceptional positions in the upper base plane described above.

Consequently every reduced arrangement of the system arises by choosing, in succession, points at which these successive minima are attained. In the ordinary case the arrangement is unique. If equalities among the minima or among the boundary conditions occur, several reduced parallelepipeds may appear; but they are precisely those obtained by the stated alternative choices of the points realizing the successive minima.

It follows in particular that if a reduced parallelepiped is such that every segment which is required not to exceed certain others is in fact strictly smaller than them, then the corresponding system has no second reduced arrangement. Conversely, a second reduced arrangement can occur only when one of the limiting equalities holds.

Thus, when a system admits more than one reduced parallelepiped, knowledge of one arrangement suffices to find all the others. The first, non-singular case occurs always and only when the reduced parallelepiped given by this arrangement is of such a nature that all the lines which must not be exceeded by others are in fact strictly smaller: all face diagonals are greater than the corresponding sides, and all body diagonals of the parallelepiped are greater than its edges.

§. 6.

We now translate the results of the preceding paragraph to ternary forms. For uniformity, and to avoid useless distinctions, suppose that by interchanging or changing the signs of the undetermined elements–which does not remove the form from its class–each ternary form

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy \tag{7}$$

has first been put into a shape such that $a \leq b \leq c$; secondly, among the coefficients a', b', c', unless all three are non-zero and negative, none has the negative sign; and thirdly, if $b = c$, then c', disregarding its sign, is not greater than b', if $a = b$, then b' is not greater than a', and if $a = b = c$, then neither c' is greater than b' nor b' greater than a'. It is easy to see that these requirements can be satisfied in only one way. Their introduction has the advantage that, just as before every ternary form corresponded to a completely determined parallelepiped, now to every parallelepiped there belongs an analytic expression whose coefficients are completely determined also in their order and in their signs.

Under this convention we call the form (1), for which $a \le b \le c$, *reduced* when it corresponds to a reduced parallelepiped. Since the diagonals of the faces must not be smaller than the sides of those faces, one has

$$a \pm 2c' + b \ge b, \qquad a \pm 2b' + c \ge c, \qquad b \pm 2a' + c \ge c.$$

Put $\sigma = -1$ when a', b', c' are all three negative, and otherwise $\sigma = 1$. These conditions are equivalent to

$$a \ge 2c'\sigma, \qquad a \ge 2b'\sigma, \qquad b \ge 2a'\sigma, \tag{8}$$

and only when equality occurs will one of the diagonals in the corresponding parallelogram be equal to a side. The conditions concerning the body diagonals give

$$a + b + c + 2a'\delta + 2b'\varepsilon + 2c'\delta\varepsilon \ge c \qquad (\delta = \pm 1,\ \varepsilon = \pm 1),$$

where the signs in $\delta = \pm 1$, $\varepsilon = \pm 1$ are arbitrary.

First take the case in which none of a', b', c' is negative. Considering the four sign combinations and the condition that, if a and b are equal, then $b' \le a'$, one sees at once that the preceding inequality is then automatically satisfied by virtue of the conditions already obtained. The limiting case of equality, in which a body diagonal becomes equal to the edge $\sqrt{c}$, can occur only once, and only when one of b', c' is zero and when, at the same time, among the conditions (2), both the one involving the other of the two quantities b', c' and the one involving a' are limiting equalities. If a', b', c' are negative, the inequality is always satisfied, and without a limiting equality except when $\delta = \varepsilon = 1$. Thus one must add the new condition

$$a + b + 2a' + 2b' + 2c' \ge 0, \tag{9}$$

where again the lower sign refers to a diagonal becoming equal to the edge $\sqrt{c}$.

If the conditions (2), and also (3) when a', b', c' are negative, are satisfied without equality in any of the inequalities, then in the class to which the form belongs there is no second distinct reduced form, whether with or without equality in the defining conditions. This follows from the end of the preceding paragraph: the corresponding point system can then be divided according to only one reduced parallelepiped. The situation is different when not all conditions hold with strict inequality. Then several reduced forms may occur in the same class, and they can be derived from a given one. It suffices to show this in a principal case; we choose the case $b < c$.

Assume first $a > 2c'\sigma$. Then only the direction of the edge $\sqrt{c}$ can be changed, namely if in the plane of the upper base face there are one or more further points whose distance from the vertex is $\sqrt{c}$. If $\xi, \eta, 1$ are the values of the elements corresponding to such a point, then, when the third edge is directed toward it, all coefficients remain unchanged except a', b', which become respectively

$$a' + c'\xi + b\eta, \qquad b' + a\xi + c'\eta,$$

as is seen easily and almost without calculation.

By the preceding enumeration, the values of ξ, η satisfying the condition are, when $a = 2b'\sigma$,

$$\xi = -\sigma, \qquad \eta = 0;$$

when $b = 2a'\sigma$,

$$\xi = 0, \qquad \eta = -\sigma;$$

when simultaneously $a = 2b'$, $b = 2a'$, $c' = 0$,

$$\xi = -1, \qquad \eta = -1;$$

and when a', b', c' are negative and satisfy

$$a + b + 2a' + 2b' + 2c' = 0,$$

then

$$\xi = 1, \qquad \eta = 1.$$

Corresponding to these four hypotheses, a', b' are transformed into

$$a' - c'\sigma,\ b' - a\sigma (= -b'); \qquad -a',\ b' - c'\sigma; \qquad -a',\ -b'; \qquad a' + b + c',\ a + b' + c'.$$

The third case, and in general the assumption $c' = 0$, may be disregarded, since it gives a new form which, after the prescribed changes of sign at the beginning of this paragraph, is plainly identical with the original one. In each of the three remaining cases one obtains, after making the necessary sign changes, a new reduced form in the same class, unless it coincides with the original. If two of the hypotheses hold simultaneously, two such forms are obtained. This completes the enumeration, since all three hypotheses clearly cannot hold simultaneously. If, still under the assumption $b < c$, one had $a = 2c'\sigma$, then, if $a < b$, the second edge in the base face would have had to be turned; if $a = b$, the first edge could also have been brought into the original position of the second; and this new position, or these two new positions, of the first two edges would have had to be combined with all directions of the third, including the original one.

The inconvenience that in singular cases several reduced forms may occur in the same class can easily be removed by adding to the general definition certain secondary conditions for such cases. They arise, for example, if one requires that the last coefficient c', when it is not completely determined, have the smallest numerical value of which it is capable among the reduced forms of the class, and then proceeds similarly with b'. To illustrate this, consider among the singular cases just treated the case $b < c$, where none of the three conditions (2) holds with the lower sign, but the three negative

values a', b', c' satisfy

$$a + b + 2a' + 2b' + 2c' = 0.$$

As noted above, c' is then determined, and there are only two reduced forms. If a' and b' are the values of the fourth and fifth coefficients in one of them, then in the other they are $a' + b + c'$, $a + b' + c'$, or, since these latter values are evidently positive while c' is negative and therefore z must be replaced by $-z$ to satisfy the sign convention, rather

$$-(a' + b + c'), \qquad -(a + b' + c').$$

As is natural, substituting these values for a', b' again gives the equation

$$a + b + 2a' + 2b' + 2c' = 0,$$

and the original values a', b' arise from them in the same way as they arose from a', b'. Thus the fifth coefficient can have only the two negative values b' and $-(a + b' + c')$, whose sum is $-a - c'$. Hence, if one adds to the defining conditions the further condition

$$-b' \leq \frac{1}{2}(a + c'),$$

the class will contain only one reduced form.

To conclude the memoir we derive from our principles a beautiful theorem found by Seeber by induction and proved by Gauss in the often mentioned review. According to this theorem, in a reduced form the product of the first three coefficients is not greater than twice the absolute value of the determinant.

Since the absolute value of the determinant is equal to the square of the volume of the parallelepiped corresponding to the form, the inequality to be proved, in the notation used in §. 5, 3^{o}, is

$$abc \leq 2\Delta h,$$

where Δ denotes the square of the base area. Put

$$c = b + t,$$

where $t \geq 0$. From the inequality obtained in §. 5, 3^{o},

$$h \geq c - \varrho = b - \varrho + t,$$

after multiplying it by 2Δ, subtract the equation

$$abc = ab^2 + abt.$$

This gives

$$2\Delta h - abc \geq 2\Delta(b - \varrho) - ab^2 + (2\Delta - ab)t.$$

But by the inequality proved at the end of §. 4, $2\Delta(b-\varrho)-ab^2$ is non-negative, and $2\Delta-ab \geq \frac{1}{2}ab$ is positive. The theorem follows.

VI.

On the Determination of Mean Values in Number Theory

G. Lejeune Dirichlet

Proceedings of the Royal Prussian Academy, 1849, pp. 69–83.

On the determination of mean values in number theory.

[Read in the Academy of Sciences on 9 August 1849[1].]

Although the functions considered in number theory are almost never representable by analytic expressions and appear to proceed quite irregularly, a greater regularity nevertheless emerges in their mean values the further one follows their sequence. That is, there are certain simple expressions which represent the progress of the mean values with continuously increasing accuracy, just as one curve approaches the course of another curve of which it is the asymptote. Toward the end of the fifth section of the *Disquisitiones arithmeticae* one finds several highly remarkable expressions of this kind, relating to the theory of quadratic forms. Since these interesting results, which are only communicated there incidentally and without proof, have not yet been proved, and since in general no methods have been known for treating similar questions, I began several years ago to look for suitable means. From that earlier work, however, I gave nothing to the public except a few new results[2], because there appeared to me the prospect of essentially simplifying the treatment of such problems by continued efforts, and in particular of making it independent of the integral calculus. Other investigations then drew me away from this subject for a longer time. After later taking it up again, I became convinced that in many cases one arrives at the asymptotic expression for the mean value by quite elementary considerations founded on a very simple transformation of series. For the moment I restrict myself to a series of problems for which the indicated means alone is sufficient. In a subsequent paper I shall concern myself with more difficult problems whose solution requires combining the transformation just mentioned with other auxiliary means.

1.

To put the transformation on which the solution of the problems treated in this paper chiefly rests in its proper light, it seems appropriate to begin at once with one of the simplest questions, to carry its solution as far as can be done without that transformation, and then to complete it with its help.

Let $f(n)$ denote the number of divisors of the integer n, and consider the problem of determining the so-called summatory term of this function, that is, the sum

$$f(1) + f(2) + \cdots + f(n) = F(n).$$

If s is an integer $\leq n$, then in as many terms of our sum there will be a unit corresponding to the divisor s as there are multiples of s among the numbers $1, 2, \ldots, n$. But the number of

[1] The corresponding notice in the Academy report of 1849, p. 218, reads: "Mr. Dirichlet read on the determination of mean values in number theory." K.

[2] Cf. pp. 351 and 372 of volume I of this edition of G. Lejeune Dirichlet's works. K.

these multiples is $[n/s]$, if, as everywhere in what follows, we use square brackets to denote the greatest integer contained in the bracketed value. It follows that

$$F(n) = \sum_1^n \left[\frac{n}{s}\right],$$

where the summation sign refers to s. This equation immediately gives an approximate determination, that is, an asymptotic expression for $F(n)$. Since n/s exceeds the integer $[n/s]$ by less than one unit, one has, up to an error that cannot exceed n,

$$F(n) = n\sum_1^n \frac{1}{s}.$$

But

$$\sum_1^n \frac{1}{s} = \log n + C + \frac{1}{2n} + \cdots,$$

where $C = 0.5772156\ldots$. Since, however, the last expression for $F(n)$ is already affected by an error of order n, only the first term of the infinite series is to be retained; and one sees that the equation

$$F(n) = n\log n$$

is exact only up to an error of first order, for which, incidentally, one can easily give a bound that it cannot exceed. Whether the function, which is of order $n\log n$, contains in its asymptotic expression a term of order n with constant coefficient, or, in other words, whether

$$\frac{1}{n}F(n) - \log n$$

approaches a fixed limit as n increases, cannot be decided in this way.

2.

The transformation already mentioned, with which we now wish to occupy ourselves, rests on the simple observation that, while the terms of the sequence

$$\left[\frac{n}{1}\right], \quad \left[\frac{n}{2}\right], \quad \ldots, \quad \left[\frac{n}{s}\right], \quad \ldots,$$

which stops with the nth term, at first decrease very rapidly, from a certain point on each term is either equal to the following one or exceeds it by one unit. This will plainly be the case as soon as the difference

$$\frac{n}{s} - \frac{n}{s+1} = \frac{n}{s(s+1)}$$

has become equal to one or to a proper fraction. Thus if one determines the least integer μ satisfying

$$\mu(\mu+1) \geq n,$$

the mentioned property will occur at the latest from the μth term inclusive. Since for our purpose it is quite immaterial whether one more or one fewer term is drawn into the intended transformation, we shall, for greater simplicity, choose for μ the integer that is either equal to $\sqrt{n}$ or, when this is irrational, lies immediately above it. Thus

$$\mu^2 \geq n,$$

and consequently $\mu(\mu+1) > n$. Put

$$\left[\frac{n}{\mu}\right] = \nu.$$

Then, as is easy to see, ν can differ from μ by at most two units. Let t be any of the numbers $\nu, \nu-1, \dots, 2, 1$. It is then easy to determine the index s of the term furthest from the beginning for which $[n/s]$ has the value t, and after which a term with value $t-1$ follows. The desired index is given by the double condition

$$\frac{n}{s} \geq t, \qquad \frac{n}{s+1} < t,$$

and hence

$$s \leq \frac{n}{t}, \qquad s+1 > \frac{n}{t}, \qquad \text{that is,} \quad s = \left[\frac{n}{t}\right].$$

Now consider the sum

$$\left[\frac{n}{1}\right]\varphi(1) + \left[\frac{n}{2}\right]\varphi(2) + \cdots + \left[\frac{n}{s}\right]\varphi(s) + \cdots + \left[\frac{n}{p}\right]\varphi(p),$$

where p is, of course, assumed to be greater than μ and not greater than n, or rather consider only that part of the sum which extends from the μth term onward:

$$\left[\frac{n}{\mu}\right]\varphi(\mu) + \cdots + \left[\frac{n}{s}\right]\varphi(s) + \cdots + \left[\frac{n}{p}\right]\varphi(p).$$

We may transform this by combining into partial sums those terms for which $[n/s]$ has the same value, and then adding all the partial sums.

For greater uniformity, omit from the first partial sum the term corresponding to $s = \mu$, join it to the already separated first $\mu - 1$ terms, and put

$$\left[\frac{n}{p}\right] = q.$$

Then the partial sums extend respectively from

$s = \mu$	excluded to $s = [n/\nu]$	included, with corresponding value $[n/s] = \nu$,	
$s = [n/\nu]$	to $s = [n/(\nu-1)]$		$\nu - 1$,
$s = [n/(\nu-1)]$	to $s = [n/(\nu-2)]$		$\nu - 2$,
$\cdots$			$\cdots$,
$s = [n/(q+1)]$	to $s = p$		q.

Let $\psi(s)$ denote the summatory term of the function $\varphi(s)$, so that

$$\psi(s) = \sum_{h=1}^{h=s} \varphi(h).$$

The values of the partial sums are then

$$\nu\left(\psi\left[\frac{n}{\nu}\right] - \psi(\mu)\right), \quad (\nu-1)\left(\psi\left[\frac{n}{\nu-1}\right] - \psi\left[\frac{n}{\nu}\right]\right),$$

$$(\nu-2)\left(\psi\left[\frac{n}{\nu-2}\right] - \psi\left[\frac{n}{\nu-1}\right]\right), \quad \dots, \quad q\left(\psi(p) - \psi\left[\frac{n}{q+1}\right]\right),$$

and their combination gives the expression

$$-\nu\psi(\mu) + \psi\left[\frac{n}{\nu}\right] + \psi\left[\frac{n}{\nu-1}\right] + \cdots + \psi\left[\frac{n}{q+1}\right] + q\psi(p).$$

Thus we obtain the general transformation formula

$$\sum_1^p \left[\frac{n}{s}\right]\varphi(s) = q\psi(p) - \nu\psi(\mu) + \sum_1^\mu \left[\frac{n}{s}\right]\varphi(s) + \sum_{q+1}^\nu \psi\left[\frac{n}{s}\right]. \tag{a}$$

The case that occurs most frequently in the application of this equation is that in which $p = n$ and consequently $q = 1$. Under this assumption one obtains

$$\sum_1^n \left[\frac{n}{s}\right]\varphi(s) = -\nu\psi(\mu) + \sum_1^\mu \left[\frac{n}{s}\right]\varphi(s) + \sum_2^\nu \psi\left[\frac{n}{s}\right]. \tag{b}$$

As one sees, the advantage of this transformation is that by it a series whose number of terms is n, and which contains in each term an expression of the form $[n/s]$, is reduced to two others in which the number of terms still containing $[n/s]$ is lowered to order $\sqrt{n}$. A similar transformation can still be carried out if, in the expression $[n/s]$, the denominator s is replaced by a function of s increasing with s. Since such generality is superfluous for our present purpose, we restrict ourselves to the special form of series just considered.

3.

Returning now to the problem treated in Article 1, we obtain, when in equation (b) $\varphi(s) = 1$ and hence $\psi(s) = s$, the formula

$$F(n) = \sum_1^n \left[\frac{n}{s}\right] = -\mu\nu + \sum_1^\mu \left[\frac{n}{s}\right] + \sum_1^\nu \left[\frac{n}{s}\right].$$

If in the two sums one puts n/s in place of $[n/s]$, the error is only of order $\sqrt{n}$; and since likewise

$$n\sum_1^\mu \frac{1}{s} = n\sum_1^\nu \frac{1}{s} = \frac{1}{2}n\log n + Cn, \qquad \mu\nu = n,$$

it follows, exactly up to an error of order $\sqrt{n}$, that

$$F(n) = n\log n + (2C-1)n.$$

From the asymptotic expression just found for $F(n)$ one can now easily derive an expression for the mean value of the number of divisors. Write the equation, for greater clarity, in the form

$$F(n) = n\log n + (2C-1)n + \xi\sqrt{n},$$

where ξ is unknown but, for however large n may be, remains numerically below a fixed bound that could easily be given. Replace n by $n+s$, with ξ becoming ξ', and divide the difference of the two equations by s. One then obtains for the arithmetic mean of the numbers of factors corresponding to the s numbers $n+1, n+2, \ldots, n+s$:

$$\log(n+s) + 2C - 1 + \frac{n}{s}\log\left(1+\frac{s}{n}\right) + \frac{\xi'\sqrt{n+s} - \xi\sqrt{n}}{s}.$$

If one now lets $s/\sqrt{n}$ increase beyond every bound, the last term tends to zero; that is, the difference between the mentioned arithmetic mean and the expression formed from the remaining terms becomes smaller than any assignable magnitude. Combining the assumption already made with the further one that n/s also exceeds every bound, one obtains, for the mean corresponding to this double condition, the very simple asymptotic value

$$\log n + 2C.$$

This also remains valid when n, instead of denoting the number immediately preceding the sequence of s numbers over which the mean is taken, coincides with one of these numbers. For if m is one of them, then $n+t=m$ with $t \leq s$, and under the above assumption the difference $\log m - \log n$ tends to zero.

4.

The equation obtained above,

$$\sum_1^n \left[\frac{n}{s}\right] = n \log n + (2C-1)n,$$

in which the error is of order $\sqrt{n}$, gives occasion for an observation at which we shall pause for a moment. Since, on the other hand,

$$\sum_1^n \frac{n}{s} = n \log n + Cn$$

where the error for every n does not exceed a fixed bound, one has, with an error of order $\sqrt{n}$,

$$\sum_1^n \left(\frac{n}{s} - \left[\frac{n}{s}\right]\right) = (1-C)n.$$

Thus the arithmetic mean of the values of the difference

$$\frac{n}{s} - \left[\frac{n}{s}\right],$$

corresponding to $s = 1, 2, \ldots, n$, is equal to $1-C$ and hence is less than $\frac{1}{2}$. It is therefore natural to conjecture that, when n is divided successively by the numbers just named, the case in which the remainder lies below half the divisor will occur more often than the opposite case, in which it is equal to or exceeds half the divisor. We shall test the correctness of this conjecture and try to determine the ratio in which the numbers $1, 2, \ldots, n$ divide into two groups in this respect.

The number s gives the first or second case according as

$$\frac{n}{s} - \left[\frac{n}{s}\right] < \frac{1}{2} \quad \text{or} \quad \geq \frac{1}{2}.$$

Accordingly one has, respectively,

$$\left[\frac{2n}{s}\right] - 2\left[\frac{n}{s}\right] = 0 \quad \text{or} \quad \left[\frac{2n}{s}\right] - 2\left[\frac{n}{s}\right] = 1.$$

Therefore the number of integers contained in the second group is given by

$$\sum_1^n \left[\frac{2n}{s}\right] - 2\sum_1^n \left[\frac{n}{s}\right],$$

whose second term has already been determined. For the determination of the first one uses equation (a), in which n is replaced by $2n$ and $p=n$, $q=2$, $\varphi(s)=1$, $\psi(s)=s$ are put. Thus one obtains

$$\sum_1^n \left[\frac{2n}{s}\right] = 2n - \mu\nu + \sum_1^\mu \left[\frac{2n}{s}\right] + \sum_3^\nu \left[\frac{2n}{s}\right].$$

Since μ is the integer immediately above $\sqrt{2n}$ and $\nu = [2n/\mu]$, $\mu\nu$ differs from $2n$ only by a quantity of order $\sqrt{n}$, and the first two terms cancel. The value of the first sum is, still neglecting terms of order $\sqrt{n}$, given by

$$\sum_1^\mu \left[\frac{2n}{s}\right] = 2n\sum_1^\mu \frac{1}{s} = 2n(\log\mu + C) = n\log 2n + 2Cn.$$

The same value would also hold for the second sum if the two terms corresponding to $s = 1$ and $s = 2$ were not missing. Therefore, to obtain $\sum_1^n [2n/s]$, one must double the expression just found and subtract

$$\left[\frac{2n}{1}\right] + \left[\frac{2n}{2}\right] = 3n.$$

Thus the number of integers in the second group is found to be

$$(\log 4 - 1)n,$$

and for the first group

$$(2 - \log 4)n.$$

For large n, therefore, the number of divisors having the first property is to the number having the second as $2 - \log 4$ is to $\log 4 - 1$.

5.

Now consider the function $f(n)$ which expresses the sum of the divisors of n, or rather, first again as in No. 1, the sum formed from it,

$$F(n) = f(1) + f(2) + \cdots + f(n).$$

By arguments quite similar to those used there, one obtains

$$F(n) = \sum_1^n s\left[\frac{n}{s}\right].$$

Application of equation (b) gives

$$F(n) = -\frac{1}{2}(\mu^2 + \mu)\nu + \sum_1^\mu s\left[\frac{n}{s}\right] + \frac{1}{2}\sum_1^\nu \left(\left[\frac{n}{s}\right]^2 + \left[\frac{n}{s}\right]\right).$$

To see the order of magnitude of the quantities that must be neglected as not completely determinable, first consider the first part of the last sum. Put

$$\left[\frac{n}{s}\right] = \frac{n}{s} - \varepsilon,$$

where ε is a positive proper fraction depending on n and s. Then

$$\sum_1^\nu \left[\frac{n}{s}\right]^2 = \sum_1^\nu \frac{n^2}{s^2} - 2\sum_1^\nu \frac{n}{s}\varepsilon + \sum_1^\nu \varepsilon^2.$$

The second and third terms, whose orders cannot exceed $n\log n$ and $\sqrt{n}$, respectively, must be omitted because of the analytically inexpressible fraction ε occurring in them. The second part of the sum, namely $\sum_1^\nu [n/s]$, which we have already determined and which is of order $n\log n$, is likewise not to be taken into account, although this part can be specified exactly down to and including the first order. Further, neglecting the first order,

$$(\mu^2+\mu)\nu = n^{3/2}, \qquad \sum_1^{\mu} s\left[\frac{n}{s}\right] = n^{3/2};$$

hence, exactly up to an error of order $n\log n$,

$$F(n) = \frac{1}{2}n^{3/2} + \frac{1}{2}n^2\sum_1^{\nu}\frac{1}{s^2}.$$

But by a known formula,

$$\sum_1^{\nu}\frac{1}{s^2} = \frac{\pi^2}{6} - \frac{1}{\nu} + \frac{1}{2\nu^2} - \cdots,$$

and therefore, excluding the first order,

$$\frac{1}{2}n^2\sum_1^{\nu}\frac{1}{s^2} = \frac{\pi^2}{12}n^2 - \frac{1}{2}n^{3/2}.$$

Consequently one finally obtains

$$F(n) = \frac{\pi^2}{12}n^2,$$

where the error does not exceed order $n\log n$.

Using the expression just found to determine the mean value of $f(n)$, one finds for this mean value, namely for

$$\frac{1}{s}\left(f(n+1)+\cdots+f(n+s)\right),$$

the asymptotic expression

$$\frac{\pi^2}{6}n,$$

provided that the simultaneous growth of s and n is understood in such a way that

$$\frac{n}{s} \quad \text{and} \quad \frac{s}{\log n}$$

exceed every finite bound. This result, however, has an essentially different meaning from the one found at the end of No. 3. There the difference between the true mean value and its asymptotic expression became smaller than any given magnitude; here this holds only for the ratio of this difference to the true or approximate mean value.

6.

In the problems treated so far, to which it seems superfluous to add others solved in quite the same way, the function to be approximately determined had directly the form of the series (a). In other cases this function is given by an equation containing such a series, in whose general term the function to be determined occurs; thus one knows only a recurrence relation between successive values of the function. The simplest example of this kind is given by the function $\varphi(n)$, which expresses the number of integers in the sequence $1, 2, \ldots, n$ relatively prime to n and which plays so large a role in number theory. It is well known that this function is expressed by

$$\varphi(n) = n\left(1-\frac{1}{a}\right)\left(1-\frac{1}{b}\right)\left(1-\frac{1}{c}\right)\cdots,$$

where $a, b, c, \ldots$ denote the distinct prime numbers dividing n. This formula, however, is not suited to our investigation, and we must choose a different starting point. We begin with the familiar equation

$$\sum \varphi(d) = n,$$

where the summation sign extends over all divisors d of n. Adding this equation and the similar ones for $n-1, n-2, \ldots, 1$, the term $\varphi(s)$ on the left, where $s \leq n$,

occurs as many times as there are multiples of s in the sequence $1, 2, \ldots, n$, that is, $[n/s]$ times. Thus

$$\sum_1^n \left[\frac{n}{s}\right] \varphi(s) = \frac{1}{2}n^2 + \frac{1}{2}n.$$

Before going further, let us return for a moment to formula (b) and observe that, for some problems, including the one treated in No. 5, it gives a shortcut when the transformation embraces the whole series, although this sacrifices the advantage, indispensable in other cases, of lowering the number of terms to a smaller order. To obtain the modified formula, observe that, if t denotes quite generally one of the numbers $1, 2, 3, \ldots, n$, this value is not always found among the terms

$$\left[\frac{n}{1}\right], \quad \left[\frac{n}{2}\right], \quad \ldots, \quad \left[\frac{n}{s}\right], \quad \ldots, \quad \left[\frac{n}{n}\right],$$

since these terms decrease very rapidly at the beginning of the sequence; nevertheless there is always one and only one term that is not greater than t and is followed by another whose value is smaller than t. As before, the index s of this term must satisfy

$$\frac{n}{s} \geq t, \qquad \frac{n}{s+1} < t,$$

whence

$$s = \left[\frac{n}{t}\right].$$

Thus in general $[n/s] = t$ from $s = [n/(t+1)]$ excluded up to $s = [n/t]$ included; and the sum of all terms in which $[n/s]$ has the value t is

$$\left(\psi\left[\frac{n}{t}\right] - \psi\left[\frac{n}{t+1}\right]\right) t.$$

This expression vanishes and remains correct when no such terms exist, that is, when

$$\left[\frac{n}{t}\right] = \left[\frac{n}{t+1}\right].$$

Combining the values of the expression for $t = 1, 2, \ldots, n$, and observing that for $t = n$ the term $\psi\left[n/(t+1)\right]$ is plainly to be reduced to zero, one obtains

$$\sum_1^n \left[\frac{n}{s}\right] \varphi(s) = \sum_1^n \psi\left[\frac{n}{s}\right]. \tag{c}$$

This transformation is the one we shall now use in our problem. With its help our preceding equation becomes

$$\sum_1^n \psi\left[\frac{n}{s}\right] = \frac{1}{2}n^2 + \frac{1}{2}n.$$

It is soon clear that the asymptotic expression for $\psi(n)$ will have the form αn^2, where α is a constant. One could at first leave this constant undetermined in the following proof, in which case the value $\alpha = 3/\pi^2$ would emerge in the course of the development. Since this value is also easy to foresee, however, we shall for brevity at once consider the expression $\frac{3}{\pi^2}n^2$. Whatever the still unknown function $\psi(n)$ may be, we may set in general

$$\psi(n) = \frac{3}{\pi^2}n^2 + \xi\chi(n),$$

where ξ also depends on n and the function $\chi(n)$ is arbitrary except for the restriction that it always remain positive, increase with its argument, and not vanish at $n = 1$. Suppose $\chi(n)$ has been chosen in the indicated way, and then substitute all values from $n = 1$ to $n = N$ into our equation. The corresponding values of ξ will all be finite. Let A be the greatest numerical value so obtained for ξ. With this assumed, we bring our equation into the form

$$\psi(n) = -\sum_2^n \psi\left[\frac{n}{s}\right] + \frac{1}{2}n^2 + \frac{1}{2}n$$

and consider the values of n lying between $N+1$ and $2N$. Since $[n/s]$ in the sum does not exceed the bound N, the part of our sum arising from substituting the second term of the expression assumed above for $\psi(n)$ is numerically smaller than

$$A\sum_2^n \chi\left(\frac{n}{s}\right).$$

The part arising from the first term, namely

$$\frac{3}{\pi^2}\sum_2^n \left[\frac{n}{s}\right]^2,$$

becomes, if one again writes $[n/s] = n/s - \varepsilon$,

$$\frac{3n^2}{\pi^2}\sum_2^n \frac{1}{s^2} - \frac{6n}{\pi^2}\sum_2^n \frac{\varepsilon}{s} + \frac{3}{\pi^2}\sum_2^n \varepsilon^2.$$

Since

$$\sum_2^n \frac{1}{s^2} = \frac{\pi^2}{6} - 1 + \tau,$$

where τ is of order $1/n$, and since the two other sums cannot exceed the orders $\log n$ and n, respectively, one obtains an equation

$$\psi(n) = \frac{3}{\pi^2}n^2 + \xi,$$

in which ξ is numerically smaller than

$$Pn\log n + A\sum_2^n \chi\left(\frac{n}{s}\right),$$

where P is a sufficiently large constant independent of N. Comparing this result, valid from $n = N+1$ to $n = 2N$, with the expression assumed above for $\psi(n)$,

$$\frac{3}{\pi^2}n^2 + \xi\chi(n),$$

shows that for this new interval the greatest numerical value of ξ does not exceed the maximum, in that interval, of the expression

$$\frac{Pn\log n}{\chi(n)} + \frac{A}{\chi(n)}\sum_2^n \chi\left(\frac{n}{s}\right).$$

If one now gives the previously undetermined function $\chi(n)$ the form of a positive power n^δ, then the expression multiplied by A becomes

$$\sum_2^n \frac{1}{s^\delta} < \sum_2^\infty \frac{1}{s^\delta} = q,$$

and δ can be chosen between 1 and 2 so that the constant q is a proper fraction. On the other hand, N can be chosen so large that for $n \geq N$,

$$\frac{Pn \log n}{n^\delta} < k,$$

where the constant k is arbitrarily small. If A' denotes the greatest numerical value of ξ occurring between $n = N + 1$ and $n = 2N$, then

$$A' < Aq + k.$$

Since k can be made arbitrarily small as N increases, while A does not decrease and q remains constant, for sufficiently large N one has

$$A' < A.$$

That is, the maximum A originally valid up to $n = N$ is also valid up to $2N$, and for the same reason also up to $4N, 8N, \ldots$, and therefore quite generally.

Thus

$$\psi(n) = \frac{3}{\pi^2} n^2$$

with an error that cannot exceed order n^δ, where the constant δ exceeds, however slightly, the value γ given by the equation

$$\sum_2^\infty \frac{1}{s^\gamma} = 1.$$

For the mean value of $\varphi(n)$ this gives the expression

$$\frac{6}{\pi^2} n,$$

whose meaning is clear from the foregoing.

7.

As a final example we choose the function $\varphi(n)$ which denotes the number of all possible decompositions of n into two factors without a common divisor, and which, as is well known, has the expression 2^ρ, where ρ is the number of distinct prime divisors of n. Put again

$$\sum_1^n \varphi(s) = \psi(n).$$

Then $\psi(n)$ stands in a simple relation to the function $F(n)$ treated in Articles 1 and 3. Indeed, $F(n)$ plainly expresses the number of pairs of integers x, y satisfying $xy \leq n$, whereas $\psi(n)$ denotes the number of pairs of relatively prime integers ξ, η for which likewise $\xi\eta \leq n$. Divide the pairs x, y into groups, the sth containing those pairs for which s is the greatest common divisor; thus, if $x = \xi s$, $y = \eta s$, the numbers ξ and η are relatively prime. Dividing the inequality by s^2 gives

$$\xi\eta \leq \frac{n}{s^2} \quad \text{or} \quad \xi\eta \leq \left[\frac{n}{s^2}\right].$$

Conversely, every pair of relatively prime integers ξ, η satisfying this condition gives, by multiplication by s, two integers x, y with greatest common divisor s and with $xy \leq n$. Hence the number of pairs contained in the sth group is expressed by $\psi\left[n/s^2\right]$. Thus one obtains the equation

$$\sum \psi\left[\frac{n}{s^2}\right] = F(n) = n\log n + (2C-1)n,$$

where the sum extends from $s = 1$ to $s = [\sqrt{n}]$, and by the above the neglected term does not exceed order $\sqrt{n}$. It is consequently easy to see that the asymptotic expression for $\psi(n)$ will have the form

$$\alpha n\log n + \beta n,$$

where α and β are two constants still to be determined. If, as in the preceding article, one sets in our equation

$$\psi(n) = \alpha n\log n + \beta n + \xi n^{\delta},$$

and chooses α and β so that the terms of orders $n\log n$ and n cancel, namely

$$\alpha = \frac{6}{\pi^2}, \qquad \beta = \frac{6}{\pi^2}\left(\frac{12C'}{\pi^2} + 2C - 1\right),$$

where C has its former meaning and

$$C' = \sum_2^{\infty} \frac{\log s}{s^2},$$

then by arguments entirely analogous to those developed in the preceding article one finds that ξ, however large n may be, remains under a fixed bound, provided only that $\delta > \frac{1}{2}\gamma$, where γ is understood in the above sense. From the expression thus obtained for $\psi(n)$,

$$\alpha n\log n + \beta n,$$

there follows for $\varphi(n)$, in the sense determined by what has just been said, the mean value

$$\frac{6}{\pi^2}\left(\log n + \frac{12C'}{\pi^2} + 2C\right).$$

The question just treated is connected with the mean value given in Article 301 of the *Disquisitiones arithmeticae* for the number of *genera* corresponding to a negative determinant $-n$. According to known theorems, the expression for the number of *genera*, corresponding to the five linear forms

$$n = 8h, \quad n = 8h+4, \quad n = 4h+2, \quad n = 4h+1, \quad n = 4h+3,$$

is respectively

$$\varphi(n), \quad \tfrac{1}{2}\varphi(n), \quad \tfrac{1}{2}\varphi(n), \quad \varphi(n), \quad \tfrac{1}{2}\varphi(n).$$

On the other hand, by obvious modifications of the preceding considerations, whose execution we leave to the reader, one can determine the approximate value of $\psi(n)$ when n has a prescribed linear form, and the sum $\sum \varphi(s) = \psi(n)$ is no longer extended over all the numbers $1, 2, \ldots, n$, but only over the numbers of this form contained in that sequence; from this one can then derive the mean value of $\varphi(n)$. For the linear forms listed above,

$$n = 8h, \quad n = 8h+4, \quad n = 4h+2, \quad n = 4h+1, \quad n = 4h+3,$$

one finds as the mean value of $\varphi(n)$, if for abbreviation one puts

$$\log n + \frac{12C'}{\pi^2} + 2C = \Delta,$$

the following five expressions:

$$\frac{8}{\pi^2}\left(\Delta - \frac{8}{3}\log 2\right), \quad \frac{8}{\pi^2}\left(\Delta - \frac{2}{3}\log 2\right), \quad \frac{8}{\pi^2}\left(\Delta + \frac{1}{3}\log 2\right),$$

$$\frac{4}{\pi^2}\left(\Delta + \frac{4}{3}\log 2\right), \quad \frac{4}{\pi^2}\left(\Delta + \frac{4}{3}\log 2\right).$$

By the foregoing, these expressions, multiplied respectively by $1, \frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}$, give the mean numbers of the *genera* of determinant $-n$ for the previously enumerated linear forms. Since among every eight consecutive numbers, respectively $1, 1, 2, 2, 2$ are contained in these forms, the sum of our expressions multiplied successively by

$$1 \cdot \tfrac{1}{8}, \quad \tfrac{1}{2} \cdot \tfrac{1}{8}, \quad \tfrac{1}{2} \cdot \tfrac{2}{8}, \quad 1 \cdot \tfrac{2}{8}, \quad \tfrac{1}{2} \cdot \tfrac{2}{8}$$

is, that is,

$$\frac{4}{\pi^2}\left(\Delta - \frac{1}{2}\log 2\right),$$

the desired mean number of *genera* corresponding to the determinant $-n$, when this determinant is thought of generally, that is, no longer restricted to a particular linear form. This agrees with the result given at the cited place. That the same expression also holds for the positive determinant n follows easily from the fact that the mean values of $\varphi(n)$ corresponding to the linear forms $4h+1$ and $4h+3$ coincide, and that, on the other hand, the exceptions occurring for square determinants are plainly without influence on the final result.

VII.

On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

By G. Lejeune Dirichlet.

Abhandlungen der Königlich Preussischen Akademie der Wissenschaften, 1850, pp. 99–116.

On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

[Read in the Academy of Sciences on 28 November 1850.[1]]

According to a beautiful theorem first stated and proved by Gauss,[2] every surface can be covered with an infinitely thin mass layer whose potential at each point of the surface assumes an arbitrarily prescribed value, provided that this value varies continuously over the whole extent of the surface. The actual determination of the mass distribution which effects this, however, is at the present state of science possible only for a few special surfaces; among these, as Gauss already observed, is the whole spherical surface.

Let R denote the radius of the sphere, r the distance of any point in space from the centre, θ the angle between r and a fixed straight line, and φ the angle in the surface between the plane determined by this line and r and a fixed plane. The potential value V, prescribed on the entire surface and expressed by θ and φ, may then be expanded in the known spherical functions. Let

$$V = \sum X_n$$

be this expansion, where the summation sign extends from $n = 0$ to $n = \infty$, and let

$$\rho = \sum Y_n$$

be the analogous expansion of the density ρ to be determined. With the help of the latter, the potential v at any point in space can be represented.[3] Of the two expressions for it, one valid in the interior and the other in the exterior space, either would suffice for our purpose; the former is

$$v = 4\pi R \sum \frac{1}{2n+1} \left(\frac{r}{R}\right)^n Y_n.$$

Since this expression remains valid up to $r = R$, in which case v becomes V, and since the same function is capable of only one expansion in spherical functions, comparison with the first series

[1] The Academy report for 1850, p. 464, states: "Mr. Dirichlet gave a lecture on a new expression for the mass distribution on a spherical surface when the potential at every point of the surface is to receive an arbitrarily prescribed value." K.

[2] General theorems concerning the attractive and repulsive forces acting in inverse ratio to the square of the distance, by C. F. Gauss.

[3] Mécanique céleste, Book III, No. 13.

gives

$$Y_n = \frac{2n+1}{4\pi R} X_n,$$

and hence

$$\rho = \frac{1}{4\pi R} \sum (2n+1) X_n.$$

In an earlier paper[4] it was shown that every function prescribed arbitrarily over the whole spherical surface, that is, from $\theta = 0$, $\varphi = 0$ to $\theta = \pi$, $\varphi = 2\pi$, can be developed convergently in spherical functions, provided only that it nowhere becomes infinite. The convergence of the series for V, which is the starting point of the solution just described, is therefore beyond doubt. The matter is different for the series $\sum Y_n$ assumed for the density ρ and then determined by comparison with it. Since the existence of a completely determined mass distribution corresponding to the prescribed potential value V is compatible with the density becoming infinite at particular points or along particular lines, it remains entirely uncertain in all such cases whether the series preserves its convergence at all places where the density remains finite and whether it actually represents the density. It seemed to me not uninteresting to submit this question to closer examination, for which my earlier paper supplied the necessary aids, especially since this investigation allowed one to expect a simplification of the expression just found for ρ. That expression involves a fourfold infinite operation: a double summation and likewise a double integration. That the latter cannot be removed from the expression is clear, since the density at each point must depend on all potential values prescribed, up to continuity, over the whole extent of the surface. On the other hand, it has turned out that the density can always be represented without series by a double integral, which in many cases can even be reduced to a single integral.

1.

Put

$$V = f(\theta, \varphi).$$

It is known that for the general term X_n one has

$$X_n = \frac{2n+1}{4\pi} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi',$$

where

$$\cos\omega = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi - \varphi')$$

and where $P_n(\cos\omega)$ denotes the coefficient of α^n in the expansion of the radical expression

$$\frac{1}{\sqrt{1 - 2\alpha\cos\omega + \alpha^2}}.$$

The double integration extends from

$$\theta' = 0,\ \varphi' = 0 \qquad \text{to} \qquad \theta' = \pi,\ \varphi' = 2\pi.$$

If the divisor R is omitted, or equivalently if the radius of the sphere is set equal to unity, the general term of the series representing ρ becomes

$$\frac{(2n+1)^2}{(4\pi)^2} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi'.$$

[4] Sur les séries dont le terme général dépend de deux angles, et qui servent à exprimer des fonctions arbitraires entre des limites données. Crelle's Journal, vol. XVII. Vol. I, p. 283 of this edition of G. Lejeune Dirichlet's Werke. K.

It will plainly be enough to examine this series for the case $\theta = 0$, that is, for the pole p of the spherical polar coordinates θ, φ, since, as in the earlier paper, the result found for the point p may be transferred immediately to any other point m of the surface. Then

$$\cos\omega = \cos\theta',$$

and $P_n(\cos\omega)$ is independent of φ'. Put

$$\frac{1}{2\pi}\int_0^{2\pi} f(\theta', \varphi')\, d\varphi' = F(\theta').$$

Thus $F(\theta')$ denotes the mean value of the potential V on the circle described about p as centre with spherical radius θ'. Writing γ instead of θ', in agreement with the notation of the earlier paper to which we shall often refer, the general term becomes

$$\frac{(2n+1)^2}{8\pi}\int_0^{\pi} F(\gamma)P_n(\cos\gamma)\sin\gamma\, d\gamma.$$

If $(2n+1)^2$ is decomposed into its components $4n^2$, $4n$, 1, then the general term splits into three others, and consequently the series itself into three partial series. Since the convergence of the second and third of these series has already been shown in the earlier paper, we have only to consider the first, whose general term is

$$\frac{1}{2\pi}n^2\int_0^{\pi} F(\gamma)P_n(\cos\gamma)\sin\gamma\, d\gamma.$$

Taking the sum of the first $n+1$ terms, expressing $P_n(\cos\gamma)$ by means of a definite integral according to equation (3) of the earlier paper,[5] and reversing the order of the two integrations, one obtains

$$\frac{1}{\pi^2}\int_0^{\pi}(\cos\psi + 4\cos 2\psi + \cdots + n^2\cos n\psi)\Pi(\psi)\, d\psi,$$

where, as before,

$$\Pi(\psi) = \sin\frac{\psi}{2}\int_0^{\psi} F(\gamma)\sin\gamma\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\int_{\psi}^{\pi} F(\gamma)\sin\gamma\frac{d\gamma}{E},$$

$$\Delta = \sqrt{2(\cos\gamma - \cos\psi)}, \qquad E = \sqrt{2(\cos\psi - \cos\gamma)}.$$

Since the integrals contained in $\Pi(\psi)$ are functions of ψ which remain continuous from $\psi = 0$ to $\psi = \pi$ (by §3 of the earlier paper), the same property belongs to the function $\Pi(\psi)$ itself.

The present investigation also requires discussion of the derivative $\Pi'(\psi)$. In forming it, because of the denominators Δ and E contained in the integrals, a preliminary transformation by integration by parts is necessary. Taking into account that $F(\gamma)$, from its origin as the continuous potential value $f(\theta, \varphi)$, is itself continuous, this double operation gives

$$\Pi'(\psi) = (F(0) - F(\pi))\sin\psi + \frac{1}{2}\cos\frac{\psi}{2}\int_0^{\psi} F'(\gamma)\Delta\, d\gamma + \frac{1}{2}\sin\frac{\psi}{2}\int_{\psi}^{\pi} F'(\gamma)E\, d\gamma$$
$$+ \sin\frac{\psi}{2}\sin\psi\int_0^{\psi} F'(\gamma)\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\sin\psi\int_{\psi}^{\pi} F'(\gamma)\frac{d\gamma}{E}.$$

We now assume that $F'(\gamma)$ remains finite everywhere from $\gamma = 0$ to $\gamma = \pi$. Then all components of $\Pi'(\psi)$, and hence $\Pi'(\psi)$ itself, will be finite and continuous from $\psi = 0$ to $\psi = \pi$. For the first three terms this is evident; for the last two, arguments very similar to those cited above show that the integrals contained in them retain this double property so long as, in the first, $\psi < \pi$

[5] Vol. I, p. 291 of this edition of G. Lejeune Dirichlet's Werke. K.

and, in the second, $\psi > 0$ is assumed. For $\psi = \pi$ and $\psi = 0$ these integrals may respectively take infinite values, but, as is easy to see, their order cannot exceed

$$\log\left(\frac{1}{\cos\frac{\psi}{2}}\right) \qquad \text{and} \qquad \log\left(\frac{1}{\sin\frac{\psi}{2}}\right),$$

so that the terms themselves become zero.

For what follows one must also know the second derivative $\Pi''(\psi)$ for the special value $\psi = 0$. Since $\Pi'(0) = 0$, the desired value is most easily found by letting ψ become infinitely small in the quotient $\frac{1}{\psi}\Pi'(\psi)$. Thus one obtains

$$\Pi''(0) = F(0) - F(\pi) + \frac{1}{2}\int_0^\pi F'(\gamma)\sin\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma,$$

or, if the third term is integrated by parts,

$$\Pi''(0) = F(0) - \frac{1}{2}F(\pi) - \frac{1}{4}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.$$

If, in addition to the finiteness of $F'(\gamma)$ just assumed and the consequent continuity of $\Pi'(\psi)$, a definite finite value also exists for $\Pi''(0)$, or what is the same, if the second integral has such a value, then our series will always converge and its sum will be easy to state. Indeed, for shortness put

$$X(\psi) = \frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}} = \frac{1}{2} + \cos\psi + cos2\psi + \cdots + \cos n\psi.$$

The above expression for the sum of the first $n+1$ terms is then

$$-\frac{1}{\pi^2}\int_0^\pi \Pi(\psi)X''(\psi)\,d\psi.$$

Since the boundary term emerging after twice integrating by parts,

$$\Pi(\psi)X'(\psi) - \Pi'(\psi)X(\psi),$$

is continuous and vanishes at both limits, this operation gives

$$-\frac{1}{\pi^2}\int_0^\pi \Pi''(\psi)\frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}}\,d\psi.$$

By a known theorem, this expression tends, as n increases, even when $\Pi''(\psi)$ becomes infinite for one or more non-zero values of ψ, to the limit

$$\begin{aligned}-\frac{1}{2\pi}\Pi''(0) = -\frac{1}{2\pi}F(0) + \frac{1}{4\pi}F(\pi) + \frac{1}{8\pi}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma \\ -\frac{1}{4\pi}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.\end{aligned}$$

Adding to this value those of the two other series, as found in the earlier paper, gives for the density at p

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma\right].$$

It remains to examine how the series expressing the density behaves, as regards convergence, in the case where the integral

$$\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma$$

and therefore also the expression just found for ρ retains a definite finite value, while the function $F'(\gamma)$ becomes infinite for one or more non-zero values of γ. According to the above, the condition for convergence then consists solely in this: the function $\Pi''(\psi)$ derived from $F'(\gamma)$ must remain finite and continuous from $\psi = 0$ to $\psi = \pi$. A more exact consideration of the way $\Pi''(\psi)$ is formed shows that, if the becoming infinite of $F'(\gamma)$ occurs in such a way that for every value c at which $F'(\gamma)$ takes an infinitely large value, $\sqrt{\varepsilon}F'(c \pm \varepsilon)$ is itself infinitely small for infinitely small ε, then the continuity of $\Pi''(\psi)$ obtains just as in the previously examined case of an everywhere finite function $F'(\gamma)$, and with it the convergence of the series representing the density. In contrast, in general divergence occurs when the condition just stated is no longer fulfilled. Although the proof of this result presents no essential difficulty, it would require too much space and would be of too little interest to carry it out here. For the secure use of the series it is enough to know, from the preceding discussion, the only cases in which the convergence of the series can cease. Moreover, an occasion will arise below to demonstrate the divergence actually occurring in such a case by means of a very simple example.

2.

Since the preceding derivation of the finite expression found for ρ loses its validity when the series ceases to converge, a proof of this expression applicable to all cases is still needed.

According to a known theorem, the derivative $\frac{\partial v}{\partial r}$, when θ and φ are kept constant and r is allowed to approach unity, tends to two different limits according as $r < 1$ or $r > 1$ is assumed throughout. If these limiting values are denoted respectively by K and L, then the density at the corresponding point of the surface is determined by

$$\rho = \frac{1}{4\pi}(K - L).$$

Between the limits K, L and the potential value V on the surface there is a simple relation, so that only one of the two limiting values has to be found. Namely, if v and v_1 are the potential values for two points on the same radius vector, their distances from the centre, r and r_1, satisfying

$$rr_1 = 1,$$

then, as is readily seen,

$$v_1 = \frac{1}{r_1}v,$$

and therefore

$$\frac{\partial v_1}{\partial r_1} = -\frac{1}{r_1^2}v + \frac{1}{r_1}\frac{\partial v}{\partial r}\frac{\partial r}{\partial r_1} = -\frac{1}{r_1^2}v - \frac{1}{r_1^3}\frac{\partial v}{\partial r}.$$

Thus one sees that

$$L = -V - K$$

and

$$\rho = \frac{1}{4\pi}(V + 2K).$$

To determine K it suffices to use the expression for v valid in the interior space:

$$v = \frac{1 - r^2}{4\pi} \iint \frac{f(\theta', \varphi') \sin\theta' \, d\theta' \, d\varphi'}{(1 - 2r\cos\omega + r^2)^{3/2}}.$$

This is easily proved without developing in series. It is enough to observe that this expression, as r approaches its upper limit 1, passes by a known and often treated theorem into $f(\theta, \varphi) = V$, and that on the other hand it satisfies the partial differential equation first set up by Laplace for v, since

$$\frac{1-r^2}{(1-2r\cos\omega+r^2)^{3/2}},$$

considered as a function of the polar coordinates r, θ, φ, is a particular integral of this equation. In the case $\theta = 0$, the expression takes, writing γ again instead of θ' and keeping the earlier notation, the simple form

$$v = \frac{1}{2}(1-r^2)\int_0^\pi \frac{F(\gamma)\sin\gamma\, d\gamma}{(1-2r\cos\gamma+r^2)^{3/2}}.$$

If one differentiated the expression in this form, the limiting value of $\frac{\partial v}{\partial r}$ would be difficult to determine. It is therefore expedient first to perform an integration by parts. One obtains

$$2v = \left(\frac{1}{r}+1\right)F(0) - \left(\frac{1}{r}-1\right)F(\pi) + \left(\frac{1}{r}-r\right)\int_0^\pi \frac{F'(\gamma)\, d\gamma}{(1-2r\cos\gamma+r^2)^{1/2}},$$

where passage to the limit after differentiation now presents no further difficulty and gives

$$K = -\frac{1}{2}F(0) + \frac{1}{2}F(\pi) - \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\, d\gamma.$$

Taking into account that for the point p,

$$V = F(0),$$

it follows that

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\, d\gamma\right],$$

which agrees with the formula derived from the series.

3.

Transferring the result found for the point corresponding to $\theta = 0$ to an arbitrary point m, one obtains the following general rule for determining the density.

By a first integration, find for the circle described on the surface about m as centre with arbitrary spherical radius λ the mean value of the given potential V. Denote this mean by $\varphi(\lambda)$. Then the density sought is given by

$$\rho = \frac{1}{4\pi}\left[\varphi(\pi) - \int_0^\pi \frac{\varphi'(\lambda)}{\sin\frac{\lambda}{2}}\, d\lambda\right].$$

One may also remark that, by the meaning of the function $\varphi(\lambda)$, the special value $\varphi(\pi)$ denotes the potential value at the point antipodal to m. It hardly needs saying that the function $\varphi(\lambda)$ will be different for every point m, or, in other words, that besides λ it will also contain the two quantities which determine the position of m on the surface.

It is perhaps not superfluous, however, to forestall an error which can easily arise from an inattentive consideration of the result just stated. At first sight, because the denominator under the integral sign vanishes at the lower limit, one is tempted to suppose that the integral remains finite only for special positions of the point m, but in general becomes infinite; in fact, precisely the reverse is the case. First, by the continuity of $\varphi(\lambda)$ it is easy to see that the part of the integral extending from any positive value δ, however small, to the upper limit π always remains determinate and finite, however often $\varphi'(\lambda)$ itself may become infinite in this interval. That the same is also true, apart from singular cases, for the part extending from 0 to δ has its reason in the fact that for small values of λ, $\varphi'(\lambda)$ is generally of the same first order as the denominator. If the potential value V is represented as a function $\chi(\lambda, \psi)$ of spherical polar coordinates λ, ψ whose pole is the point m, so that

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\lambda, \psi)\, d\psi,$$

then the property just mentioned does not appear only because $\chi(\lambda, \psi)$ is not arbitrary, but must satisfy the special conditions of being periodic with respect to ψ and becoming independent of ψ for $\lambda = 0$. One is immediately convinced of the assertion made above if one projects the spherical surface onto an arbitrary plane, whose points are referred to two rectangular axes t, u, and then expresses V at each point by the coordinates t, u of its projection. This representation has the disadvantage that, in order to cover the whole surface, two functions of t and u must be considered; but this disadvantage disappears for our purpose, which requires consideration only of an arbitrarily small surface element containing the point m. Taking the tangent plane at m as the plane of projection, m as the origin of coordinates, and putting $V = \chi(t, u)$, this new function has to satisfy no condition other than continuity. Since evidently one may take

$$t = \sin\lambda \cos\psi, \qquad u = \sin\lambda \sin\psi,$$

the function $\varphi(\lambda)$ assumes the form

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\sin\lambda\cos\psi, \sin\lambda\sin\psi)\, d\psi.$$

In this form the occurrence of the above property is evident, provided the partial derivatives of $\chi(t, u)$ up to and including the second order, for the case where $t = 0$ and $u = 0$ simultaneously, retain definite finite values. Moreover, for the finiteness of ρ it is not even necessary that $\varphi'(\lambda)$ be of the first order for small values of the variable; it is enough that the order of $\varphi'(\lambda)$ agree with that of some positive power of λ.

As with the becoming infinite of ρ and the consequent divergence of the series, so also with the case where the series ceases to converge while the density remains finite. This case is even more restricted than the one just discussed, as is easily seen by closer consideration of the conditions on which it depends.

4.

In order to illustrate the actual occurrence of divergence of the series for ρ in the exceptional cases indicated above by a simple example, suppose that the potential value V does not contain the angle φ and is represented by $f(\theta)$. Then, as is known, the spherical-function series take a simpler form, and one has

$$V = \frac{1}{2} \sum (2n+1) A_n P_n(\cos\theta), \qquad \rho = \frac{1}{8\pi} \sum (2n+1)^2 A_n P_n(\cos\theta),$$

where the general coefficient A_n is given by

$$A_n = \int_0^{\pi} f(\theta) P_n(\cos\theta) \sin\theta\, d\theta.$$

Put

$$f(\theta) = \sqrt{\cos\theta} \quad \text{as long as } 0 < \theta < \frac{1}{2}\pi, \qquad f(\theta) = 0 \quad \text{when } \theta \text{ lies between } \frac{1}{2}\pi \text{ and } \pi.$$

Then the requirement of continuity is satisfied, and the integral

$$A_n = \int_0^{\frac{1}{2}\pi} P_n(\cos\theta)\sqrt{\cos\theta}\sin\theta\,d\theta$$

has to be determined. Since its evaluation by means of the known expressions for $P_n(\cos\theta)$ does not immediately give the result in the simplest form, the following way is preferable.

From

$$\sum P_n(\cos\theta)\alpha^n = \frac{1}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

the required integral is the coefficient of α^n in the expansion of

$$\int_0^{\frac{1}{2}\pi} \frac{\sqrt{\cos\theta}\sin\theta\,d\theta}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

where the proper fraction α may be regarded as positive. By the substitution $t=\sqrt{\cos\theta}$ and by carrying out the integration, one obtains

$$2\int_0^1 \frac{t^2\,dt}{\sqrt{1-2\alpha t^2+\alpha^2}} = \sqrt{\frac{1}{8\alpha^3}}\left((1+\alpha^2)\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} - (1-\alpha)\sqrt{2\alpha}\right).$$

For convenience in developing the expression, put

$$z = \arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}}.$$

Then differentiating gives

$$\frac{dz}{d\alpha} = \frac{1+\alpha}{1+\alpha^2}\sqrt{\frac{1}{2\alpha}} = \frac{1}{\sqrt{2}}(\alpha^{-\frac{1}{2}}+\alpha^{\frac{1}{2}}-\alpha^{\frac{3}{2}}-\alpha^{\frac{5}{2}}+\cdots),$$

and, on integrating and observing that z vanishes with α,

$$\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} = \sqrt{2}\left(\alpha^{\frac{1}{2}}+\frac{1}{3}\alpha^{\frac{3}{2}}-\frac{1}{5}\alpha^{\frac{5}{2}}-\frac{1}{7}\alpha^{\frac{7}{2}}+\cdots\right).$$

Substitution gives for the required expansion

$$-\frac{1}{2}\left(\left(\frac{1}{-1}-\frac{1}{3}\right)-\left(\frac{1}{1}-\frac{1}{5}\right)\alpha-\left(\frac{1}{3}-\frac{1}{7}\right)\alpha^2+\left(\frac{1}{5}-\frac{1}{9}\right)\alpha^3+\cdots\right),$$

where the signs inside the parentheses form the four-term period

$$+\;-\;-\;+,$$

which may be represented by $(-1)^{\frac{1}{2}n(n+1)}$. Thus

$$A_n = -(-1)^{\frac{1}{2}n(n+1)}\frac{2}{(2n-1)(2n+3)},$$

and the preceding series take the form

$$V = -\sum(-1)^{\frac{1}{2}n(n+1)}\frac{2n+1}{(2n-1)(2n+3)}P_n(\cos\theta),$$

$$\rho = -\frac{1}{4\pi}\sum(-1)^{\frac{1}{2}n(n+1)}\frac{(2n+1)^2}{(2n-1)(2n+3)}P_n(\cos\theta).$$

From the remarks in the preceding article, one sees without difficulty in the present case that the derivative $\varphi'(\lambda)$ becomes infinite in such a way that the condition stated at the end of Article 1 is not fulfilled only when $\lambda = \frac{1}{2}\pi$ and the point m corresponds to $\theta = 0$ or $\theta = \pi$; in these cases, however, $\varphi'(\lambda)$ is of order λ for small values of λ, and ρ therefore retains a definite finite value. One also sees that the density becomes infinite only at the equator, or for $\theta = \frac{1}{2}\pi$, since then $\varphi'(\lambda)$ is of order λ^{-1} for small λ. In these three cases the series therefore ceases to converge. This is immediately verified if one takes into account that $P_n(\cos\theta)$ has the values 1 and $(-1)^n$ respectively for $\theta = 0$ and $\theta = \pi$, while for $\theta = \frac{1}{2}\pi$ it vanishes when n is odd and, when n is even, equals

$$\frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n}(-1)^{\frac{n}{2}},$$

which, as is known, is of order $n^{-\frac{1}{2}}$ for large n.

In the first two cases, where ρ retains a definite finite value, the series has the character of an oscillating one: among every four consecutive terms, two have positive and two negative sign, and their terms tend to unity as a limit. In the third case, all terms have the same sign and yield an infinitely large sum.

5.

Although the method by which we have just determined the coefficient A_n is no longer applicable if, while retaining the assumption $f(\theta) = 0$ for the second part of the interval, one takes $f(\theta) = \cos^k\theta$ in the first part, where k denotes an arbitrary positive constant, the very simple form of the result found for the special value $k = \frac{1}{2}$ leads one to expect something similar in the more general case. Since the simplest expression for A_n is somewhat hidden, we pause briefly to find it. Poisson, who in one of his papers[6] chose precisely the function just defined as an example of development in spherical functions, gives the coefficient A_n the form

$$A_n = \frac{1 \cdot 3 \cdots (2n-1)}{1 \cdot 2 \cdots n}\left(\frac{1}{k+n+1} - \frac{n(n-1)}{2(2n-1)}\frac{1}{k+n-1} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)}\frac{1}{k+n-3} - \cdots\right).$$

This is the form in which the coefficient is obtained directly when the usual expression for $P_n(\cos\theta)$ is used in the expansion. It does not reveal the considerable simplification of which it is capable, and that simplification would not easily be suspected unless one were led to it by a special case such as the one above. The following very short, though not elementary, route leads to the simplest expression for A_n.

By §1 of the earlier paper,[7] the two integrals

$$\frac{2}{\pi}\int_0^\gamma \frac{\cos n\psi \cos\frac{\psi}{2}}{\sqrt{2(\cos\psi - \cos\gamma)}}\,d\psi, \qquad -\frac{2}{\pi}\int_0^\gamma \frac{\sin n\psi \sin\frac{\psi}{2}}{\sqrt{2(\cos\psi - \cos\gamma)}}\,d\psi$$

are respectively the coefficients of α^n in the expansions of

$$\frac{1}{2} + \frac{1}{2}\frac{1+\alpha}{\sqrt{1 - 2\alpha\cos\gamma + \alpha^2}}, \qquad -\frac{1}{2} + \frac{1}{2}\frac{1-\alpha}{\sqrt{1 - 2\alpha\cos\gamma + \alpha^2}}.$$

Adding gives

$$P_n(\cos\gamma) = \frac{2}{\pi}\int_0^\gamma \frac{\cos(n+\frac{1}{2})\psi}{\sqrt{2(\cos\psi - \cos\gamma)}}\,d\psi.$$

[6] Connaissance des temps pour l'an 1829.

[7] Vol. I, pp. 292–294 of this edition of G. Lejeune Dirichlet's Werke. K.

Multiplying this equation by $\cos^k \gamma \sin\gamma \, d\gamma$, then integrating from $\gamma = 0$ to $\gamma = \frac{1}{2}\pi$ and reversing the order of integrations on the right, yields

$$A_n = \frac{2}{\pi} \int_0^{\frac{1}{2}\pi} d\psi \, \cos(n + \tfrac{1}{2})\psi \int_\psi^{\frac{1}{2}\pi} \frac{\cos^k \gamma \sin\gamma}{\sqrt{2(\cos\psi - \cos\gamma)}} \, d\gamma.$$

If one introduces a new variable t by $\cos\gamma = t\cos\psi$, the two integrations separate:

$$A_n = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{t^k}{\sqrt{1-t}} \, dt \cdot \int_0^{\frac{1}{2}\pi} \cos^{k+\frac{1}{2}} \psi \cos(n + \tfrac{1}{2})\psi \, d\psi.$$

Now, by a known formula,

$$\int_0^{\frac{1}{2}\pi} \cos^{p-1} \psi \cos q\psi \, d\psi = \frac{\Gamma(p)}{\Gamma\left(\frac{1}{2}(1+p+q)\right)\Gamma\left(\frac{1}{2}(1+p-q)\right)} \cdot \frac{\pi}{2^p},$$

where the constant p must be positive, and the transcendental function $\Gamma(l)$, when the argument l becomes negative, is to be understood by the relation $\Gamma(l+1) = l\Gamma(l)$, or, what is the same thing, by Gauss's definition, which comprises positive and negative arguments uniformly. By this formula and by the known representation of a Eulerian integral of the first kind by three of the second, our equation becomes

$$A_n = \frac{\Gamma(k+1)}{\Gamma\left(\frac{1}{2}(k+n+3)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)} \cdot \frac{\sqrt{\pi}}{2^{k+1}},$$

or

$$A_n = \frac{k(k-1)(k-2)\cdots(k-n+1)}{\frac{1}{2}(k+n+1)\left(\frac{1}{2}(k+n+1)-1\right)\cdots\left(\frac{1}{2}(k+n+1)-n\right)} \cdot \frac{\Gamma(k-n+1)}{\Gamma\left(\frac{1}{2}(k-n+1)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)} \cdot \frac{\sqrt{\pi}}{2^{k+1}}.$$

If for the second factor one substitutes its known value $2^{k-n}\pi^{-\frac{1}{2}}$, this gives

$$A_n = \frac{k(k-1)(k-2)\cdots(k-n+1)}{(k+n+1)(k+n-1)(k+n-3)\cdots(k-n+1)},$$

where the factors in the denominator decrease by two units.

It hardly needs mention that this simple expression can be verified with the greatest ease. For instance, it may be decomposed into partial fractions in order to recognize its agreement with the expression cited above. If one does not wish to use the latter for this verification, one may use instead the equation

$$(n+1)A_{n+1,k} = (2n+1)A_{n,k+1} - nA_{n-1,k},$$

which follows immediately from the relation between every three consecutive coefficients in the expansion of $P_n(\cos\gamma)$.

To simplify the expression still further, one distinguishes whether n is even or odd and obtains respectively

$$A_n = \frac{k(k-2)\cdots(k-n+2)}{(k+1)(k+3)\cdots(k+n+1)}, \qquad A_n = \frac{(k-1)(k-3)\cdots(k-n+2)}{(k+2)(k+4)\cdots(k+n+1)}.$$

With the help of these expressions and the known formulae which give the approximate values of factorials containing a very large number of factors, one may easily verify that the special case $k = \frac{1}{2}$ examined above is exactly the limiting case for the divergence of the series occurring at $\theta = 0$ and $\theta = \pi$, and that the divergence ceases as soon as $k > \frac{1}{2}$. The situation is different for the value $\theta = \frac{1}{2}\pi$: at that place the divergence and the infinitely large value of the density persist as long as $k > 1$ is not assumed.

6.

We shall now make one application of the finite expression found for the density to a special case. Let V again be regarded as a mere function of θ, which may be put in the form $f(\cos\theta)$. Then ρ too depends only on the spherical distance θ of the point m from the pole p. On the circle described about m as centre with spherical radius λ, the fundamental formula of trigonometry gives

$$V = f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi),$$

where ψ denotes the angle between an arbitrary radius λ and the one directed from m to p. Since this expression has the same value for ψ and $-\psi$, the integral expressing the mean value $\varphi(\lambda)$ may be taken from $\psi = 0$ to $\psi = \pi$ and then doubled. Thus

$$\varphi(\lambda) = \frac{1}{\pi}\int_0^{\pi} f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi.$$

Whenever $\varphi'(\lambda)$ can be represented without an integral sign, which is especially the case when $f(z)$, or more generally $f'(z)$, is a rational function of z, whether the form of that function remains the same throughout the whole interval between $z = -1$ and $z = +1$ or changes in places without violating continuity, ρ can be reduced to a single quadrature or even expressed without any integral sign.[8]

Among the cases belonging here we consider only the one in which

$$f(\cos\theta) = \cos\theta$$

as long as $\theta < \frac{1}{2}\pi$, and

$$f(\cos\theta) = 0$$

when $\theta > \frac{1}{2}\pi$. The integral then extends only over that part of the interval between $\psi = 0$ and $\psi = \pi$ in which

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi$$

assumes positive values. Suppose first that $\theta < \frac{1}{2}\pi$; the case $\theta > \frac{1}{2}\pi$ is easily reduced to this. A simple discussion of the preceding expression shows that, so long as λ is less than $\frac{1}{2}\pi - \theta$, the integral is to be taken from $\psi = 0$ to $\psi = \pi$; for values of λ between $\frac{1}{2}\pi - \theta$ and $\frac{1}{2}\pi + \theta$, the limits of the integral are 0 and the angle ψ_1 lying between 0 and π determined by

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi_1 = 0;$$

finally, for $\lambda > \frac{1}{2}\pi + \theta$, the integral vanishes, since the preceding expression is then always negative. Thus, in the three intervals just distinguished, the function $\varphi(\lambda)$ is

$$\cos\theta\cos\lambda, \qquad \frac{1}{\pi}\int_0^{\psi_1}(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi, \qquad 0.$$

In the second case the actual integration is postponed until after differentiation, in order to shorten the calculation. On differentiating, the term arising from the variability of the upper limit ψ_1 vanishes by the equation determining ψ_1, and one obtains the following three expressions for $\varphi'(\lambda)$:

$$-\cos\theta\sin\lambda, \qquad \frac{1}{\pi}(-\psi_1\cos\theta\sin\lambda + \sin\psi_1\sin\theta\cos\lambda), \qquad 0.$$

[8] In the general case, where V contains both angles θ, φ, the reduction of ρ to a single quadrature always takes place if, putting $x = \cos\theta$, $y = \sin\theta\cos\varphi$, $z = \sin\theta\sin\varphi$, one can represent V by a function $f(x, y, z)$ of the quantities x, y, z whose three first partial derivatives are rational, integral or fractional functions of x, y, z; again the form of $f(x, y, z)$ may be different in different parts of the surface. This is a result remarkable for its great range.

These formulas also show in passing that, at least so long as m remains on the hemisphere for which they apply, $\varphi'(\lambda)$ does not become infinite and, for a small λ for which the first formula holds, remains of the first order. In this respect there is an exception only for $\theta = \frac{1}{2}\pi$: then the two outer intervals disappear and $\varphi'(\lambda)$ everywhere has the value $\frac{1}{\pi}\cos\lambda$, which for small values of λ is of order zero; hence an infinitely large density occurs at the equator.

According to the expressions found for $\varphi'(\lambda)$, the integral contained in the general expression for ρ splits into two others, extending respectively from 0 to $\frac{1}{2}\pi - \theta$ and from $\frac{1}{2}\pi - \theta$ to $\frac{1}{2}\pi + \theta$. The first has the simple value

$$-4\cos\theta\sin\left(\frac{1}{4}\pi - \frac{1}{2}\theta\right).$$

The second, consisting of two parts, is, before taking the limits into account,

$$\frac{\sin\theta}{\pi}\int\frac{\sin\psi_1\cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda - \frac{2\cos\theta}{\pi}\int\psi_1\cos\frac{1}{2}\lambda\,d\lambda.$$

Integration by parts in the last term gives it the form

$$\frac{\sin\theta}{\pi}\int\frac{\sin\psi_1\cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda + \frac{4\cos\theta}{\pi}\int\frac{\partial\psi_1}{\partial\lambda}\sin\frac{1}{2}\lambda\,d\lambda - \frac{4}{\pi}\psi_1\cos\theta\sin\frac{1}{2}\lambda.$$

Since the equation determining ψ_1 gives, at the two limits, respectively $\psi_1 = \pi$ and $\psi_1 = 0$, the term free of the integral sign contributes, on passing to the limits, a value opposite to that found above for the first integral. Thus only the first two terms remain. Substitution of the expressions obtained from the equation for ψ_1,

$$\sin\psi_1 = \frac{\sqrt{\sin^2\theta - \cos^2\lambda}}{\sin\theta\sin\lambda}, \qquad \frac{\partial\psi_1}{\partial\lambda} = -\frac{\cos\theta}{\sin\lambda}\frac{1}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

gives them the form

$$\frac{1}{\pi}\int\frac{\cos\lambda}{\sin\lambda\sin\frac{1}{2}\lambda}\sqrt{\sin^2\theta - \cos^2\lambda}\,d\lambda - \frac{4\cos^2\theta}{\pi}\int\frac{\sin\frac{1}{2}\lambda}{\sin\lambda}\frac{d\lambda}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

where the integrals are to be taken from $\lambda = \frac{1}{2}\pi - \theta$ to $\lambda = \frac{1}{2}\pi + \theta$. Introducing as new variable

$$z = \frac{\cos\lambda}{\sin\theta},$$

the limits for it become $+1$ and -1; changing the sign, they become -1 and $+1$. Substitution in the expression for ρ, while also taking account of the vanishing of $\varphi(\pi)$, gives

$$\rho = \frac{1}{2\sqrt{2}\pi^2}\int_{-1}^{+1}\left(\left(\frac{2 - z\sin\theta}{1 - z^2\sin^2\theta}\right)\cos^2\theta - z\sin\theta\right)\frac{dz}{\sqrt{(1-z^2)(1-z\sin\theta)}}.$$

This result only appears to contain a component depending on elliptic integrals of the third kind. Since the limits are two consecutive values for which the radical vanishes, the expression, brought to the usual form, will contain only so-called complete elliptic integrals[9] and hence, by a beautiful theorem due to Legendre, can be reduced to the first and second kinds.

Finally, let us note how the case of the problem treated in the paper mentioned above, for a surface which differs only slightly from a spherical surface, can be reduced to the case of the sphere without a series expansion.

Let

$$r = 1 + \gamma z$$

[9]Fundamenta nova theoriae functionum ellipticarum, by C. G. J. Jacobi, p. 14. C. G. J. Jacobi's collected works, vol. I, p. 67. K.

be the equation of the surface, where γ is a small constant whose higher powers are to be neglected, and z denotes a function of θ and φ. Agree to denote by an added accent the value which a function of θ, φ takes when θ, φ are replaced by θ', φ'. Let ds' be the element of the surface belonging to θ', φ', and let p be the distance between the two points of the surface corresponding to the coordinate pairs θ, φ and θ', φ'. The problem requires that the equation

$$V = \int \rho' \frac{ds'}{p}$$

be satisfied for all values of θ, φ in the same range, the double integration extending over the whole surface, or from $\theta' = 0$, $\varphi' = 0$ to $\theta' = \pi$, $\varphi' = 2\pi$. If $d\sigma'$ is the element of the spherical surface corresponding to ds', cut off by the same radius vector as ds', and if q is the distance between the points on the spherical surface belonging to θ, φ and θ', φ', then the simplest geometrical considerations immediately give

$$ds' = (1 + 2\gamma z')d\sigma'$$

and

$$p = q\left(1 + \frac{1}{2}\gamma(z + z')\right).$$

Substitution of these expressions turns our equation into

$$V = \int \rho' \left(1 + \gamma\left(\frac{3}{2}z' - \frac{1}{2}z\right)\right)\frac{d\sigma'}{q},$$

or

$$V + \frac{1}{2}\gamma z \int \rho' \frac{d\sigma'}{q} = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right)\frac{d\sigma'}{q}.$$

Since, by the penultimate equation, the integral multiplied by γ in the last equation differs from V only by a quantity of order γ, V may be put for it. One obtains

$$\left(1 + \frac{1}{2}\gamma z\right)V = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right)\frac{d\sigma'}{q},$$

which reduces the problem to the case of the spherical surface. Thus one sees that the given potential value, multiplied by $1 + \frac{1}{2}\gamma z$, is to be regarded as applying to the spherical surface, and that the density determined on this assumption is then to be divided by $1 + \frac{3}{2}\gamma z$.

VIII.

On the Representability of Numbers by Three Squares.

By G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, vol. 40, pp. 228–232.

On the Representability of Numbers by Three Squares.

The theory of decomposing the integers n which are not of one of the forms $4k$, $8k+7$ into three squares without a common divisor belongs to the more complicated parts of higher arithmetic, if one wishes to develop this theory completely, that is, not merely to prove representability, but also to determine the number of all possible decompositions. This number can be expressed either by means of the binary forms of determinant $-n$, as Gauss first showed[1], or also independently of the latter[2]. Since, however, there are cases in which only representability is to be assumed, as for example in the proof given by Cauchy[3], later simplified by Legendre, of Fermat's theorem on polygonal numbers, which rests solely on the fact that every number, apart from those excluded above, is the sum of three squares, a simple proof of representability seems not to be without interest. Such a proof is to be communicated in this paper.

For this purpose one first needs the known theorem that every positive ternary form of determinant -1, that is, every expression

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy, \tag{1}$$

whose integral coefficients satisfy the equation

$$aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -1 \tag{2}$$

and, moreover, satisfy the conditions that

$$a, \quad b, \quad c, \quad bc - a'^2, \quad ac - b'^2, \quad ab - c'^2$$

are positive (of which conditions, incidentally, the first and fourth imply the others), is equivalent to the form

$$x^2 + y^2 + z^2. \tag{3}$$

To prove this theorem it suffices to verify that, for determinant -1, the expression (3) is the only reduced positive form.

If the form (1) is to be reduced, then, by the theorem proved at the end of the preceding paper[4], abc must not be greater than 2. Hence, if one assumes $a \le b \le c$, it follows at once that

$$a = b = 1.$$

[1] Disquisitiones arithmeticae, art. 229.

[2] Crelle's Journal, vol. 21, p. 155. The editor of the Werke adds: p. 496 of vol. I of this edition of G. Lejeune Dirichlet's works. K.

[3] Exercices de Mathématiques par Cauchy, année 1826, p. 265. The editor of the Werke adds: Oeuvres complètes d'Augustin Cauchy, IIe Série, Tome VI, p. 320. K.

[4] Vol. II, p. 48 of this edition of G. Lejeune Dirichlet's works. The cited paper, "Über die Reduction der positiven quadratischen Formen mit drei unbestimmten ganzen Zahlen", is printed in vol. 40 of Crelle's Journal immediately before this paper. K.

On the other hand, by the definition of reduced forms, both $2c'$ and $2b'$, apart from sign, must not be greater than a, and $2a'$ must not be greater than b; thus one obtains

$$a' = b' = c' = 0,$$

and then, from (2),

$$c = 1.$$

This being assumed, the decomposability of the positive number a will have been shown as soon as one has found a positive ternary form (1) of determinant -1 whose first coefficient is a. For since such a form is equivalent to the form (3), the latter can be transformed into it, and one obtains

$$a = \alpha^2 + \alpha'^2 + \alpha''^2,$$

where α, α', α'' are three of the nine substitution coefficients and cannot have a common divisor, since every term of the determinant formed from these nine coefficients has either α, or α', or α'' as a factor, and this determinant is equal to unity.

Thus everything comes down to the following: regarding a as given, satisfy equation (2) by five integers b, c, a', b', c', with only the single remaining condition that $bc - a'^2$ must become positive. If one takes $b' = 1$, $c' = 0$, the equation becomes

$$b = a\Delta - 1,$$

where

$$\Delta = bc - a'^2$$

is set; and it remains only to show that the positive number Δ can be chosen so that $-\Delta$ is a quadratic residue of $b = a\Delta - 1$, since under this assumption c and a' can be so determined that $a'^2 - bc = -\Delta$.

We shall now show that the condition just stated can always be satisfied by an odd value of Δ such that, if a has the form $4k + 2$, then $b = a\Delta - 1$ is an odd prime number p, and if a is odd but not of the form $8k + 7$, then b is twice such an odd prime number p.

We begin with the second case. We have the equation

$$2p = a\Delta - 1,$$

where for the moment p is not yet assumed prime, but only odd. Put $\Delta = 8t + \varepsilon$, where ε is one of the numbers 1, 3, 5, 7. One sees at once that in each of the four cases presented by a with respect to the divisor 8, Δ admits two forms with respect to this divisor, or ε admits two values, if p, as required, is to be odd. For each of the eight cases thus obtained, apply the law of reciprocity to the left side of the equation following from $2p = a\Delta - 1$,

$$\left(\frac{p}{\Delta}\right) = \left(\frac{-2}{\Delta}\right),$$

where the Legendre symbol is used in the extended sense introduced by Jacobi; put for $\left(\frac{-2}{\Delta}\right)$ its known value, and then multiply by $\left(\frac{-1}{p}\right) = \pm 1$, where ± 1 will likewise be known according to the corresponding linear form of p. One obtains the following results:

$$
\begin{aligned}
a &= 8k+1, \quad \begin{cases} \Delta = 8t+3, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+7, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = -1, \end{cases} \\
a &= 8k+3, \quad \begin{cases} \Delta = 8t+1, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+5, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = +1, \end{cases} \\
a &= 8k+5, \quad \begin{cases} \Delta = 8t+3, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+7, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = -1, \end{cases} \\
a &= 8k+7, \quad \begin{cases} \Delta = 8t+1, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = -1, \\ \Delta = 8t+5, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = -1. \end{cases}
\end{aligned}
$$

It follows that, if, as assumed, a is not of the form $8k+7$, the condition $\left(\frac{-\Delta}{p}\right) = 1$ can always be met. That in all eight cases p can also be made a prime number is clear from the expression

$$p = \frac{1}{2}(a\Delta - 1) = 4at + \frac{1}{2}(a\varepsilon - 1),$$

in which $\frac{1}{2}(a\varepsilon - 1)$ is, by the choice of ε, odd and also has no common divisor with a. This expression is therefore the general term of an arithmetic progression which necessarily contains prime numbers. Once one has a prime number p for which $\left(\frac{-\Delta}{p}\right) = 1$, $-\Delta$ is a quadratic residue of p and consequently also of $2p$, as was required.

In the case of an even a we at once put $a = 4k+2$, since one knows beforehand that for $a = 4k$ the condition cannot be satisfied. Since in the equation

$$p = a\Delta - 1$$

we assume Δ to be odd, p has the form $4s+1$, and we obtain

$$\left(\frac{-1}{\Delta}\right) = \left(\frac{p}{\Delta}\right) = \left(\frac{\Delta}{p}\right) = \left(\frac{-\Delta}{p}\right).$$

Thus Δ must be given the form $4t+1$ in order that $\left(\frac{-\Delta}{p}\right) = 1$. One obtains

$$p = 4at + a - 1,$$

and this expression can again become a prime number, for which $-\Delta$ is then a quadratic residue.

The application of the procedure just developed is not restricted to the case of determinant -1, and we add a second example for determinant -3.

If one again seeks, as above, the reduced forms for this determinant, then the condition

$$abc \leq 6$$

gives, under the assumption $a \leq b \leq c$, the value 1 for a, while b can be equal to 1 or to 2. It follows, since $2c'$ and $2b'$ must not be numerically greater than $a = 1$, that

$$b' = c' = 0,$$

and the equation by which the determinant is determined reduces to

$$a'^2 - bc = -3.$$

Now, since $2a'$ also must not be numerically greater than b, if $b = 1$ then $a' = 0$, and consequently $c = 3$. If, on the other hand, $b = 2$, then $2a' = 0$ or $2a' = \pm 2$. The first value does not satisfy the above equation, in which $bc = 4$. It remains only to set $a' = \pm 1$, whence $c = 2$. Neglecting the lower sign, which amounts only to changing the sign of z, one therefore obtains the two reduced forms

$$x^2 + y^2 + 3z^2, \qquad x^2 + 2y^2 + 2z^2 + 2yz, \tag{4}$$

so that every positive ternary form (1) of determinant -3, that is, one in which the coefficients satisfy the equation

$$aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -3, \tag{5}$$

will be equivalent to one of these forms (4). If one now examines the residues modulo 8 which the expressions (4) leave when the elements x, y, z are given all combinations of even and odd values, excluding only the case of three even values simultaneously, since the elements are always to be assumed without a common divisor, one finds that the first of the forms (4) offers all possible residues modulo 8, whereas the second expression assumes neither of the forms $8k + 5$, $4k$ for any combination. If one further considers that two equivalent forms always represent the same numbers, then, for a form of determinant -3 which represents a number $8k + 5$ or $4k$ (which is in particular always the case when one of its first three coefficients, for instance a, is such a number), one can at once conclude that it is not equivalent to the second form, and consequently that it is equivalent to the first of the forms (4).

It remains to prove that the given number a is representable by the first of the forms (4). By the preceding method, and by again putting in (5)

$$b' = 1, \qquad c' = 0, \qquad \Delta = bc - a'^2,$$

we need only show that the positive number Δ can be so determined that $b = a\Delta - 3$ has the form $8k + 5$ or $4k$, and that $-\Delta$ is at the same time a quadratic residue of b. We restrict ourselves here to the numbers a which have no common divisor with the determinant, that is, which are not divisible by 3. Put

$$\Delta = 8\Delta',$$

where Δ' is to be odd and not divisible by 3. Then b will be relatively prime to Δ' and will have the form $8k + 5$, and only the other condition remains to be fulfilled. From the equation $b = 8a\Delta' - 3$ it follows at once that

$$\left(\frac{b}{\Delta'}\right) = \left(\frac{-3}{\Delta'}\right),$$

or, if one takes account of the congruence $b \equiv 1 \pmod 4$ and applies known theorems,

$$\left(\frac{b}{\Delta'}\right) = \left(\frac{\Delta'}{b}\right) = \left(\frac{-\Delta'}{b}\right) = \left(\frac{-3}{\Delta'}\right).$$

Multiplication by $\left(\frac{8}{b}\right) = \left(\frac{2}{b}\right) = -1$ gives from this

$$\left(\frac{-\Delta}{b}\right) = -\left(\frac{-3}{\Delta'}\right),$$

and one sees that the condition $\left(\frac{-\Delta}{b}\right) = 1$ is fulfilled if

$$\Delta' = 6t - 1$$

and consequently

$$b = 48at - 8a - 3$$

is set. Since this latter expression can evidently be made equal to a prime number, it is proved that every number not divisible by 3 can be expressed by the form

$$x^2 + y^2 + 3z^2$$

in such a way that x, y, z have no common divisor. Similar considerations can be applied to the numbers divisible by 3, for whose representability an easily obtained condition is required, as well as to the numbers expressible by the second of the forms (4).

Berlin, June 1850.

IX.

On a Problem Concerning Division.

By G. Lejeune Dirichlet.

Report on the Proceedings of the Royal Prussian Academy of Sciences, 1851, pp. 20–25.

On a Problem Concerning Division.

[Communicated in the meeting of the physical-mathematical class of the Academy of Sciences on 20 January 1851.[1]]

In a paper[2] belonging to the year 1849, it was incidentally observed that, in dividing an integer n by all divisors not larger than it, the case in which the remainder lies below half the divisor occurs more often than the opposite case, in which the remainder exceeds or equals half the divisor. It was also shown there that, as n increases, the ratio of the number of divisors for which the first case occurs to their total number n tends to the limit $2 - \log 4 = 0.61370\ldots$. It seems of some interest to generalize the investigation and to determine the number h of those divisors $1, 2, \ldots, p$, where $p \le n$, for which the corresponding remainder has ratio to the divisor less than a given proper fraction α. If square brackets are used to denote the greatest integer contained in the enclosed value, so that $x - [x]$ is always zero or a positive proper fraction, then it is easy to see that the divisor s will or will not have the required property according as the difference

$$\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]$$

is equal to the positive unit or to zero. Thus one has

$$h = \sum_1^p \left(\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]\right) = \sum_1^p \left[\frac{n}{s}\right] - \sum_1^p \left[\frac{n}{s} - \alpha\right],$$

where, here and throughout what follows, the summation sign refers to s. In this form the expression for h is neither well suited to numerical computation, nor does it show how h changes for increasing values of n and p. A form serving this double purpose is obtained by the following transformation, which is also applicable in many other cases.

Let $y = f(x)$ be a function which continually decreases as the variable x increases from $x = \mu$ to $x = p$. The inverse function $x = F(y)$ will plainly have the same character and will likewise continually decrease, while the variable y increases from $y = f(p)$ to $y = f(\mu)$. Taking the constants μ and p to be integers, put, for abbreviation,

$$[f(\mu)] = \nu, \qquad [f(p)] = q,$$

and form the sequence

$$[f(\mu)],\ [f(\mu+1)],\ \ldots,\ [f(s)],\ \ldots,\ [f(p)],$$

[1] In the 1851 volume of the Academy reports the communication is introduced on p. 20 with the words: Herr Lejeune Dirichlet laid before the class the following note on a problem concerning the theory of division. In the printing on pp. 151–154 of volume 47 of Crelle's Journal (1854), Dirichlet altered the title and text somewhat. I have here used this later version almost throughout. K.

[2] On the determination of mean values in number theory. Vol. II, p. 49 of this edition of G. Lejeune Dirichlet's Works. The cited remark is found on p. 57. K.

in which each term is equal to or exceeds the following one. We shall now determine which terms of this sequence are equal to an arbitrary integer t lying between q and ν.

For this purpose first seek the completely determined index s of the term whose value is $\geq t$, while the following term is $< t$. Thus one has

$$[f(s)] \geq t, \qquad [f(s+1)] < t,$$

or, what is the same thing,

$$f(s) \geq t, \qquad f(s+1) < t,$$

and from the hypothesis made on the function $f(x)$ it follows that

$$s \leq F(t), \qquad s+1 > F(t),$$

hence

$$s = [F(t)]\,.$$

Applying this result to t and $t+1$, one sees that the value t belongs only to those terms whose index s satisfies the double condition

$$s > [F(t+1)]\,, \qquad s \leq [F(t)]\,.$$

Because the beginning and end of the sequence are prescribed, this result is modified for $t = \nu$ by requiring the first condition to be $s \geq \mu$, and for $t = q$ by replacing the second condition by $s \leq p$.

Taking this result into account, it is now easy to transform the sum

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s),$$

where $\varphi(s)$ denotes an entirely arbitrary function: first combine all terms for which $[f(s)]$ has one and the same value, and then add all the partial sums so obtained. Put

$$\sum_{1}^{s} \varphi(s) = \Psi(s).$$

Then the partial sum in which $[f(s)]$ has the value t is

$$\begin{aligned} &t\big(\Psi[F(t)] - \Psi[F(t+1)]\big), && \text{if } q < t < \nu, \\ &\nu\big(\Psi[F(\nu)] - \Psi(\mu)\big), && \text{if } t = \nu, \\ &q\big(\Psi(p) - \Psi[F(q+1)]\big), && \text{if } t = q, \end{aligned}$$

and therefore

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s) = q\Psi(p) - \nu\Psi(\mu) + \sum_{q+1}^{\nu} \Psi[F(s)].$$

Now separate, in each of the two sums contained in the expression for h given above, the first μ terms, and apply the formula just found to the remaining terms. This gives

$$\sum_{\mu+1}^{p} \left[\frac{n}{s}\right] = \sum_{q+1}^{\nu} \left[\frac{n}{s}\right] + pq - \mu\nu,$$

where $\left[\frac{n}{\mu}\right] = \nu$ and $\left[\frac{n}{p}\right] = q$, and

$$\sum_{\mu+1}^{p} \left[\frac{n}{s} - \alpha\right] = \sum_{q'+1}^{\nu'} \left[\frac{n}{s+\alpha}\right] + pq' - \mu\nu',$$

where $\left[\frac{n}{\mu}-\alpha\right]=\nu'$ and $\left[\frac{n}{p}-\alpha\right]=q'$.

For abbreviation denote $\nu-\nu'$ by δ and $q-q'$ by ε, so that δ and ε can have only the values 0 or 1. Put the last sum in the form

$$\sum_{q'+1}^{\nu'}\left[\frac{n}{s+\alpha}\right]=\sum_{q+1}^{\nu}\left[\frac{n}{s+\alpha}\right]+\varepsilon\left[\frac{n}{q+\alpha}\right]-\delta\left[\frac{n}{\nu+\alpha}\right],$$

and substitute. One obtains

$$\begin{aligned}h=\sum_{1}^{\mu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s}-\alpha\right]\right)+\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)\\ +\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon-\left(\mu-\left[\frac{n}{\nu+\alpha}\right]\right)\delta.\end{aligned}$$

Although this transformation is valid for all values of p, it is advantageous only when p is greater than $\sqrt{n}$. Under this hypothesis it is most advantageous when, as will be done below, one chooses for the number μ, which has so far been arbitrary, one of the integers adjacent to $\sqrt{n}$. As is easy to see, the number of divisions needed for the exact determination of h is then approximately

$$2\sqrt{n}-\frac{n}{p},$$

whereas the original expression required p divisions.

We shall now, under the hypothesis that p is of higher order than $\sqrt{n}$, that is, that $\frac{p}{\sqrt{n}}$ grows beyond every bound with n, seek to determine the limiting value of the ratio $\frac{h}{p}$ of the number of divisors having the required property to their total number p. In this investigation one may neglect, in the expression for h, all terms whose order is lower than that of p. Dropping the first, whose order does not exceed $\sqrt{n}$, and also the fourth, which can contain only a bounded number of units, one obtains

$$h=\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)+\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon,$$

or also, if one omits the square brackets, which plainly causes only a change whose order does not exceed $\sqrt{n}$,

$$h=n\sum_{q+1}^{\nu}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\varepsilon.$$

If the upper limit ν is changed to ∞, the sum receives the increment

$$\frac{1}{\nu+1}-\frac{1}{\nu+1+\alpha}+\frac{1}{\nu+2}-\frac{1}{\nu+2+\alpha}+\cdots<\frac{1}{\nu+1},$$

whose value, multiplied by n, likewise does not exceed the order $\sqrt{n}$. Thus one obtains

$$\lim\frac{h}{p}=\frac{n}{p}\sum_{q+1}^{\infty}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\frac{\varepsilon}{p}.$$

It remains to distinguish the case where the quotient $\frac{n}{p}$, which by hypothesis is ≥ 1, grows beyond every bound, from the case where this quotient remains finite.

In the first case the second term approaches zero, while the ratio of the sum contained in the first term to $\frac{\alpha}{q}$ tends to unity. Thus the limit of $\frac{h}{p}$ coincides with that of $\frac{n}{pq}\alpha$, that is, with α.

In the second case, where $\frac{n}{p}$, and hence also $q = \left[\frac{n}{p}\right]$, remains finite, it is expedient to bring the limiting value of $\frac{h}{p}$ given directly by the last equation into another form. Instead of the sum one introduces the difference of two others, taken from $s = 1$ to respectively $s = \infty$ and $s = q$, and then expresses the first by an integral. Our equation then becomes

$$\lim \frac{h}{p} = \frac{n}{p} \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi - \frac{n}{p} \sum_1^q \left(\frac{1}{s} - \frac{1}{s + \alpha}\right) + \left(1 - \frac{n}{p} \frac{1}{q + \alpha}\right) \varepsilon,$$

where the integral, which for every rational value of α can be represented by logarithms and circular functions, is a known transcendental function studied many times. If one sets in particular $p = n$, then $q = 1$, $q' = 0$, $\varepsilon = 1$, and the limiting expression passes into

$$\lim \frac{h}{n} = \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi.$$

With the help of the table of these transcendents given by Gauss in the paper *Disquisitiones generales circa seriem infinitam* ...[3], one can easily determine the value of α corresponding to a given value of the integral. For example, one finds that, if for half the divisors $1, 2, \ldots, n$ the ratio of the remainder to the divisor is to lie below α, then $\alpha = 0.384686\ldots$ is required.

[3]Gauss' Works, vol. III, p. 161. K.

X.

On the Composition of Binary Forms of the Second Degree.

P. G. Lejeune Dirichlet.

Commentary, Berlin, Academic Press, 1851.

On the Composition of Binary Forms of the Second Degree[1].

Several years ago, when the problem had arisen of transferring to the theory of complex integers the question of the number of classes of forms of the second degree corresponding to a given determinant, I had to set out the elements of the theory of forms from the beginning, since these had previously been developed only in the case of real integers. With the aid of certain considerations not previously used in this theory, and equally suited to real and complex integers, I succeeded in completing that exposition of the elements in a few pages[2]. There I restricted the investigation to those properties which concern equivalence of forms, transformation, and representation of numbers by forms, since these alone were required for the question to which that paper was devoted. I did not then treat the composition of forms, an argument that the illustrious Gauss, in the fifth section of the *Disquisitiones arithmeticae*, is known to have treated with the greatest generality, but by computations so lengthy that very few have been able to grasp the nature of composition, all the more because the great geometer, as he himself noted, arranged the demonstrations of the more difficult theorems synthetically in the interest of brevity, suppressing the analysis by which they had been discovered. I therefore believe I may trust that a new and wholly elementary exposition of this topic will not be unwelcome to students of analysis.

I.

Before we approach the composition of forms, a few facts, either known or easily deduced from known ones, must be stated first.

Given values $\zeta, \zeta', \zeta'', \ldots$ satisfying the congruence

$$u^2 \equiv D$$

with respect respectively to the moduli $m, m', m'', \ldots$, we shall call them concordant with one another if a root Z of the same congruence, referred to the modulus $mm'm'' \ldots$, can be found such that

$$\zeta \equiv Z \pmod{m}, \qquad \zeta' \equiv Z \pmod{m'}, \qquad \zeta'' \equiv Z \pmod{m''}, \ldots.$$

It will suffice to consider the case in which the moduli $m, m', m'', \ldots$ are odd and prime to D itself.

It is easy to see that the proposed congruences cannot be satisfied unless, with respect to each prime number dividing two or more of $m, m', m'', \ldots$, the corresponding values $\zeta, \zeta', \zeta'', \ldots$ are

[1] In the printing on pp. 155–160 of volume 47 of Crelle's Journal (1854), Dirichlet altered the text in several places. I have here used this later version almost throughout. K.

[2] *Recherches sur les formes quadratiques à coefficients et à indéterminées complexes*, in Crelle's Journal, vol. XXIV. Vol. I, p. 533 of this edition of G. Lejeune Dirichlet's Works. K.

congruent to one another. If this condition holds, then from the given values $\zeta, \zeta', \zeta'', \dots$ one knows the residues $\pi, \pi', \pi'', \dots$ of Z with respect to the distinct prime numbers $p, p', p'', \dots$ dividing the product $mm'm'' \dots$. Since the residues $\pi, \pi', \pi'', \dots$ are plainly roots of our congruence with respect respectively to the moduli $p, p', p'', \dots$, it follows from the elementary theory of congruences that there exists a root, and in fact a unique root, Z satisfying the congruence modulo $mm'm'' \dots$ and congruent to $\pi, \pi', \pi'', \dots$ modulo $p, p', p'', \dots$ respectively. At the same time it is immediate that

$$Z \equiv \zeta \pmod{m}, \qquad Z \equiv \zeta' \pmod{m'}, \qquad Z \equiv \zeta'' \pmod{m''}, \dots.$$

Since the constant term D occurring in our congruence is to retain the same value throughout what follows, we shall, for brevity, denote a given root ζ corresponding to the modulus m by the notation (m, ζ). With this notation introduced, the root Z to be deduced in the indicated way from the mutually concordant given roots $\zeta, \zeta', \zeta'', \dots$, and which we shall say is composed from them, may conveniently be denoted as follows:

$$(m, \zeta)(m', \zeta')(m'', \zeta'') \cdots = (mm'm'' \dots, Z).$$

We further observe that the roots $\zeta, \zeta', \zeta'', \dots$ are always mutually concordant if $m, m', m'', \dots$ are relatively prime to one another; and this remains true in the case that we excluded, in which $m, m', m'', \dots$ are not prime to $2D$. Indeed, then Z is already completely determined modulo $mm'm'' \dots$ by the congruences

$$Z \equiv \zeta \pmod{m}, \qquad Z \equiv \zeta' \pmod{m'}, \qquad Z \equiv \zeta'' \pmod{m''}, \dots,$$

and at the same time one has

$$Z^2 \equiv D \pmod{mm'm'' \dots}.$$

II.

For the forms of the second degree considered here,

$$ax^2 + 2bxy + cy^2,$$

whose determinant is $D = b^2 - ac$, we shall suppose the coefficients a, b, c to be free of a common divisor, since the remaining cases are easily reduced to this one. It is known that forms satisfying the stated condition constitute two orders, according as at least one of the outer coefficients a, c is odd or both are even. For brevity we shall exclude the latter case here, since the arguments applicable to the former apply, with the necessary changes, in the other as well.

If, in the given form, definite relatively prime values are assigned to the indeterminates x, y (which will always be understood), so that

$$ax^2 + 2bxy + cy^2 = m,$$

we shall say that the number m (which is always to be supposed prime to $2D$) is represented by the form. Taking integers ξ, η such that $x\eta - y\xi = 1$, it is known and easily proved that the expression

$$\zeta = (ax + by)\xi + (bx + cy)\eta$$

is a root of the congruence $u^2 \equiv D \pmod{m}$, and that the same root is always obtained in this way, however ξ and η may vary. The root (m, ζ) to which we shall say the given representation belongs can be defined in another way, more suited to our purpose. In the equation by which we have displayed ζ, multiply by y and put $x\eta - 1$ in place of the product $y\xi$; then one sees that

$$ax + by \equiv -y\zeta \pmod{m}. \tag{1}$$

This congruence fully determines ζ, provided that y is prime to the modulus m. If, however, y has with the modulus a greatest common divisor δ greater than unity, which plainly also divides a, our congruence gives only

$$\frac{ax+by}{\delta} \equiv -\frac{y}{\delta}\zeta \pmod{\frac{m}{\delta}},$$

so that the residues of ζ cannot be known in this way with respect to those prime divisors of δ, if there are any, which do not divide $\frac{m}{\delta}$. This inconvenience is easily remedied if one observes that the equations above imply

$$\zeta \equiv bx\eta, \qquad x\eta \equiv 1 \pmod{\delta},$$

and hence

$$\zeta \equiv b \pmod{\delta}. \tag{2}$$

Together with the preceding formula, this now fully determines the residues of ζ with respect to the individual divisors of m. We observe that, if ε is a common divisor of x and m, one similarly obtains

$$\zeta \equiv -b \pmod{\varepsilon}. \tag{3}$$

We add the following facts; although they are sufficiently well known, it is useful to put them in view here.

1°. If two forms are equivalent, properly (as will always be understood), and the former passes into the latter by the substitution

$$x = \alpha x' + \beta y', \qquad y = \gamma x' + \delta y',$$

where $\alpha\delta - \beta\gamma = 1$, then clearly a number m representable by one of them can also be represented by the other. It is also easy to prove that the two representations of the same number, connected by the preceding equations, belong to the same root (m, ζ). Hence the representations belonging to a given expression (m, ζ) must be referred to one whole class, and that class is unique.

2°. Conversely, one can prove that if the same number m is represented by two forms of the same determinant, and the two representations belong to the same root (m, ζ), then the forms are equivalent.

3°. Finally, it is clear that, given any root (m, ζ), there exists a class by whose forms m can be represented in such a way that the root corresponding to these representations is (m, ζ). Indeed m is plainly represented by the form

$$\left(m, \zeta, \frac{\zeta^2 - D}{m}\right),$$

whose coefficients are free of a common divisor and in which m is odd, by putting

$$x = 1, \qquad y = 0;$$

it is evident that this representation belongs to (m, ζ).

III.

With these preliminaries completed, let us approach the proposed question. Let two forms φ and φ' of the same determinant D be given, and let m and m' be any two odd integers prime to D, representable respectively by φ and φ' in such a way that the roots (m, ζ) and (m', ζ') to which these representations belong are mutually concordant. Under these hypotheses, the whole matter turns on proving that the representations of mm' belonging to the expression

$(m, \zeta)(m', \zeta')$ will always be effected by the same form, or rather are to be referred to the same class, however m and m' may vary. This is the fundamental theorem in this theory.

Suppose that the given forms (a, b, c) and (a', b', c') have been so prepared that the expressions (a, b) and (a', b') are mutually concordant. This is achieved, for example, by transforming one of the forms into an equivalent one whose first coefficient has no common divisor with the first coefficient of the other. But it must be carefully noted that the analysis developed below requires nothing except that (a, b) and (a', b') be mutually concordant; it is not necessary that a and a' be relatively prime to one another, nor that they be prime to $2D$. Denote by (aa', B) the expression arising from the composition of (a, b) and (a', b'), so that

$$B \equiv b \pmod{a}, \qquad B \equiv b' \pmod{a'}, \qquad D = B^2 - aa'C,$$

where C is an integer. It is then clear that the forms can be changed into the following equivalent forms:

$$ax^2 + 2Bxy + a'Cy^2 = \varphi, \qquad a'x'^2 + 2Bx'y' + aCy'^2 = \varphi'.$$

First multiplying these respectively by a and a', and then multiplying the resulting equations together, one obtains

$$(ax + By)^2 - Dy^2 = a\varphi, \qquad (a'x' + By')^2 - Dy'^2 = a'\varphi',$$

$$((ax + By)(a'x' + By') + Dyy')^2 - D((ax + By)y' + (a'x' + By')y)^2 = aa'\varphi\varphi'.$$

Now, observing that $D = B^2 - aa'C$, one has

$$(ax + By)(a'x' + By') + Dyy' = aa'X + BY, \tag{4}$$

where

$$X = xx' - Cyy', \qquad Y = (ax + By)y' + (a'x' + By')y = axy' + a'x'y + 2Byy'. \tag{5}$$

The last equation, divided by aa', assumes the form

$$aa'X^2 + 2BXY + CY^2 = \varphi\varphi' = \psi. \tag{6}$$

After we have displayed indefinitely the product of the forms φ and φ' by the form ψ of the same determinant D, suppose that both x, y and x', y' are assigned definite relatively prime values for which $\varphi = m$ and $\varphi' = m'$, and which satisfy the conditions indicated above. Then it remains to prove: 1° that the representation of mm' by the form ψ, which we see is supplied by equations (5) and (6), will be proper, that is, that X and Y will be free of a common divisor; and 2° that the root to which this representation belongs, say (mm', Z), will indeed be composed from the roots (m, ζ) and (m', ζ') to which the representations of m and m' are assumed to belong.

1°. To show that X and Y have no common divisor, let us start from the supposition that a prime number p divides both, and see what follows. Since p must divide one of m, m', let us suppose, as we may, that m is a multiple of p. I then say that m' too may be supposed divisible by p. Since X and Y are divisible by p, it is clear, multiplying the first of equations (5) by $-ay'$ and the second by x' and adding, that an integer divisible by p is obtained:

$$(a'x'^2 + 2Bx'y' + aCy'^2)y = m'y.$$

If one were now to suppose that m' is not divisible by p, then y, and consequently also a, would be multiples of p. But then, from

$$xx' - Cyy' = X \equiv 0 \pmod{p},$$

and from the fact that x is prime to y, it follows that $x' \equiv 0$, whence finally

$$m' = a'x'^2 + 2Bx'y' + aCy'^2$$

is seen to be a multiple of p. Since p divides both m and m', by equation (1) we have

$$ax + By \equiv -y\zeta, \qquad a'x' + By' \equiv -y'\zeta' \pmod{p}.$$

Substitution of these formulas into the second of equations (5) gives the congruence

$$(\zeta + \zeta')yy' \equiv 0 \pmod{p}.$$

If $\zeta + \zeta' \equiv 0$, then $\zeta' \equiv -\zeta$, contrary to the hypothesis that the roots ζ and ζ' are concordant, namely that $\zeta' \equiv \zeta$. It remains to see what follows from either of the suppositions $y \equiv 0$ and $y' \equiv 0$, which are entirely similar. If y, and therefore also x' by the congruence $xx' - Cyy' \equiv 0$, were divisible by p, then by equations (2) and (3) we would have

$$\zeta \equiv B, \qquad \zeta' \equiv -B \pmod{p},$$

and just as above

$$\zeta + \zeta' \equiv 0 \pmod{p}.$$

It is therefore established that X and Y are relatively prime.

2°. We now prove that the root to which the representation of mm' belongs, and which we have denoted by (mm', Z), is indeed composed from (m, ζ) and (m', ζ'). By symmetry it suffices to show that $Z \equiv \zeta$ with respect to any prime divisor p of m. Since by equation (1)

$$ax + By \equiv -\zeta y \pmod{p},$$

equation (4) and the second of equations (5) readily give the congruences

$$(-\zeta(a'x' + By') + Dy')y \equiv aa'X + BY, \qquad (-\zeta y' + a'x' + By')y \equiv Y \pmod{p}.$$

Multiplying the latter by ζ and adding it to the former, and taking account of the formula $\zeta^2 \equiv D \pmod{m}$, one obtains

$$aa'X + BY \equiv -Y\zeta \pmod{p}.$$

Comparing this congruence with

$$aa'X + BY \equiv -YZ \pmod{p},$$

one sees that

$$Z \equiv \zeta \pmod{p},$$

provided p does not divide Y. It remains to consider the case in which Y is divisible by p. In this case, by equation (2), $Z \equiv B \pmod{p}$. If now y is also a multiple of p, then similarly $\zeta \equiv B \pmod{p}$, and hence

$$Z \equiv \zeta \pmod{p}.$$

If, however, y is not divisible by p, the congruence found above gives

$$a'x' + By' - \zeta y' \equiv 0 \pmod{p},$$

from which it is easily deduced that p is a factor of the product $a'm'$. For

$$a'm' = (a'x' + By')^2 - Dy'^2 = (a'x' + By')^2 - \zeta^2 y'^2 \equiv 0 \pmod{p}.$$

Two cases must now be distinguished. First suppose that a' is not divisible by p. Since in this case m' is divisible by p, one has

$$a'x' + By' + \zeta'y' \equiv 0 \pmod{p}.$$

Comparing this formula with the preceding one gives

$$(\zeta + \zeta')y' \equiv 0 \pmod{p},$$

which leads to an absurd conclusion. For since $\zeta + \zeta'$ cannot be a multiple of p, it would follow that $y' \equiv 0 \pmod{p}$, and hence, from the equation

$$m' = a'x'^2 + 2Bx'y' + aCy'^2,$$

that $a' \equiv 0 \pmod{p}$, contrary to the hypothesis. Therefore this case cannot occur. In the remaining case, where a' is divisible by p, the congruence

$$a'x' + By' - \zeta y' \equiv 0 \pmod{p}$$

gives

$$(B - \zeta)y' \equiv 0 \pmod{p}.$$

If p does not divide y', one has

$$\zeta \equiv B \pmod{p},$$

which agrees with the congruence $Z \equiv B \pmod{p}$. If, on the other hand, p divides y', and consequently also m', then by equation (2) one has

$$\zeta' \equiv B \pmod{p},$$

and, as above, one concludes that $\zeta \equiv B \pmod{p}$, since the roots ζ and ζ' are concordant.

XI.

On the Motion of a Rigid Body in an Incompressible Fluid Medium.

By G. Lejeune Dirichlet.

Report on the Proceedings of the Royal Prussian Academy of Sciences, 1852, pp. 12–17.

On the Motion of a Rigid Body in an Incompressible Fluid Medium.[1]

It appears that up to now, for no case, however simple, has the resistance which a rigid body moved through a fluid at rest experiences from that fluid been derived from the general equations of hydrodynamics known since Euler; or, what amounts essentially to the same thing, there is no example of a purely theoretical determination of the modifications which an immovable rigid body situated in the interior of a fluid produces in the progressive motion of that fluid. A notable mathematician, very practiced in investigations of this kind,[2] has therefore been led to the opinion that, for the problem just mentioned, even in the case where the fluid is of infinite extent and the rigid body is given the simplest shape, the known integration methods do not suffice. This opinion, however, is unfounded, and the problem can be treated completely when the rigid body has the shape of a sphere, or also that of an ellipsoid. Of these two cases only the first, as the simpler one, will be discussed in this notice.

To treat the problem first in the second of the forms mentioned above, let c be the radius of the immovable sphere and let its center be the origin of rectangular axes x, y, z. Let an accelerating force σ act on the initially resting homogeneous fluid, whose density is denoted by ϱ, and suppose that at the same time t this force has everywhere the same intensity and direction, but may change arbitrarily with time, so that its components α, β, γ are given functions of t. If p denotes the pressure occurring at the point (x, y, z) after the time t, and u, v, w the three components of the velocity, set, for abbreviation,

$$r = \sqrt{x^2 + y^2 + z^2},$$

$$\lambda = \int_0^t \alpha\, dt, \qquad \mu = \int_0^t \beta\, dt, \qquad \nu = \int_0^t \gamma\, dt,$$

$$\varphi = \left(1 + \frac{c^3}{2r^3}\right)(\lambda x + \mu y + \nu z).$$

Then, for the determination of p, u, v, w, one has

$$u = \frac{\partial\varphi}{\partial x} = \left(1 + \frac{c^3}{2r^3}\right)\lambda - \frac{3c^3 x}{2r^5}(\lambda x + \mu y + \nu z),$$

$$v = \frac{\partial\varphi}{\partial y} = \left(1 + \frac{c^3}{2r^3}\right)\mu - \frac{3c^3 y}{2r^5}(\lambda x + \mu y + \nu z),$$

[1] In the 1852 volume of the Academy reports, the communication is introduced on p. 12 with the words: Herr Lejeune Dirichlet read on some cases in which the motion of a rigid body in an incompressible fluid medium can be determined theoretically. K.

[2] Résumé des leçons de Mécanique données à l'École Polytechnique by M. Navier, p. 480.

$$w = \frac{\partial \varphi}{\partial z} = \left(1 + \frac{c^3}{2r^3}\right)\nu - \frac{3c^3 z}{2r^5}(\lambda x + \mu y + \nu z),$$

$$\frac{1}{\varrho} p = T - \frac{c^3}{2r^3}(\alpha x + \beta y + \gamma z) - \frac{1}{2}(u^2 + v^2 + w^2),$$

where T denotes an arbitrary quantity depending only on t, which, as Euler already observed, remains undetermined in every problem of hydrodynamics so long as the pressure at a point of the surface of the fluid is not prescribed for every instant. This indeterminacy in the expression for p plainly has no influence on the result obtained when all elementary pressure forces acting on a closed surface are combined into a single force and a couple. For the surface of our sphere, all these elementary forces have a single resultant, passing through the center; and for its components one finds, by a very simple calculation, the values

$$\frac{2\pi}{3} c^3 \varrho\alpha, \qquad \frac{2\pi}{3} c^3 \varrho\beta, \qquad \frac{2\pi}{3} c^3 \varrho\gamma.$$

Thus for the resultant one obtains the expression

$$\frac{2\pi}{3} c^3 \varrho\sigma,$$

and at the same time it is clear that it is parallel to the force σ. Hence in the present case the pressure exerted by the moving fluid on the surface of the rigid body depends only on the accelerating force acting at each instant, and not on the velocity of the fluid. If the accelerating force ceases to act, the pressure also vanishes, λ, μ, ν become constant, and the motion immediately becomes permanent, so that u, v, w no longer depend on t. The permanent motion which then occurs is always of the same kind, if one takes account only of the ratios of the velocities and, for the paths described by the fluid particles, only of the relative position of these curves and not their absolute position in space.

All these curves are plane, and all their planes always pass through a fixed straight line, which may be regarded as the axis of the system of curves. Around this axis the curves lie symmetrically: the curves lying in one plane generate all the others by rotation about the axis. The direction of the axis, which passes through the center of the sphere, is determined by the three angles it forms with the coordinate axes, and the cosines of these angles are equal to

$$\frac{\lambda}{\sqrt{\lambda^2 + \mu^2 + \nu^2}}, \qquad \frac{\mu}{\sqrt{\lambda^2 + \mu^2 + \nu^2}}, \qquad \frac{\nu}{\sqrt{\lambda^2 + \mu^2 + \nu^2}},$$

where by λ, μ, ν one is to understand the values of these quantities after they have become constant. The equations of the curves just mentioned take a very simple form if polar coordinates are introduced. Draw any plane through the axis, and let θ denote the angle between the axis and the radius vector r drawn in that plane to the point in question. Then all the curves are contained in the equation

$$(r^3 - c^3)\sin^2\theta = \varepsilon r,$$

where the parameter ε must run through the interval from 0 to ∞ in order that the equation represent all curves lying in the plane. One sees without difficulty that for large ε these curves more and more assume the shape of straight lines parallel to the axis, while as ε decreases the form of such a curve approaches ever more closely a semicircle, which continues at its two endpoints along the prolonged diameter.

To derive from the results just indicated the simplest case of the motion of a rigid body in a resting fluid, imagine the sphere homogeneous, or at least suppose that its center of gravity coincides with its center. Let ϱ' denote the mean density of the sphere, and let an accelerating force act on all parts of it whose components are

$$-\frac{\varrho}{2\varrho'}\alpha, \qquad -\frac{\varrho}{2\varrho'}\beta, \qquad -\frac{\varrho}{2\varrho'}\gamma.$$

Then the pressure exerted by the fluid is cancelled at every instant, and the sphere may be regarded as movable without ceasing to be at rest and without modifying the motion of the fluid determined above. Now let the accelerating force with components $-\alpha, -\beta, -\gamma$ be imagined as acting everywhere on the whole mechanical system formed by the sphere and the fluid, which has the character of a so-called free system. By composition one obtains the motion arising in our initially resting system for the case where an accelerating force acts on the sphere whose components have the values

$$-\left(1+\frac{\varrho}{2\varrho'}\right)\alpha, \qquad -\left(1+\frac{\varrho}{2\varrho'}\right)\beta, \qquad -\left(1+\frac{\varrho}{2\varrho'}\right)\gamma,$$

while the fluid is driven by no force. In this motion the components of the velocity at time t for all parts of the sphere are $-\lambda, -\mu, -\nu$, whereas in the fluid, at the same time and at the point determined by the coordinates

$$x-\int_0^t \lambda\, dt, \qquad y-\int_0^t \mu\, dt, \qquad z-\int_0^t \nu\, dt,$$

the velocity components have the values

$$u-\lambda, \qquad v-\mu, \qquad w-\nu.$$

Thus the sphere, under the action of the accelerating force $\left(1+\frac{\varrho}{2\varrho'}\right)\sigma$, arbitrarily variable in direction and intensity, moves in the resisting medium exactly as it would move in empty space if, at each instant, the force were diminished, with its direction retained, in the ratio of $2\varrho'+\varrho$ to $2\varrho'$. The additional force $\frac{\varrho}{2\varrho'}\sigma$ is therefore expended in overcoming the resistance of the fluid. But this resistance does not correspond to the usual conception of the action of a fluid medium on a rigid body moving in it, according to which a resistance is already present and must be overcome even if the motion occurring at one instant is not to be altered during the next element of time. In contrast, in our case the motion of the rigid body immediately becomes rectilinear and uniform as soon as the accelerating force ceases to act. The resistance here does not depend at all on the existing motion, but only on the change of motion to be produced in the next element of time, and it is always equal to the same aliquot part of the force that would produce the same change in empty space, directed exactly opposite to it–a fact which one could hardly have expected in the case where the velocity is to be diminished.

XII.

On the First of the Proofs Given by Gauss of the Reciprocity Law in the Theory of Quadratic Residues.

By G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, vol. 47, pp. 139–150.

On the First of the Proofs Given by Gauss of the Reciprocity Law in the Theory of Quadratic Residues.

Among the numerous proofs of the fundamental theorem in the theory of quadratic congruences, the earliest, found by Gauss already in 1796 and published in the *Disquisitiones arithmeticae* (Section IV),[1] has always seemed especially remarkable to me: both because of the very simple idea on which it rests, and because, so far as I know, this proof is the only one in which the argument never leaves the domain of congruences of the second degree, to which the theorem essentially belongs, whereas the other ways of grounding it rest on principles that appear more or less foreign to this domain.[2] If this beautiful proof lacks the brevity by which some of the later ones are so highly distinguished, that defect is not inherent in the method. Rather it is due to the accidental circumstance that no notation suitable for calculation is used to express certain relations that recur frequently in this treatment; this made it necessary to distinguish eight different cases, each of which again splits into several subdivisions. By introducing the symbol first used by Legendre,[3] in the more general meaning later given to it by Jacobi, and by making a few other simplifications which nevertheless do not alter the nature of the proof, the proof contracts so much that it seems scarcely inferior to any of the others in brevity, provided one does not ignore the fact that part of the developments occurring in it remains indispensable for the theory of quadratic residues even when another method of proof is chosen for the reciprocity law. I have thought myself justified in devoting a few pages to a new presentation of this subject simplified in the indicated way, all the more because experience has repeatedly shown me how much the understanding of the earlier proof is made difficult for beginning mathematicians by the large number of cases distinguished in it.

§. 1.

In this first paragraph a few elementary propositions and definitions needed below will be stated.

According as the congruence

$$x^2 \equiv k \pmod{m},$$

in which k denotes a positive or negative integer, is possible or impossible, k is called a quadratic residue or non-residue of the modulus m, whose sign is of course immaterial. For our purpose

[1] Gauss' Works, vol. I, pp. 104–111. K.

[2] Gauss himself judged his first proof in a later paper (Comment. soc. Gott. vol. XVI, p. 70) as follows: "Sed omnes hae demonstrationes, etsi respectu rigoris nil desiderandum relinquent, quaeri videantur, e principiis nimis heterogeneis derivatae sunt, prima forsitan excepta, quae tamen per ratiocinia magis laboriosa procedit, operationibusque prolixioribus premitur."

[3] Gauss' Works, vol. II, p. 4. K.

it is permissible always to suppose that k and m have no common divisor. In the especially important case where m is an odd prime p, the symbol

$$\left(\frac{k}{p}\right)$$

will denote the positive or negative unit according as k is a quadratic residue or non-residue of p.

The well-known theorem that the product $k'k''\ldots$ is a quadratic residue or non-residue of p according as those factors $k', k'', \ldots$ that are quadratic non-residues of p occur in even or odd number is expressed, with the help of our notation, by

$$\left(\frac{k'k''\ldots}{p}\right) = \left(\frac{k'}{p}\right)\left(\frac{k''}{p}\right)\cdots.$$

For the possibility of the congruence

$$x^2 \equiv k \pmod{p^\varpi},$$

where $\varpi > 1$, the condition

$$\left(\frac{k}{p}\right) = 1$$

is plainly necessary; and one also easily proves that, once this condition is fulfilled, the congruence is soluble for every value of ϖ.

The congruence

$$x^2 \equiv k \pmod{2^\varpi}$$

behaves differently. For $\varpi = 1$ the odd number k has no condition to satisfy; if $\varpi = 2$ or $\varpi \geq 3$, the necessary and sufficient condition for solubility is respectively that k have the form

$$4\varrho + 1 \quad \text{or} \quad 8\varrho + 1.$$

If the modulus in our congruence is the product of powers of different primes, then for its possibility it is necessary and sufficient that it be soluble modulo the several prime powers.

We shall now give the symbol $\left(\frac{k}{p}\right)$, which was defined above for the case where p denotes an odd prime not dividing the positive or negative number k, a more general meaning. Suppose that the odd number m, whose sign is immaterial, has no common divisor with k and is decomposed into its equal or unequal prime factors $p', p'', p''', \ldots$, so that $m = p'p''p''' \cdots$. By the symbol $\left(\frac{k}{m}\right)$ we shall understand the product

$$\left(\frac{k}{p'}\right)\left(\frac{k}{p''}\right)\left(\frac{k}{p'''}\right)\cdots.$$

It is easy to see that, in such a symbol, one may replace k by any number congruent to it modulo m, and that for such symbols the two equations

$$\left(\frac{k}{m}\right)\left(\frac{l}{m}\right) = \left(\frac{kl}{m}\right), \qquad \left(\frac{k}{m}\right)\left(\frac{k}{m'}\right) = \left(\frac{k}{mm'}\right)$$

hold.

But it must not be overlooked that $\left(\frac{k}{m}\right)$ now no longer has the same relation to the congruence

$$x^2 \equiv k \pmod{m}$$

as before, when m was prime. If the congruence is soluble, it still follows that $\left(\frac{k}{m}\right) = 1$, since then all the factors whose product defines $\left(\frac{k}{m}\right)$ are equal to the positive unit. But the converse conclusion from the condition $\left(\frac{k}{m}\right) = 1$ to the possibility of the congruence is plainly not permitted.

Finally, still assuming m odd, note that the possibility of the congruence

$$lx^2 \equiv k \pmod{m},$$

when k and l are relatively prime to m, has as consequence the equation

$$\left(\frac{kl}{m}\right) = 1.$$

This is immediate after bringing the congruence into the form

$$(lx)^2 \equiv kl \pmod{m}.$$

§. 2.

We now come to the proper subject of this paper: the criteria by which, for a given positive or negative number k, the simple odd moduli p of which k is a quadratic residue are distinguished from those with respect to which k has the opposite behavior.

Since, by the theorem cited above, the required distinction reduces to the corresponding one for the prime factors of k, we have only to investigate the three cases in which k is equal to one of the numbers -1, 2, or to a positive odd prime q. For these three cases the desired criteria are contained in the following equations, where the odd prime p is supposed different from q and, like q, positive:

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{1}{2}(p-1)}, \tag{a}$$

$$\left(\frac{2}{p}\right) = (-1)^{\frac{1}{8}(p^2-1)}, \tag{b}$$

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{1}{2}(p-1)\frac{1}{2}(q-1)}. \tag{c}$$

According to the first of these equations, $\left(\frac{-1}{p}\right)$ is $+1$ or -1 according as p is contained in the form $4\mu+1$ or $4\mu+3$. According to the second, one has

$$\left(\frac{2}{p}\right) = 1 \quad \text{for } p = 8\mu+1, 8\mu+7,$$

and

$$\left(\frac{2}{p}\right) = -1 \quad \text{for } p = 8\mu+3, 8\mu+5.$$

According to the third, one always has

$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right),$$

except when both primes p, q have the form $4\mu+3$, in which case

$$\left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right).$$

The equations (a) and (b) are easy to prove in certain special cases: for the first, in the case where p has the form $4\mu+3$, so that $\left(\frac{-1}{p}\right)$ must be -1. If this did not hold for all primes $4\mu+3$, let p be the least for which $\left(\frac{-1}{p}\right) = 1$. Then one can set

$$e^2 + 1 = ph,$$

and if, as can always be done, $e < p$ is chosen even, then also $h < p$ and h is of the form $4\mu + 3$. Hence h has a prime factor $r < p$ of the same form $4\mu + 3$, for which the equation $e^2 + 1 = ph$ gives

$$\left(\frac{-1}{r}\right) = 1,$$

contrary to our supposition.

Secondly, to prove that always

$$\left(\frac{2}{p}\right) = -1$$

when p is contained in one of the forms $8\mu + 3$, $8\mu + 5$, suppose that there are primes of one of these forms for which the theorem fails, and let p be the least. Then one may again set

$$e^2 - 2 = ph.$$

If e is chosen odd and also smaller than p, then, since

$$e^2 - 2 = 8\mu + 7,$$

the number h will have the form $8\mu + 5$ or $8\mu + 3$ according as p has the form $8\mu + 3$ or $8\mu + 5$, and it will also be smaller than p. Since no number of the form $8\mu + 3$ or $8\mu + 5$ can be formed from prime factors all contained in the forms $8\mu + 1$, $8\mu + 7$, there is therefore a prime factor r of h that has the form $8\mu + 3$ or $8\mu + 5$, and for which, by the preceding equation, contrary to our supposition, $\left(\frac{2}{r}\right) = 1$.

The proof, to be conducted in an entirely similar way, that

$$\left(\frac{-2}{p}\right) = -1$$

when p has one of the forms $8\mu + 5$, $8\mu + 7$ will be omitted for brevity. If this latter result is put, for $p = 8\mu + 7$, into the form

$$\left(\frac{2}{p}\right)\left(\frac{-1}{p}\right) = -1,$$

and if we note that, by what was proved above, since $8\mu + 7$ is a special case of the form $4\mu + 3$,

$$\left(\frac{-1}{p}\right) = -1,$$

then for primes $p = 8\mu + 7$ one obtains

$$\left(\frac{2}{p}\right) = 1.$$

Thus theorem (b) has now been proved for all primes not contained in the form $8\mu + 1$.

§. 3.

Before we can undertake to prove generally the equations stated in the preceding paragraph, a few consequences are to be drawn from those equations, provisionally assumed correct. We first make the following observation.

If one wants to determine, for a product of odd factors r,

$$R = \Pi r,$$

the remainder modulo 4, and writes each factor r in the form $(r-1)+1$, then in multiplying one may omit all terms in which first parts of these binomials are multiplied together. Thus

$$R \equiv 1 + \Sigma(r-1) \pmod 4,$$

or $\frac{1}{2}(R-1)$ and $\Sigma\frac{1}{2}(r-1)$ are of the same parity, that is, either both even or both odd.

In the same way, from $R^2 = \Pi r^2$, since every odd square r^2 has the form $8\mu + 1$, it follows that

$$R^2 \equiv 1 + \Sigma(r^2-1) \pmod{64},$$

and hence again that $\frac{1}{8}(R^2-1)$ and $\Sigma\frac{1}{8}(r^2-1)$ have the same parity.

With the help of this observation it is now easy to derive from the equations above the following more general equations of the same form, in which P and Q denote any two positive odd numbers with no common divisor:

$$\left(\frac{-1}{P}\right) = (-1)^{\frac{1}{2}(P-1)}, \tag{a'}$$

$$\left(\frac{2}{P}\right) = (-1)^{\frac{1}{8}(P^2-1)}, \tag{b'}$$

$$\left(\frac{Q}{P}\right)\left(\frac{P}{Q}\right) = (-1)^{\frac{1}{2}(P-1)\frac{1}{2}(Q-1)}. \tag{c'}$$

Putting $P = \Pi p$, where p denotes each equal or unequal prime factor contained in P, applying theorem (a) to each p, and multiplying all the resulting equations together, one obtains

$$\left(\frac{-1}{P}\right) = (-1)^{\Sigma\frac{1}{2}(p-1)},$$

which passes into equation (a') when the exponent is replaced by the number of the same parity $\frac{1}{2}(P-1)$. The correctness of (b') is seen in the same way.

To prove (c'), decompose also Q into its simple factors q. The left-hand side can then be represented as a product of expressions of the form $\left(\frac{q}{p}\right)\left(\frac{p}{q}\right)$, where each p is combined with each q. Replacing every such expression by its value given by formula (c) gives

$$\left(\frac{Q}{P}\right)\left(\frac{P}{Q}\right) = (-1)^{\Sigma\frac{1}{2}(p-1)\frac{1}{2}(q-1)}.$$

The summation sign here embraces all combinations p, q. The sum is therefore the product of the factors

$$\Sigma\frac{1}{2}(p-1) \quad \text{and} \quad \Sigma\frac{1}{2}(q-1),$$

for which one may substitute the numbers of the same parity

$$\frac{1}{2}(P-1) \quad \text{and} \quad \frac{1}{2}(Q-1),$$

and thus obtains (c').

Multiplying (a') and (c') gives

$$\left(\frac{-Q}{P}\right)\left(\frac{P}{Q}\right) = (-1)^{\frac{1}{2}(P-1)\frac{1}{2}(Q+1)}$$

or

$$\left(\frac{-Q}{P}\right)\left(\frac{P}{-Q}\right) = (-1)^{\frac{1}{2}(P-1)\frac{1}{2}(-Q-1)}.$$

Thus equation (c') remains valid when it is applied to the positive number P and the negative number $-Q$, and in that case it is a consequence of (a') and of the original equation (c'). Conversely, the latter is plainly a consequence of (a') and of equation (c') applied to the combination $P, -Q$. The same observation naturally applies also to equations (a) and (c), which are contained in those presently under consideration.

§. 4.

To prove the theorems (a) and (c) in general, suppose that both hold up to, but not including, an arbitrary positive odd prime q: that is, the first holds for every positive odd prime less than q, and the second for every two such primes. If one can derive from this assumption that (a) holds for q, and that (c) holds for every combination p, q with p a positive odd prime less than q, then both theorems, since they are plainly true for the first primes, are proved in general.

It is easy to see that the required verification reduces to the following two points.

First, if for a suitably chosen sign in $\varpi = \pm p$ one has

$$\left(\frac{\varpi}{q}\right) = 1,$$

then, according to equation (c) applied to the combination q, ϖ, one must have

$$\left(\frac{q}{\varpi}\right) = \left(\frac{\varpi}{q}\right)(-1)^{\frac{1}{2}(q-1)\frac{1}{2}(\varpi-1)} = (-1)^{\frac{1}{2}(q-1)\frac{1}{2}(\varpi-1)}.$$

Second, when q has the form $4\mu + 1$ and $\left(\frac{p}{q}\right) = -1$, it remains to derive $\left(\frac{q}{p}\right) = -1$, and at the same time to show that this second case occurs for every q of the form $4\mu + 1$, that is, that among the primes $p < q$ there is always at least one which is a quadratic non-residue of q.

If q has the form $4\mu + 3$, then (a) has already been proved for q; hence, by the observation at the end of the preceding paragraph, it is immaterial for which of the combinations p, q and $-p, q$ one proves equation (c). Since

$$\left(\frac{-1}{q}\right) = -1,$$

it follows that

$$\left(\frac{-p}{q}\right) = -\left(\frac{p}{q}\right),$$

so one of these combinations is always contained in the first case.

If, on the other hand, q has the form $4\mu + 1$, then from the assumption

$$\left(\frac{p}{q}\right) = 1$$

the first case (with $\varpi = p$) gives

$$\left(\frac{q}{p}\right) = 1,$$

and from the assumption

$$\left(\frac{p}{q}\right) = -1$$

the second gives

$$\left(\frac{q}{p}\right) = -1,$$

as required. That moreover

$$\left(\frac{-1}{q}\right) = 1$$

then follows as follows. For a prime p belonging to the second case one has

$$\left(\frac{p}{q}\right) = -1, \qquad \left(\frac{q}{p}\right) = -1.$$

If now

$$\left(\frac{-1}{q}\right) = -1,$$

and consequently

$$\left(\frac{-p}{q}\right) = 1,$$

then the first case would give

$$\left(\frac{q}{p}\right) = 1,$$

which contradicts the result derived from the second case.

Before carrying out the two points just indicated, one must note that the hypothesis made at the beginning of this paragraph, together with the derivation in the preceding paragraph of (a') and (c') from (a) and (c), plainly implies the correctness of (c') for any two odd relatively prime numbers, provided they are not both negative and all prime factors contained in them are less than q.

§. 5.

Under the condition of the first case,

$$\left(\frac{\varpi}{q}\right) = 1,$$

one can set

$$e^2 - \varpi = qf.$$

In this equation, as can always be done, choose e even and smaller than q; then f is odd, positive[4] (since the numerical value of ϖ is smaller than q), and also smaller than q. The equation requires a separate treatment according as f is or is not divisible by ϖ.

I. If f is not divisible by ϖ, then by §. 1 one has

$$\left(\frac{\varpi}{f}\right) = 1, \qquad \left(\frac{qf}{\varpi}\right) = \left(\frac{q}{\varpi}\right)\left(\frac{f}{\varpi}\right) = 1.$$

Multiplication and application of (c') to the combination f, ϖ gives

$$\left(\frac{q}{\varpi}\right) = (-1)^{\frac{1}{2}(f-1)\frac{1}{2}(\varpi-1)}.$$

On the other hand, from the equation above, in which e is even, it follows that

$$\frac{1}{2}(f-1) \quad \text{and} \quad \frac{1}{2}(q-1) + \frac{1}{2}(\varpi+1)$$

have the same parity. Substituting the latter for the former in the exponent and omitting $\frac{1}{4}(\varpi^2 - 1)$ as even gives, as required,

$$\left(\frac{q}{\varpi}\right) = (-1)^{\frac{1}{2}(q-1)\frac{1}{2}(\varpi-1)}.$$

II. If f contains the factor ϖ, put

$$e = \varpi\varepsilon, \qquad f = \varpi\varphi,$$

[4]In what follows, Latin letters denote essentially positive numbers, whereas Greek letters denote numbers that may be positive or negative.

so that ϖ and φ have the same sign. From the resulting equation

$$\varpi\varepsilon^2 - 1 = q\varphi$$

one obtains

$$\left(\frac{\varpi}{\varphi}\right) = 1, \qquad \left(\frac{-q\varphi}{\varpi}\right) = \left(\frac{q}{\varpi}\right)\left(\frac{-\varphi}{\varpi}\right) = 1.$$

Multiplication and application of (c') to the numbers $\varpi, -\varphi$, of which only one is negative, gives

$$\left(\frac{q}{\varpi}\right) = (-1)^{\frac{1}{2}(\varpi-1)\frac{1}{2}(\varphi+1)},$$

or, since $\frac{1}{2}(q-1)$ and $\frac{1}{2}(\varphi+1)$ plainly have the same parity,

$$\left(\frac{q}{\varpi}\right) = (-1)^{\frac{1}{2}(q-1)\frac{1}{2}(\varpi-1)}.$$

§. 6.

The hypothesis occurring in the second case,

$$\left(\frac{p}{q}\right) = -1, \qquad q = 4\mu + 1,$$

does not, as in the first case, immediately allow one to write down an equation. One must first prove the existence of an auxiliary prime p', smaller than q, satisfying

$$\left(\frac{q}{p'}\right) = -1.$$

If q has the form $8\mu + 5$, this presents no difficulty: then $q - 2$ has the form $8\mu + 3$ and consequently has a prime factor p' of one of the forms $8\mu + 3$, $8\mu + 5$, smaller than q, for which $q \equiv 2 \pmod{p'}$ and hence $\left(\frac{q}{p'}\right) = \left(\frac{2}{p'}\right)$, or, by §. 2, $\left(\frac{q}{p'}\right) = -1$.

It is not so easy to show the existence of p' when q has the form $8\mu + 1$. The need to provide this proof for that case was perhaps the greatest difficulty Gauss had to overcome in his first foundation of the reciprocity theorem. He reached it by a very sharp argument, which essentially amounts to the following.

Let $2m + 1 < q$, and suppose that q is a quadratic residue of all odd primes not exceeding $2m+1$. Since $q \equiv 8\mu+1$, by §. 1 the congruence $x^2 \equiv q$ is then soluble for every modulus which, besides an arbitrary power of 2, contains only odd prime factors not exceeding $2m + 1$. These conditions are satisfied by the product

$$1 \cdot 2 \cdot 3 \cdots (2m + 1) = M,$$

and so one may set

$$k^2 \equiv q \pmod{M},$$

with k chosen positive. Then

$$(q - 1^2)(q - 2^2) \cdots (q - m^2) \equiv (k^2 - 1^2)(k^2 - 2^2) \cdots (k^2 - m^2) \pmod{M}.$$

Now the right-hand side, when multiplied by the factor k, which is relatively prime to M, can be written as the continuous product

$$(k + m)(k + m - 1) \cdots (k - m).$$

This product is a multiple of M, as is easily shown arithmetically, and also follows from the fact that, when divided by M, it represents a number of combinations. Hence the left-hand side must also be divisible by M. Writing the quotient of this division in the form

$$\frac{1}{m+1}\cdot\frac{q-1^2}{(m+1)^2-1^2}\cdot\frac{q-2^2}{(m+1)^2-2^2}\cdots\frac{q-m^2}{(m+1)^2-m^2},$$

one obtains an obvious contradiction if m is chosen to be the integer immediately below $\sqrt{q}$, since the quotient is then a product of proper fractions. It has tacitly been assumed that this choice of m satisfies the condition $2m+1<q$ on which the argument is based. This is indeed the case, since $2m+1<2\sqrt{q}+1$, and $2\sqrt{q}+1$ is plainly smaller than q for all primes $8\mu+1$, the least of which is 17. Thus it is proved that there is always a prime p' smaller than $2m+1$, and therefore smaller than q, for which

$$\left(\frac{q}{p'}\right)=-1.$$

Also note that for our auxiliary prime p' one has

$$\left(\frac{p'}{q}\right)=-1,$$

because the assumption $\left(\frac{p'}{q}\right)=1$ would, by the preceding paragraph, imply $\left(\frac{q}{p'}\right)=1$.

We now pass to showing that from

$$\left(\frac{p}{q}\right)=-1,$$

where $q=4\mu+1$, it always follows that

$$\left(\frac{q}{p}\right)=-1.$$

We may regard p as different from the auxiliary prime p', since for the latter the simultaneous equations

$$\left(\frac{p'}{q}\right)=-1,\qquad\left(\frac{q}{p'}\right)=-1$$

have already been established. With their help, our hypothesis can be represented by

$$\left(\frac{pp'}{q}\right)=1,$$

and the conclusion to be drawn by

$$\left(\frac{q}{pp'}\right)=1.$$

From the first of these equations one can set

$$e^2-pp'=q\varphi,$$

where e is to be even and smaller than q. Then φ is odd and, since $p<q$ and $p'<q$, numerically smaller than q. There are three cases according as φ is divisible by none of the primes p, p', by exactly one of them, or by both at once. Since the equation above and the relation to be derived from it are symmetric in p and p', it makes no difference in the treatment of the second case whether p or p' is taken as a factor of φ.

I. If φ is divisible by neither p nor p', then from $e^2 - pp' = q\varphi$ one obtains

$$\left(\frac{pp'}{\varphi}\right) = 1, \qquad \left(\frac{q\varphi}{pp'}\right) = \left(\frac{q}{pp'}\right)\left(\frac{\varphi}{pp'}\right) = 1.$$

Multiplication and application of (c') give

$$\left(\frac{q}{pp'}\right) = (-1)^{\frac{1}{2}(pp'-1)\frac{1}{2}(\varphi-1)}.$$

But from the preceding equation, since $q = 4\mu + 1$, the numbers $\frac{1}{2}(pp'-1)$ and $\frac{1}{2}(\varphi - 1)$ have opposite parity–one is even. Therefore

$$\left(\frac{q}{pp'}\right) = 1$$

must hold.

II. By symmetry we may suppose φ is divisible by p'. Put

$$\varphi = p'\psi, \qquad e = p'g.$$

Then $e^2 - pp' = q\varphi$ becomes

$$p'g^2 - p = q\psi,$$

where ψ is divisible by neither p nor p'. From this last equation one gets

$$\left(\frac{pp'}{\psi}\right) = 1, \qquad \left(\frac{p'q\psi}{p}\right) = 1, \qquad \left(\frac{-pq\psi}{p'}\right) = 1.$$

By multiplication,

$$\left(\frac{q}{pp'}\right) = \left(\frac{pp'}{\psi}\right)\left(\frac{\psi}{pp'}\right)\left(\frac{p'}{p}\right)\left(\frac{-p}{p'}\right).$$

Applying (c') to the two combinations pp', ψ and $p', -p$ gives

$$\left(\frac{q}{pp'}\right) = (-1)^{\frac{1}{2}(pp'-1)\frac{1}{2}(\psi-1)+\frac{1}{2}(p+1)\frac{1}{2}(p'-1)}.$$

Replace $\frac{1}{2}(\psi - 1)$ by the number $\frac{1}{2}(p + 1)$, which has the same parity by the preceding equation, and replace $\frac{1}{2}(pp' - 1)$ by $\frac{1}{2}(p - 1) + \frac{1}{2}(p' - 1)$. Then the exponent has the value

$$\frac{1}{2}(p+1)(p'-1) + \frac{1}{4}(p^2 - 1),$$

which is plainly even. Hence

$$\left(\frac{q}{pp'}\right) = 1.$$

III. In the third case put

$$\varphi = pp'\psi, \qquad e = pp'g.$$

Then $e^2 - pp' = q\varphi$ becomes

$$pp'g^2 - 1 = q\psi,$$

whence

$$\left(\frac{pp'}{\psi}\right) = 1, \qquad \left(\frac{-q\psi}{pp'}\right) = \left(\frac{q}{pp'}\right)\left(\frac{-\psi}{pp'}\right) = 1,$$

and then

$$\left(\frac{q}{pp'}\right) = (-1)^{\frac{1}{2}(pp'-1)\frac{1}{2}(\psi+1)}.$$

Since $\frac{1}{2}(\psi + 1)$ is plainly even, it follows again that

$$\left(\frac{q}{pp'}\right) = 1.$$

Thus equations (a) and (c), and the equations (a') and (c') derived from them, have been proved in general.

§. 7.

It remains to fill the case of theorem (b) left open above, for primes of the form $8\mu+1$.

Let q denote any prime of this form, and suppose that the theorem already holds for all primes of the same form smaller than q, or, what by §. 2 is the same thing, for all primes smaller than q. If from this hypothesis the equation $\left(\frac{2}{q}\right)=1$ can be deduced, the theorem will hold without restriction. This equation will be proved by deriving a contradiction from the assumption $\left(\frac{2}{q}\right)=-1$.

Choose an auxiliary prime p, smaller than q, such that

$$\left(\frac{p}{q}\right)=-1.$$

Then, under the assumption just made,

$$\left(\frac{2p}{q}\right)=1,$$

and consequently one may set

$$e^2-2p=q\varphi,$$

where, taking e odd and smaller than q, φ will likewise be odd and, apart from sign, smaller than q. It remains to distinguish whether φ is or is not divisible by p.

I. In the first case one immediately obtains

$$\left(\frac{2p}{q\varphi}\right)=\left(\frac{2}{q}\right)\left(\frac{2}{\varphi}\right)\left(\frac{p}{q\varphi}\right)=1,\qquad\left(\frac{q\varphi}{p}\right)=1,$$

and then

$$\left(\frac{2}{q}\right)=\left(\frac{2}{\varphi}\right)\left(\frac{p}{q\varphi}\right)\left(\frac{q\varphi}{p}\right)=\left(\frac{2}{\varphi}\right)(-1)^{\frac{1}{2}(p-1)\frac{1}{2}(q\varphi-1)}.$$

Now equation (b'), in which the sign of P is immaterial and which follows from (b), is applicable to the number φ, all of whose prime factors satisfy equation (b). Thus

$$\left(\frac{2}{\varphi}\right)=(-1)^{\frac{1}{8}(\varphi^2-1)}.$$

Because of the assumption $\left(\frac{2}{q}\right)=-1$, the last equation can therefore be written in the form

$$-1=(-1)^{\frac{1}{8}[2(p-1)(q\varphi-1)+\varphi^2-1]}.$$

The expression in brackets plainly changes by a multiple of 16 if $q\varphi-1$ and φ are replaced by other numbers congruent to them modulo 8. But from $e^2-2p=q\varphi$, since

$$e^2\equiv 1,\qquad q\equiv 1\pmod 8,$$

one immediately obtains

$$q\varphi-1\equiv-2p,\qquad\varphi\equiv 1-2p\pmod 8.$$

Thus the bracketed expression is congruent modulo 16 to

$$-4p(p-1)+(1-2p)^2-1,$$

hence to zero. Consequently the right-hand side of our equation is equal to the positive unit, in contradiction with the left-hand side.

II. If φ is divisible by p, put

$$\varphi = p\psi, \qquad e = pg,$$

so that $e^2 - 2p = q\varphi$ becomes

$$pg^2 - 2 = q\psi.$$

From this equation and from the assumption $\left(\frac{2}{q}\right) = -1$ one has

$$\left(\frac{2p}{q\psi}\right) = -\left(\frac{2}{\psi}\right)\left(\frac{p}{q\psi}\right) = 1, \qquad \left(\frac{-2q\psi}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{-q\psi}{p}\right) = 1.$$

Then, as above,

$$-1 = (-1)^{\frac{1}{8}[2(p-1)(q\psi+1)+p^2+\psi^2-2]}.$$

Replacing again $q\psi + 1$ and ψ by the numbers congruent to them modulo 8 in consequence of the preceding equation, namely $p-1$ and $p-2$, one sees that the expression in brackets in the exponent is congruent modulo 16 to

$$2(p-1)^2 + p^2 + (p-2)^2 - 2 = 4(p-1)^2,$$

hence to zero. This gives the same contradiction as before.

XIII.

Simplification of the Theory of Binary Quadratic Forms of Positive Determinant.

G. Lejeune Dirichlet

Abhandlungen der Königl. Preussischen Akademie der Wissenschaften, 1854, pp. 99–115.

Simplification of the Theory of Binary Quadratic Forms of Positive Determinant.

Read at the Academy of Sciences on 13 July 1854.[1]

The more the higher arithmetic has gained in scope through Gauss's epoch-making work and through later investigations, the more desirable it becomes that access to this beautiful branch of analysis should be made as easy as possible by simplifying its elementary part. With this aim I have, in several earlier papers, prefaced my investigations with new proofs of the known propositions required for them. The present paper has a similar simplification in view, and is devoted to the theory of quadratic forms of positive determinant. In its previous shape this theory requires, as is well known, quite detailed discussions; the following treatment shows how they may be removed.

I begin with some remarks on continued fractions. Although their essential content is not new, they must be put in the form needed for the application made here.

§. 1.

A finite or infinite continued fraction such as

$$\alpha + \cfrac{1}{\beta + \frac{1}{\gamma + \cdots}}$$

will be denoted below by

$$(\alpha, \beta, \gamma, \ldots).$$

We shall consider only continued fractions all of whose terms are integers, except of course the last term in the case where the expansion has not been carried to an end and where that term, as a complete quotient, may have any value. Of special importance for arithmetical investigations are the continued fractions whose terms are all positive except possibly the first, for which the value zero is also allowed. A positive irrational number ω can be expressed in this form in only one way. We call this expression of ω, or, when ω is negative, the expression of its absolute value with a preceding minus sign, the normal continued-fraction expansion of ω.

We first have to derive the normal expansion of an irrational number ω from a continued fraction of the form

$$\omega = (\alpha, \beta, \ldots, \mu, \nu, p, q, r, \ldots, u, v, \ldots),$$

in which all terms are positive only from p onward, including p itself. This can be done by a sequence of transformations which leave unchanged all terms following a sufficiently remote u.

[1] The notice in the Academy report for 1854, p. 384, reads: "Mr. Dirichlet read: On a property of continued fractions and its use for simplifying the theory of quadratic forms of positive determinant." K.

The number of new terms which finally replace $\alpha, \beta, \ldots, \nu$ differs from the number of these old terms by an even or an odd number according as ω is positive or negative.

First suppose that ν is not the first term. Then μ, ν and some immediately following terms may be replaced, all other terms being left unchanged, by new terms whose number differs by an even number, and all of which are positive except possibly the first, which may be zero or negative. The cases are as follows. If $\nu = 0$, the three terms $\mu, 0, p$ are replaced by the single term $\mu + p$. If $\nu = -n$, one distinguishes

$$n > 1, \qquad n = 1,\ p > 1, \qquad n = 1,\ p = 1,$$

and uses, respectively,

$$(\mu, -n, p, q, \ldots) = (\mu - 1, 1, n - 2, 1, p - 1, q, \ldots),$$
$$(\mu, -1, p, q, \ldots) = (\mu - 2, 1, p - 2, q, \ldots),$$
$$(\mu, -1, 1, q, r, s, \ldots) = (\mu - q - 2, 1, r - 1, s, \ldots).$$

The change in the number of terms is respectively $2, 0, -2$. If one of the non-negative differences

$$n - 2, \quad p - 1, \quad p - 2, \quad r - 1$$

becomes zero, the zero and the two adjacent positive terms are replaced by a single term equal to their sum.[2]

By repeating this procedure one makes all terms from the second onward positive. If the first term is then not negative, the operation is finished. If instead the first term is $-a$, so that

$$\omega = (-a, b, c, d, \ldots),$$

then, according as $b > 1$ or $b = 1$, one writes

$$\omega = -(a - 1, 1, b - 1, c, \ldots), \qquad \text{or} \qquad \omega = -(a - 1, c + 1, d, \ldots).$$

This again gives the result just asserted.

§. 2.

I. Let two quantities ω, Ω and integers $\alpha, \beta, \gamma, \delta$, with $\alpha \neq 0$, satisfy

$$\omega = \frac{\gamma + \delta\Omega}{\alpha + \beta\Omega}, \qquad \alpha\delta - \beta\gamma = 1.$$

Then one can always form an equation

$$\omega = (\lambda, m, \ldots, r, \sigma, \Omega)$$

in which, among the integers $\lambda, m, \ldots, r, \sigma$, only the first and the last can be zero or negative; the intermediate terms, if any occur, are positive and are even in number.

Since the signs of $\alpha, \beta, \gamma, \delta$ may all be changed at the same time, suppose $\alpha > 0$. If $\alpha = 1$, then immediately

$$\omega = \frac{\gamma + (\beta\gamma + 1)\Omega}{1 + \beta\Omega} = (\gamma, \beta, \Omega).$$

[2] Lagrange already observed that negative terms can be removed from a continued fraction; but the equation he gave for this purpose, which is the first of the equations above, is insufficient, because when $p = 1$ it introduces a new negative term. Applying the same equation again returns one to the original continued fraction. Oeuvres de Lagrange, t. 2 p. 624.

If $\alpha > 1$, expand γ/α in the ordinary way, choosing all remainders positive:

$$\frac{\gamma}{\alpha} = (\lambda, m, \ldots, r).$$

Only λ can be zero or negative, and the number of terms $m, \ldots, r$ may be assumed even, since an end term $r > 1$ may, if necessary, be replaced by $(r-1, 1)$. Let the convergents be

$$\frac{\lambda}{1}, \quad \frac{\lambda m + 1}{m}, \quad \ldots, \quad \frac{\varphi}{f}, \quad \frac{\gamma}{\alpha}.$$

The last has denominator α, and the standard relation is

$$\alpha\varphi - \gamma f = 1.$$

Comparison with $\alpha\delta - \beta\gamma = 1$ gives

$$\beta = \alpha\sigma + f, \qquad \delta = \gamma\sigma + \varphi$$

with an integer σ. Thus δ/β is annexed to the sequence of convergents by the new term σ, and

$$\omega = (\lambda, m, \ldots, r, \sigma, \Omega).$$

II. We shall need the special case in which $\alpha, \beta, \gamma, \delta$ are all positive and also

$$\gamma \geq \alpha, \qquad \delta > \gamma.$$

Then λ and σ are positive. If $\alpha = 1$, this is immediate, for then $\lambda = \gamma$ and $\sigma = \beta$. If $\alpha > 1$, then λ is the integer immediately below γ/α, hence positive. Also $\sigma > 0$; for from

$$\delta = \gamma\sigma + \varphi$$

one would obtain $\delta = \varphi < \gamma$ if $\sigma = 0$, and a negative δ if $\sigma < 0$. Denoting the positive numbers λ, σ by l, s, we have in this special case

$$\frac{\gamma}{\alpha} = (l, m, \ldots, r), \qquad \frac{\delta}{\beta} = (l, m, \ldots, r, s), \qquad \omega = (l, m, \ldots, r, s, \Omega),$$

where all terms $l, m, \ldots, r, s$ are positive and even in number.

§. 3.

We now turn to the quadratic forms

$$ax^2 + 2bxy + cy^2 = (a, b, c),$$

all of which are to have the same positive determinant

$$D = b^2 - ac.$$

The positive integer D is arbitrary except that it must not be a square. Hence the outer coefficients a, c are never zero, and once D, the middle coefficient, and one of the two outer coefficients are given, the other outer coefficient, and therefore the form itself, is determined.

To each form (a, b, c) we attach the equation

$$a + 2b\omega + c\omega^2 = 0,$$

whose roots are written

$$\omega = \frac{-b \mp \sqrt{D}}{c}$$

with the unchanged third coefficient c as denominator. According to the upper or lower sign, these are called the first and second roots of the form. A form is completely determined by its determinant and either one of its roots: if the same value is a first root, or a second root, of both (a, b, c) and (A, B, C), then

$$\frac{-b \mp \sqrt{D}}{c} = \frac{-B \mp \sqrt{D}}{C},$$

and the irrationality of $\sqrt{D}$ gives $B = b$, $C = c$.

Two forms

$$ax^2 + 2bxy + cy^2, \qquad AX^2 + 2BXY + CY^2$$

will be called equivalent only in the proper sense: there exists a substitution

$$x = \alpha X + \beta Y, \qquad y = \gamma X + \delta Y,$$

with

$$\alpha\delta - \beta\gamma = 1$$

which transforms the first form into the second. Substitutions of determinant -1 are excluded throughout.

I. If two forms are equivalent, then their roots of the same name, ω and Ω, and the coefficients of the substitution satisfy

$$\omega = \frac{\gamma + \delta\Omega}{\alpha + \beta\Omega}.$$

Equivalently,

$$\Omega = \frac{\gamma - \alpha\omega}{\beta\omega - \delta}.$$

After substituting the value of ω and rationalising the denominator, the right hand side becomes

$$\frac{-M \mp \sqrt{D}}{N},$$

where

$$M = a\alpha\beta + b(\alpha\delta + \beta\gamma) + c\gamma\delta, \qquad N = a\beta^2 + 2b\beta\delta + c\delta^2.$$

These are precisely the second and third coefficients B, C of the transformed form.

II. Conversely, if this equation holds for a pair of roots of the same name and if $\alpha, \beta, \gamma, \delta$ satisfy $\alpha\delta - \beta\gamma = 1$, then the two forms are equivalent and the indicated substitution carries the first into the second. For one obtains

$$\frac{-B \mp \sqrt{D}}{C} = \frac{-M \mp \sqrt{D}}{N},$$

hence $B = M$, $C = N$.

III. We shall often use neighbouring forms, that is, forms

$$(a, b, a'), \qquad (a', b', a'')$$

such that

$$b + b' \equiv 0 \pmod{a'}.$$

They are always equivalent. Applying to the first form the substitution

$$\begin{pmatrix} 0 & 1 \\ -1 & \delta \end{pmatrix}$$

one obtains first coefficient a' and second coefficient $-b - a'\delta$, which equals b' when

$$\delta = -\frac{b + b'}{a'}.$$

For the corresponding roots

$$\omega = \delta - \frac{1}{\omega'}, \qquad \omega' = -\frac{1}{\omega - \delta}.$$

§. 4.

A form (a, b, c) is called reduced if, of the two roots

$$\frac{-b - \sqrt{D}}{c}, \qquad \frac{-b + \sqrt{D}}{c},$$

the first has absolute value greater than 1, the second has absolute value less than 1, and the two roots have opposite signs. Then $b > 0$ and $b < \sqrt{D}$; hence $-ac = D - b^2 > 0$, so a and c have opposite signs. The sign of the first root is the sign of a and is opposite to that of c.

If (a, b, c) is reduced, so is (c, b, a); each root of one is the reciprocal of the opposite-name root of the other.

There are only finitely many reduced forms of a given determinant D. They are obtained by taking each positive $b < \sqrt{D}$, finding all positive and negative factors c of $D - b^2$ whose absolute values lie between $\sqrt{D} + b$ and $\sqrt{D} - b$, and then putting

$$a = -\frac{D - b^2}{c}.$$

Let (a, b, a') be a reduced form. Among its right neighbours (a', b', a'') there is exactly one reduced form. If the neighbour is to be reduced, its first root ω' must have sign opposite to the first root ω of (a, b, a'). In

$$\omega' = -\frac{1}{\omega - \delta},$$

choose δ so that ω' is an improper fraction and has sign opposite to ω. Equivalently, $\omega - \delta$ must be a proper fraction with the same sign as ω. There is exactly one such integer: the integer adjacent to ω on the same side of ω as zero. It is nonzero, has the sign of ω, and its absolute value is immediately below $|\omega|$.

This neighbour is indeed reduced. If ω now denotes the second root in the same equation, then ω' has the sign of $-\delta$; also $\omega - \delta$ is improper, hence ω' is proper and has the sign required for the second root.

For the middle coefficient of this unique right neighbour set, for the first root,

$$\omega - \delta = \sigma,$$

where σ is a proper fraction with the same sign as ω. From

$$b' = -b - a'\delta$$

we get

$$b' = \sqrt{D} + a'\sigma.$$

Since σ has sign opposite to a', b' lies between $\sqrt{D}$ and $\sqrt{D} \mp a'$, with the upper or lower sign according as a' is positive or negative. Together with

$$b' \equiv -b \pmod{a'},$$

this determines b' without ambiguity. Similarly there is a unique reduced left neighbour $('a, 'b, a)$.

§. 5.

Starting from a reduced form φ_0, let φ_1 be the reduced form immediately to its right, let φ_2 be the right neighbour of φ_1, and so on; do the same to the left. One obtains an infinite two-sided sequence of equivalent reduced forms

$$\ldots, \varphi_{-2}, \varphi_{-1}, \varphi_0, \varphi_1, \varphi_2, \ldots.$$

Because only finitely many reduced forms have the same determinant, forms in this sequence repeat. Since the first coefficients alternate in sign, identity can occur only at an even difference of indices. If two forms are identical, every pair of forms standing the same distance to the same side of them is also identical.

Let φ_{2n} be the first form after φ_0 identical with φ_0. Then

$$\varphi_0, \varphi_1, \ldots, \varphi_{2n-1}$$

are all distinct and form a period. Thus

$$\varphi_\mu = \varphi_\nu \quad \Longleftrightarrow \quad \mu \equiv \nu \pmod{2n}.$$

The same criterion holds for the equality of corresponding roots ω_μ, ω_ν.

Let δ_ν be the greatest integer contained in the absolute value of the first root ω_ν, taken with the sign of ω_ν. Then

$$\omega_\nu = \delta_\nu - \frac{1}{\omega_{\nu+1}}.$$

The congruence of indices implies equality of the δ's, though not conversely.

Choose the index so that forms of even index have positive first coefficient. Then the signs of ω_ν and δ_ν are $(-1)^\nu$. Write

$$\delta_\nu = (-1)^\nu k_\nu,$$

with $k_\nu > 0$. The numbers k_ν have period $2n$, and

$$\omega_\nu = (-1)^\nu (k_\nu, k_{\nu+1}, \ldots, k_{2n+u-1}, k_\nu, k_{\nu+1}, \ldots).$$

For the second root ω'_ν,

$$\frac{1}{\omega'_\nu} = (-1)^{\nu-1} (k_{\nu-1}, k_{\nu-2}, \ldots, k_{\nu-2n}, k_{\nu-1}, k_{\nu-2}, \ldots).$$

Thus the period for the reciprocal of the second root is the reverse of the period for the first root.

The period $k_0, k_1, \ldots, k_{2n-1}$ has no shorter even period. Otherwise ω_0 and ω_{2m}, hence also φ_0 and φ_{2m}, would coincide with $2m < 2n$.

All reduced forms of a given determinant can be partitioned into periods of this sort: form a period from any reduced form, and, if any reduced forms remain, repeat the operation until none are left.

§. 6.

We now decide whether two given forms are equivalent. Since every form can be reduced and forms in the same period are equivalent, it remains only to decide whether forms in different periods can be equivalent.

Choose in each period a form with positive first coefficient:

$$\varphi_0 = (a, b, c), \qquad \Phi_0 = (A, B, C),$$

and write their first roots in normal continued fractions

$$\omega_0 = (k_0, k_1, k_2, \ldots), \qquad \Omega_0 = (K_0, K_1, K_2, \ldots).$$

Suppose the forms are equivalent, and that a substitution of determinant 1 takes the first into the second. Then

$$\omega_0 = \frac{\gamma + \delta\Omega_0}{\alpha + \beta\Omega_0}.$$

Here α cannot be zero: otherwise $\gamma = \pm 1$ and $A = c$, contrary to the sign assumptions. Hence by §. 2

$$\omega_0 = (\lambda, m, \ldots, r, \sigma, K_0, K_1, K_2, \ldots).$$

Let the number of inserted terms be $2g$. When this continued fraction is brought to normal form, the number of terms up to a sufficiently remote unchanged term K_ν changes, since $\omega_0 > 0$, by an even number $2h$. Thus for all sufficiently large ν,

$$K_\nu = k_{2g+2h+u}.$$

Replacing ν by $2Ni + \nu$, where $2Ni$ is a large multiple of the period $2N$ of the K-sequence, and reducing $2g + 2h + 2Ni$ modulo $2n$ to $2m$, gives for all $\nu \geq 0$

$$K_\nu = k_{2m+u}.$$

Therefore

$$\Omega_0 = \omega_{2m}, \qquad \Phi_0 = \varphi_{2m}.$$

The second form lies in the period of the first. Thus forms from different periods cannot be equivalent.

§. 7.

It remains to indicate how the rest of the theory is modified in the points that depend on the proof just given.

The operations which decide equivalence give one substitution taking one form to the other. It remains to derive all such substitutions from one of them. This reduces to describing all substitutions which carry a form into itself. We may suppose that the coefficients of the form are coprime, since removing a common factor does not change these substitutions.

Let (a, b, c) be primitive, and let σ be the positive greatest common divisor of $a, 2b, c$. Then σ is 1 or 2, and the latter occurs only when $D = b^2 - ac$ has the form $4h + 1$. All substitutions carrying the form into itself are given by

$$\alpha = \frac{t - bu}{\sigma}, \qquad \beta = -\frac{cu}{\sigma}, \qquad \gamma = \frac{au}{\sigma}, \qquad \delta = \frac{t + bu}{\sigma},$$

where t, u run through all integer solutions of

$$t^2 - Du^2 = \sigma^2.$$

The complete solution of this indeterminate equation is easily obtained from its solution in the least positive numbers.

For a reduced form the transformation problem can be solved directly. Suppose $a > 0$ and consider substitutions whose four coefficients are positive. Let

$$\omega = (k_0, k_1, \ldots, k_{2n-1}, k_0, k_1, \ldots)$$

be the normal periodic continued fraction for the first root. If γ/α and δ/β are successive convergents, the second corresponding to the last term k_{2n-1} of a period, then

$$\alpha\delta - \beta\gamma = 1, \qquad \omega = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

and hence the corresponding substitution carries the form into itself.

Conversely, let $\alpha, \beta, \gamma, \delta$ be the coefficients of such a positive substitution. Then

$$\beta\omega^2 + (\alpha - \delta)\omega - \gamma = 0.$$

Since one root lies between 1 and ∞ and the other between -1 and 0, the left hand side is negative at $\omega = 1$ and positive at $\omega = -1$. Consequently

$$\gamma - \alpha > \beta - \delta, \qquad \delta - \gamma > \alpha - \beta,$$

and hence

$$\gamma \geq \alpha, \qquad \delta > \gamma.$$

All hypotheses of §. 2, II, are therefore satisfied:

$$\frac{\gamma}{\alpha} = (l, m, \ldots, r), \qquad \frac{\delta}{\beta} = (l, m, \ldots, r, s), \qquad \omega = (l, m, \ldots, r, s, \omega).$$

Substitution of the periodic expansion of ω shows that $l, m, \ldots, r, s$ consists of one or more complete periods, and that the two fractions are successive convergents ending at the end of a period. The smallest positive substitution corresponds to the end of the first period.

The link with the Pell-type equation is as follows. From the formulas for α and δ,

$$\alpha\delta = \frac{t^2 - b^2u^2}{\sigma^2} = \frac{t^2 - Du^2}{\sigma^2} - \frac{acu^2}{\sigma^2} = 1 - \frac{acu^2}{\sigma^2}.$$

Since $-ac > 0$, α and δ have the same sign, which is the sign of t because

$$\alpha + \delta = \frac{2t}{\sigma}.$$

Similarly, β and γ, unless both zero, have the sign of u. Thus the positive substitutions just considered arise from positive t, u; because β and γ increase with u, the substitution expressed in the smallest numbers gives the least positive values of t, u, namely

$$t = \frac{\sigma}{2}(\delta + \alpha), \qquad u = \frac{\sigma}{2b}(\delta - \alpha).$$

XIV.

Simplification of the Theory of Binary Quadratic Forms of Positive Determinant.

G. Lejeune Dirichlet

Liouville, Journal de Mathématiques pures et appliquées, Series II, Volume II, pp. 353–376.

Simplification of the Theory of Binary Quadratic Forms of Positive Determinant.

Read at the Academy of Sciences on 13 July 1854.[1]

The more the higher arithmetic has gained in scope through Gauss's epoch-making work and through later investigations, the more desirable it becomes that access to this beautiful branch of analysis should be made as easy as possible by simplifying its elementary part. With this aim I have, in several earlier papers, prefaced my investigations with new proofs of the known propositions required for them. The present paper has a similar simplification in view, and is devoted to the theory of quadratic forms of positive determinant. In its previous shape this theory requires, as is well known, quite detailed discussions; the following treatment shows how they may be removed.

I begin with some remarks on continued fractions. Although their essential content is not new, they must be put in the form needed for the application made here.

§. 1.

A finite or infinite continued fraction such as

$$\alpha + \cfrac{1}{\beta + \frac{1}{\gamma + \cdots}}$$

will be denoted below by

$$(\alpha, \beta, \gamma, \ldots).$$

We shall consider only continued fractions all of whose terms are integers, except of course the last term in the case where the expansion has not been carried to an end and where that term, as a complete quotient, may have any value. Of special importance for arithmetical investigations are the continued fractions whose terms are all positive except possibly the first, for which the value zero is also allowed. A positive irrational number ω can be expressed in this form in only one way. We call this expression of ω, or, when ω is negative, the expression of its absolute value with a preceding minus sign, the normal continued-fraction expansion of ω.

We first have to derive the normal expansion of an irrational number ω from a continued fraction of the form

$$\omega = (\alpha, \beta, \ldots, \mu, \nu, p, q, r, \ldots, u, v, \ldots),$$

in which all terms are positive only from p onward, including p itself. This can be done by a sequence of transformations which leave unchanged all terms following a sufficiently remote u.

[1] The notice in the Academy report for 1854, p. 384, reads: "Mr. Dirichlet read: On a property of continued fractions and its use for simplifying the theory of quadratic forms of positive determinant." K.

The number of new terms which finally replace $\alpha, \beta, \ldots, \nu$ differs from the number of these old terms by an even or an odd number according as ω is positive or negative.

First suppose that ν is not the first term. Then μ, ν and some immediately following terms may be replaced, all other terms being left unchanged, by new terms whose number differs by an even number, and all of which are positive except possibly the first, which may be zero or negative. The cases are as follows. If $\nu = 0$, the three terms $\mu, 0, p$ are replaced by the single term $\mu + p$. If $\nu = -n$, one distinguishes

$$n > 1, \qquad n = 1,\ p > 1, \qquad n = 1,\ p = 1,$$

and uses, respectively,

$$(\mu, -n, p, q, \ldots) = (\mu - 1, 1, n - 2, 1, p - 1, q, \ldots),$$
$$(\mu, -1, p, q, \ldots) = (\mu - 2, 1, p - 2, q, \ldots),$$
$$(\mu, -1, 1, q, r, s, \ldots) = (\mu - q - 2, 1, r - 1, s, \ldots).$$

The change in the number of terms is respectively $2, 0, -2$. If one of the non-negative differences

$$n - 2, \quad p - 1, \quad p - 2, \quad r - 1$$

becomes zero, the zero and the two adjacent positive terms are replaced by a single term equal to their sum.[2]

By repeating this procedure one makes all terms from the second onward positive. If the first term is then not negative, the operation is finished. If instead the first term is $-a$, so that

$$\omega = (-a, b, c, d, \ldots),$$

then, according as $b > 1$ or $b = 1$, one writes

$$\omega = -(a - 1, 1, b - 1, c, \ldots), \qquad \text{or} \qquad \omega = -(a - 1, c + 1, d, \ldots).$$

This again gives the result just asserted.

§. 2.

I. Let two quantities ω, Ω and integers $\alpha, \beta, \gamma, \delta$, with $\alpha \neq 0$, satisfy

$$\omega = \frac{\gamma + \delta\Omega}{\alpha + \beta\Omega}, \qquad \alpha\delta - \beta\gamma = 1.$$

Then one can always form an equation

$$\omega = (\lambda, m, \ldots, r, \sigma, \Omega)$$

in which, among the integers $\lambda, m, \ldots, r, \sigma$, only the first and the last can be zero or negative; the intermediate terms, if any occur, are positive and are even in number.

Since the signs of $\alpha, \beta, \gamma, \delta$ may all be changed at the same time, suppose $\alpha > 0$. If $\alpha = 1$, then immediately

$$\omega = \frac{\gamma + (\beta\gamma + 1)\Omega}{1 + \beta\Omega} = (\gamma, \beta, \Omega).$$

[2]Lagrange already observed that negative terms can be removed from a continued fraction; but the equation he gave for this purpose, which is the first of the equations above, is insufficient, because when $p = 1$ it introduces a new negative term. Applying the same equation again returns one to the original continued fraction. Oeuvres de Lagrange, t. 2 p. 624.

If $\alpha > 1$, expand γ/α in the ordinary way, choosing all remainders positive:

$$\frac{\gamma}{\alpha} = (\lambda, m, \dots, r).$$

Only λ can be zero or negative, and the number of terms $m, \dots, r$ may be assumed even, since an end term $r > 1$ may, if necessary, be replaced by $(r-1, 1)$. Let the convergents be

$$\frac{\lambda}{1}, \quad \frac{\lambda m + 1}{m}, \quad \dots, \quad \frac{\varphi}{f}, \quad \frac{\gamma}{\alpha}.$$

The last has denominator α, and the standard relation is

$$\alpha\varphi - \gamma f = 1.$$

Comparison with $\alpha\delta - \beta\gamma = 1$ gives

$$\beta = \alpha\sigma + f, \qquad \delta = \gamma\sigma + \varphi$$

with an integer σ. Thus δ/β is annexed to the sequence of convergents by the new term σ, and

$$\omega = (\lambda, m, \dots, r, \sigma, \Omega).$$

II. We shall need the special case in which $\alpha, \beta, \gamma, \delta$ are all positive and also

$$\gamma \geq \alpha, \qquad \delta > \gamma.$$

Then λ and σ are positive. If $\alpha = 1$, this is immediate, for then $\lambda = \gamma$ and $\sigma = \beta$. If $\alpha > 1$, then λ is the integer immediately below γ/α, hence positive. Also $\sigma > 0$; for from

$$\delta = \gamma\sigma + \varphi$$

one would obtain $\delta = \varphi < \gamma$ if $\sigma = 0$, and a negative δ if $\sigma < 0$. Denoting the positive numbers λ, σ by l, s, we have in this special case

$$\frac{\gamma}{\alpha} = (l, m, \dots, r), \qquad \frac{\delta}{\beta} = (l, m, \dots, r, s), \qquad \omega = (l, m, \dots, r, s, \Omega),$$

where all terms $l, m, \dots, r, s$ are positive and even in number.

§. 3.

We now turn to the quadratic forms

$$ax^2 + 2bxy + cy^2 = (a, b, c),$$

all of which are to have the same positive determinant

$$D = b^2 - ac.$$

The positive integer D is arbitrary except that it must not be a square. Hence the outer coefficients a, c are never zero, and once D, the middle coefficient, and one of the two outer coefficients are given, the other outer coefficient, and therefore the form itself, is determined.

To each form (a, b, c) we attach the equation

$$a + 2b\omega + c\omega^2 = 0,$$

whose roots are written

$$\omega = \frac{-b \mp \sqrt{D}}{c}$$

with the unchanged third coefficient c as denominator. According to the upper or lower sign, these are called the first and second roots of the form. A form is completely determined by its determinant and either one of its roots: if the same value is a first root, or a second root, of both (a, b, c) and (A, B, C), then

$$\frac{-b \mp \sqrt{D}}{c} = \frac{-B \mp \sqrt{D}}{C},$$

and the irrationality of $\sqrt{D}$ gives $B = b$, $C = c$.

Two forms

$$ax^2 + 2bxy + cy^2, \qquad AX^2 + 2BXY + CY^2$$

will be called equivalent only in the proper sense: there exists a substitution

$$x = \alpha X + \beta Y, \qquad y = \gamma X + \delta Y,$$

with

$$\alpha\delta - \beta\gamma = 1$$

which transforms the first form into the second. Substitutions of determinant -1 are excluded throughout.

I. If two forms are equivalent, then their roots of the same name, ω and Ω, and the coefficients of the substitution satisfy

$$\omega = \frac{\gamma + \delta\Omega}{\alpha + \beta\Omega}.$$

Equivalently,

$$\Omega = \frac{\gamma - \alpha\omega}{\beta\omega - \delta}.$$

After substituting the value of ω and rationalising the denominator, the right hand side becomes

$$\frac{-M \mp \sqrt{D}}{N},$$

where

$$M = a\alpha\beta + b(\alpha\delta + \beta\gamma) + c\gamma\delta, \qquad N = a\beta^2 + 2b\beta\delta + c\delta^2.$$

These are precisely the second and third coefficients B, C of the transformed form.

II. Conversely, if this equation holds for a pair of roots of the same name and if $\alpha, \beta, \gamma, \delta$ satisfy $\alpha\delta - \beta\gamma = 1$, then the two forms are equivalent and the indicated substitution carries the first into the second. For one obtains

$$\frac{-B \mp \sqrt{D}}{C} = \frac{-M \mp \sqrt{D}}{N},$$

hence $B = M$, $C = N$.

III. We shall often use neighbouring forms, that is, forms

$$(a, b, a'), \qquad (a', b', a'')$$

such that

$$b + b' \equiv 0 \pmod{a'}.$$

They are always equivalent. Applying to the first form the substitution

$$\begin{pmatrix} 0 & 1 \\ -1 & \delta \end{pmatrix}$$

one obtains first coefficient a' and second coefficient $-b - a'\delta$, which equals b' when

$$\delta = -\frac{b + b'}{a'}.$$

For the corresponding roots

$$\omega = \delta - \frac{1}{\omega'}, \qquad \omega' = -\frac{1}{\omega - \delta}.$$

§. 4.

A form (a, b, c) is called reduced if, of the two roots

$$\frac{-b - \sqrt{D}}{c}, \qquad \frac{-b + \sqrt{D}}{c},$$

the first has absolute value greater than 1, the second has absolute value less than 1, and the two roots have opposite signs. Then $b > 0$ and $b < \sqrt{D}$; hence $-ac = D - b^2 > 0$, so a and c have opposite signs. The sign of the first root is the sign of a and is opposite to that of c.

If (a, b, c) is reduced, so is (c, b, a); each root of one is the reciprocal of the opposite-name root of the other.

There are only finitely many reduced forms of a given determinant D. They are obtained by taking each positive $b < \sqrt{D}$, finding all positive and negative factors c of $D - b^2$ whose absolute values lie between $\sqrt{D} + b$ and $\sqrt{D} - b$, and then putting

$$a = -\frac{D - b^2}{c}.$$

Let (a, b, a') be a reduced form. Among its right neighbours (a', b', a'') there is exactly one reduced form. If the neighbour is to be reduced, its first root ω' must have sign opposite to the first root ω of (a, b, a'). In

$$\omega' = -\frac{1}{\omega - \delta},$$

choose δ so that ω' is an improper fraction and has sign opposite to ω. Equivalently, $\omega - \delta$ must be a proper fraction with the same sign as ω. There is exactly one such integer: the integer adjacent to ω on the same side of ω as zero. It is nonzero, has the sign of ω, and its absolute value is immediately below $|\omega|$.

This neighbour is indeed reduced. If ω now denotes the second root in the same equation, then ω' has the sign of $-\delta$; also $\omega - \delta$ is improper, hence ω' is proper and has the sign required for the second root.

For the middle coefficient of this unique right neighbour set, for the first root,

$$\omega - \delta = \sigma,$$

where σ is a proper fraction with the same sign as ω. From

$$b' = -b - a'\delta$$

we get

$$b' = \sqrt{D} + a'\sigma.$$

Since σ has sign opposite to a', b' lies between $\sqrt{D}$ and $\sqrt{D} \mp a'$, with the upper or lower sign according as a' is positive or negative. Together with

$$b' \equiv -b \pmod{a'},$$

this determines b' without ambiguity. Similarly there is a unique reduced left neighbour $('a, 'b, a)$.

§. 5.

Starting from a reduced form φ_0, let φ_1 be the reduced form immediately to its right, let φ_2 be the right neighbour of φ_1, and so on; do the same to the left. One obtains an infinite two-sided sequence of equivalent reduced forms

$$\ldots, \varphi_{-2}, \varphi_{-1}, \varphi_0, \varphi_1, \varphi_2, \ldots.$$

Because only finitely many reduced forms have the same determinant, forms in this sequence repeat. Since the first coefficients alternate in sign, identity can occur only at an even difference of indices. If two forms are identical, every pair of forms standing the same distance to the same side of them is also identical.

Let φ_{2n} be the first form after φ_0 identical with φ_0. Then

$$\varphi_0, \varphi_1, \ldots, \varphi_{2n-1}$$

are all distinct and form a period. Thus

$$\varphi_\mu = \varphi_\nu \quad \Longleftrightarrow \quad \mu \equiv \nu \pmod{2n}.$$

The same criterion holds for the equality of corresponding roots ω_μ, ω_ν.

Let δ_ν be the greatest integer contained in the absolute value of the first root ω_ν, taken with the sign of ω_ν. Then

$$\omega_\nu = \delta_\nu - \frac{1}{\omega_{\nu+1}}.$$

The congruence of indices implies equality of the δ's, though not conversely.

Choose the index so that forms of even index have positive first coefficient. Then the signs of ω_ν and δ_ν are $(-1)^\nu$. Write

$$\delta_\nu = (-1)^\nu k_\nu,$$

with $k_\nu > 0$. The numbers k_ν have period $2n$, and

$$\omega_\nu = (-1)^\nu (k_\nu, k_{\nu+1}, \ldots, k_{2n+u-1}, k_\nu, k_{\nu+1}, \ldots).$$

For the second root ω'_ν,

$$\frac{1}{\omega'_\nu} = (-1)^{\nu-1}(k_{\nu-1}, k_{\nu-2}, \ldots, k_{\nu-2n}, k_{\nu-1}, k_{\nu-2}, \ldots).$$

Thus the period for the reciprocal of the second root is the reverse of the period for the first root.

The period $k_0, k_1, \ldots, k_{2n-1}$ has no shorter even period. Otherwise ω_0 and ω_{2m}, hence also φ_0 and φ_{2m}, would coincide with $2m < 2n$.

All reduced forms of a given determinant can be partitioned into periods of this sort: form a period from any reduced form, and, if any reduced forms remain, repeat the operation until none are left.

§. 6.

We now decide whether two given forms are equivalent. Since every form can be reduced and forms in the same period are equivalent, it remains only to decide whether forms in different periods can be equivalent.

Choose in each period a form with positive first coefficient:

$$\varphi_0 = (a, b, c), \qquad \Phi_0 = (A, B, C),$$

and write their first roots in normal continued fractions

$$\omega_0 = (k_0, k_1, k_2, \ldots), \qquad \Omega_0 = (K_0, K_1, K_2, \ldots).$$

Suppose the forms are equivalent, and that a substitution of determinant 1 takes the first into the second. Then

$$\omega_0 = \frac{\gamma + \delta\Omega_0}{\alpha + \beta\Omega_0}.$$

Here α cannot be zero: otherwise $\gamma = \pm 1$ and $A = c$, contrary to the sign assumptions. Hence by §. 2

$$\omega_0 = (\lambda, m, \ldots, r, \sigma, K_0, K_1, K_2, \ldots).$$

Let the number of inserted terms be $2g$. When this continued fraction is brought to normal form, the number of terms up to a sufficiently remote unchanged term K_ν changes, since $\omega_0 > 0$, by an even number $2h$. Thus for all sufficiently large ν,

$$K_\nu = k_{2g+2h+u}.$$

Replacing ν by $2Ni + \nu$, where $2Ni$ is a large multiple of the period $2N$ of the K-sequence, and reducing $2g + 2h + 2Ni$ modulo $2n$ to $2m$, gives for all $\nu \geq 0$

$$K_\nu = k_{2m+u}.$$

Therefore

$$\Omega_0 = \omega_{2m}, \qquad \Phi_0 = \varphi_{2m}.$$

The second form lies in the period of the first. Thus forms from different periods cannot be equivalent.

§. 7.

It remains to indicate how the rest of the theory is modified in the points that depend on the proof just given.

The operations which decide equivalence give one substitution taking one form to the other. It remains to derive all such substitutions from one of them. This reduces to describing all substitutions which carry a form into itself. We may suppose that the coefficients of the form are coprime, since removing a common factor does not change these substitutions.

Let (a, b, c) be primitive, and let σ be the positive greatest common divisor of $a, 2b, c$. Then σ is 1 or 2, and the latter occurs only when $D = b^2 - ac$ has the form $4h + 1$. All substitutions carrying the form into itself are given by

$$\alpha = \frac{t - bu}{\sigma}, \qquad \beta = -\frac{cu}{\sigma}, \qquad \gamma = \frac{au}{\sigma}, \qquad \delta = \frac{t + bu}{\sigma},$$

where t, u run through all integer solutions of

$$t^2 - Du^2 = \sigma^2.$$

The complete solution of this indeterminate equation is easily obtained from its solution in the least positive numbers.

For a reduced form the transformation problem can be solved directly. Suppose $a > 0$ and consider substitutions whose four coefficients are positive. Let

$$\omega = (k_0, k_1, \ldots, k_{2n-1}, k_0, k_1, \ldots)$$

be the normal periodic continued fraction for the first root. If γ/α and δ/β are successive convergents, the second corresponding to the last term k_{2n-1} of a period, then

$$\alpha\delta - \beta\gamma = 1, \qquad \omega = \frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

and hence the corresponding substitution carries the form into itself.

Conversely, let $\alpha, \beta, \gamma, \delta$ be the coefficients of such a positive substitution. Then

$$\beta\omega^2 + (\alpha - \delta)\omega - \gamma = 0.$$

Since one root lies between 1 and ∞ and the other between -1 and 0, the left hand side is negative at $\omega = 1$ and positive at $\omega = -1$. Consequently

$$\gamma - \alpha > \beta - \delta, \qquad \delta - \gamma > \alpha - \beta,$$

and hence

$$\gamma \geq \alpha, \qquad \delta > \gamma.$$

All hypotheses of §. 2, II, are therefore satisfied:

$$\frac{\gamma}{\alpha} = (l, m, \ldots, r), \qquad \frac{\delta}{\beta} = (l, m, \ldots, r, s), \qquad \omega = (l, m, \ldots, r, s, \omega).$$

Substitution of the periodic expansion of ω shows that $l, m, \ldots, r, s$ consists of one or more complete periods, and that the two fractions are successive convergents ending at the end of a period. The smallest positive substitution corresponds to the end of the first period.

The link with the Pell-type equation is as follows. From the formulas for α and δ,

$$\alpha\delta = \frac{t^2 - b^2u^2}{\sigma^2} = \frac{t^2 - Du^2}{\sigma^2} - \frac{acu^2}{\sigma^2} = 1 - \frac{acu^2}{\sigma^2}.$$

Since $-ac > 0$, α and δ have the same sign, which is the sign of t because

$$\alpha + \delta = \frac{2t}{\sigma}.$$

Similarly, β and γ, unless both zero, have the sign of u. Thus the positive substitutions just considered arise from positive t, u; because β and γ increase with u, the substitution expressed in the smallest numbers gives the least positive values of t, u, namely

$$t = \frac{\sigma}{2}(\delta + \alpha), \qquad u = \frac{\sigma}{2b}(\delta - \alpha).$$

Addition to this paper, by the author.

In order that the reader need not turn back to the writings cited in the note concerning §. 7, we shall prove in a few words the formulas recalled at the point to which that note refers. The necessary and sufficient conditions for the substitution

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

to transform the form (a, b, c) into itself are plainly expressed by

$$\alpha\delta - \beta\gamma = 1, \qquad a(\alpha^2 - 1) + 2b\alpha\gamma + c\gamma^2 = 0, \qquad a\alpha\beta + 2b\beta\gamma + c\gamma\delta = 0,$$

the third of these having been simplified by replacing $\alpha\delta$ with $\beta\gamma + 1$. Adding the last two equations after multiplying them respectively by δ and $-\gamma$, and then after multiplying them by $-\beta$ and α, one obtains the equivalent system

$$\alpha\delta - \beta\gamma = 1, \qquad a(\alpha - \delta) + 2b\gamma = 0, \qquad a\beta + c\gamma = 0.$$

Since $a \neq 0$, if we put $\gamma = a\psi$, the rational quantity ψ is completely determined by γ, and the last two equalities may be replaced by

$$\delta - \alpha = 2b\psi, \qquad \beta = -c\psi, \qquad \gamma = a\psi.$$

Now $\delta - \alpha$, β, and γ are integers, and if the greatest common divisor of $a, 2b, c$ is denoted by σ, ψ must necessarily be of the form u/σ, where u is an integer. Thus

$$\delta - \alpha = \frac{2bu}{\sigma}, \qquad \beta = -\frac{cu}{\sigma}, \qquad \gamma = \frac{au}{\sigma}.$$

Substitution in

$$(\delta + \alpha)^2 = (\delta - \alpha)^2 + 4\alpha\delta = (\delta - \alpha)^2 + 4\beta\gamma + 4$$

and multiplication by σ^2 give

$$[\sigma(\delta + \alpha)]^2 = 4[(b^2 - ac)u^2 + \sigma^2] = 4(Du^2 + \sigma^2).$$

It follows that $\sigma(\delta + \alpha)$ is even. Writing

$$\sigma(\delta + \alpha) = 2t,$$

with t an integer, the last equation takes the form

$$t^2 - Du^2 = \sigma^2,$$

and the integers $\alpha, \beta, \gamma, \delta$ are expressed by

$$\alpha = \frac{t - bu}{\sigma}, \qquad \beta = -\frac{cu}{\sigma}, \qquad \gamma = \frac{au}{\sigma}, \qquad \delta = \frac{t + bu}{\sigma}.$$

Conversely, every integral solution (t, u) of this equation gives integral values of $\alpha, \beta, \gamma, \delta$, and these values satisfy the equations that express the sufficient conditions for the corresponding substitution to transform (a, b, c) into itself. The integrality is immediate when $\sigma = 1$; when $\sigma = 2$ it follows from the fact that D and b are odd in that case, so that t and u are both even or both odd. The remaining assertion is verified by direct substitution. (Toul, 14 August 1857.)

Extract from a letter of Dirichlet to Liouville.

On looking over the paper recently translated by Hoüel, I noticed that it could be made more complete without lengthening it, by using a remark that I have had occasion to make in my course. The last two paragraphs of §. 4 are to be replaced by the following.

To determine the coefficient b', observe that the reciprocal value of the first of the roots belonging to the form (a', b', a'') must be a quantity σ whose absolute value is less than unity and whose sign is that of a'. Hence

$$\sigma = \frac{-b' + \sqrt{D}}{a'}, \qquad b' = \sqrt{D} - a'\sigma.$$

Thus b' lies between $\sqrt{D}$ and $\sqrt{D} - (a')$, where (a') denotes the absolute value of a'. This condition, together with

$$b' \equiv -b \pmod{a'},$$

determines b' uniquely and easily.

One proves in exactly the same way that there always exists a unique reduced form contiguous to the proposed one on the left; this also follows from the preceding case by the earlier observation.

We finish the paragraph by showing that one can always obtain a reduced form equivalent to any given form (a, b, a'). To this end form the sequence

$$(a, b, a'), \quad (a', b', a''), \quad (a'', b'', a'''), \ldots,$$

each form being obtained from the preceding one by the process just indicated. The forms so obtained will eventually all be reduced, and this will occur at the latest when one reaches a form whose first coefficient does not exceed the third in absolute value. That point cannot fail to occur, since the positive integers

$$(a),\ (a'),\ (a''), \ldots$$

cannot decrease indefinitely.

It remains only to prove that a form

$$(A, B, C),$$

with $(A) \leq (C)$ and with (B) between $\sqrt{D}$ and $\sqrt{D} - (A)$, is reduced. By the last condition,

$$B = \sqrt{D} - A\sigma, \qquad \sigma = \frac{-B + \sqrt{D}}{A}, \qquad \frac{1}{\sigma} = \frac{-B - \sqrt{D}}{C},$$

where σ has absolute value less than one and the sign of A. From the last two equations and from $(A) \leq (C)$ it follows that the roots

$$\frac{-B - \sqrt{D}}{C}, \qquad \frac{-B + \sqrt{D}}{C}$$

of the form have, in absolute value, the first greater than one and the second less than one. Since the first root is thus numerically greater than the second, B is necessarily positive; then the next-to-last equation, in which σ and A have the same sign, gives $B < \sqrt{D}$. Consequently the two roots have opposite signs. Q.E.D.

XV.

On a Property of Quadratic Forms of Positive Determinant.

By M. G. Lejeune Dirichlet

English translation

On a Property of Quadratic Forms of Positive Determinant.

Communicated in the session of the physical-mathematical class of the Academy of Sciences on 16 July 1855.

Since the new theorem which forms the subject of this lecture is closely connected with the theory of the equation

$$t^2 - Du^2 = 1, \tag{1}$$

some remarks must first be made about this equation. Here the given positive integer D is not to be equal to a square, and we shall have to consider only those solutions which are expressed by positive integers t, u.

Let T, U be the smallest values satisfying (1). Then, as is well known, all the solutions of the kind just mentioned are obtained from the formula

$$(T + U\sqrt{D})^n = t_n + u_n\sqrt{D}$$

by assigning to the integer n all positive values. Among these solutions there are infinitely many in which u_n is divisible by an arbitrary positive number S, and the exponents n for which this circumstance occurs are the successive multiples of the smallest such exponent N. Indeed, put

$$D' = DS^2$$

and form the new equation

$$t'^2 - D'u'^2 = 1.$$

It is then clear that every solution of this latter equation gives, when one puts

$$t = t', \qquad u = Su',$$

a solution of (1) in which u is divisible by S, and conversely. This proves the assertion and also shows that N is determined by the equation

$$(T + U\sqrt{D})^N = T' + U'\sqrt{D'},$$

where T', U' denote the smallest values of t', u'.

One can specify the exponent N without knowing T', U', as soon as, for each of the distinct prime factors p contained in S, one knows the smallest value ν for which u_ν is divisible by p, and also the exponent δ of the highest power of p by which u_ν is divisible. It is not hard to see that, if e denotes an arbitrary number divisible by p^ε, and by no higher power of p, under the assumption just made $p^{\delta+\varepsilon}$ will be the highest power of p occurring in $u_{\nu e}$. Therefore the necessary and sufficient condition for $u_{\nu e}$ to be divisible by p^α is that e be divisible by $p^{\alpha-\delta}$, where of course the difference $\alpha - \delta$, if it becomes negative, is to be replaced by zero.

Now distinguish by indices the various prime factors p contained in S, together with their corresponding values ν, δ, and write

$$S = p_1^{\alpha_1} p_2^{\alpha_2} \cdots .$$

Then, by what has just been said, N is the least common multiple of the numbers

$$\nu_1 p_1^{\alpha_1 - \delta_1}, \qquad \nu_2 p_2^{\alpha_2 - \delta_2}, \qquad \ldots .$$

It follows at once that, if without admitting new primes into S one lets all the exponents $\alpha_1, \alpha_2, \ldots$ grow beyond every bound, the quotient S/N will soon reach a fixed value no longer depending on $\alpha_1, \alpha_2, \ldots$.

The property just proved has an interesting consequence for the theory of quadratic forms of positive determinant. Add to the preceding assumptions the further assumption that D contains no square factor. Let h denote the number of classes into which the forms belonging to determinant D are divided, and let h' have the analogous meaning for the determinant $D' = DS^2$. Then, as was proved in an earlier paper, one has the equation

$$h' = h \, \frac{\log(T + U\sqrt{D})}{\log(T' + U'\sqrt{D})} \, SR,$$

where, as regards the factor R, it is to be observed that it depends on the primes $p_1, p_2, \ldots$, but not on the exponents $\alpha_1, \alpha_2, \ldots$. Since this equation may also be put in the form

$$h' = h \, \frac{S}{N} R,$$

it follows that from every determinant D infinitely many other determinants DS^2 can be derived, all of which have the same number of classes. By a suitable choice of D and of the primes $p_1, p_2, \ldots$, one can arrange that this unvarying number of classes agrees with the number of genera. This proves the correctness of Gauss's conjecture that the series of positive determinants which possess only one class in each corresponding genus does not break off, whereas the negative determinants having the same property seem to exist only in finite number [*Disquisitiones arithmeticae*, art. 304].

G. Lejeune Dirichlet's Werke. II.

XVI.

On a Property of Quadratic Forms with Positive Determinant.

By G. Lejeune Dirichlet

English translation

On a Property of Quadratic Forms with Positive Determinant.

Extract from the proceedings of the Berlin Academy (July 1855), freely translated by the author.

Since the property of quadratic forms which I propose to present in this Note is intimately connected with the theory of the equation

$$t^2 - Du^2 = 1, \tag{1}$$

I shall begin with a few remarks on this equation. In it the given positive integer D is assumed not to be a square, and we shall have to consider only solutions expressed in positive integers t and u.

It is known that, if T, U denote the smallest positive integers satisfying the proposed equation, all positive solutions of the equation are given by the formula

$$(T + U\sqrt{D})^n = t_n + u_n\sqrt{D},$$

where n is to be assigned successively the integer values from $n = 1$ up to $n = \infty$. Among the values thus obtained for u_n, there are infinitely many which are divisible by any given positive integer S, and it is easy to prove that if N denotes the smallest index n for which this condition holds, the other admissible values of n form the sequence of successive multiples of N. To see this, let $D' = DS^2$, and consider the equation

$$t'^2 - D'u'^2 = 1, \tag{2}$$

which has the same form as equation (1). One sees that by putting

$$t = t', \qquad u = Su',$$

every solution of equation (2) gives a solution of equation (1) for which u is divisible by S, and conversely. The assertion made above is therefore justified, and one also recognizes that the integer N is given by the equation

$$(T + U\sqrt{D})^N = T' + U'\sqrt{D'},$$

where T' and U' denote the smallest positive values of t' and u'.

The exponent N can be specified without first determining T' and U'. It suffices to know, for each prime divisor p of S, the smallest index ν such that u_ν is divisible by p, and at the same time to know the highest power p^δ of p which divides u_ν. Under the two assumptions just made, one can immediately indicate the exponent of the highest power of p contained in $u_{\nu e}$, where e is an arbitrarily chosen integer. The exponent in question is $\delta + \varepsilon$, where ε, which may also reduce to zero, denotes the highest power of p which divides e.

To recognize the truth of what has just been stated, let θ and η be two integers satisfying equation (1), and let p^k, with $k \neq 0$, be the highest power of p contained in η. Deriving a new solution from this one by means of the formula

$$(\theta + \eta\sqrt{D})^m = t + u\sqrt{D},$$

we shall have

$$u = \eta\left(m\theta^{m-1} + \frac{m(m-1)(m-2)}{1\cdot 2\cdot 3}\theta^{m-3}\eta^2 D + \cdots\right).$$

It is manifest, since θ is not divisible by p, that the highest power of p contained in u will be p^k if m is not divisible by p, and will be p^{k+1} if $m = p$. Since raising to an arbitrary power can be reduced to a combination of the two preceding special cases, the result stated above is established.

From the property just proved, we see that the necessary and sufficient condition for $u_{\nu e}$ to be divisible by an arbitrary power p^α, where $\alpha > 0$, is that e have the factor $p^{\alpha-\delta}$, the exponent $\alpha - \delta$, if it becomes negative, being replaced by zero.

Let us now use indices to denote the unequal prime numbers p contained in S, and the corresponding values of ν and δ, and put

$$S = p_1^{\alpha_1} p_2^{\alpha_2} \cdots .$$

The required exponent N, according to what precedes, will be the least common multiple of the numbers

$$\nu_1 p_1^{\alpha_1-\delta_1}, \qquad \nu_2 p_2^{\alpha_2-\delta_2}, \qquad \ldots .$$

It is easy to see that, if without introducing new prime factors into S one lets all the exponents $\alpha_1, \alpha_2, \ldots$ increase indefinitely, the quotient S/N will soon cease to vary, while retaining a value independent of $\alpha_1, \alpha_2, \ldots$.

The theorem just established for equation (1) leads to a remarkable consequence concerning the theory of quadratic forms with positive determinant. To the preceding assumptions, add that D should have no square divisor. Let h denote the number of classes into which the quadratic forms with common determinant D are distributed, and give h' the analogous meaning relative to the determinant $D' = DS^2$. We then have the equation

$$h' = h\,\frac{\log(T + U\sqrt{D})}{\log(T' + U'\sqrt{D'})}\,SR,$$

established in my memoir *Researches on Various Applications of Infinitesimal Analysis to the Theory of Numbers*, part two. As for the quantity R which enters into it and which we have not had to mention above, it will suffice for our purpose to recall that this quantity does in fact depend on the primes $p_1, p_2, \ldots$, but in no way on the exponents $\alpha_1, \alpha_2, \ldots$. Since our equation can be put in the form

$$h' = h\,\frac{S}{N}R,$$

we conclude from the preceding result that from every positive determinant D one can derive infinitely many other determinants of the form DS^2, all having the same number of classes. By choosing the integer D and the primes $p_1, p_2, \ldots$ suitably, one can arrange, in infinitely many ways, that the invariable number of classes coincide for this entire infinite series of determinants with the number of genera. This confirms the conjecture stated by the illustrious author of the *Disquisitiones arithmeticae* (see art. 304), according to which the positive determinants possessing a single class in each of the corresponding genera form an infinite sequence. This is an analytical phenomenon all the more remarkable because the negative determinants enjoying the same property appear to be only finite in number, and indeed very limited. If one may rely on an induction carried very far – up to 10000, if I am not mistaken – first by Euler, who dealt with these same numbers under the name *numeri idonei*, that is, numbers suitable for finding very large prime numbers, at a time when the theory of quadratic forms, founded by Lagrange, did not yet exist, and later by Gauss, the negative determinants in question would number only 65, and the greatest of these determinants, disregarding the sign, would have the value 1848.

G. Lejeune Dirichlet's Werke. II.

XVII.

On a Theorem Concerning Series.

By M. G. Lejeune Dirichlet

English translation

On a Theorem Concerning Series.

The theorem which I am about to establish in a new way is the same one which has served me as a lemma in several preceding memoirs. The proof which I gave of it in the first of these memoirs assumes a certain condition which, although it is satisfied in most of the applications one can make of the lemma, is not indispensable, as I have already remarked elsewhere and as will also be seen in what follows.

I begin with a very simple special case. Let a and b denote positive constants and let ρ likewise be a positive variable. Then

$$\lim_{\rho \to 0} \rho \sum_{n=0}^{n=\infty} \frac{1}{(b+na)^{1+\rho}} = \frac{1}{a},$$

where the limit refers to the indefinite decrease of ρ. To prove this special case, consider the integral

$$\int_b^\infty \frac{dx}{x^{1+\rho}} = \frac{1}{\rho b^\rho}$$

as the area under a curve whose abscissa is x. Since this curve approaches more and more closely the x-axis, if one draws parallels to this axis through the endpoints of the ordinates corresponding to $x = b, b+a$, and so on, one obtains exterior and interior rectangles for the area. It follows at once that the sum whose limit is in question lies between the two quantities

$$\frac{1}{ab^\rho} \quad \text{and} \quad \frac{1}{ab^\rho} + \frac{\rho}{b^{1+\rho}},$$

whose common limit is $1/a$.

Now to pass to the general theorem, let

$$k_1, \quad k_2, \quad k_3, \quad \ldots, \quad k_n, \quad \ldots$$

be positive constants arranged in increasing order of magnitude, so that $k_n \leq k_{n+1}$. Suppose that these constants are such that, if t denotes a positive continuous variable and T the number of terms of our sequence which do not exceed t, the ratio

$$\frac{T}{t}$$

converges to the finite limit α as t increases indefinitely. Under this assumption, I say that the same quantity α is also the limit of the sum

$$\rho \sum_{n=1}^{n=\infty} \frac{1}{k_n^{1+\rho}} \tag{1}$$

for an infinitely small value of ρ.

By the assumed condition, if δ denotes a number as small as desired, one can choose another number τ large enough so that, whenever t exceeds τ, one always has

$$\alpha - \delta < \frac{T}{t} < \alpha + \delta.$$

This done, let N be the value of T corresponding to $t = \tau$, and consider first only those terms of our series whose index $n > N$. Such a term k_n can present two cases, according as one has

$$k_n < k_{n+1} \quad \text{or} \quad k_n = k_{n+1}.$$

In the first case, the value of T corresponding to $t = k_n$ is n, and one has

$$\alpha - \delta < \frac{n}{k_n} < \alpha + \delta.$$

In the second case, let $k_{m'}$ be the last among the terms following k_n which is still equal to k_n, and, among the preceding terms, let k_m be the first whose value is smaller than that of k_n. If now we let t increase starting from $t = k_m$, T remains constantly equal to m until t reaches the value $k_{m'}$, and our ratio T/t thereby approaches $m/k_{m'}$ indefinitely, while always remaining greater than $\alpha - \delta$. Hence

$$\frac{m}{k_{m'}} > \alpha - \delta.$$

But by the first case we also have

$$\frac{m'}{k_{m'}} < \alpha + \delta,$$

and moreover

$$m < n < m', \qquad k_n = k_{m'}.$$

We therefore conclude, as before, that

$$\alpha - \delta < \frac{n}{k_n} < \alpha + \delta.$$

Putting this double inequality in the form

$$(\alpha - \delta)^{1+\rho} \frac{\rho}{n^{1+\rho}} < \frac{\rho}{k_n^{1+\rho}} < (\alpha + \delta)^{1+\rho} \frac{\rho}{n^{1+\rho}},$$

and summing from $n = N + 1$ to $n = \infty$, we obtain

$$(\alpha - \delta)^{1+\rho} \rho \sum_{n=N+1}^{n=\infty} \frac{1}{n^{1+\rho}} < \rho \sum_{n=N+1}^{n=\infty} \frac{1}{k_n^{1+\rho}} < (\alpha + \delta)^{1+\rho} \rho \sum_{n=N+1}^{n=\infty} \frac{1}{n^{1+\rho}}.$$

But the expression $\rho \sum 1/n^{1+\rho}$ falls under the special case examined above when one puts

$$b = N + 1, \qquad a = 1,$$

and consequently has unity as its limit. Thus the part of our sum (1) which extends from $n = N + 1$ to $n = \infty$ will eventually remain between $\alpha - \varepsilon$ and $\alpha + \varepsilon$, where ε is only slightly larger than δ and therefore, like δ, arbitrarily small. At the same time the first part, which has only the finite number N of terms, plainly converges to zero. It follows that the limit of expression (1) is indeed α, Q.E.D.

XVIII.
On the equation $t^2 + u^2 + v^2 + w^2 = 4m$

Translated from the French

Liouville, Journal de Mathématiques pures et appliquées, Série II, Tome I, 1856, p. 210–214.

On the equation $t^2 + u^2 + v^2 + w^2 = 4m$.

[Extract from a letter of M. Lejeune Dirichlet to M. Liouville.]

Permit me, dear friend, to return for a moment to the conversation we had recently about Jacobi's beautiful theorem on the number of decompositions of an integer into four squares, a theorem which that illustrious geometer first drew from his elliptic series and for which he later gave an arithmetical proof. Having better recalled my memories, I shall present to you, with more development and especially in somewhat better order, what the other day I could only indicate. As I was saying to you, I have always experienced a certain difficulty, which I believe has also occurred to other persons, when I wanted to reproduce this beautiful proof without having the author's text before my eyes and without recurring to the operations on series of which Jacobi's proof, as he himself warns, is only the translation. This circumstance led me long ago to seek a new proof of the theorem in question on other principles; but I shall not speak of it here, since that proof will find its natural place in a work whose writing has occupied me for some time. For the moment, my only object is to simplify Jacobi's proof and especially to make it easier to retain, by bringing into full light the arithmetical, or rather algebraic, fact which forms its principal foundation.

To avoid useless repetitions, I shall note that all the integers which we denote by Latin letters are positive and odd, except those denoted by x and x', which are not subject to this restriction.

As for the lemma which serves as Jacobi's point of departure and which defines the number of solutions of the equation

$$t^2 + u^2 = 2p,$$

p being supposed given, I shall not reproduce here the very simple demonstration that Jacobi gave of it, all the more since this lemma falls within a general proposition of which I have had occasion to speak elsewhere. It is known that one must perform all decompositions

$$p = ad$$

of p into two factors, and that the number in question is equal to the excess of the number of cases in which the first factor a is of the form $4\nu + 1$ over the number of cases in which a is of the form $4\nu + 3$. Thus, agreeing to make correspond to each decomposition a unit δ, positive or negative according to these two cases, the number in question will be expressed by the sum

$$\sum \delta.$$

This being established, let us try to determine the number of solutions of the equation

$$t^2 + u^2 + v^2 + w^2 = 4m, \tag{1}$$

in which m denotes a given odd integer. To obtain all these solutions, one must set, in all possible ways,

$$4m = 2p + 2q \quad \text{or} \quad 2m = p + q,$$

and then solve, for each pair p, q, the two equations

$$t^2 + u^2 = 2p, \qquad v^2 + w^2 = 2q$$

in the most general manner.

If, first considering a fixed pair p, q, we make correspond to each decomposition

$$q = bc$$

a unit $\varepsilon = \pm 1$, which depends on b in the same manner as δ depends on a, the number of solutions of equation (1) corresponding to this pair p, q will be

$$\sum \delta \cdot \sum \varepsilon,$$

or also

$$\sum \eta,$$

the terms of this sum corresponding to the different combinations that can be formed with the decompositions

$$p = ad, \qquad q = bc,$$

and η being equal to the positive or negative unit according as the residues of a and b, when divided by 4, are or are not equal; equivalently, according as $a - b$ is or is not divisible by 4. With this understood, it is manifest that the number of all solutions of our equation (1) will again be expressed by the sum

$$\sum \eta,$$

whose terms will always depend in the same manner on a and b, but will now correspond to all combinations which a and b present in the complete solution of the equation

$$2m = ad + bc. \tag{2}$$

Before going further, it is useful to make a very simple remark which will be useful to us later: among the two even integers

$$a - b \quad \text{and} \quad c + d$$

there is always one, and only one, which is divisible by 4. Indeed, if it were otherwise, one would have

$$\text{either } a \equiv b,\ c \equiv -d, \quad \text{or } a \equiv -b,\ c \equiv d \pmod 4,$$

and it would follow that $ad + bc \equiv 0 \pmod 4$. Thus η may be defined otherwise by saying that η is equal to -1 or $+1$, according as $c + d$ is or is not divisible by 4.

Let us now distribute the solutions of equation (2) into two classes, putting into the first those for which $a = b$, and into the second all the others. Since the latter are associated two by two and result one from the other by exchanging simultaneously a with b, and d with c, which does not change the value of η, one sees that the number H of solutions of equation (1) can be put in the form

$$H = \sum \eta' + 2 \sum \eta'',$$

where the terms η' of the first sum, which moreover are all equal to the positive unit, correspond to the solutions of equation (2) for which $a = b$, and the terms η'' of the second correspond to the solutions of the same equation that satisfy the condition $a > b$.

This being done, we shall first consider the solutions of (2) for which $a > b$. Put

$$a' = c(x+1) + d(x+2), \qquad c' = a(x+1) - b(x+2),$$
$$b' = cx + d(x+1), \qquad d' = -ax + b(x+1).$$

Then identically

$$a'd' + b'c' = ad + bc.$$

It follows that every solution of equation (2) supplies infinitely many others by means of the indeterminate integer x; but it remains to see to what restriction this integer must be subject in order that a', b', c', d' be odd and positive, and that in addition $a' > b'$ as in the primitive solution. Now each of the pairs

$$x,\ x+1 \quad \text{and} \quad x+1,\ x+2$$

is composed of two integers, one even and the other odd, so the first condition is always fulfilled. As for the second, since

$$c' = (a-b)(x+1) - b, \qquad d' = b - (a-b)x,$$

one sees that it completely determines the integer x: $(a-b)x$ must be the multiple of the positive even number $a-b$ immediately below the odd number b. One also sees, since x is not negative, that a' and b' will be positive and that, moreover, $a' > b'$.

Having assured ourselves that the expressions given above always furnish a unique solution subject to the same conditions as the primitive solution, let us see to what solution we shall be led if we take the solution a', b', c', d' in turn as point of departure. For this, in the expressions

$$c'(x'+1) + d'(x'+2), \qquad a'(x'+1) - b'(x'+2),$$
$$c'x' + d'(x'+1), \qquad -a'x' + b'(x'+1),$$

we must give to the indeterminate x' the value, wholly determined by what precedes, which makes these expressions positive. On the other hand, the equations set above give by their solution

$$a = c'(x+1) + d'(x+2), \qquad c = a'(x+1) - b'(x+2),$$
$$b = c'x + d'(x+1), \qquad d = -a'x + b'(x+1),$$

whose first members are positive. Thus x' coincides with x, and we are brought back to the primitive solution a, b, c, d.

If now we observe that to our two solutions so linked with one another, and which are mutually deduced one from the other by a uniform procedure, there always correspond opposite values of η'', as follows from the equation

$$a - b = c' + d'$$

and from the remark made above, we conclude that $\sum \eta''$ reduces to zero, and we simply have

$$H = \sum \eta'.$$

To evaluate this sum, all of whose terms η' are equal to the positive unit, everything reduces to finding the number of solutions of the equation

$$a(d+c) = 2m.$$

Now if a is a fixed divisor of m, $m/a = e$ will also be such a divisor, and one will have to solve the equation

$$d + c = 2e,$$

whose number of solutions is e. Hence, since a, and consequently also e, must successively be made equal to all divisors of m, the number H that was to be determined coincides with the sum of the divisors of m.

XIX.

A new proof of a proposition concerning the theory of quadratic forms

Translated from the French

Liouville, Journal de Mathématiques pures et appliquées, Série II, Tome I, 1857, p. 273–276.

A new proof of a proposition concerning the theory of quadratic forms.

The proposition of which I wish to give a new and very simple proof belongs to a theory presented in detail in Gauss's *Disquisitiones arithmeticae*; I may therefore refer to that work for the terminology I shall have to use, and pass immediately to the object to be accomplished.

> Given an improper substitution
> $$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$
> by which the quadratic form $(a, b, c) = f$, whose determinant is supposed different from zero, is changed into itself, one can always obtain a new form belonging to the same class, and for which twice the middle coefficient is a multiple of the first coefficient.

The supposition in the statement immediately supplies, besides the equation

$$\alpha\delta - \beta\gamma = -1,$$

the following three:

$$a(\alpha^2 - 1) + 2b\alpha\gamma + c\gamma^2 = 0,$$
$$a\alpha\beta + 2b\alpha\delta + c\gamma\delta = 0,$$
$$a\beta^2 + 2b\beta\delta + c(\delta^2 - 1) = 0,$$

where the second has been simplified by substituting $\alpha\delta + 1$ for $\beta\gamma$. Taking the sum of the first two after multiplying them respectively by $-\delta$ and γ, and then the sum of the last two after multiplying by β and $-\alpha$, one obtains the new equations

$$a[\alpha(\beta\gamma - \alpha\delta) + \delta] = a(\alpha + \delta) = 0,$$

$$c[\delta(\beta\gamma - \alpha\delta) + \alpha] = c(\alpha + \delta) = 0.$$

It follows that, if a and c do not vanish simultaneously, then

$$\alpha + \delta = 0.$$

But this last equation also holds in the excepted case; for, since in that case b is different from zero, one has at the same time

$$\alpha\gamma = 0, \qquad \alpha\delta = 0, \qquad \beta\delta = 0,$$

equations compatible with

$$\alpha\delta - \beta\gamma = -1$$

only if one supposes simultaneously

$$\alpha = 0, \qquad \delta = 0.$$

It is therefore proved that our improper substitution is always of the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix},$$

where

$$\alpha^2 + \beta\gamma = 1.$$

Let us pause for a moment at the particular case in which the third coefficient γ of this substitution vanishes. Our equations then give

$$\alpha = -\delta = \pm 1, \qquad 2b = \pm a\beta.$$

Thus, in this case, the given form f already has the required property. It follows that, if γ is not zero, everything reduces to obtaining a form equivalent to f, and transformable into itself by an improper substitution whose third coefficient vanishes.

For this purpose let

$$\begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

be the most general proper substitution; that is, let λ, μ, ν, ρ be four integers subject to the sole condition

$$\lambda\rho - \mu\nu = 1.$$

Let φ denote the form into which f is changed when this substitution is applied to it. The inverse substitution

$$\begin{pmatrix} \rho & -\mu \\ -\nu & \lambda \end{pmatrix},$$

which is also proper, will change φ into f.

Now compose the three substitutions

$$\begin{pmatrix} \rho & -\mu \\ -\nu & \lambda \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

in the order in which we have just written them. Since the first changes φ into f, the second changes f into f, and the third changes f into φ, the composed substitution, which is moreover evidently improper, changes the form φ into itself. The question is thus reduced to making the third coefficient of our composed substitution equal to zero.

To obtain this coefficient, first compose the first two substitutions, and write only the third and fourth coefficients of the substitution that results from this first composition. These coefficients are

$$\gamma\lambda - \alpha\nu, \qquad -\alpha\lambda - \beta\nu.$$

It follows that the coefficient to be obtained is

$$(\gamma\lambda - \alpha\nu)\lambda - (\alpha\lambda + \beta\nu)\nu = \gamma\lambda^2 - 2\alpha\lambda\nu - \beta\nu^2.$$

Since γ is not zero by hypothesis, we may, before setting our expression equal to zero, multiply it by γ. We then have

$$\gamma(\gamma\lambda^2 - 2\alpha\lambda\nu - \beta\nu^2) = (\gamma\lambda - \alpha\nu)^2 - (\alpha^2 + \beta\gamma)\nu^2 = (\gamma\lambda - \alpha\nu)^2 - \nu^2 = 0,$$

and consequently

$$[\gamma\lambda - (\alpha + 1)\nu][\gamma\lambda - (\alpha - 1)\nu] = 0.$$

Thus, if after reducing the ratio $(\alpha \pm 1)/\gamma$, where the ambiguous sign is arbitrary, to its simplest expression λ/ν, one derives from the integers λ, ν two integers μ, ρ satisfying

$$\lambda\rho - \mu\nu = 1,$$

then the substitution

$$\begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

applied to the form f will change that form into another equivalent form φ having the required property.

Toul, 15 August 1857.

XX.
Investigations on a problem of hydrodynamics

Translated from the German

Nachrichten von der G. A. Universität und der Königl. Gesellschaft der Wissenschaften zu Göttingen, Jahrg. 1857, Nr. 14, August 10, S. 205–207.

Investigations on a problem of hydrodynamics.

[Extract from a memoir presented to the Royal Society on 31 July 1857.]

Although the fundamental equations of hydrodynamics, in the form in which one finds them in textbooks, have been known since Euler, and in another form, due to Lagrange, since the appearance of the first edition of the *Mécanique analytique*, nevertheless in all cases investigated up to now in which the fluid changes its shape in the course of the motion, one has not derived the complete solution of the problem from the fundamental equations, but has confined oneself to an approximate determination of the motion. Prompted by the circumstance just mentioned to attempt to treat a problem of the indicated kind in complete rigor, the author of the memoir, given the difficulty of the subject, whose resolution requires the integration of a system of nonlinear partial differential equations, was prepared for such an attempt to succeed only under the simplest hypotheses on the conditions determining the motion of the fluid. He was therefore not a little surprised when, after several fruitless efforts, a case presented itself which is not to be counted among the simpler ones, insofar as the mutual attraction of the elements of the fluid is taken into account, and in which, notwithstanding this circumstance, the motion can be recognized not only in its general character, but also, under certain restrictions, determined in its details.

The conditions of this case of motion, which is treated in the memoir, and the general result relating to it, are expressed in the following proposition:

> If a homogeneous incompressible fluid which at its surface undergoes a constant pressure, or one varying only with time, initially has the shape of an ellipsoid; and if, moreover, the initial motion is decomposable into two simpler motions, a rotation of the fluid like that of a rigid body about an axis passing through the center of mass, and a second motion changing the relative position of the elements, in which the velocities of those elements, resolved perpendicular to three fixed planes meeting at right angles at the center, are proportional to the distances from the planes, then the fluid, in the motion arising as a consequence of such an initial state, will at every later time also have the shape of an ellipsoid concentric with the original one, but whose axes in general change with time both in direction and in magnitude. Of the instantaneous motion occurring at any given time, the same holds as was assumed for the original motion; that is, it is decomposable into two simpler motions of the kind just defined, but so that both the axis of rotation and the three mutually perpendicular planes to which these component motions refer also in general assume a different position at every instant.

For complete knowledge of the motion whose general character has just been indicated, it is necessary to determine 9 functions of time, which are defined by just as many equations, one finite equation and 8 differential equations of the second order. For the latter one can in general set up 7 first integrals; these, however, are not sufficient for the complete solution of the problem, that is, for its reduction to quadratures, unless the initial shape and the state of motion occurring in the first instant, which contains 8 arbitrary elements, are subjected to further restrictions.

Among the cases reducible to quadratures, the simplest, and the only one that can be mentioned here, is the case in which the mass originally has the form of a spheroid and no initial velocities are present. The motion then consists of isochronous oscillations in which the fluid, passing through the spherical shape, alternately assumes the form of a prolate and of an oblate ellipsoid.

XXI.
Notice concerning Jacobi's scientific estate

Translated from the German

Crelle, Journal für die reine und angewandte Mathematik, Bd. 42, S. 91–92.

Notice concerning Jacobi's scientific estate.

Soon after Jacobi's lamented death, Frau Professor Jacobi, with a trust that honored me, placed in my hands the whole scientific estate of my unforgettable friend. In order first to obtain a general overview of it, it was necessary to arrange the great mathematician's numerous manuscripts by subject. This task, which I undertook jointly with Jacobi's friends here, Messrs. Borchardt and Joachimsthal, was not without difficulty, since the manuscripts, probably as a consequence of repeated changes of residence, were in great disorder, and the separate quires or leaves that belonged together, usually without pagination, often had to be laboriously sought out from different bundles. As soon as this preliminary work is completed, in the execution of which we have been assisted by Messrs. Kummer and Rosenhain, who are presently here, the manuscripts are to be distributed among the friends of the deceased who have declared themselves ready for this, for the purpose of a detailed examination.

In the preliminary arrangement it has been found that only little is completely ready for print. In most cases there are repeated treatments of the same or closely related subjects, which evidently, although every indication of date is lacking in the manuscripts, belong to very different times. These treatments will have to be reviewed and compared with the greatest care, in order to determine which of the earlier ones have become superfluous through the later ones, or what is to be taken from the former and inserted at a suitable place in the later, so that nothing essential of the creations of the great, indefatigable investigator may be lost to science. The papers suitable for publication will be printed in the present journal, in which, with the exception of the two special works *Fundamenta nova theoriae functionum ellipticarum* and *Canon arithmeticus*, almost all of Jacobi's works first appeared, and are later to be collected, as he himself had already begun to do by publishing the first volume of his works (Berlin, G. Reimer, 1846).

Besides publishing the papers written by Jacobi himself, his friends intend to bring before the public the most important of the university lectures that he gave in Königsberg and here. To all who take an interest in the progress of the mathematical sciences it is known what influence Jacobi exerted, also in his vocation as a university teacher, to which he always devoted himself with special love and with the rarest success, on the great upswing that mathematical studies have taken during the last quarter century in our German fatherland. If the knowledge of higher analysis is now spread among us to a degree as at no earlier time, and if numerous younger mathematicians are extending and enriching the science in the most diverse directions, then he has had the greatest share in so gratifying a phenomenon. Almost all have been his pupils; rarely did a budding talent escape his attention, and no one, once he had recognized it, lacked his furthering counsel and encouraging participation.

So that such a successful teaching activity, to which death set so early a limit, may at least retain the after-effect which the printed word, in the absence of the living spoken word, is able to produce, the friends of the deceased will publish in print his most important lectures in exact reproduction. They are convinced that no more effective means of education can be placed in the hands of the younger followers of science than the lectures of a creative mind who had made it his particular task, in all difficult investigations, to introduce his hearers into the course of thought of discovery.

Since Jacobi always delivered his lectures quite freely and without using a written draft, his estate contains, apart from a few brief notes, nothing in his own hand relating to his lectures. On the other hand, there are found in it very accurate transcripts of his most important lectures, made by several of his most distinguished hearers, which he had carefully collected for years, partly in order later to facilitate thereby the preparation of his lectures, and partly to use them in the preparation of textbooks which he intended to publish. With the help of these transcripts and others of equal accuracy that have been put at their disposal for this purpose, Jacobi's friends will be able to reproduce his most important lectures with great fidelity. Among these lectures, which will appear in individual volumes, the following may preliminarily be named: 1. those on the theory of elliptic functions; 2. those on the division of the circle and its applications to number theory; 3. those on analytic mechanics; and finally 4. those on the general theory of curves and surfaces. For some others of smaller scope, the decision whether they are to be printed remains reserved.

Berlin, August 1851.

G. Lejeune Dirichlet’s Werke. II.

XXII.
Memorial Address on Carl Gustav Jacob Jacobi

Translated from the German

Abhandlungen der Königlich Preussischen Akademie der Wissenschaften von 1852, pp. 1–27.

Memorial Address on Carl Gustav Jacob Jacobi.

Memorial Address on Carl Gustav Jacob Jacobi [Delivered in the Academy of Sciences on 1 July 1852[1].]

In undertaking to describe the scientific achievements of the greatest mathematician who, since Lagrange, has belonged to our body as a resident member, I feel vividly the difficulties of the task. I must set forth the full significance of the creations of a man who, with a strong hand, intervened in almost every region of a science which through two thousand years of labour has grown to immeasurable extent; who wherever he directed his creative mind brought important and often deeply hidden truths to light; and who, by introducing new fundamental ideas into science, raised mathematical speculation in more than one direction to a higher level. Only the conviction that a debt of gratitude is owed for such services rendered to science and to those who cultivate it can silence the reservations called forth in me by the consciousness of my inadequacy; for on whom could the fulfilment of this duty fall more than on me, who, like all my colleagues in the field, was so essentially advanced by Jacobi’s scientific productions and who, besides, owed no less instruction to my many years of close intercourse with the great investigator.

Carl Gustav Jacob Jacobi was born on 10 December 1804 in Potsdam, where his father was a well-to-do merchant. His first instruction in the ancient languages and in the elements of mathematics was given by his maternal uncle, Mr. Lehmann, who had less to teach the active boy than to guide him. Under this intelligent guidance Jacobi made such rapid progress that, not yet twelve years old, he entered the second class of the Potsdam Gymnasium and after only half a year was admitted to the first. He remained there for a full four years, since he could not properly attend the university before completing his sixteenth year. The mathematical instruction, which was treated wholly as a matter of memory, could not suit the young senior pupil. His relation to the teacher was therefore for some time very disagreeable; in the end it improved, because the teacher had the good sense to let the unusual pupil go his own way and to permit him to occupy himself with Euler’s Introductio while the other pupils recited elementary propositions that had been laboriously learned. How far Jacobi’s intellectual development had already advanced is shown by the attempt he made at about this time to solve equations of the fifth degree, an attempt which he later mentioned in one of his papers.

[1]The corresponding note in the Academy report for 1852 reads on p. 433: ”Hereupon Mr. Dirichlet delivered a memorial address on the deceased member of the Academy, the mathematician Jacobi.” K.

Many of those who later acquired a great name first tested their powers on this problem; and it is indeed easy to understand what attraction this problem must have exercised on an awakening talent as long as its impossibility had not yet been proved. To the fame which so many fruitless efforts had given this investigation there was added the special circumstance that the problem, belonging to a domain which borders immediately on the elements, seemed accessible without a large store of preliminary knowledge.

At the university here Jacobi divided his time among philosophical, philological, and mathematical studies. As a participant in the exercises of the philological seminar he attracted the attention of our colleague Böckh, the head of that institute, who became very fond of the young man because of his keen and distinctive mind and distinguished him by special goodwill. He seems to have attended few mathematical lectures, since at that time they had too elementary a character at this university to advance Jacobi essentially; he was already familiar with some of the chief works of Euler and Lagrange. All the more zealously he surveyed the mathematical literature, seeking in particular to gain a general overview of the great scientific treasures contained in the academy collections.

Jacobi's nature was not satisfied with the mere gathering of information. He felt the need to become fully master of the things with which he occupied himself. After about two years of university study he recognized the necessity of making a decision and of renouncing either philology or mathematics. Since the decision he made had important consequences not only for him but also for the science to which from then on he devoted himself exclusively, one will gladly hear from him the reasons that determined his choice. He wrote on this subject to the uncle already mentioned: "Although for a time I occupied myself seriously with philology, I did at least succeed in gaining a glimpse into the inner splendour of ancient Hellenic life, so that I could not give up its further investigation without a struggle. But for the present I must give it up entirely. The immense colossus which the labours of Euler, Lagrange, and Laplace have called forth demands the most immense force and exertion of thought if one is to penetrate into its inner nature, and not merely rummage about its exterior. A drive that will not rest or pause compels one to become master of this structure, so that one need not fear at every moment to be crushed by it, until one stands above and can survey the whole work. Only then is it possible to work calmly and properly on the perfection of its individual parts, and to carry the great whole further according to one's powers, once one has grasped its spirit."

For his doctoral dissertation Jacobi chose a subject already often treated, the decomposition of algebraic fractions. In it he first proved remarkable formulae which Lagrange had given without proof in the memoirs of our Academy; he then passed to a new kind of decomposition, which unlike the one exclusively considered until then is not completely determinate; and he concluded the paper with investigations on the transformation of series, in which a new principle already becomes noticeable, one of which he made repeated use in later works.

Immediately after his promotion Jacobi habilitated at the university and gave a lecture on the theory of curved surfaces and curves in space. According to the testimony of one of his hearers at that time, his teaching talent must already have been highly developed when he first appeared, and he must have known how to treat his subject with great clarity and in a way very stimulating for his audience. The twenty-one-year-old lecturer also showed very early maturity of judgement in that, undisturbed by the discredit into which the method of the infinitely small had at that time fallen through a great authority, he followed precisely this method in his exposition and sought, with the best success, to convince his hearers that the suspected method differs from the strict method of the ancients only in its abbreviated form, but becomes indispensable precisely through this form in all more complex questions.

The attention Jacobi began to attract caused the highest educational authority to ask him to continue his teaching activity provisionally as a Privatdocent in Königsberg, where the professorship of mathematics, just then vacant, gave him greater prospects of promotion than did Berlin. On moving to Königsberg it was an important event for Jacobi to become personally acquainted with the great astronomer Bessel, and for the first time to see, close at hand and in a field so nearly related to his own, a genius. The daily sight of the fiery zeal of this extraordinary man exercised the mightiest influence even on Jacobi, though he had been accustomed from his earliest youth to demand the greatest efforts from himself; he often mentioned this influence later with gratitude.

It was a fortunate circumstance for Jacobi's literary career that its beginning coincided with the founding of the mathematical journal through whose publication our colleague Crelle acquired so great and lasting a merit, not only for the dissemination but also for the enlivening of the study of science. Jacobi, who belonged to the earliest collaborators of the journal, remained faithful to it until his death; and, apart from the two separate works Fundamenta nova theoriae functionum ellipticarum and Canon arithmeticus, almost all his other works first appeared in Crelle's Journal.

Jacobi's first papers already show him as a fully mature mathematician, whether, as in the essays on Gauss's new method for the approximate determination of integrals and on Pfaff's method for the integration of partial differential equations, he was considering known theories from a new point of view and essentially simplifying them, or whether he was treating problems not yet solved and arriving at new results. Among works of the latter kind, two should be especially mentioned here: a paper of a few pages in which he teaches a hitherto unknown fundamental property of the remarkable function first introduced into science by Legendre, which plays so great a role in all later general investigations on attraction; and another paper on cubic residues. The latter contains only theorems without proofs, but these theorems are of such a kind that they cannot be the result of induction and leave no doubt that Jacobi already then possessed new and fruitful principles in the scientific field that Gauss had opened to mathematical speculation a quarter of a century earlier, and that belongs as much to higher algebra as to the theory of numbers. This is also confirmed by a later publication in which he expressly mentions that he had already at that time communicated these principles to Gauss by letter.

Jacobi was then drawn away from the further pursuit of this subject by another work: his investigations on elliptic functions, which were soon to bring him so great a fame and to assign him a place among the foremost mathematicians of the age. The young mathematician, who had already tried his powers successfully in so many directions, seemed for a longer time to be not favoured by fortune in the theory of elliptic functions. One of his friends, who one day found him strikingly out of spirits, asked the reason for this mood and received the answer: "You see me just on the point of sending this book, Legendre's Exercices, back to the library; I have had decided bad luck with it. When I have otherwise studied an important work, it has always stirred me to thoughts of my own and something has always fallen to me from it. This time I have come away quite empty and have not been inspired by the slightest idea."

If his own thoughts were somewhat long in coming in this case, they appeared later all the more abundantly - so abundantly that, together with Abel's simultaneous thoughts, they produced an unexpected extension and the complete transformation of one of the most important branches of analysis. Since progress here proceeded at the same time from two different sides, it is necessary, beside Jacobi's investigations, to mention Abel's simultaneous works. Though independent of one another in origin, the discoveries of both later interlocked so closely that the exposition of one would be scarcely comprehensible without consideration of the other.

The theory of elliptic functions, with which the names of Abel and Jacobi are forever connected, does not in its beginnings reach back beyond the first half of the preceding century. An Italian mathematician of unusual acuteness, Count Fagnano from the Papal States, made the remarkable discovery that the integral which expresses the arc of the curve then much studied by mathematicians under the name of the lemniscate possesses properties similar to those of the simpler integral that represents a circular arc; and that, for example, between the limits of two integrals of this kind, one of which is equal to twice the value of the other, there is a simple algebraic connection. Thus an arc of the lemniscate, though a transcendent of higher kind, can nevertheless, like a circular arc, be doubled or halved by a geometrical construction[2]. Euler found, a few years later, the true source of this and of other similar properties in a theorem that belongs among the finest enrichments which science owes to that great investigator.

According to Euler's theorem, a certain integral, more general than that considered by Fagnano and in our present terminology called an elliptic integral of the first kind, depends on its limit in such a way that two such integrals with arbitrary limits can always be united into a third whose limit is a simple algebraic combination of the limits of the former, just as the sine of an arc composed of two parts can be formed algebraically from the sines of its parts. But the elliptic integral is more general than the one that expresses a circular arc. Brought to its simplest form, it depends not, as that one does, merely on its limit, but also on another magnitude contained in the function, the so-called modulus. Euler's theorem gave only relations among integrals with the same modulus. The first example of a connection among integrals that differ in their moduli was offered by a later transformation due to Landen and, in a somewhat different form, to Lagrange. Such a transformation, combined with Euler's theorem, leads in particular to a theorem which includes Fagnano's duplication as a special case.

The theory as Abel and Jacobi found it presented several highly enigmatic phenomena for whose clarification the principles then known did not suffice. Thus, to mention only one of these phenomena, it had been found that the degree of the equation by means of which the division of an elliptic integral into an arbitrary number of equal parts is effected is the square of the number of parts. Since even the division by two already leads to an equation of the fourth degree, the problem of division, notwithstanding Fagnano's discovery, seemed to belong to the class of questions that are algebraically insoluble. In the same way, the transformations that had become known through Landen and Lagrange appeared to stand isolated, and there was no consciousness before Abel and Jacobi of any general principle by which they might be connected.

[2]Fagnano's theorem is found in the *Giornale de' Letterati d'Italia*, tomo 29, Anno 1718. F.

The first fundamental idea by which the theory was transformed was the inversion of the elliptic integral. Earlier mathematicians had viewed the integral as a function of its limit. Abel and Jacobi recognized that the inverse function, the limit regarded as a function of the integral, must be made the proper object of study. This simple change of viewpoint opened the way to the true analogy with the circular functions. A second idea, common to Abel and Jacobi, was the introduction of imaginary quantities. Jacobi later often repeated that this introduction of the imaginary was the true key to the nature of the elliptic functions. Indeed, when one considers how widely Euler had already used imaginary quantities in the theory of the trigonometric functions, it would have been surprising had this thought escaped him entirely. Yet in the new domain it required a special insight. Abel and Jacobi, by introducing the imaginary into the functions obtained by inversion from the elliptic integral of the first kind - those which we now call elliptic functions - recognized that these functions share at once in the nature of the circular functions and in that of exponential quantities, and that while the former are periodic only for real and the latter only for imaginary values of the argument, elliptic functions unite both kinds of periodicity.

Placed by these fundamental ideas on new ground, Abel and Jacobi directed their investigations to two different regions of the theory. Abel's activity turned to the problems concerning the multiplication and division of elliptic integrals. By means of the principle of double periodicity he penetrated deeply into the nature of the roots of the equation on which division depends, and arrived at the wholly unexpected discovery that the general division of the elliptic integral with arbitrary limit can always be performed algebraically, that is, by mere extraction of roots, as soon as the special division of the so-called complete integrals is assumed already carried out. This special division seems possible only for special moduli, among which the simplest is the one corresponding to the lemniscate. Carrying out the solution for this case, he showed that the division of the whole lemniscate is completely analogous to the division of the circle and can be achieved by geometrical construction in precisely the same cases in which, according to Gauss's beautiful theory given twenty-five years earlier, the circle admits such a division.

A noteworthy historical circumstance attaches to this last work of Abel. In the introduction to the last section of the Disquisitiones arithmeticae, which is devoted to the division of the circle, Gauss had remarked in passing that the same principle on which his cyclotomy rests is also applicable to the division of the lemniscate. In fact, the Gauss principle according to which the roots of the equation to be solved are to be brought into a cycle so that each depends on the preceding one in the same way lies essentially at the foundation of Abel's paper on the division of the lemniscate. But while, for the division of the circle, long-known properties of trigonometric functions sufficed to arrange the roots according to Gauss's principle, in the case of the lemniscate an insight into the nature of the roots was required for a similar arrangement, indeed even to recognize the possibility of such an arrangement; and only the principle of double periodicity could provide that insight. Gauss's earlier remark therefore became, through Abel's paper, an incontrovertible testimony that, far in advance of his time, he had already at the beginning of the century recognized the principle of the double period. This testimony, however, first became intelligible through Abel's later work and therefore detracts nothing from the right of Abel and Jacobi to this discovery.

Besides the results just mentioned relating to division, Abel's investigations had another, no less important consequence. Letting the multiplier become infinite in the formulae by which he had represented the elliptic functions of a multiple argument by the functions of the simple argument, he obtained remarkable expressions for the elliptic functions in the form of infinite series and also as quotients of infinite products - a discovery perhaps of even greater importance for analysis than Abel's demonstration of the algebraic solvability of the equations for division.

At the same time as Abel was carrying out these beautiful investigations, Jacobi was no less successfully occupied in another part of the same field. The substitution mentioned above, by which an elliptic integral is transformed into an integral of the same form, had until then been the only one of its kind. Legendre had, it is true, found a second transformation of elliptic integrals not long before Jacobi turned to this subject; but this second transformation, with which Legendre considered the subject closed, was not then known in Germany. It therefore required rare acuteness to infer from a visible ring the existence of an infinite chain, and equally great boldness to set oneself the task of recognizing the nature of that chain.

A fortunate induction, in which the subtle and wholly new idea played an essential role of considering transformation and multiplication from a common point of view and the latter as a special case of the former, led Jacobi to the conjecture that rational functions of every degree are capable of transforming an elliptic integral into an integral of the same form. This conjecture was at once confirmed, for it appeared that the number of arbitrary coefficients available in each degree sufficed to satisfy all conditions that had to be fulfilled if the transformed integral was to agree in form with the original one. Yet even if so simple a consideration could hardly leave any doubt concerning the possibility of the matter, a great step still had to be taken in order to recognize the inner analytical nature of the fractional expressions suited to transformation. What sort of difficulties had to be overcome here, and by what ingenious considerations Jacobi overcame them, cannot be set out here; nor am I permitted to enumerate all the important consequences that followed from the completely solved problem. I mention only the remarkable result of this investigation, that multiplication can always be composed of two transformations.

While Abel and Jacobi were thus perfecting the theory simultaneously in two different directions, it seemed as though fate wished to divide evenly between the young rivals the honour of the advance to be accomplished. For the way in which each soon afterwards carried further the other's invention left no doubt that either one, had the other not anticipated him in part of the work, would have accomplished the whole advance by himself. Jacobi had proceeded in his investigations from the assumption that in the transformation the original variable is rationally expressed by the new one. Abel treated the problem under the more general hypothesis that some algebraic equation holds between the two variables, and reached the result that the problem thus generalized can always be reduced to the case which Jacobi had treated so completely.

Jacobi intervened no less successfully in the theory of general division given by Abel. The manner in which Abel had solved the problem showed indeed that the roots are always algebraically expressible, but for their actual representation it required the formation of certain symmetric combinations of roots, which could be carried out only in each particular case. From a new principle, to be mentioned more closely in a moment, Jacobi derived the final expressions for the roots, valid for every degree and formed directly from the data of the problem. These expressions had, moreover, a greater simplicity of form than Abel's. When Jacobi published the result of this work in a brief note, he hoped to astonish Abel by the perfection of the solution of the division problem; but this hope remained unfulfilled. Abel had just died, scarcely twenty-seven years old, less than two years after the publication of his first works on elliptic functions. Death had set so early a limit to the brilliant career of this profound and comprehensive mind.

Jacobi's further investigations on elliptic transcendents, as well as the one just mentioned, arose from a thought which, because of its consequences, must perhaps be assigned the first place among his conceptions. It was the idea of introducing into analysis, as independent transcendents, the infinite products by whose quotients Abel had expressed the elliptic functions. Once he had succeeded in representing these products, which are all of the same nature and may be regarded as special cases of a single transcendent, in the form of series, he recognized a function that had already presented itself to French mathematicians in investigations of mathematical physics, but had been little noticed there and of which only one property had been observed. Jacobi subjected it to a deeply penetrating investigation, explored its analytical nature, and then introduced it into the theory of integrals of the second and third kind. This led not only to the recognition of the inner connection among already known but isolated properties of these integrals, but also to the important discovery that the integrals of the third kind, which depend on three elements, can be expressed by means of the new transcendent, which contains only two.

In Jacobi's later exposition of the whole theory, as he was accustomed to give it in his lectures, the consideration of this function formed the starting point. The whole doctrine thereby gained not only a surprising degree of simplicity and transparency; this reversed order is also noteworthy because it became the model for other investigations to be mentioned later. If one considers that the new function now rules the whole domain of elliptic transcendents, that Jacobi derived important theorems of higher arithmetic from its properties, and that it plays an essential role in many applications - among which I mention here only the representation by means of this transcendent of the rotational motion that forms one of Jacobi's last and finest works - one must assign to this function the next place after the elementary transcendents long received into science. Strikingly, so important a function has as yet no other name than the transcendent Theta, after the accidental symbol with which it first appears in Jacobi; mathematicians would only fulfil a duty of gratitude if they agreed to attach Jacobi's name to it, in order to honour the memory of the man among whose finest discoveries belongs the first recognition of the inner nature and high significance of this transcendent.

Abel's works mentioned above are not the only first-rank achievement of this outstanding mathematician; they are not even the most important of his achievements. His greatest discovery is contained in a theorem that bears his name and wholly bears the stamp of his extraordinary spirit, whose characteristic feature was to treat the questions of science in the most comprehensive generality. The Euler theorem already described - I speak here of the principle itself, not of the consequences drawn from it, which were daily extended - then formed, in the domain to which it belongs, the boundary of science, beyond which only conjecture could reach. Abel's theorem suddenly opened an immeasurable region to inquiry. Legendre called Abel's theorem a monument more lasting than bronze, and Jacobi designated the same theorem, as it appears in simple generality, as the greatest discovery of the century.

In its original form the theorem asserted that a sum of any number of elliptic integrals can always be reduced to a certain fixed number of integrals of the same kind. Abel's thought went far beyond elliptic integrals and applied to the integrals of arbitrary algebraic functions. It is not possible here to indicate even in outline the whole range of this theorem and the problems to which it gives rise. This work has already begun, and Jacobi himself had the greatest share in it. Since a problem to be solved by Abel's theorem leads to the inversion of several integrals at once, and since this inversion would be impossible by the old means, one needed again a guiding function analogous to the elliptic theta-function. Jacobi indicated the way. Later, taking his treatment of elliptic functions as their model, two younger mathematicians of unusual talent, Goepel and Rosenhain, founded their beautiful works on the consideration of infinite series and thereby carried into the new theory the same kind of illuminating principle that Jacobi had introduced into the old one.

Although in the preceding account of Jacobi's discoveries in the field of elliptic and Abelian transcendents I have had to confine myself to the most essential points, this account has already grown to a length that forces me to touch only briefly on his remaining achievements. Yet even these brief indications will show how widely his mind ranged. Already above I spoke of his investigations on cyclotomy and on the form that Gauss first gave to the solution of binomial equations. In some of these results he met with the great mathematician Cauchy, who was occupied at the same time with similar researches and mentioned this circumstance when he published extracts from his works during Jacobi's first stay in Paris.

From a beautiful theorem derived from cyclotomy, also found by Cauchy, according to which all primes that, on division by a given prime or by four times it, leave the remainder one are represented, when raised to a certain power whose exponent depends only on the latter prime, by the so-called principal quadratic form having the negative of the given prime as determinant, Jacobi drew the conjecture that this exponent must agree with the number of distinct quadratic forms corresponding to that determinant. Since this conjecture was confirmed in all numerical examples, he did not hesitate to publish the observation in a brief note. I believe I ought to mention here, following Jacobi's oral communication, the hitherto unknown origin of this result as a remarkable example of acute induction, although the rigorous proof of it does not seem capable of being based on cyclotomy, but appears to require essentially different principles drawn from integral calculus and the theory of series, principles that were introduced into science only later.

Gauss's second memoir on biquadratic residues, published in 1832, made an epoch by the profound idea of treating complex integers in higher arithmetic just as real ones are treated, and by the reciprocity law established there, which holds in the theory of biquadratic residues between two complex primes. This memoir gave Jacobi occasion to resume his earlier investigations, and he succeeded in deriving with great simplicity from cyclotomy the beautiful theorem of Gauss just mentioned and a similar theorem relating to cubic residues. Although Jacobi completely wrote down these investigations and others connected with them, which I cannot even indicate, in the years 1836 to 1839, he never came to publish them in print. His delay arose from the wish to give some of his results greater extension, for which, occupied by so many other works, he did not find the necessary leisure. A part of his researches, and especially the proofs of the reciprocity theorems already mentioned, have nevertheless become known to some German mathematicians through transcripts of the lectures which he gave in Königsberg in the winter of 1836-37 on cyclotomy and its application to number theory.

Jacobi discovered another highly productive source for higher arithmetic in the theory of elliptic functions, from which he derived beautiful theorems on the number of decompositions of a number into two, four, six, and eight squares, and others concerning numbers that are contained simultaneously in several quadratic forms. These important enrichments of science are a fruit of the above-mentioned introduction of the Jacobi function into the theory of elliptic transcendents.

Jacobi repeatedly occupied himself with the reduction and evaluation of double and multiple integrals. I mention here especially the simple method by which he reduced the determination of the surface area of a triaxial ellipsoid to elliptic integrals of the first and second kind, a reduction which Legendre, to whom it belongs among his finest achievements, had succeeded in obtaining only with the aid of very hidden properties of integrals of the third kind. In another paper belonging here Jacobi extended Euler's addition theorem to double integrals, and soon afterwards observed how Abel's theorem too was capable of a similar extension.

Only a part of Jacobi's works on this chapter of integral calculus was published. A large memoir on the attraction of ellipsoids, although almost completed for a long time, remained unprinted and became known only through a few occasional notes. When he was occupied with this problem he also came upon the beautiful theorem found by Poisson at about the same time, according to which the attraction exerted on a point in exterior space by an infinitely thin shell bounded by two concentric, similar, and similarly situated ellipsoidal surfaces can be represented without integral signs. Jacobi never publicly mentioned this circumstance, although he could have appealed to the testimony of several mathematicians to whom he had communicated the theorem before the first notice of Poisson's paper appeared.

Connected with these investigations is another work of Jacobi's that must not be passed over here because of its surprising result. Maclaurin, as is well known, first showed that a homogeneous liquid mass can, while preserving its external shape, rotate uniformly about a fixed axis if that shape is that of an ellipsoid of revolution. This beautiful result was later completed by d'Alembert and Laplace through the demonstration that to each value of the angular velocity, if it lies below a certain limit, there correspond two and only two such ellipsoids. Lagrange seems first to have thought of the possibility that a triaxial ellipsoid too might satisfy the conditions of permanence; at least, in his analytical mechanics, when treating this question, he starts from the more general assumption of three unequal axes. But without carrying the investigation through he believes he may conclude from two apparently insufficient equations that the equilibrium is impossible in this more general case. A similar hasty conclusion by an author of a well-known textbook first aroused Jacobi's suspicion. On closer consideration of those two equations he soon found, to his own and surely to every mathematician's great surprise, that a triaxial ellipsoid also satisfies the conditions of equilibrium. Thus a new branch was grafted on the old theory of rotating fluid masses.

The occasion given to Jacobi in his investigations on the attraction of ellipsoids by Poisson's theorem led him to other researches in geometry. Among them are his works on surfaces of the second degree and on the geodesic lines of an ellipsoid. The latter problem is connected with the calculus of variations and with analytical mechanics; Jacobi recognized that the path of a point moving freely on an ellipsoid under no forces is represented by geodesics, and he thereby brought into contact the geometry of surfaces, mechanics, and the theory of elliptic functions. Similar connections guided him in his treatment of Hamilton's great discovery on the integration of the equations of dynamics. It was characteristic of Jacobi that he did not rest content with using a result, but sought the simple thought lying behind it and then brought the whole matter into a form in which the calculation became transparent.

To Jacobi's most important investigations also belong those on Pfaff's problem, that is, on the integration of a differential expression with several variables. This problem had remained wholly fruitless until Jacobi freed it from an unnecessary complication and showed that, instead of the sequence of systems whose successive integration Pfaff had demanded, the treatment of the first alone makes all the rest superfluous; the first step of the older method therefore already leads completely to the goal. The improvement which Jacobi made in the calculus of variations has a similar character. For the existence of a maximum or a minimum the vanishing of the first variation is necessary, but this condition alone is not sufficient; the nature of the second variation decides whether a maximum, a minimum, or neither occurs. According to the theory as Jacobi found it, after the integrations demanded by the vanishing of the first variation, new integrations had to be performed in order to discuss the second variation. Jacobi showed that the former involve the latter, so that here too the complete solution of the problem is already given with the completion of the first step.

If it is the ever more prominent tendency of modern analysis to set thoughts in the place of calculation, there nevertheless remain certain domains in which calculation retains its rights. Jacobi, who advanced that tendency so essentially, also achieved admirable things in these domains by virtue of his mastery of technique. To this class belong his papers on the transformation of homogeneous functions of the second degree, on elimination, on the simultaneous values satisfying a number of algebraic equations, on the inversion of series, and on the theory of determinants. In the last-named chapter we owe to him a developed theory of the expressions he called functional determinants. Pursuing far the analogy of these expressions with differential quotients, he arrived at a general principle which he called the principle of the last multiplier, and which, in almost all integration problems occurring in applications, gives the means of carrying out the last integration by specifying a priori the integrating factor required for it.

The influence that Jacobi exercised on the progress of science would appear only incompletely if I did not mention his activity as a public teacher. It was not his task to transmit anew what was finished and handed down. His lectures moved entirely outside the realm of textbooks and embraced only those parts of science in which he himself had appeared as a creator; and with him that meant that they offered the richest abundance of variety. His lectures were not distinguished by the sort of clarity that may also fall to intellectual poverty, but by a clarity of a higher kind. Above all he sought to set forth the guiding ideas that underlie each theory. Removing everything that bore the appearance of artificiality, he developed the solution of problems so naturally before his hearers that they could form the hope of being able to create something similar. As he knew how to treat the most difficult subjects, he could rightly encourage his hearers with the assurance that in his lectures they would have to appropriate only quite simple thoughts.

The success of so unusual a mode of teaching, as I have just described it and as only a creative mind can command, was truly extraordinary. If now in Germany knowledge of the methods of analysis is spread to a degree unknown in any earlier time, and if numerous younger mathematicians extend and enrich science in every direction, Jacobi has had the most essential share in so gratifying a phenomenon.

Almost all were his pupils; rarely did a budding talent escape his attention; and no one, once he had recognized it, lacked his promoting counsel and his encouraging participation.

I have just tried to present Jacobi as inventor and in his activity as teacher. If I now venture to describe how he appeared outside the scientific sphere to those who stand far from the mathematical sciences, then I must indicate as the basic trait of his nature that he lived wholly in the world of thoughts, and that in him thinking had become, not something that even most significant people require a special effort to undertake, but a habitual state and as it were a second nature. When anything in life or in science had once attracted his attention, he did not rest until he had worked it up into thoughts of his own; and with this uninterrupted intellectual activity there was united in him so rare a memory that he could at once bring before himself and command everything with which he had once occupied himself.

The inexhaustible store of knowledge and of his own thoughts that stood at Jacobi's disposal at every moment, a rare intellectual mobility by which he knew how to adapt himself to every age and every capacity, and a characteristically humorous mode of expression that sharply designated things, gave the great mathematician an unusual significance also in social intercourse. This was heightened by his readiness to treat scientific questions extempore. This readiness sprang from the innermost essence of his nature, which found its true satisfaction in overcoming difficulties; and for him there was therefore a quite special attraction in making scientific results intelligible by simple considerations even to those who lacked the preliminary knowledge that seemed indispensable for them. Only he had to be convinced, in order to make such an attempt, that those with whom he spoke had a real interest in the matter.

Where, on the other hand, he thought he noticed thoughtless curiosity, or heard definite opinions pronounced with self-satisfaction by people who had never imposed on themselves the hard work of thinking for themselves, his patience left him, and he usually brought the conversation to an end by an ironic and often sharply dismissive remark. He was often reproached for having shown himself too conscious of his intellectual power on such occasions. But those who judged him so might perhaps have changed their opinion had they known the price for which he had acquired the right to such a consciousness. A letter from the year 1824, from a time when Jacobi was still completely unknown and therefore could have had no interest in describing his intellectual struggles in exaggerated colours, contains the following passage, which I communicate here literally as a remarkable contribution to the characterization of the extraordinary man. Jacobi had then just turned twenty and had for about a year been occupied exclusively with mathematical studies.

"It is hard work that I have done, and hard work in which I am engaged. It is not diligence and memory that lead here to the goal; here they are the most subordinate servants of pure thought in motion. But stubborn, brain-rending reflection demands more strength than the most enduring diligence. If, therefore, through constant exercise of this reflection I have gained some power in it, one should not think that it has become easy for me through some happy natural gift. Hard, hard work I have to endure, and the anxiety of reflection has often shaken my health mightily. The consciousness, to be sure, of the power gained gives the finest reward for the work, and at the same time the encouragement to continue and not grow slack. Thoughtless people, to whom that work and that consciousness are therefore wholly foreign, seek to spoil this consolation, which alone can prevent one from letting courage sink on the difficult path, by making hateful under the name of conceit or arrogance the consciousness of being one's own free self - for only in the movement of thought is the human being free and with himself. Everyone who carries within himself the idea of a science cannot do otherwise than estimate things according to the way the human mind reveals itself in them. By this great measure, much must therefore appear insignificant to him which may seem quite praiseworthy to others. Thus people have often accused me of arrogance too; or, as they have most beautifully praised me when they meant to utter blame, I am proud toward everything lower and humble only toward the higher. But that infinite measure which one applies to the world within and without prevents all overestimation of oneself, because one always has before one's eyes the infinite goal and one's limited power. In that pride and that humility I shall always strive to persist, indeed to become ever prouder and ever humbler."

That it was no mere phrase in Jacobi when he said that he estimated things according to the manner in which the human mind reveals itself in them, and that he really treated everything that did not touch the world of thoughts, if not with indifference then with equanimity, he showed in the most difficult situations of his life. This truly philosophical equanimity was revealed most admirably when the misfortune befell him of losing the entire fortune inherited from his father - a loss that might have been all the more painful to him because he had been married for ten years and had to care for a numerous family. Whoever saw him then, when he had hurried to assist with counsel and deed his mother, who had suffered a similar loss, could perceive not the slightest change in his mood. He spoke with the same interest as always about scientific things and complained only that the unexpected journey had torn him away from an investigation that was then occupying him vividly.

I have already mentioned how Jacobi's cult of thought made itself known in his recognition of Abel's great discovery. He showed a similar sense for everything intellectually significant, and the saying of an ancient writer does not apply to him, that people really admire only what they believe they themselves could accomplish. His recognition embraced the whole intellectual domain; and in his science Jacobi's joy over another's invention was all the more lively the more that invention differed in stamp from his own creations. It was a natural impulse in him in such a case to strengthen the expression of his approval by the admission that he himself would never have had that thought.

It remains only to complete, in a few words, what I mentioned above concerning Jacobi's external circumstances. When he began to make known his investigations on elliptic functions, he was still a Privatdocent. The admiration that his discoveries aroused among all those competent to judge such matters had the consequence that he was at once promoted to extraordinary professor and soon afterwards to ordinary professor.

In speaking of the reception that Abel's and Jacobi's discoveries - for here the names of both are inseparable - found among all specialists, I cannot refrain from mentioning by name the man who, through his many years of research, was especially called to appreciate the unexpected advance in its full significance. Legendre, who had so often accused his contemporaries of lack of interest and not long before had expressed his regret that his favourite science, abandoned by all others, had by him alone after forty years of labour, as he believed, been brought to completion, greeted Abel's and Jacobi's discoveries, which carried the theory far beyond the limits that seemed to him to be set by the nature of the subject itself, with such warm and indeed enthusiastic recognition that it is difficult to say whom such recognition honoured more: the young mathematicians to whom it came at the entrance of their career, or the noble old master who, almost at the goal, showed himself capable of such warmth of feeling.

A no less honourable distinction followed soon afterwards when the Paris Academy, although it had opened no prize competition on the theory of elliptic functions, recognized Abel's and Jacobi's works as the most important discovery of the time by awarding one of its great mathematical prizes and dividing it between Jacobi and Abel's heirs. I must restrict myself here to mentioning the proofs of recognition that marked Jacobi's entrance into the scientific career. The limits set for me do not allow me to enumerate all the distinctions that were later bestowed on him in so rich a measure, and whose mention ought not to be absent from a detailed biography.

Soon after Jacobi published in 1829 his Fundamenta nova theoriae functionum ellipticarum, which contains only a part of his investigations on this subject, he made his first extended journey abroad. He took the route through Göttingen in order to make Gauss's personal acquaintance, and then turned to Paris, where he stayed several months and where at that time, besides Legendre - with whom he had for some time been in close correspondence and for whom he always preserved great reverence - Fourier, Poisson, and other outstanding mathematicians who survived Jacobi were gathered.

A second journey abroad was undertaken by Jacobi, who since 1831 had been married to a woman of outstanding intellectual formation, only again in 1842, in the company of his wife. The occasion for this journey was so honourable for him that I cannot leave it unmentioned. To the enlightened statesman who then stood at the head of the administration in the province of Prussia, it seemed desirable in the interest of science that Bessel and Jacobi should once accept the invitation often addressed to them to take part in the learned assembly held yearly in England. He therefore submitted to the king the request that the costs of such a journey be granted; His Majesty deigned with royal munificence to comply with that request.

Soon after Jacobi's return from this journey, symptoms of an unfortunately incurable illness appeared. For a long time he hovered in the greatest danger. When this danger had finally for the moment been removed, his physicians declared that, to strengthen him, a longer stay in a southern climate was necessary. This medical opinion placed Jacobi in no small embarrassment, but the embarrassment was not of long duration. As soon as the situation became known through our colleague Alexander von Humboldt, whose weighty mediation is never absent where the honour of science and the welfare of its representatives are concerned, to His Majesty the King, a new act of royal generosity assigned a considerable sum for a journey to Italy.

The mild climate of Rome, where Jacobi spent the winter, had so beneficial an effect on him that those who saw him there, far from recognizing a convalescent in him, had to marvel at his truly extraordinary activity. During the five months of his stay there he not only wrote, besides several smaller essays that appeared in a scientific journal in Rome itself, an important and very extensive memoir intended for Crelle's Journal; he also undertook the comparison of the manuscripts of Diophantus preserved in the Vatican, with whom he had for some time occupied himself assiduously.

After returning to his fatherland he was transferred from Königsberg to Berlin, where the at least relatively milder climate seemed to threaten his health less. Without belonging here to the university, he had only the obligation to give lectures insofar as this would be compatible with the care that his state of health so greatly required. His literary activity during his residence here hardly fell behind that of the best Königsberg period, as is shown by the papers written here in about six years, which fill two substantial quarto volumes.

At the beginning of the year 1851 he had an attack of influenza to endure. Since, however, he recovered quickly and began again to work with great zeal, his friends might allow themselves the hope that he would long be preserved to them and to science, when on 11 February he suddenly fell ill again. His condition at once aroused the greatest concern; and when after several days it was recognized that he had been seized by smallpox, which on the ground undermined by his old illness showed the most malignant character, every hope vanished. On 18 February, at eleven o'clock in the evening, eight days after falling ill, he succumbed without struggle.

Jacobi's scientific career comprises exactly a quarter century, thus a far shorter period than that of most earlier mathematicians of the first rank, and scarcely half the time over which Euler's activity extended. With Euler he has the greatest similarity, not only through versatility and fertility, but also in that all the resources of science were always present to him and at his command at every moment.

Death, which so early and so suddenly took him away from work in possession of his full power, did not grant to science the great enrichments that it could still expect from Jacobi's never tiring activity. In saying this I do not speak merely on the assumption that in such a mind creative power could be extinguished only together with physical life; I also have before my eyes a series of almost completed works to which he himself could in a short time - perhaps during printing, as he liked so much to do in his last years - have put the finishing touch, and which must now be brought to light by his friends as fragments in imperfect form.

Even during his illness, scarcely four days before his death, he lamented the misfortune that had ruled over many of his larger works, interrupted by illness or domestic calamity. "When later," he added sadly, "I returned to work, I preferred to begin something new rather than take up again investigations that awakened such sad memories in me. But I see that I must no longer delay giving to the public those older works to which I have devoted so large a part of my best strength, if they are still to intervene fruitfully in the course of science. Fortunately, only a very short time is needed for this, which, I hope, will not be denied me."

XXIII.

Excerpts from the Monthly Reports of the Academy of Sciences in Berlin

Monthly Reports of the Academy of Sciences in Berlin, 1837–1853.

Excerpts from the Monthly Reports of the Academy of Sciences in Berlin.

On the representation of arbitrary functions by definite integrals, with special application to the function P_n which occurs in the attraction of spheroids.

(Monthly Report 1837, p. 79.)[1]

On some problems which require the determination of an unknown function under the integral sign.

(Monthly Report 1843, p. 152.)[2]

[1]Only the title; cf. vol. I of this edition of G. Lejeune Dirichlet's Works, p. 285 and following. F.

[2]Only the title. F.

Remarks on Kummer's proof of Fermat's theorem, concerning the impossibility of $x^\lambda - y^\lambda = z^\lambda$ for an infinite number of prime numbers λ.

(Monthly Report 1847, pp. 139–141.)

As regards the second of the two assumptions on which Mr. Kummer's ingenious proof rests, its correctness for each particular value of λ can be tested with the help of the general theory of complex units, of which some indications were given in the March report of 1846 and which will be published in one of the next issues of Crelle's Journal[3]. According to the theory just mentioned, for every λ one can set up the general expression for all complex units composed from λ-th roots of unity, and the formation of this expression presents no difficulty other than that of a numerical calculation whose complication increases rapidly with λ. Once this expression is known, containing $\frac{\lambda-3}{2}$ fundamental units raised to indeterminate powers, it can soon be decided without great difficulty whether the condition stated in assumption (B.) is fulfilled; that is, whether the expression can be congruent, modulo λ, to a real number only when all exponents are divisible by λ.

Assumption (A.) concerns a theory which has the greatest analogy with that of quadratic forms. Just as not every number m for which the congruence

$$\xi^2 \equiv D \pmod{m}$$

is possible is always contained in the form $x^2 - Dy^2$, but in general a limited number of essentially different quadratic forms exists by which all such numbers m can be represented, so analogous relations occur between higher congruences and corresponding higher forms. Consider, for example, the congruence

$$\frac{\xi^\lambda - 1}{\xi - 1} \equiv 0 \pmod{m},$$

with respect to which Euler already completely determined the moduli m for which it is possible. These numbers m are likewise not always of the form

$$\varphi(\alpha)\varphi(\alpha^2)\ldots\varphi(\alpha^{\lambda-1}),$$

where

$$\varphi(\alpha) = x_0 + \alpha x_1 + \alpha^2 x_2 + \cdots + \alpha^{\lambda-2} x_{\lambda-2},$$

and $x_0, x_1, \ldots, x_{\lambda-2}$ denote indeterminate integers; but there always exist finitely many forms which, like the preceding one, can be decomposed into linear factors and which, together with it, express all numbers m for which Euler's congruence is solvable. After the representability of all units by $\frac{\lambda-3}{2}$ fundamental units had been recognized, which here is the analogue of the general solution of Fermat's equation

$$x^2 - Dy^2 = 1,$$

it was natural to try to pursue the analogy between quadratic forms and these higher forms further, and in particular to determine the number of the latter by similar means to those by which the same question had earlier been settled in the theory of quadratic forms. This investigation, to which Mr. Kummer refers at the beginning of his communication, was in fact brought to completion about

[3]The paper appeared in vol. 40 of the Journal, pp. 130–138. F.

three years ago with the help of a new principle not needed in the determination of the number of forms of the second degree, and led to a result whose form seems remarkable and which is as simple as could ever be expected in a question covering forms of all degrees. The expression that I found for the number of forms of degree $(\lambda-1)$, which, as the analogy with quadratic forms led one to expect, contains the above-mentioned $\frac{\lambda-3}{2}$ fundamental units, gives, as soon as these are known, the means of testing assumption (A.) by a fairly simple numerical calculation. Whether it will be possible, however, to derive something general about the connection between the two assumptions from the way in which the fundamental units enter into the expression for the number of forms, and to test the conjecture expressed by Mr. Kummer at the end of his letter, namely that the first assumption always involves the second, I do not venture to decide for the present. Such a decision can only be the result of a careful discussion of the expression for the number of forms, undertaken from the point of view just described.

Report on Ehrenzeller's solution of numerical equations by series.

(Monthly Report, 18 February 1850, p. 36.)

Mr. Dirichlet delivered a report on the writing, assigned to the class, of Mr. Ehrenzeller of St. Gallen, dated the 21st of December, containing a solution of numerical equations by series. The procedure agrees with the extension of Newton's approximation method to the term of the second order in the expansion according to Taylor's theorem, and therefore cannot lay claim to novelty.

On a new derivation of two arithmetical theorems from a common source.

(Monthly Report 1853, p. 300.)[4]

[4]Only the title. F.

II. SECTION,

CONTAINING

POSTHUMOUS WORKS.

XXIV.

On the biquadratic character of the number "two"

By G. Lejeune Dirichlet

From a letter of Dirichlet to Mr. Stern in Göttingen, Crelle's Journal for Pure and Applied Mathematics, vol. 57, pp. 187–188.

On the biquadratic character of the number "two".

Let p be a prime number of the form $4n+1$ and

$$p = a^2 + b^2,$$

where a is odd. From this equation and from the equation following from it,

$$2p = (a+b)^2 + (a-b)^2,$$

one obtains at once, using Legendre's generalized symbol,

$$\left(\frac{p}{a}\right) = 1, \qquad \left(\frac{2p}{a+b}\right) = 1$$

or

$$\left(\frac{p}{a+b}\right) = \left(\frac{2}{a+b}\right),$$

and then, by known theorems,

$$\left(\frac{a}{p}\right) = 1, \qquad \left(\frac{a+b}{p}\right) = (-1)^{\frac{1}{8}[(a+b)^2-1]},$$

or, what is the same thing,

$$a^{\frac{1}{2}(p-1)} \equiv 1, \qquad (a+b)^{\frac{1}{2}(p-1)} \equiv (-1)^{\frac{1}{8}[(a+b)^2-1]} \pmod{p}.$$

On the other hand, from

$$(a+b)^2 \equiv a^2 + b^2 + 2ab \equiv 2ab \pmod{p}$$

one obtains, by raising to the power $\frac{1}{4}(p-1)$,

$$(a+b)^{\frac{1}{2}(p-1)} \equiv 2^{\frac{1}{4}(p-1)} a^{\frac{1}{4}(p-1)} b^{\frac{1}{4}(p-1)} \pmod{p},$$

or, if one sets

$$b \equiv af$$

and takes account of the two last congruences,

$$(-1)^{\frac{1}{8}[(a+b)^2-1]} = (-1)^{\frac{1}{8}(p-1+2ab)} \equiv 2^{\frac{1}{4}(p-1)} f^{\frac{1}{4}(p-1)} \pmod{p}.$$

Now, since

$$a^2 + b^2 = p, \qquad b^2 \equiv a^2 f^2,$$

therefore

$$f^2 \equiv -1 \quad (\text{mod.}\, p),$$

and hence

$$(f^2)^{\frac{1}{8}(p-1+2ab)} = f^{\frac{1}{4}(p-1)} \cdot f^{\frac{1}{2}ab} \equiv 2^{\frac{1}{4}(p-1)} \cdot f^{\frac{1}{4}(p-1)} \quad (\text{mod.}\, p).$$

Dividing by $f^{\frac{1}{4}(p-1)}$ gives

$$2^{\frac{1}{4}(p-1)} \equiv f^{\frac{1}{2}ab} \quad (\text{mod.}\, p),$$

which is transformed into the result given by Gauss in § 24 of his *theoria residuorum biquadraticorum* if one multiplies by $a^{\frac{1}{2}ab}$ and again writes b for af.

Göttingen, 21 January 1857.

Dirichlet XXV: Investigations on a Problem of Hydrodynamics

G. Lejeune Dirichlet; prepared from the Nachlass by R. Dedekind

English translation, Werke Band II, pp. 263–302

XXV. Investigations on a Problem of Hydrodynamics

G. Lejeune Dirichlet
Prepared from Dirichlet's papers by R. Dedekind
From the eighth volume of the Transactions of the Royal Society of Sciences at Göttingen.

Preface

A few remarks must be made in advance about the completion and publication of this memoir, which by the author's last will was entrusted to me. The hydrodynamical problem treated here, whose solution dates from the winter of 1856–1857, was first presented in brief outline at the close of Dirichlet's lectures on partial differential equations in July 1857. At the same time the principal result of the whole investigation was published in a short notice in the *Nachrichten* of the Royal Society of Sciences. The complete exposition was delayed, partly because the author wished to investigate the object still further in its details, and partly because he was occupied with other work, until sudden illness and premature death made completion impossible.

Among the papers that referred to this subject and that came into my hands on 21 July 1859 there was first a manuscript executed with such care that it could be given to the printer without the least alteration. It is only very much to be regretted that even in this fragment the introduction, which was devoted to the discussion of some general properties of the fundamental equations of hydrodynamics, was left unfinished. Besides this manuscript, which in the following arrangement extends almost to the end of §3, there was a large number of separate papers with formulas rapidly written down and without text; their meaning, however, was easy to recognize. For the most part they were repetitions of what was already presented, and only rarely did one of them give a point of departure for a further development.

With the aid of these papers it was not difficult to find the seven first-order integral equations mentioned in the preliminary notice; they are found in §5 below. Moreover, many passages pointed to the case treated in §8, although nowhere was there a discussion of it; I have tried in §6 to connect it with the other case investigated in §7, which, because of its simplicity, was also communicated in the preliminary notice. Furthermore, as may be seen from the notes that have all been added by me, some passages of the manuscript occasioned computations more laborious than difficult. Since their results may be useful for future work, I have included them in the memoir, and they form §4. After deriving them, I used them in some further investigations; the results, insofar as they have succeeded up to now, are communicated in the final paragraph.

I do not conceal from myself that, despite all the care and affection spent on the work, many things could have been carried out more completely and better. But I did not wish to delay publication still longer, all the less because I may trust that this last work of the great thinker, who was not granted the opportunity to put the master's hand to the exposition himself, will be appreciated even in its imperfect form.

Zurich, 10 November 1859.
R. Dedekind.

Introduction

In establishing the general equations by which the motion of fluid bodies is determined, one may begin from two different points of view. According to the first way of conceiving the subject, one sets oneself the task of determining, for an arbitrary place (x, y, z) and an arbitrary time t, the state of the moving mass: its density, its pressure, and the three components of its velocity. These five quantities are then to be determined as functions of the four variables x, y, z, t. To this point of view correspond the fundamental equations of hydrodynamics that occur in all textbooks and that Euler first set up. Euler's equations also form the basis of a great memoir that Lagrange published more than twenty years later in the same academic collection, and from which he later formed, with some additions, the section of his *Mécanique analytique* devoted to hydrodynamics.

The most important of these additions stands at the beginning of that section and concerns a treatment essentially different from Euler's. Lagrange seeks to follow the motion of each element of the fluid; that is, he determines the coordinates x, y, z, the pressure, and the density of this element by means of its initial coordinates a, b, c and the time t elapsed since the beginning of the motion. Remarkably, however, Lagrange makes no use of the equations corresponding to this point of view. After observing that they are somewhat complicated, he transforms them into Euler's equations and then adds that the latter, because of their greater simplicity, are preferable for the solution of special problems.

Dirichlet's objection is that the greater formal simplicity of Euler's equations is more than counterbalanced by a hidden difficulty. In Euler's form the coordinates x, y, z are not independent variables in the proper sense of the word, since the extent in which they are valid is the region occupied at the given instant by the moving mass; that region is itself determined by the whole preceding motion. We now know how essential a part of a problem is the domain assigned to functions of several variables that are defined by partial differential equations and boundary conditions. Euler's form is therefore at its best when the fluid mass preserves the same exterior shape during the motion, or when a fixed solid body moves in an infinite fluid. The problem considered here is of the opposite type: the boundary is free and changes with the motion.

Dirichlet also points out an error, protected by great authority and therefore repeated in textbooks. Euler had observed that his equations are greatly simplified if, for the whole duration of the motion, both the three velocity components and the three force components are the partial derivatives, with respect to the coordinates, of functions of those coordinates. Lagrange added the important assertion that this condition on the velocity components holds automatically for all time if it holds initially and if the force components satisfy the same condition at each time. Dedekind notes that the manuscript breaks off just here. The correction intended is essentially this: the property belongs not to a fixed absolute region of space, but to the material mass that is being followed. If initially, in a region occupied by the fluid, $u\,dx + v\,dy + w\,dz$ is a complete differential, then the corresponding statement holds later in the region that contains the same material elements.

§1. Lagrange's equations and Dirichlet's linear ansatz

We restrict ourselves to the homogeneous case and put the density equal to one. Lagrange's fundamental equations are, in Dirichlet's notation,

$$\begin{aligned}
\left(\frac{d^2x}{dt^2} - X\right)\frac{\partial x}{\partial a} + \left(\frac{d^2y}{dt^2} - Y\right)\frac{\partial y}{\partial a} + \left(\frac{d^2z}{dt^2} - Z\right)\frac{\partial z}{\partial a} + \frac{\partial p}{\partial a} &= 0,\\
\left(\frac{d^2x}{dt^2} - X\right)\frac{\partial x}{\partial b} + \left(\frac{d^2y}{dt^2} - Y\right)\frac{\partial y}{\partial b} + \left(\frac{d^2z}{dt^2} - Z\right)\frac{\partial z}{\partial b} + \frac{\partial p}{\partial b} &= 0,\\
\left(\frac{d^2x}{dt^2} - X\right)\frac{\partial x}{\partial c} + \left(\frac{d^2y}{dt^2} - Y\right)\frac{\partial y}{\partial c} + \left(\frac{d^2z}{dt^2} - Z\right)\frac{\partial z}{\partial c} + \frac{\partial p}{\partial c} &= 0,\\
\det\left(\frac{\partial(x,y,z)}{\partial(a,b,c)}\right) &= 1.
\end{aligned}\tag{1}$$

Here a, b, c are the initial coordinates of an arbitrary element, x, y, z its coordinates at the time t, p the pressure exerted on it, and X, Y, Z the components of the accelerating force. The last equation expresses the incompressibility of the fluid.

The force to be considered is the attraction of the whole mass on itself, with elementary attraction inversely proportional to the square of the distance. If V is the potential of the fluid at the interior point (x, y, z), and if ε is the constant measuring the attraction between two unit masses at unit distance, then

$$X = \varepsilon\frac{\partial V}{\partial x}, \qquad Y = \varepsilon\frac{\partial V}{\partial y}, \qquad Z = \varepsilon\frac{\partial V}{\partial z}.$$

After substitution, the first three equations of (1) become

$$\begin{aligned}
\frac{d^2x}{dt^2}\frac{\partial x}{\partial a} + \frac{d^2y}{dt^2}\frac{\partial y}{\partial a} + \frac{d^2z}{dt^2}\frac{\partial z}{\partial a} - \varepsilon\frac{\partial V}{\partial a} + \frac{\partial p}{\partial a} &= 0,\\
\frac{d^2x}{dt^2}\frac{\partial x}{\partial b} + \frac{d^2y}{dt^2}\frac{\partial y}{\partial b} + \frac{d^2z}{dt^2}\frac{\partial z}{\partial b} - \varepsilon\frac{\partial V}{\partial b} + \frac{\partial p}{\partial b} &= 0,\\
\frac{d^2x}{dt^2}\frac{\partial x}{\partial c} + \frac{d^2y}{dt^2}\frac{\partial y}{\partial c} + \frac{d^2z}{dt^2}\frac{\partial z}{\partial c} - \varepsilon\frac{\partial V}{\partial c} + \frac{\partial p}{\partial c} &= 0.
\end{aligned}\tag{2}$$

The investigation is confined to the assumption that x, y, z, as functions of a, b, c, t, contain the three material coordinates only linearly. Dirichlet uses "linear" in the strict homogeneous sense: there is no term independent of a, b, c. Thus

$$\begin{aligned}
x &= la + mb + nc,\\
y &= l'a + m'b + n'c,\\
z &= l''a + m''b + n''c,
\end{aligned}\tag{3}$$

where the nine coefficients depend only on t. Incompressibility gives

$$\det \begin{pmatrix} l & m & n \\ l' & m' & n' \\ l'' & m'' & n'' \end{pmatrix} = 1. \tag{4}$$

For $t = 0$, x, y, z coincide with a, b, c; hence $l = m' = n'' = 1$, while the other six coefficients vanish.

Differentiating (3) with respect to time gives the velocity components

$$\begin{aligned} u &= \frac{dx}{dt} = \dot{l}a + \dot{m}b + \dot{n}c, \\ v &= \frac{dy}{dt} = \dot{l}'a + \dot{m}'b + \dot{n}'c, \\ w &= \frac{dz}{dt} = \dot{l}''a + \dot{m}''b + \dot{n}''c. \end{aligned} \tag{5}$$

Their initial values are not arbitrary; differentiating the determinant condition and putting $t = 0$ gives

$$\dot{l}_0 + \dot{m}'_0 + \dot{n}''_0 = 0.$$

Dirichlet then assumes that the fluid initially fills an ellipsoid whose centre is the origin and whose axes coincide with the coordinate axes:

$$\frac{a^2}{A^2} + \frac{b^2}{B^2} + \frac{c^2}{C^2} = 1. \tag{6}$$

Solving (3) for a, b, c gives

$$\begin{aligned} a &= \lambda x + \lambda' y + \lambda'' z, \\ b &= \mu x + \mu' y + \mu'' z, \\ c &= \nu x + \nu' y + \nu'' z, \end{aligned} \tag{7}$$

where, since the determinant is one, $\lambda, \mu, \nu, \dots$ are just the cofactors of the matrix in (3). Substitution in (5) gives the equation of the boundary at time t:

$$\frac{(\lambda x + \lambda' y + \lambda'' z)^2}{A^2} + \frac{(\mu x + \mu' y + \mu'' z)^2}{B^2} + \frac{(\nu x + \nu' y + \nu'' z)^2}{C^2} = 1. \tag{8}$$

Thus an initially ellipsoidal boundary remains an ellipsoid, concentric with the original one. More is true: particles that initially form a concentric, similar, and similarly situated ellipsoid retain the analogous relation to the instantaneous boundary.

It remains to show that (3) can be made to satisfy the equations (2). The potential of a homogeneous ellipsoid at an interior point is a constant plus a quadratic function of the coordinates. Therefore, after expressing the current ellipsoid in arbitrary axes and returning to the material coordinates, Dirichlet writes

$$V = H - La^2 - Mb^2 - Nc^2 - 2L'bc - 2M'ca - 2N'ab, \tag{9}$$

where H, L, M, N, L', M', N' are functions of time. Since the pressure is to be constant on the boundary, its dependence on a, b, c must be through the same quadratic form that defines the boundary. With P denoting the boundary pressure, one has

$$p = P + \sigma \left(1 - \frac{a^2}{A^2} - \frac{b^2}{B^2} - \frac{c^2}{C^2}\right), \tag{10}$$

where σ depends only on time.

Substitution gives ten equations for the ten unknown functions $l, m, n, l', m', n', l'', m'', n'', \sigma$:

$$
\begin{aligned}
l\ddot{l} + l'\ddot{l}' + l''\ddot{l}'' &= -2\varepsilon L + \frac{2\sigma}{A^2}, & m\ddot{m} + m'\ddot{m}' + m''\ddot{m}'' &= -2\varepsilon M + \frac{2\sigma}{B^2}, \\
n\ddot{n} + n'\ddot{n}' + n''\ddot{n}'' &= -2\varepsilon N + \frac{2\sigma}{C^2}, & m\ddot{n} + m'\ddot{n}' + m''\ddot{n}'' &= -2\varepsilon L', \\
n\ddot{m} + n'\ddot{m}' + n''\ddot{m}'' &= -2\varepsilon L', & n\ddot{l} + n'\ddot{l}' + n''\ddot{l}'' &= -2\varepsilon M', \\
l\ddot{n} + l'\ddot{n}' + l''\ddot{n}'' &= -2\varepsilon M', & l\ddot{m} + l'\ddot{m}' + l''\ddot{m}'' &= -2\varepsilon N', \\
m\ddot{l} + m'\ddot{l}' + m''\ddot{l}'' &= -2\varepsilon N', & \det\begin{pmatrix} l & m & n \\ l' & m' & n' \\ l'' & m'' & n'' \end{pmatrix} &= 1.
\end{aligned}
\tag{11}
$$

Dirichlet observes that one could eliminate σ, but he retains the symmetric form.

§2. Determination of the accelerations and sign of pressure

Although the system just displayed satisfies the boundary and field equations and has the same number of equations as unknowns, it is still necessary to show that it actually determines a motion. Given at some instant finite known values of the nine coefficients and their first derivatives, the equations must determine the second derivatives. Solving the three equations containing $\ddot{l}, \ddot{m}, \ddot{n}$, and doing the same for the two other rows, gives each second derivative in the form $e\sigma + f$, where e and f are finite functions of the known data. Substitution in the differentiated determinant equation gives an equation $e'\sigma + f' = 0$. The coefficient e' is a sum of squares of quantities which cannot all vanish because of the determinant condition. Hence σ, and then all accelerations, are determined.

The pressure condition is a further physical restriction. Formula (9) shows that the factor in parentheses takes all values between zero and one inside the fluid. If σ is ever negative, the exterior pressure P must be at least $|\sigma|$, otherwise the pressure inside would become negative. Only when σ is never negative may the motion occur in empty space without external pressure.

Dirichlet also determines the initial value of σ. At $t = 0$, adding the first three equations in (10) and using the potential of the initial ellipsoid gives a formula in which σ_0 is expressed as a sum of squared differences of the initial velocity coefficients, together with the gravitational contribution. If the velocity matrix is initially symmetric, i.e. if the entries symmetric about the diagonal are equal, then σ_0 is positive. The proof that it remains positive in that case is postponed to §5.

§3. Decomposition of a linear instantaneous motion

To obtain a simple intuition for the motion considered in §1, Dirichlet decomposes the instantaneous linear velocity field into two simpler motions. After substituting the inverse transformation (6) in (4), the velocity components have the form

$$u = gx + hy + kz, \quad v = g'x + h'y + k'z, \quad w = g''x + h''y + k''z. \tag{12}$$

The incompressibility condition gives

$$g + h' + k'' = 0.$$

Every linear velocity field can be decomposed into a strain and a rotation. For the strain, after a suitable change to axes ξ, η, ζ, the components take the diagonal form

$$p = a\xi, \qquad q = b\eta, \qquad r = c\zeta,$$

with $a + b + c = 0$ in the incompressible case. For a pure rotation around an axis through the origin the velocity is

$$u_1 = \gamma y - \beta z, \qquad v_1 = \alpha z - \gamma x, \qquad w_1 = \beta x - \alpha y. \tag{13}$$

Conversely, the sum of these two fields is any prescribed linear velocity field, and the decomposition is unique. Dirichlet justifies this by reducing the quadratic form associated with the symmetric part of the linear field to principal axes. The skew-symmetric part is the rotation. The restriction $a+b+c = 0$ is exactly the preservation of volume.

§4. Auxiliary formulas

Dirichlet next records the formulas that had only been indicated before. Let an ellipsoid, referred to arbitrary rectangular axes, be written

$$Sx^2 + S'y^2 + S''z^2 + 2Tyz + 2T'zx + 2T''xy < 1. \tag{14}$$

Let F be the quadratic form on the left and let F^* denote its adjoint form. If

$$\Delta(s) = \sqrt{\det \begin{pmatrix} 1 + sS & sT'' & sT' \\ sT'' & 1 + sS' & sT \\ sT' & sT & 1 + sS'' \end{pmatrix}},$$

then the potential for an interior point is obtained by a single integral whose integrand is rational in s and divided by $\Delta(s)$. This is the same result one obtains by first passing to the principal axes and then returning to the original axes; the method of the discontinuous factor gives it more directly.

For the present problem the coefficients of this quadratic form depend on the combinations

$$P = l^2 + l'^2 + l''^2, \quad Q = m^2 + m'^2 + m''^2, \quad R = n^2 + n'^2 + n''^2,$$
$$P' = mn + m'n' + m''n'', \quad Q' = nl + n'l' + n''l'', \quad R' = lm + l'm' + l''m''.$$

These six quantities satisfy the determinant relation

$$PQR - PP'^2 - QQ'^2 - RR'^2 + 2P'Q'R' = 1. \tag{15}$$

The coefficients L, M, N, L', M', N' of the potential are then given by definite integrals in s involving P, Q, R, P', Q', R'. The source writes these formulas explicitly; the important structural point is that the motion enters the potential only through the six quadratic combinations just displayed, and not separately through the nine entries of the transformation matrix.

The same section writes the coefficients of the velocity field (11) in terms of the transformation matrix. In modern matrix notation, if

$$M(t) = \begin{pmatrix} l & m & n \\ l' & m' & n' \\ l'' & m'' & n'' \end{pmatrix},$$

then

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \dot{M} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \dot{M}M^{-1} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Thus the matrix of coefficients $(g, h, k; g', h', k'; g'', h'', k'')$ is $\dot{M}M^{-1}$. Since $\det M = 1$, its trace is zero. The instantaneous rotations about the coordinate axes are

$$\alpha = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \quad \beta = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right), \quad \gamma = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right).$$

These formulas are used both in the determination of σ and in the integral equations.

§5. Seven first integrals

Seven first-order integrals are now set up. Three of them are obtained directly from (10) by subtracting pairs of equations with the same right-hand side. In component form they express the constancy of the vorticity combination carried by the material coordinates. In modern notation the skew-symmetric matrix

$$\mathcal{L} = MD\dot{M}^T - \dot{M}DM^T, \qquad D = \mathrm{diag}(A^2, B^2, C^2), \tag{16}$$

has constant entries. Dirichlet gives the three scalar component equations, and then rewrites them in terms of the spatial velocity derivatives

$$\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \qquad \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \qquad \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.$$

It follows immediately that the axis of instantaneous rotation is always formed by the same elements of the fluid mass. If the three displayed quantities vanish at one time, so that there is no rotation, then they vanish throughout the motion.

Since the acting forces come only from the mutual attraction of the elements of the fluid, the principle of areas gives three further integrals:

$$\int (yw - zv)\, d\tau = \text{const.}, \quad \int (zu - xw)\, d\tau = \text{const.}, \quad \int (xv - yu)\, d\tau = \text{const.}. \tag{17}$$

To compute them, one returns to the original ellipsoid and uses

$$\int a^2\, d\tau = \frac{MA^2}{5}, \quad \int b^2\, d\tau = \frac{MB^2}{5}, \quad \int c^2\, d\tau = \frac{MC^2}{5},$$

while the mixed integrals $\int ab\, d\tau$, $\int bc\, d\tau$, $\int ca\, d\tau$ vanish. This gives Dirichlet's three explicit component formulas involving l, m, n and their derivatives.

The seventh integral is the principle of living force. In the present notation it is

$$\int \left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right] d\tau = \text{constant} + 2\int V\, d\tau. \tag{18}$$

The left-hand side becomes a quadratic form in $\dot{l}, \dot{m}, \ldots, \dot{n}''$; the self-potential integral is evaluated from the ellipsoidal formula. If the current ellipsoid is referred to its principal axes and has semiaxes $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, then Dirichlet obtains the same integral by computing

$$\int V\, d\tau$$

directly over that ellipsoid. The result agrees with the expression obtained from the formulas of §4. The calculation also shows that the symmetry assumptions made later are compatible with the full differential equations.

§6. The rotationally symmetric case

Because the differential equations (10) are highly complicated, a complete solution can be expected only under special initial assumptions. Dirichlet first assumes complete symmetry, both of shape and of initial velocity, about a fixed axis. Thus the initial ellipsoid is a spheroid of revolution. Taking the symmetry axis to be the c-axis gives $B = A$. The initial velocity must be invariant under every rotation of the (a, b)-plane. This leads to the form

$$u = ga + hb, \qquad v = -ha + gb, \qquad w = -2gc. \tag{19}$$

The two parts of the instantaneous motion are therefore a rotational part about the symmetry axis and a strain that expands or contracts in the meridian planes.

The same symmetry persists for all time. The linear transformation therefore has the form

$$x = la + mb, \qquad y = -ma + lb, \qquad z = n''c, \tag{20}$$

with incompressibility

$$(l^2 + m^2)n'' = 1. \tag{21}$$

The boundary remains a spheroid of revolution. The velocity components, when written in spatial coordinates, again show a meridian strain together with a rotation about the axis. The angular velocity about the z-axis is proportional to

$$l\dot{m} - m\dot{l}.$$

Substitution in the fundamental equations reduces the nine equations to three independent equations for l, m, n'' and σ, together with (20). One first integral is

$$l\dot{m} - m\dot{l} = \frac{\gamma}{n''}, \tag{22}$$

where γ is constant. Hence the angular velocity is proportional to the length of the axis of rotation. The equation (20) allows l, m to be eliminated. Introducing the ratio

$$\alpha = \frac{n''}{D}, \tag{23}$$

of the axis of rotation to the radius D of the sphere of equal volume, and putting

$$f(\alpha) = \int_0^\infty \frac{ds}{(\alpha^2 + s)\sqrt{(1+s)^2(\alpha^2 + s)}},$$

Dirichlet transforms the equations into a one-dimensional system for $\alpha(t)$. The constants of integration are those fixed by the initial shape and initial velocity.

For later discussion he records the known properties of f. Explicit evaluation gives different elementary forms according as $\alpha < 1$ or $\alpha > 1$; both give $f(1) = 2$. As α tends to zero or infinity, $f(\alpha)$ tends to zero. The function has a single maximum at $\alpha = 1$. Also, the function $\alpha^2 f'(\alpha)$ has a numerical maximum approximately 0.2246, a number already known from the theory of uniformly rotating fluid masses.

§7. Motion without rotation

Dirichlet next considers the special case in which initially, and therefore always, no rotation occurs: $\gamma = 0$. Suppose first that the fluid starts from rest. Then the equations reduce to

$$\left(1 + \frac{1}{2\alpha^3}\right)\left(\frac{d\alpha}{dt}\right)^2 = 2\varepsilon\big(f(\alpha) - f(\alpha_0)\big), \tag{24}$$

up to Dirichlet's normalizing constants. If $\alpha_0 = 1$, the initial shape is a sphere and the solution remains the sphere. If $\alpha_0 < 1$, so that the initial spheroid is oblate, there is a second root $\alpha_1 > 1$ of

$$f(\alpha_1) = f(\alpha_0).$$

The quantity α runs periodically through the interval from α_0 to α_1 and back. The motion consists of isochronous oscillations in which the fluid passes through the spherical shape and alternately assumes the form of a prolate and an oblate ellipsoid.

If the spheroid does not start from rest, the same character persists as long as the initial velocity is below a certain bound depending on the initial form. If that bound is exceeded, $\dot{\alpha}$ does not change sign after a finite time. With positive initial $\dot{\alpha}$, the spheroid lengthens without bound; with negative initial $\dot{\alpha}$, it flattens without bound. In all of these nonrotating cases, however, the quantity σ never becomes negative. Therefore these motions are physically possible without an imposed exterior pressure.

§8. Motion with rotation

Assume now that $\gamma \neq 0$, so that rotation is present throughout the motion. From the properties of f there is a unique δ with $0 < \delta < 1$ satisfying the equation corresponding to the balance of the rotational term and the gravitational term. Dirichlet introduces

$$\psi(\alpha) = \frac{\gamma^2}{\alpha^2} + \varepsilon f(\alpha),$$

again up to the constants used in the source. This function decreases from $\alpha = 0$ to δ, reaches a negative minimum, and then increases without bound. The equation of motion may be written in the form

$$\left(1 + \frac{1}{2\alpha^3}\right)\left(\frac{d\alpha}{dt}\right)^2 = \psi(\alpha_0) + K - \psi(\alpha), \tag{25}$$

with $K \geq 0$ determined by the initial velocity.

It follows first that α always remains below an assignable finite limit: even the smallest initial rotation prevents unlimited elongation of the spheroid. Three cases remain.

1) The constant has the minimum value. Then $K = 0$, $\alpha_0 = \delta$, and the spheroid keeps its form while rotating uniformly about its short axis. This is the case first treated by Maclaurin. The square of the angular velocity must not exceed the known numerical limit $0.2246\ldots$. For each value below this bound there are two corresponding spheroids, which coincide when the limiting value is reached.

2) The constant is between the minimum and zero. The equation $\psi(\alpha) = \psi(\alpha_0) + K$ has two positive roots $\alpha' < \alpha''$, with $0 < \alpha' < \delta$. The variable α oscillates periodically between these two values. At the minimum value the rotation is too small, and at the maximum value too large, for the fluid to maintain the instantaneous form.

3) The constant is positive. Then the same equation has only one positive root, and either from the beginning or after a finite time the spheroid begins to flatten more and more without bound. In the rotating cases, if the rotation at the instant of greatest elongation is sufficiently large, the physical possibility of the motion may require an exterior pressure.

§9. Further ellipsoidal motions and Jacobi's theorem

The preceding cases have the special property that the three quantities denoted in §4 by P', Q', R' remain zero throughout the motion. Dirichlet and Dedekind therefore ask whether other motions with the same property are possible. A careful analysis gives two further classes compatible with the fundamental equations. One of them is still expressible by a diagonal transformation of the axes but is not reducible to quadratures merely by using the integral of living force.

The second class yields Jacobi's beautiful result. A triaxial ellipsoid whose axes A, B, C satisfy a certain definite integral relation can rotate uniformly about the smallest axis. In suitable coordinates the motion is

$$x = a\cos kt + b\sin kt, \qquad y = -a\sin kt + b\cos kt, \qquad z = c, \tag{26}$$

with the appropriate scaling of axes and with k fixed by the same integral condition. This is the Jacobi ellipsoid: a homogeneous self-gravitating fluid ellipsoid can rotate like a rigid body about its centre only when the axis of rotation is fixed and coincides with a principal axis, the Maclaurin and Jacobi cases being the corresponding possibilities.

The geometrical meaning of the equations $P' = Q' = R' = 0$ is that the material elements originally lying on the three coordinate axes, that is, on the principal axes of the ellipsoid, continue throughout the motion to fill three mutually perpendicular straight lines. Since linear transformations preserve conjugate diameters, the actual meaning is that the three principal axes of the ellipsoid are always formed by the same material elements of the fluid mass.

A related hypothesis is that the directions of the three principal axes remain fixed. In the notation of §4 this is expressed by $T = T' = T'' = 0$. It holds in the rotationally symmetric case already treated and in one of the additional cases just mentioned. It also gives a third case, a remarkable counterpart to Jacobi's theorem. Every triaxial ellipsoid that satisfies Jacobi's condition can preserve its external shape and position while an internal motion of its elements takes place:

$$x = a\cos kt + b\frac{A}{B}\sin kt, \qquad y = -a\frac{B}{A}\sin kt + b\cos kt, \qquad z = c. \tag{27}$$

Each particle describes the ellipse

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} = \frac{a^2}{A^2} + \frac{b^2}{B^2}, \qquad z = c,$$

and it does so exactly as if it were isolated and attracted toward the centre of its orbit by a force proportional to distance, whose measure for unit distance is k^2.

Demonstration of a Theorem of Abel

[Note by M. Lejeune Dirichlet communicated by M. Liouville.]

G. Lejeune Dirichlet, *Werke*, Band II, pp. 303–306.

The point is to prove that if the series

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots$$

is convergent and has sum A, then the sum of the series

$$a_0 + a_1\rho + a_2\rho^2 + \cdots + a_n\rho^n + \cdots ,$$

which will be convergent *a fortiori* when the variable ρ is taken positive and less than unity, will tend to the limit A when ρ is made to tend indefinitely to unity.

One day, in conversation with my excellent and much-lamented friend Lejeune Dirichlet, I told him that I found it rather difficult to set out (and even to understand) the proof which Abel had given of this important theorem. Dirichlet at once began writing before my eyes, solely in order to help me, the following note, which has been of great assistance to me and which the public will be grateful to have. The method of proof found there admits numerous applications and has often been useful to me in my lectures at the Collège de France.

I transcribe Dirichlet's note textually, without adding anything to it and, of course, without changing anything.

"From the assumed convergence of the series

$$A = a_0 + a_1 + a_2 + \cdots + a_n + \text{etc.},$$

it follows that the sum

$$s_n = a_0 + a_1 + \cdots + a_n$$

always remains numerically below a certain constant k, and converges to the limit A when n increases indefinitely. Now consider the series

$$S = a_0 + a_1\rho + a_2\rho^2 + \cdots + a_n\rho^n + \text{etc.},$$

the quantity ρ being assumed positive and less than unity. Replacing a_0, a_1, a_2, etc. in it by $s_0, s_1 - s_0, s_2 - s_1$, etc., it takes the form

$$S = s_0 + (s_1 - s_0)\rho + (s_2 - s_1)\rho^2 + \cdots + (s_n - s_{n-1})\rho^n + \text{etc.}$$

and then, by ordering otherwise,

$$S = (1-\rho)(s_0 + s_1\rho + s_2\rho^2 + \cdots + s_n\rho^n + \cdots).$$

This transposition involves no difficulty, since it reduces to adding to the sum of the first $n+1$ terms the term $-s_n\rho^{n+1}$, which vanishes for $n = \infty$.

"Let us now see to what limit S converges when the positive variable $\varepsilon = 1 - \rho$ becomes infinitely small. For this purpose decompose S into two parts, one comprising the first n terms and the other all the following terms, and let n grow as ε decreases, but slowly enough that the limit of $n\varepsilon$ is zero. The first part

$$(1-\rho)(s_0 + s_1\rho + \cdots + s_{n-1}\rho^{n-1}),$$

being numerically less than $n\varepsilon k$, converges to zero. As for the second,

$$(1-\rho)(s_n\rho^n + s_{n+1}\rho^{n+1} + \cdots),$$

one may give it the form

$$P(1-\rho)(\rho^n + \rho^{n+1} + \cdots) = P\rho^n = P(1-\varepsilon)^n,$$

where P denotes a value lying between the greatest and the smallest of the quantities $s_n, s_{n+1}, \ldots$. Now all these latter quantities converge to the limit A, and therefore so does P. On the other hand, the factor $(1-\varepsilon)^n$, by virtue of the hypothesis made above, evidently converges to unity. Thus it is proved that the limit of S, when the positive variable $\rho < 1$ approaches unity indefinitely, is precisely the sum A of the series considered at the beginning."

I do not think that anyone can now think of asking for further clarification.

Appendix

Intervening source divider in Werke Band II.

G. Lejeune Dirichlet, *Werke*, Band II, pp. 307–308.

A P P E N D I X.

Memorial Address on Gustav Peter Lejeune Dirichlet

E. E. Kummer

Read at the public session of the Royal Academy of Sciences on 5 July 1860.

Source: G. Lejeune Dirichlet, *Werke*, Band II, pp. 309–344; appendix item XXVII.

It is not yet ten years since the three men to whom our German fatherland owes a new flowering of the mathematical sciences - Gauss, Jacobi, and Dirichlet - were still alive and still active, working to renew and confirm, with new splendour, the old fame for deep knowledge of the most abstract mathematical truths, as well as of those mathematical truths concretely realized in nature, which Kepler and Leibniz above all had won for the German nation. At that time our Academy had the good fortune to possess two of these outstanding men as active members, Jacobi and Dirichlet. Personally bound by friendship, they stimulated and advanced one another by the free communication of their deep mathematical thoughts, and exercised the most lasting influence on the general development of the mathematical sciences. Jacobi's early death was the first irreparable loss suffered by the science that had unfolded to such bloom in our country. The importance of the creations of that rarely gifted investigator, and the eminent position which he will occupy for all time in the history of mathematics, were described by Dirichlet in the memorial address delivered here eight years ago today, so profoundly and so truly that he thereby erected the fairest and worthiest monument to the memory of the departed. When, four years after Jacobi, the aged Gauss passed from life in the undisputed fame of being the first mathematician of his time, this great general loss had for our Academy the further deplorable consequence that Dirichlet, as the only worthy successor to that great man, was called to Göttingen and left the number of its resident members.

The Academy, unable either to avert or to replace this loss, preserved by electing him one of its ordinary non-resident members the right still to regard Dirichlet as its own. For his special colleagues he remained the living centre of their investigations and labours until death set an end to his life and work. Our Academy, to which Dirichlet belonged for twenty-seven years, and in whose writings his imperishable masterpieces are deposited, has the right to regard the scientific fame of this great mathematician as its own; and above all others it has therefore the duty to preserve his memory and to pay him the last academic honour in a public memorial address. The veneration which I myself have always felt for the departed, the friendship which he bestowed on me and preserved for more than twenty years, and the close relation in which my own studies stand to his scientific work, have moved me to take the floor here before this highly distinguished assembly, in order to describe the great scientific significance of his masterpieces and at the same time, in a few strokes, to sketch the image of his life and of his character, which was noble and pure like his writings. I know that I shall be able to solve the task set before me only very imperfectly: not only for the substantive reason that the true significance of the intellectual creations of great men can often be properly recognized and appreciated only in the later historical course of science, where they not seldom become starting-points of richly unfolding theories, but also because of my own weakness, for which I must ask your kind indulgence.

Gustav Peter Lejeune Dirichlet was born on 13 February 1805 in Düren. His father, who held there the office of post director, a gentle, obliging, and lovable man, and his mother, still living now at a very advanced age, a spirited and finely cultivated woman, gave the boy, endowed by nature with more than ordinary gifts, a very careful upbringing. He received his first instruction in an elementary school; when this was no longer found sufficient for him, he attended a private school, and in addition received special instruction in Latin so that he might later attend a gymnasium. His great preference for mathematics appeared very early.

When he was not yet twelve years old, he used his pocket money to buy mathematical books, with which he occupied himself very diligently, especially in the evenings. When people told him that he could not possibly understand them, he answered: I read them until I understand them. His parents wished him to become a merchant; but when he showed decided aversion to that path, they yielded and in 1817 sent him to the gymnasium at Bonn.

Through the friendly communication of Professor Elvenich of Breslau, who at that time, as a student in Bonn, lived in the same house as the young Dirichlet and had been urgently recommended by the anxious parents to supervise and guide him, I am able to give the following faithful and lively account of the boy, then about thirteen years old. In his behaviour he distinguished himself very favourably by propriety and good manners; and the unselfconsciousness and openness of his whole nature made all who had to do with him sincerely well disposed toward him. His diligence was orderly, yet directed chiefly to mathematics and history. He studied even when he had no schoolwork to do, for even then his active mind was always occupied with worthy objects of reflection. Great historical events, notably the French Revolution, and public affairs interested him in a high degree; and he judged these and other matters with an independence unusual for his youth, from the standpoint of a liberal cast of thought, perhaps the fruit of his upbringing at home. Everything coarse and ignoble was always repugnant to him. Games and other youthful amusements, however, had almost no attraction for him, while he loved sociable conversation and took part gladly and animatedly, especially in discussions of politics and historical subjects. In general his mind, whose prominent quality was sharp discernment, moved in a much higher sphere than is usual at that age.

Dirichlet remained at the Bonn gymnasium only two years and then exchanged it for the Jesuit Gymnasium in Cologne, which his parents preferred for reasons unknown to me. Here his teacher in mathematics was Georg Simon Ohm, later famous for the discovery of the law of electrical resistance that bears his name. Under Ohm's instruction, and by his own diligent study of mathematical works, Dirichlet made very considerable progress in that science and acquired an unusual breadth of knowledge. Yet he by no means neglected the other disciplines, and he passed through the course of the gymnasium very rapidly, so that as early as 1821, when he was only sixteen, he received the leaving certificate for the university and returned home to discuss with his parents the choice of his future vocation. It was natural that they should meet his decision to study mathematics with earnest admonition to secure his livelihood in the world by a more practical study, for which they proposed jurisprudence. He replied to them modestly but firmly that, if they required it, he would obey them, but that he could not abandon his favourite study and would at least devote the nights to it. The parents, as sensible as they were tender, then yielded to their son's decided wish.

Mathematical study at the Prussian and other German universities was then in a sorry state. The lectures, which rose only a little above the domain of elementary mathematics, were in no way fitted to satisfy the urge toward deeper knowledge which animated the young Dirichlet; and apart from the one great name of Gauss there was in Germany no other that could have exerted a special attraction on him. In France, by contrast, and especially in Paris, mathematics still stood in its full bloom. There a circle of men whose great names will shine for all time in the history of the mathematical sciences worked, in research and teaching, at their living development and dissemination. There still lived the great Laplace, to whom his Celestial Mechanics secured undisputed first rank, and who was still working on the completion of that work and on a supplement to his theory of probability. Legendre, restlessly active into old age, was perfecting his theory of elliptic functions by the discovery of a new transformation of them and preparing the third edition of his work on number theory. Fourier, who had recently completed his mathematical theory of heat, gathered around him a select circle of the most talented young mathematicians for scientific and cheerful conversations. Poisson enriched mechanics and mathematical physics by a series of the most valuable memoirs. Cauchy was then laying the foundation for an essential improvement and transformation of the whole of analysis, by stricter methods and by the introduction of imaginary variables. These men, and besides them a considerable number of other scientific notabilities, some of whom are still living, worked together to make Paris the most brilliant seat of the mathematical sciences.

Correctly appreciating these circumstances, Dirichlet recognized that Paris was the place where he could expect the greatest gain for his mathematical studies. Since his parents, who from earlier times still had connections with some friendly families in Paris, readily consented, in May 1822 he entered this high school of the mathematical sciences, with the joyful consciousness that he could now devote himself entirely to his favourite study. He attended lectures at the Collège de France and at the Faculté des Sciences, where he had Lacroix, Biot, Hachette, and Francoeur as teachers. An attempt to obtain permission to attend lectures at the École Polytechnique as a guest failed because the Prussian chargé d'affaires in Paris, without special authorization from Minister von Altenstein, would not undertake to obtain permission from the French ministry.

In addition to hearing lectures and thinking through the material presented in them, Dirichlet devoted his time to the strenuous study of the foremost mathematical writings, and among these above all to Gauss's work on higher arithmetic, the Disquisitiones Arithmeticae. This work exercised a much more important influence on his whole mathematical education and direction than his other Parisian studies. He not only studied it through once or several times; throughout his life he never ceased, by repeated reading, to make again present to himself the abundance of deep mathematical thoughts contained in it. For that reason it was never set up on his bookshelf, but had its permanent place on the table at which he worked. One can infer how much effort it must have cost him to work his way into this extraordinary book from the fact that more than twenty years after it appeared none of the living mathematicians of the time had completely studied and understood it; and that even Legendre, who had devoted a large part of his life to higher arithmetic, had to confess in the second edition of his number theory that he would gladly have enriched his work with Gauss's results, but that the methods of that author were so peculiar that he could not have reproduced them without the greatest detours or without taking on the role of a mere translator. Dirichlet was the first not only to understand this work completely, but also to open it up for others, making fluid and transparent the rigid methods behind which deep thoughts lay concealed, and in many main points replacing them by simpler, more genetic ones, without yielding the least in the perfect rigour of the proofs. He was also the first, going beyond it, to reveal a rich treasure of still deeper secrets of number theory.

Dirichlet's outward life in the first year of his stay in Paris was extremely simple and withdrawn. His studies, interrupted only once by an attack of smallpox, occupied him completely, and his social intercourse was limited to a few houses to which he had been recommended and to a few young Germans who were there for their studies. In the summer of 1823, however, a change occurred that was of the greatest importance for his general cultivation. General Foy, a man of spirit and many-sided education, distinguished no less by the eminent position he held as head of the opposition in the Chamber of Deputies and as one of its most celebrated orators than by his brilliant military career, and whose house was one of the most respected and sought after in Paris, was then looking for a young man as teacher for his children, chiefly to instruct them in German language and literature. Through the mediation of a friend of the Dirichlet family, M. Larchet de Chamont, our Dirichlet was recommended to him. At their first personal meeting, the young man's open and modest manner made so favourable an impression on the general that he immediately offered him the position of teacher to his children, with a respectable salary and with such slight duties that he still had enough free time to continue the studies he had begun.

Dirichlet accepted with joy, not only because he was thereby placed in a position no longer to cost his parents anything, but especially because he expected much benefit for his outward worldly education, in which by his own judgment he was still very backward, from residence in the house of so broadly cultivated and distinguished a man. In this new position he felt extraordinarily content and happy, since Monsieur and Madame Foy showed him everywhere the greatest friendliness and courtesy and regarded him as a member of their own family. The instruction of the children, the eldest of whom was an eleven-year-old girl, cost him little effort; and the general's wife, who under his guidance zealously and with the best success took up again the German she had practised in childhood but had since completely forgotten, repaid his effort in an equally pleasant and useful way by helping him, through exercises in reading French, to lose the foreign accent in his pronunciation. The general, however, exercised the greatest influence on him, by the living example of an energetic, noble, and finely cultivated man; and this influence extended not merely to Dirichlet's outward cultivation, his habits and inclinations, but also to his way of thinking and acting and to his general views of life. It was also of great importance for his whole life that the general's house, a meeting-point of the chief notabilities in art and science of the French capital, and a place where the great political questions were discussed by the most respected members of the Chamber, questions whose nearest and provisional solution came in 1830, first gave him the opportunity to see life on a large scale and to take part in it.

All these new impressions that he absorbed, and the occupations and diversions connected with his position, did not turn Dirichlet away from his mathematical studies. On the contrary, precisely in this period he worked with intense diligence on his first published paper, Mémoire sur l'impossibilité de quelques équations indéterminées du cinquième degré. The subject of this memoir is most closely related to Fermat's assertion that the sum of two powers with the same exponent can never be equal to a power with the same exponent when these powers are higher than the second. This theorem, of which Fermat stated that he could prove it, but which more than 150 years after Fermat, at the time when Dirichlet occupied himself with it, had despite the strenuous efforts of Euler and Legendre been proved no farther than for the third and fourth powers, cannot indeed claim special value as an isolated item taken out of its scientific context. But it had acquired an unusually high importance because, as an apparently near yet distant goal set up in the then still quite unknown region of forms of higher degree, it had a most decisive influence on the direction taken by number theory in its historical development. Pursuing in his work the direction indicated by Fermat's theorem, Dirichlet considered the sum of two fifth powers, about which nothing had yet been established, and set himself the problem somewhat more generally: to investigate in what cases such a sum could not be equal to a given multiple of a fifth power. He levelled and secured the path of the investigation by some new theorems on the most general solution of the problem of making a quadratic form equal to a power number, and arrived at the proof of the impossibility of a whole class of equations of the fifth degree. Fermat's equation for fifth powers, one of which would necessarily have to be divisible by five, is included in this class only in the case where that one is also even; the other case, where it is odd, was shortly afterwards likewise shown to be impossible by Legendre, who followed a little further the path opened by Dirichlet. Thus Dirichlet shares with Legendre the honour of having advanced by a whole stage the proof of this historically remarkable theorem.

Not only the new results gained in one of the most difficult parts of number theory, but also the concision and sharpness of the proofs and the exceptional clarity of the exposition secured this first work of Dirichlet a brilliant success. The Paris Academy, to which he presented it, allowed him to read it at the session of 11 June 1825; and already at the next session, on the 18th of the same month, Messrs. Lacroix and Legendre, as commissioners, gave such a favourable report that the Academy resolved to include it in the collection of memoirs of foreign scholars. Dirichlet's reputation as an outstanding mathematician was thereby established; and as a young man from whom a great future could be expected he was thereafter not only admitted into the highest scientific circles of Paris but sought out by them. He thereby also entered into closer relations with several of the most respected members of the Paris Academy. Among them two must especially be mentioned: Fourier, who exerted an influence on the direction of his scientific researches, and Alexander von Humboldt, who exerted a significant influence on the later shaping of his outward life.

Fourier, who from the time of his youth, when he had taken an active part in the founding of the École Normale and the École Polytechnique, still preserved undiminished his enthusiasm for living scientific communication, and for whom it was an inner need to communicate orally what he had beautifully and greatly discovered, found in Dirichlet a young man to whom he could entirely open his mathematical heart and by whom he was not merely admired but also completely understood. He therefore drew him into the circle of distinguished young mathematicians whom he was accustomed to gather around himself, and with whom he then discussed his theory of heat, the new analytical methods invented for it, and various more general scientific subjects and questions in his own lively and attractive manner. In this circle, to which among others Sturm belonged, who shortly afterwards became generally famous through his theorem on the roots of algebraic equations, Dirichlet received manifold stimulation. In particular, Fourier awakened his interest in mathematical physics, in which he later worked with significant success; likewise Fourier series and integrals, which first received their true scientific foundation through Dirichlet's rigorous methods, occupy a not insignificant place in his later works.

Alexander von Humboldt, who was then living in Paris, had earlier heard General Foy praise Dirichlet as an outstanding mathematician, yet without attaching special weight to this praise, since it did not come from the mouth of a specialist. Only when Dirichlet, in consequence of the favourable reception that the Academy had granted to his memoir, paid his visit to Humboldt, did he become more closely acquainted with him. Humboldt received him with the most distinguished kindness and courtesy, and together with respect for his talent and scientific competence bestowed on him the liveliest personal sympathy and affection, which from then on he continually preserved and actively showed. Already during this first visit Dirichlet, in the course of conversation, let it be known that he intended later to return to his fatherland; Humboldt, who seized upon this thought with joy, strengthened him in his resolve by assuring him that there, with the extremely small number of distinguished German mathematicians, he could not fail, whenever he desired it, to obtain a very good position. Under the circumstances then prevailing, when negotiations for Gauss's call to Berlin, continued for several years, had just had to be abandoned because a few hundred thalers were lacking, this assurance was not so easy to fulfil; and soon afterwards Humboldt's tireless activity and all his influence were required to make it even approximately true.

The death in November 1825 of his highly honoured patron General Foy, and the influence of Alexander von Humboldt, who soon afterwards left Paris and moved to Berlin, brought Dirichlet to a mature decision to return to his fatherland. He addressed to Minister von Altenstein a request for a position suitable for him, which Humboldt undertook to support and to make effective by his influence; and in the autumn of 1826 he returned to his parents in Düren to await the result there.

While he was working there on a new memoir, to which I shall shortly return, Humboldt pursued the matter of his appointment with the liveliest zeal. He intervened personally with Minister von Altenstein to obtain for Dirichlet an extraordinary professorship at a Prussian university, with a salary of six to eight hundred thalers; he drew the most respected members of our Academy into his interest, so that their recommendations might support his own; and he frequently sent Dirichlet reports and good advice about what he should do on his side. But through all these efforts, even supported by Gauss through a letter addressed to our colleague Encke and handed by him to the Royal Ministry, no more could be attained than the assurance of 400 thalers a year as a fixed remuneration, so that he might habilitate at Breslau as a Privatdozent. Since the fixed remuneration secured him for the time being a modest livelihood, and since he could rely on Humboldt to continue his efforts to obtain for him a more suitable position, he accepted without hesitation. Meanwhile the philosophical faculty of the University of Bonn had created him Doctor of Philosophy honoris causa, which substantially eased his habilitation at a university.

On his journey to Breslau he chose the route through Göttingen in order to make the personal acquaintance of Gauss, and visited him on 18 March 1827. I have not been able to ascertain more detailed information about this meeting. A letter addressed to his mother from that time says only that Gauss received him very kindly and that the personal impression made by this great man was much more favourable than he had expected.

In Breslau, according to the statutes of the philosophical faculty there, he was to obtain the venia docendi by holding a trial lecture and colloquium before the faculty, writing a dissertation, and publicly defending it in Latin. For these performances, insofar as they concerned his science, he was more than equal to the task. He also knew very well how to write clearly and correctly in Latin on a scientific subject. But he had never devoted his time, consecrated to higher scientific ends, to acquiring the outward facility of speaking Latin; the empty formality of the Latin disputation was therefore highly disturbing and unpleasant to him. To his great satisfaction, but to the great regret of some gentlemen of the faculty there, after he had given his trial lecture on the irrationality of the number pi, the Royal Ministry entirely dispensed him from the public disputation. And so that he might immediately begin his lectures, without which the subject of mathematics there would have been almost wholly unrepresented, he received at the same time permission to submit his Latin habilitation dissertation afterwards.

The success of his teaching activity during the three semesters in which he lectured in Breslau was not significant. The students there, unwilling to go beyond the narrow circle of mathematical ideas and thoughts that had previously been transmitted to them in lectures, could not easily accustom themselves to his manner of teaching, which was unfamiliar to them; and his modest, even somewhat shy, appearance was not suited to impose on them. In Breslau generally, Dirichlet was gladly seen and sought in all social circles as a finely cultivated and lovable young man, but as a mathematician he was held in low esteem in comparison with his predecessor, who had written a textbook of analytic geometry. Since he himself never spoke of himself or of his own scientific merits, and had no literary following to do this for him, he did not attain there that local or provincial fame which in more restricted circles is more effective than the general recognition of the first men of science.

During his Breslau stay Dirichlet wrote two papers, both prompted by Gauss's memoir on biquadratic residues. In April 1825 the Göttinger gelehrte Anzeigen had brought a brief announcement of a series of papers on biquadratic residues and their reciprocity laws which Gauss intended to publish. The first had already been presented to the Göttingen Society of Sciences, but appeared only three years later. This announcement, giving some of Gauss's results whose proofs were to rest on a wholly new principle of number theory, aroused in the highest degree the curiosity of both Jacobi and Dirichlet. Each sought, by a quite different path, to penetrate Gauss's secret, and each succeeded in finding in this domain of higher power residues an abundance of new theorems, although the new principle, consisting in the introduction of complex integers, still remained concealed from them. For the already published Gaussian theorems, which contained the complete solution of the problem of specifying all primes for which the number two is a biquadratic residue or nonresidue, Dirichlet found astonishingly simple proofs by the known methods of number theory. In the same way he also solved the more general question for all arbitrarily given primes, so that only one step remained to the complete reciprocity law for biquadratic residues, a step that became possible only through the new Gaussian principle. In the other work then published, which Dirichlet wrote in Latin and submitted to the philosophical faculty as his habilitation dissertation, he gave an example, then quite new, of forms of arbitrary high degree whose divisors have certain linear forms. The results of this work may now be regarded as special cases of the general theorems on the divisors of the norm forms of the complex numbers formed from roots of unity.

To give a measure of the value then attributed especially to the first of these two writings, I cite the judgments of Bessel and Fourier upon it. Bessel wrote of it in a letter to Humboldt: who would have thought that genius would succeed in reducing something that seemed so difficult to such simple considerations? The name Lagrange could stand over the paper and no one would notice the incorrectness. Fourier, however, placed Dirichlet's achievement even higher than the great discoveries of Jacobi and Abel in the theory of elliptic functions, of which, to be sure, he had until then had knowledge only through Legendre's praises, which he considered exaggerated. In a letter addressed to Dirichlet, and in oral remarks made to the above-mentioned friend of the Dirichlet family, M. Larchet de Chamont, from which this judgment is taken, he also expressed the lively wish that Dirichlet might return to Paris, because he was called to occupy very soon one of the first places in the Academy there.

Meanwhile Alexander von Humboldt had obtained Dirichlet's appointment as extraordinary professor at the University of Breslau and was now working to win him for the university here and for the Academy, but first of all to draw him to Berlin. Since a mathematical teaching position about to become free at the General War School offered the appropriate opportunity, Humboldt seized it and recommended Dirichlet most urgently for it to General von Radowitz and to the Minister of War. They could not, however, decide at once to entrust the position to him definitively, probably because at the time, being only twenty-three years old, he may have seemed to them too young for it. It was therefore arranged with Minister von Altenstein that Dirichlet should at first receive a year's leave in order to take over the teaching at the War School provisionally. In the autumn of 1828 he came to Berlin to assume this new position.

The mathematical lectures which he had to give there before officers of about his own age gave him much pleasure. Association with cultivated military men, to which he had been accustomed from his time in General Foy's house, suited him very well; and since during that period he had also made thorough studies in modern military history, this common interest connected him with his listeners outside mathematics as well. Only later, after he had formed at the university here a large circle of listeners who followed him with lively scientific interest into the highest regions of mathematics, where he preferred to move, did the wish arise in him to be released from teaching at the War School; this wish then became one of the chief motives for his move to Göttingen.

Soon after his arrival in Berlin Dirichlet also took the necessary steps to be allowed to lecture at the university here. As professor at another university he did not have that right; nothing remained to him but to habilitate once more as Privatdozent, and in this sense he addressed his petition to the philosophical faculty. The faculty, however, dispensed him from the habilitation requirements in view of his otherwise proven scientific competence; and so he initially lectured here under the legal title of a Privatdozent. His definitive transfer as extraordinary professor to the university here took place only in 1831; a few months later he was elected by our Academy to ordinary membership. In the same year he married Rebecca Mendelssohn-Bartholdy, a granddaughter of Moses Mendelssohn. Remarkably, Alexander von Humboldt himself had an involuntary share even in this, insofar as it was he who first introduced Dirichlet into the famous house of his parents-in-law, distinguished by spirit and artistic sense.

For a longer time the further events of his life now recede before the significance of the scientific works that Dirichlet produced during the twenty-seven years of his life here. In describing them, which is now my task, I shall try to present in larger and more general outlines the advances of science contained in them, considering them not individually by the time of their origin but grouped according to their contents and to the underlying ideas.

Dirichlet's purely analytical works on infinite series and definite integrals, and on the functions appearing in these forms, originally arose from his study of mathematical physics and especially of Fourier's theory of heat. In applying these general forms to physical problems, however, he could not content himself with using them as ready-made tools; and since closer examination soon showed him that even in the most important points they still lacked strict scientific foundation, he directed his work first to securing these foundations. The series progressing according to the sines and cosines of multiples of an arc, which Fourier had used with the most outstanding success to represent arbitrary functions occurring in the theory of heat, had up to then, in all cases where the function to be expanded does not become infinite, shown the remarkable property of always being convergent. Yet even Cauchy's efforts had not succeeded in proving this generally and rigorously. The path taken here by that mathematician, famous not less for the rigour than for the originality of his methods, namely to examine only the relative magnitudes of the individual terms of these series and to base his conclusions on that, did not lead to true knowledge of this hidden property, but only quite close to it; for the convergence of these series in certain cases also depends on the particular way in which their positive and negative terms cancel one another. For this reason Dirichlet, returning to the original concept of convergence of infinite series, investigated the limiting value reached by the sum of a number of terms when this number is assumed to increase without bound, and he completely fathomed this question by the exact determination of the limiting value of a simple definite integral, which, because of its many applications, has since been counted among the foundations of the theory of definite integrals.

By the same method and with the same means Dirichlet also carried out the more general and more complicated investigation of the convergence of the expansion of an arbitrary function of two independent variables ordered according to spherical functions. It was moreover necessary only to choose the expression of the spherical functions by definite integrals in such a suitable way that the sum of an indeterminate number of the first terms of this expansion, whose coefficients are given as double integrals, became as simple as possible and appeared in a form in which its limiting value could easily be determined by means of the limiting value already found for that simple integral.

Dirichlet found not only the special theory of these two kinds of series expansions, but also the general theory of infinite series, still unfounded in some essential points. It was indeed known that divergent series have no determined values; but the rules and conclusions valid for finite series were still transferred to infinite series in too naive a way, and no one had ever thought that even the most elementary rule according to which every algebraic sum is independent of the order of its parts might cease to be correct for sums consisting of infinitely many parts, until Dirichlet showed that there is a class of convergent series with positive and negative terms which can take other values, and even become divergent, when only the order of their terms is changed. The deeper insight thereby gained into the nature of infinite expressions also became decisive for the treatment of definite integrals; for Dirichlet's observation, applied to multiple integrals whose upper limits are infinite, showed that there is likewise a whole class of such integrals in which changes in the order of integration can produce quite different values.

Dirichlet treated the general theory of definite integrals with special predilection in his lectures. In them he united into a connected whole, by appropriate ordering and method and while excluding all external aids not belonging to this theory itself, results previously scattered as isolated items. In addition, he enriched this discipline by inventing a new and distinctive method of integration. Its main idea consists in introducing a discontinuous factor so that the limits within which the integrations have to remain can be overstepped in such a way that arbitrary other, though wider, and especially infinitely wide, limits can be taken instead of the given ones without changing the value of the integral. In applying this method to the attraction of ellipsoids and to the evaluation of a new multiple integral he also showed that, handled with skill, it is capable of giving the solutions of certain difficult problems by a simpler path than the other known integration methods.

While occupied with these analytical works, Dirichlet never ceased also to keep in mind the great problems of his favourite discipline, number theory, and to meditate on their solution. In his mind, always striving toward unity, he could not allow these two spheres of thought to stand side by side without inquiring into their inner relations, in which he sought, and in fact found, knowledge of many deeply hidden properties of numbers. His applications of analysis to number theory that arose from this differ essentially from all earlier attempts of the kind in that in them analysis is made serviceable to number theory in such a way that it no longer merely happens to throw off some isolated results for it, but must with necessity yield the solutions of certain general classes of problems of arithmetic still wholly inaccessible by other paths. These Dirichlet methods are epoch-making for number theory in a way similar to Descartes's applications of analysis to geometry. They would have to be recognized, just as analytical geometry is, as the creation of a new mathematical discipline if they extended uniformly not merely to certain classes but to all problems of number theory.

The first application that Dirichlet made of his new method concerns the very simple theorem that every arithmetic progression whose terms do not all have a common factor contains infinitely many primes. Because of its quite elementary character this theorem is of great importance in many arithmetical investigations, and in particular it had been used merely as an unproved result in Legendre's first attempted proof of the law of quadratic reciprocity. Euler's distinctive way of concluding, from the transformation of a product containing only primes into a divergent infinite series, that the number of all primes is infinite, gave Dirichlet the occasion for the more general application of infinite series and infinite products. In seeking to arrange these analytical aids in accordance with the purpose of his investigation, he arrived at the fundamental theorem of his new method, which determines the limiting value of a general series of powers of positive decreasing quantities whose common exponent approaches the limit one. The application to the proof of the theorem on arithmetic progressions requires the development of a certain group of infinite products into infinite series, and then the task is to prove that the product of these infinite series becomes infinitely large when the common power exponent of all terms approaches the limiting value one. Since the first factor of this product, according to the fundamental theorem just mentioned, necessarily becomes infinitely large, it remains only to show that the product of all the other factors cannot be zero. Although these infinite series can be summed in finite form by means of logarithms and circular arcs, the complete settlement of this important point, especially for the case where the common difference of the arithmetic progression is a composite number, presented very considerable difficulties, which Dirichlet in the first treatment of the investigation could overcome only by very complicated and indirect considerations. But precisely this difficulty became the occasion for a second application of analysis to number theory, which contains one of his most important and brilliant discoveries: the determination of the number of classes of quadratic forms for every given determinant. The difficulty of the first investigation was disposed of by this second one in the simplest way, for it showed of itself that the product in question cannot be zero unless the number of classes of quadratic forms would also have to be zero.

The determination of this class number rests on considering the doubly infinite sum whose general term is unity divided by a power of a quadratic form, the exponent being taken infinitely near to one. This sum, which extends over all integral values of the two indeterminate variables, with certain restrictions, and over all non-equivalent forms of the same determinant, is determined in two different ways: once by taking together the representative forms of all classes, and the other time by considering them individually. Since the sum receives the same value for each of these classes, the total sum is obtained as such a single sum multiplied by the number of classes. The form in which the expression for the class number thus obtained finally presents itself is essentially different for negative and for positive determinants, and in both cases reveals a surprising connection of the class number with quite different regions of number theory: for negative determinants, with quadratic residues and nonresidues; and for positive determinants, with the two solutions of Pell's equation distinguished above all others, namely the fundamental solution containing the smallest numbers and the solution arising from the theory of division of the circle, which Dirichlet first gave in a short paper on the solution of Pell's equation by circular functions. Since then no deeper insight has been gained into the connection of these apparently quite heterogeneous objects with the class number and with one another, because in general no method other than Dirichlet's exists that would be capable of solving such difficult questions.

Although the fame of this great discovery belongs to Dirichlet alone, insofar as he drew it entirely from his own mind, a certain share that Jacobi and Gauss have in it must nevertheless not be left unmentioned. From comparing certain theorems of cyclotomy and of the composition of quadratic forms, Jacobi had already, some years earlier, more guessed than derived the class number for forms whose determinant is a negative prime; and since a series of numerical examples confirmed his conjecture, he had also published it. But this could have had no influence on Dirichlet's discovery, because according to his method he could not aim to prove a determinate result in a finished form, but only to find it, and in such a way that nothing whatsoever about its form could be predetermined. Gauss, however, as the papers he left behind have shown, had long been in possession of the complete expression for the class number for negative determinants, which he had found not by induction but probably by a method similar to Dirichlet's. That he never published this result, nor a whole series of the most important and brilliant discoveries later made by Abel and Jacobi and found in his writing desk, seems to have its reason not only in an inexplicable peculiarity of this extraordinary man, but probably also in the fact that the methods he used may not have satisfied him in all points, and that he preferred to give nothing rather than something defective.

Dirichlet extended the application of his method to the theory of quadratic forms not to the class number alone but also to all the related questions concerning the division of the classes into genera and orders. In this way he made accessible by a new path the most interesting, but because of the difficulty of the methods the hardest to understand, section of Gauss's Disquisitiones Arithmeticae. Moreover, according to principles similar to those used for arithmetic progressions, he also proved for quadratic forms the theorem that every form whose three coefficients have no common factor represents infinitely many prime numbers.

When, later, in a special memoir, he treated the theory of quadratic forms from the viewpoint that the coefficients and the variables of the form are to be considered as complex integers arising from the decomposition of the sum of two squares into imaginary factors, he obtained from the application of his method to them several new and surprising results. I must especially emphasize two of them, which are of special importance because they open deeper views into the most hidden properties of forms of higher degree. The class number of quadratic forms with complex coefficients is expressed by series which, as Dirichlet unfortunately did not develop fully but only indicated, can be summed not by circular functions but by lemniscatic functions. These therefore stand here in the same relation to the solutions of the corresponding Pell equation, or more generally to the units, as the circular functions stand to the units of non-complex quadratic forms. Since the forms considered here can also be conceived as decomposable forms of the fourth degree with four variables, this result points in general to an intimate, still unexplored connection in which certain forms of higher degree stand to definite, and indeed periodic, transcendental functions of analysis. Dirichlet further found that in the special case where the determinant is a non-complex number, the class number of these complex quadratic forms consists of two factors, both of which separately express the class numbers of non-complex forms of the same determinant, one for the negative determinant and the other for the positive. This example first revealed the more general nature of these expressions, which recurs in all class numbers of forms of higher degree subsequently determined: namely that they consist of two essentially different integral factors, one determined solely by the units, the other by power residues with respect to the determinant.

For those decomposable forms of higher degree whose linear factors contain no irrationalities other than roots of unity with prime exponent, Dirichlet determined the class number during his stay in Italy, but unfortunately he published nothing of this work.

Finally, one must also mention here the interesting and new results which Dirichlet obtained from the application of his method to the determination of mean values, or asymptotic laws, for the integral functions appearing everywhere in number theory and seeming to progress quite irregularly. They concern the question, already treated earlier by Euler, Legendre, and Gauss, of the frequency with which primes occur in the natural sequence of numbers; also the mean values indicated by Gauss for the number of classes of quadratic forms and the number of their genera; and in addition several functions occurring in the elements of number theory and relating to divisors and residues of numbers. Remarkably, in precisely this type of investigation, for which analytical treatment appears especially suited, Dirichlet's continued efforts succeeded in many cases in replacing analytical methods by purely arithmetical ones, and in this way in obtaining some further new and surprising results. I shall mention only one: when a given number is divided by all smaller numbers, the remainders that are less than half the divisor occur on average much more frequently than those that are greater.

The investigations mentioned above concerning certain forms of higher degree with several variables led Dirichlet to the general theory of decomposable forms of all degrees, which is essentially identical with the general theory of the complex numbers introduced into science by Gauss. What he published on this important subject is indeed limited to a few sketch-like brief communications in the monthly reports of our Academy; nevertheless, besides the general guiding ideas, they contain the solution of the principal fundamental difficulties. In particular, the theorems on the units and the chief points in the proofs of them are stated so completely that this part of the theory could be worked out wholly in the spirit of its author. The significant benefit that would accrue to science from a complete treatment of this general theory can already be clearly recognized from what Dirichlet gave of it; for in it one finds again the most essential and beautiful properties of the relevant special theories, notably also of quadratic forms, which in the generality corresponding to their nature do not appear in more complicated shape, but, purified of the inessential determinations that adhere to everything special, can first be recognized in their true simplicity and greatness.

The lectures on number theory which Dirichlet introduced at the university here, and indeed first introduced at German universities in general, also caused him to devote special diligence to the more elementary parts of this discipline, and especially to the simplification of Gaussian methods and proofs. Among the works belonging here, which he not merely communicated orally to his students but also published elsewhere when occasion arose, I first mention the new treatments of two Gaussian proofs of the law of quadratic reciprocity: namely, the first proof given in the Disquisitiones, which even in Dirichlet's clear and appropriate treatment is inferior in simplicity to other proofs of this theorem and has only this in its favour, that it requires no aids other than those lying within the theory of quadratic residues itself; and the fourth Gaussian proof, derived from the summation of certain finite series whose absolute value can be stated very easily, while the whole difficulty lies in determining the associated sign, which Dirichlet mastered by summing these series by means of definite integrals. Further to be emphasized is the new treatment, published as an academic occasional writing, of the doctrine of composition of quadratic forms, in which he derived in the simplest way the theorems worked out in Gauss only by a hard-to-manage apparatus of formulas, by going to the essence of the matter and considering, instead of the forms, the numbers to be represented by them. The already mentioned work on the theory of quadratic forms for complex numbers can also be counted here, insofar as its simple methods are everywhere applicable to ordinary quadratic forms as well, whereby it at the same time takes the place of a simple and thorough systematic presentation of this elementary theory. Finally there belong here the new proofs of the theorems on the number of decompositions of numbers into four and into three squares, and the general reduction of positive quadratic forms with three variables, the latter first carried out by Seeber in an extremely complicated manner. In general, one recognizes in the methods by which Dirichlet simplified number theory in these works and made it more accessible that they are drawn chiefly from the thorough study of the more general theories. The proofs of the theorems therefore rest not on special and accidental determinations, but throughout on the essential properties of the number-theoretic concepts concerned; and in the special case they thereby mediate at the same time the recognition of the general.

Dirichlet's works in mathematical physics and mechanics originally proceeded from Fourier's theory of heat, which, as already remarked, was also the source of his first analytical works. He published, however, only one work concerning the theory of heat itself: a rigorous and simple solution of the problem, already treated by Fourier, of determining the heat state of an infinitely thin rod for which the temperatures at both ends are given as functions of time.

Later, the questions and mathematical-physical investigations stimulated and carried out by Gauss came to predominate in his interest; and he chose especially the theory of forces acting inversely as the squares of the distances as the object of his researches, on which he also gave special lectures at the university. Of the two memoirs he published in this field, one gives the solution of the problem of finding the density with which an infinitely thin layer of mass must be laid upon a spherical surface so that the potential at every point of it has a given continuously varying value. Gauss had shown that this problem has a determinate solution for every surface and that for the spherical surface this solution is also analytically executable. It was especially necessary here to investigate the convergence of the expression for the density developed according to spherical functions, which follows easily from the corresponding expression of the given potential value. That convergence did not follow directly from Dirichlet's above-mentioned memoir on the convergence of series ordered according to spherical functions, since the density could also be infinite at places. The exact investigation of this point gives the result that the convergence of this series in fact does not take place generally without exception, but rather is subject to certain conditions whose absence indeed causes the series to diverge. The final result is then simplified by carrying out the summations so that it requires no infinite operation other than a double integration. The second short memoir belonging here concerns the potential as such and contains in this respect a new definition of it, since Dirichlet shows that the known equation among the second partial differential quotients, together with certain conditions of continuity and finiteness satisfied by the potential and its first differential quotients, determines it in such a way that no analytic function other than the potential satisfies all these conditions. Accordingly every expression found for a potential can be tested and verified a posteriori by differentiation. This new way of defining analytic functions by means of continuity conditions has since been raised by Dirichlet's successor in Göttingen, Professor Riemann, to a special principle of analysis, which has already proved extraordinarily fruitful in his works and seems destined to found a new epoch in the direction recently pursued by analysis, in which the solution of its problems is forced less by calculation than by thought.

The investigation of the stability of equilibrium, in which Dirichlet first rigorously proved that to every maximum of the force function there really corresponds a position of stable equilibrium, was carried out in a similar spirit. By using not the analytical rules for the determination of maxima of functions but only the original concept of a maximum, it attained not only the exceptionless general validity lacking in all earlier proofs but also a wonderful simplicity and clarity.

In his investigations on the motion of fluids Dirichlet gave the first example of a truly executed integration of the general differential equations of hydrodynamics: namely, for the case in which in an infinite mass of fluid, originally at rest, a sphere moves under arbitrary accelerating forces in a constant direction and by its motion sets the fluid itself in motion. He found in this connection the very remarkable result, contradicting ordinary ideas about the resistance of fluids, that the resistance which the sphere has to suffer in its motion depends not on its velocity itself but only on the increase of that velocity; so that when the accelerating forces cease to act on the sphere, the resistance also disappears, and the sphere makes a constant motion in a straight line in the fluid.

Finally, there must be mentioned the memoir on a problem of hydrodynamics, which was also Dirichlet's last work. It gives another example, not merely approximate but rigorous, of the integration of the general hydrodynamic equations, in the determination of the motion of a fluid under the assumption that the individual particles of mass, in their motion, continually preserve a certain condition of affinity and that the original form of the fluid is that of an ellipsoid. Dirichlet proves that such a motion is indeed possible and that throughout its whole duration the fluid retains the shape of an ellipsoid, with the same centre but with variable position and magnitude of the principal axes. In the special case where the ellipsoid is one of revolution, all integrations can be completely reduced to quadratures, and the fluid makes isochronous oscillations, alternately taking the form of a prolate and an oblate ellipsoid.

Before I now return from considering Dirichlet's scientific works to describing his life, I must emphasize a general observation suggested by comparing them with Jacobi's works. Since these two men worked contemporaneously for a quarter-century at the further development of the mathematical sciences, and stood in close personal friendship and active scientific intercourse with one another, it is a very striking phenomenon that their writings, although they often concern the same special disciplines, show almost no immediate points of contact. The special objects of their researches are, with very few insignificant exceptions, entirely different, and there are scarcely any examples even of one using the results of the other for his own investigations. This lack of relation in their writings cannot be sufficiently explained solely by the difference of the starting-points and directions of their mathematical studies and labours; rather it is grounded in the fact that both deliberately avoided encroaching upon those regions in which each recognized the superiority of the other, and that they sought to avoid even the appearance of rivalry.

The first personal acquaintance between Dirichlet and Jacobi was formed in 1829, when Jacobi came from Königsberg to Berlin to visit his relatives and friends. On a journey they undertook together to Halle, and from there in the company of Mr. W. Weber to Thuringia, they became better acquainted. Since Jacobi often spent his vacation time in Berlin, they later had opportunity for more intimate scientific and friendly intercourse. Afterwards, when Jacobi was seized by a dangerous illness and, on the advice of physicians, had to seek the milder climate of Italy for his recovery, Dirichlet, who had long intended a journey to Italy, seized this opportunity to spend a winter in Rome together with Jacobi, and in the autumn of 1843 set out there with his whole family. Since at the same time our colleagues Mr. Steiner and Mr. Borchardt also spent that winter in Rome, German mathematics was at that time represented there in a very brilliant and many-sided fashion. Dirichlet remained in Italy a year and a half, extended his journey also to Sicily, and spent the next winter in Florence. On his return he found Jacobi in Berlin, for in the meantime, by the grace and munificence of His Majesty the King, Jacobi had been given leave from Königsberg and called here, so that, without occupying a definite office, he could care for his health and live wholly for science.

The common interest in knowing truth and promoting the mathematical sciences remained the firm foundation of the friendly relation in which Jacobi and Dirichlet lived together here.

They saw one another almost daily and discussed with one another more general or more special scientific questions, whose spirited treatment, precisely because of the difference of the standpoints from which both surveyed the whole field of the mathematical sciences, always retained a new and living interest. Jacobi, who through the wonderful abundance of his mind no less than through the depth of his mathematical researches and the brilliance of his discoveries knew how to secure everywhere the recognition due him, enjoyed at that time a much more widespread reputation than Dirichlet, who did not possess the art of making himself count, and whose writings, chiefly treating only the most difficult problems of science, had a less extensive circle of readers and admirers. No one recognized this disproportion between Dirichlet's outward recognition and his scientific importance more correctly than Jacobi; and no one else was at the same time more skilful and more active in equalizing it and in procuring for his friend, even in wider circles, the recognition he deserved. It is chiefly to Jacobi's activity also that Dirichlet was preserved for our Academy when, in 1846, the government of Baden intended to win him for the University of Heidelberg. Two letters which he addressed in this matter to Alexander von Humboldt and to His Majesty the King give, in a few strong and apt strokes, a vivid presentation of Dirichlet's scientific greatness and of the irreplaceable loss that the exact sciences in Prussia, the Academy, the University, and especially Jacobi himself, would suffer if Dirichlet were to leave our fatherland.

This threatening loss, happily averted at that time, confronted us nine years later, after Jacobi and Gauss had passed away, all the more painfully. The University of Göttingen, which for half a century had enjoyed the fame of possessing the first among all living mathematicians, strove eagerly to maintain this fame further by calling Dirichlet to Gauss's position, and first addressed to him the question whether, and under what conditions, he might be inclined to accept such a call. Dirichlet had here a sphere of activity such as he could hardly expect to find again at another university; he enjoyed in high degree the veneration of his listeners and the respect and affection of his colleagues, and besides was bound to Berlin by close family ties. The only thing that made a change in his situation desirable to him was that teaching at the War School fragmented the powers that he would gladly have devoted wholly to the university and to science. It was therefore his lively wish to be released from the position at the War School and to have this loss of income covered by the university. Since the call to Göttingen offered him the opportunity to attain his purpose in one way or another, he declared in answer to the inquiry addressed to him that he would obey an official call from the Royal Hanoverian government unless, before its arrival, his position here were changed in accordance with his wishes. His friends, to whom he communicated this, did not omit to inform the Royal Ministry of it, so that timely precautions might be taken to avert the threatened loss from the university here and from the Academy. But Minister von Raumer did not wish to make an immediate decision; he wanted first to await an official step by the Royal Hanoverian government. That government soon sent Dirichlet his formal call through his friend Professor Weber in Göttingen, who delivered it personally. When now Minister von Raumer, in order to keep him here, offered him even more than he had wished, it was too late. Since Dirichlet now considered himself bound by his previous declaration, no advantages or considerations could determine him otherwise.

In the autumn of 1855 he moved from here to Göttingen. There he arranged for himself, entirely to his liking, his own pleasantly situated house with a garden. The quiet of the smaller town, which he had not enjoyed since his youth, compensated him sufficiently for the outward amenities of metropolitan life in Berlin. He also found there men of like mind to whom he could attach himself more closely; and his scientific and general intellectual importance, combined with the unpretentiousness and honourableness of his whole nature, soon won him the same general esteem that he had enjoyed here. At the university he did not indeed find as large a circle of listeners as the one he had left here; but his reputation as a teacher, acknowledged no less than his scientific reputation, drew to Göttingen many young men striving for higher training in the mathematical sciences, and some of the most distinguished academic teachers there also became his eager listeners, so that the success of his lectures there was in proportion no less than here. Since his mathematical researches, which always lay closest to his heart, were also favoured by the greater leisure he enjoyed, he felt very satisfied in his new position.

In the summer of 1858, after the close of his lectures, he travelled to Switzerland and stayed in Montreux on Lake Geneva, less for rest than to work out there a memorial address on Gauss to be delivered in the Göttingen Society of Sciences, and a memoir for that society's transactions. When he had almost completed the hydrodynamical memoir mentioned above, he was suddenly seized by an acute heart disease and hastened at once back to his family in Göttingen, where he arrived mortally ill. The art of the physicians and the loving care of his family succeeded in happily averting the immediate danger to his life; but he could scarcely raise himself again somewhat from his sickbed, and for his hoped-for complete recovery still needed the greatest rest of body and mind, when his wife, suddenly struck by apoplexy, died after a few hours, without his being able to see her once more. This unexpected blow turned his illness again for the worse, and after severe sufferings he succumbed to it on 5 May 1859.

As a human being Dirichlet was no less distinguished by his noble character than in science by the depth and solidity of his mind. The honourableness that filled his whole nature and appeared pure and unclouded in all his actions sprang from the high moral cultivation of his mind and heart, and for that reason was directed not to outward honour but everywhere only to true inner honour, whose exact and strict measure he bore within himself. Ambition, which seeks outward recognition and finds its satisfaction more in appearance than in essence, was wholly foreign to him. Even the scientific honours from learned bodies that were bestowed upon him in the richest and highest measure he valued chiefly insofar as he could see in them the approval of connoisseurs and competent judges; they had no influence on the clear judgment which he exercised with complete impartiality concerning the value of his own achievements.

As in science, so also in his whole life the love of truth was the moral foundation of his thinking and acting. It pressed back the activity of imagination in him, kept him free from prejudices and self-deceptions, and allowed him full satisfaction only where he could arrive at exact and completely secure knowledge. Truth in symbolic form corresponded less to his nature; truths that announced themselves as results of philosophical speculation appeared to him in general suspect. He used to say of philosophy that it was an essential defect of it that it had no unsolved problems, as mathematics does; that it therefore was conscious of no definite boundary within which it had really investigated truth and beyond which, for the time being, it must modestly confess that it knew nothing. The greater the claims philosophy made to omniscience, the less perfectly clear recognized truth he believed he could entrust to it, since from his own experience in the field of his science he knew how difficult the knowledge of truth is and what effort and labour it costs to advance it even one step.

A certain shyness that had belonged to Dirichlet in his youth had, in maturer age, ennobled itself into true inner modesty; yet it still showed itself in many respects as natural reserve, especially in the fact that he very unwillingly appeared in public, did not like to take the floor in larger assemblies, and never delivered speeches unless it was for him an unavoidable duty. In general he never thrust himself forward, neither with his person nor with his opinions and judgments, but was reserved even where his judgment as an expert was requested, because precisely in such cases he proceeded with the greatest conscientiousness and gave his definite judgment only after considering all sides. To his disposition, directed more toward knowledge than toward practical activity, every striving after outward influence was foreign. In his outward relations in life he in fact never made any influence effective except that which a noble and spirited man involuntarily and immediately exercises in the circles to which he belongs.

In social and friendly intercourse Dirichlet everywhere showed the genuine humanity that is grounded in general respect for the personality of human beings and in freely allowing their peculiarities and convictions. He had an open eye for the good sides of others and preferred to seek these out rather than to dwell on their weaknesses and defects, which he never made the object of self-satisfied mockery and opposed only when they betrayed a lack of honourable disposition. The same humanity he showed in his extensive scientific intercourse with the most important and capable mathematicians at home and abroad, an intercourse he preferred to maintain personally rather than by letter, because letter-writing was not pleasant to him, while on journeys he gladly visited his acquaintances and was often visited by them. He always showed a very lively participation in the achievements of others, gladly entered in conversation into their special scientific interests, and instructed by communicating the higher points of view from which he surveyed the questions at hand, without ever making the superiority of his mind be felt in an oppressive way.

The most capable among the younger German mathematicians had almost all formerly been Dirichlet's listeners, and esteemed him not merely as their teacher, to whom they owed the best part of their mathematical education, but were also always attached to him with true love and veneration. How highly he for his part knew how to value the love of his pupils, and how above all he recognized it as the highest reward of his teaching activity, he expressed shortly before his death in a beautiful and worthy way. After one of the last severe attacks of his illness, when he felt somewhat freer again, he expressed the wish to see once more one of his dearest friends and former pupils. The latter, being informed of this, immediately travelled to him and had the good fortune, during two days in which the illness had somewhat abated, to see and speak with his beloved teacher. At parting Dirichlet said to him: It is truly rewarding to be a professor when one earns such love.

The success of his teaching activity, externally measured by the number of listeners, especially in the later period of his academic work, was as considerable as probably no teacher at a German university can show in the field of higher mathematics. He owed it to no didactic artifices, nor to the gift of a brilliant delivery, but solely to the inner clarity of his mind, by virtue of which he knew how to grasp and present even the most difficult subjects in their simple truth. In doing so he spared his listeners no exertion of thought necessary for complete knowledge of the subject, but he gladly spared them and himself lengthy and time-consuming calculations by replacing them, wherever possible, by simple thoughts. If one measures the success of his teaching by the scientific competence of the younger mathematicians who were his students and who owe chiefly to him their mathematical education, then only Jacobi's activity can in general be regarded as equal to his, and perhaps in one respect even more highly valued: Jacobi founded a special mathematical school that continues to work in his spirit and sense, whereas Dirichlet's students pursue more individually different directions.

His own scientific direction was so closely connected with the peculiarity of his mind and character that it could not become the common property of a school. He did not love the much-trodden and already levelled roads of science; rather he took pleasure in choosing as the object of his reflection and work the principled difficulties that those roads usually avoid. When he had fathomed them, he did not linger in spinning out the consequences of the results obtained, but preferred to work on from them into the depths, where he found new difficulties to overcome. For this reason his writings are not extensive and consist mostly of smaller memoirs in which he treats definite problems of science and completely fathoms them. Especially characteristic of his scientific direction is also the perfect rigour and evidence of the methods and proofs by which he establishes his results. This property, although it corresponds only to a demand lying in the essence of mathematics itself, is nevertheless found even among the greatest mathematicians only rarely in complete purity. In the field of analysis it first came into force through Gauss, and since then has so little become common property that even Jacobi's writings show its absence at certain places, as Jacobi himself openly admitted.

That Dirichlet kept himself and his writings free from such defects he owed chiefly to the love of pure and perfectly secure truth that was proper to him, and in addition to the way in which he worked and to the care with which he composed his writings. The clarity and definiteness of his thinking and the unusual power of his memory, by virtue of which he kept once thought and investigated matters perfectly present to himself at any time, made the use of the pen while working almost entirely dispensable to him. Nor did he need special quiet or leisure for it; on walks, on journeys, at musical entertainments, and generally in all situations where he did not himself need to speak or act, he could continue his deep speculations with the same success as at his writing desk. As an example I may mention that he fathomed the solution of a difficult problem of number theory, with which he had struggled in vain for a long time, in the Sistine Chapel in Rome while listening to the Easter music usually performed there. When he had found important results, he devoted the greatest diligence to finding, through investigation of their connections on all sides with one another and with related theorems, the simplest method of derivation most appropriate to the nature of the subject. Only after he had succeeded in this did he proceed to written elaboration, to which he usually resolved with difficulty, but which he then carried out with the greatest care.

Of the results that Dirichlet worked out in the last years of his life, only little has been preserved for science, because he postponed their written elaboration too long. Among the papers he left behind there was found no mathematical manuscript except the one hydrodynamical memoir that recently appeared in the Transactions of the Göttingen Society, edited by Professor Dedekind, to whom Dirichlet himself had still entrusted its completion. From what he occasionally communicated to individual friends about the subjects of his research, it appears that among other things he had completed in his head a full theory of ternary indeterminate forms of the second degree; that he had succeeded in carrying the approximation of the asymptotic laws for a kind of number-theoretic functions, on which the determination of the frequency of primes depends, a whole degree further; and that he had found a mathematically completely rigorous proof of the stability of the world system. Of a great and especially valuable discovery from the last period of his life, namely a wholly new general method for the treatment and solution of problems of mechanics, he spoke only once in the summer of 1858 to one of his friends, Mr. Kronecker, with whom he stood in the most intimate scientific and friendly intercourse. He himself attached quite special weight to this discovery and asked Mr. Kronecker for the time being to speak of it to no one. Kronecker therefore communicated to his friends only after Dirichlet's death what he had learned from him about it: namely, that this method was not directed toward reducing the integrations of the relevant differential equations to quadratures, because this means, by which Jacobi had attempted to obtain the solution of mechanical problems, was too limited; that his procedure consisted rather in a stepwise approximation, in which each new step at the same time afforded a more complete and more exact insight into the nature of the motions determined by the conditions of the problem; and finally that the theory of small oscillations offered a certain support for finding this method.

I will give words to the lament over these losses to science, perhaps not to be replaced for a long time and whose magnitude can be sufficiently measured from the indications still available, only by recalling the saying which Dirichlet himself uttered in the memorial address on Jacobi concerning that man's unfinished works: Death, which took him away from his labour too early, did not grant such great enrichments to science.

Memoir on the Geometry of the Hindus

by M. Chasles[1]

Mr. Chasles takes issue with the prevailing opinion that the geometrical part of the works of Brahmagupta and of the later Bhascara is to be regarded as a complete presentation of Indian geometry, and he wishes to see in it a special theory, namely that of the cyclic quadrilateral. One must indeed admit that the propositions contained in these writings do not form a connected chain ascending from the ultimate axioms; yet nothing seems to me to prove that such a strict proof, advancing step by step, as we admire to so high a degree among the Greeks, was felt in India as the same need. It may rather appear more appropriate to the imaginative spirit of the Orient that truths accessible to immediate intuition should not be further dissected and reduced to simpler ones. Whereas, for example, among the Greeks the proportionality of the sides in equiangular triangles appears as the result of a long series of inferences, the same truth in the Orient may have been regarded as given by intuition and therefore as needing no further reduction. To every person, even one who had never occupied himself with mathematical reflections, it is in fact evident that every figure can be enlarged or reduced to any degree without the dimensions occurring in it changing their ratios. This reduction of propositions to what is immediately intuitive seems to me a very remarkable peculiarity of Indian geometry. As a particularly striking example of this kind one may cite the proof, the second one, which Bhascara gives for the Pythagorean theorem. One need only cast one's eyes on his figures in order to be convinced at a glance of a theorem which in Euclid requires a quantity of preparations. A second peculiarity of Indian geometry, which is in a certain sense opposed to the one just mentioned and contains the germ of modern mathematics, is the use of the arithmetical element, which the Greeks carefully kept away from their presentations, even if probably not from their investigations. In the Orient, geometry and arithmetic appear as completely intermingled and serve one another as supports.

Even if Indian geometry, so far as it is present in the works named, cannot be compared either in extent or in inner beauty with the grand structure of the Greek school, it is nevertheless not lacking in results which seem not to have been known to the Greeks and which are to be ascribed especially to the free use of calculation. Among these Mr. Chasles singles out, as having hitherto been wholly overlooked, the elegant formula by which the area of a quadrilateral inscribed in a circle is determined from its four sides. The analogous formula for the triangle, which earlier writers had regarded as a discovery peculiar to the Orient, is found, as Mr. Chasles seems first to show, naturally in geometrical clothing, in the writings of the elder Heron.

It is highly remarkable (*Mémoire*, p. 12) that in Brahmagupta the correct formula for the triangle and the quadrilateral is preceded by the statement of another, incorrect procedure for determining the area, and that this other procedure, which Brahmagupta expressly designates as false, was in use both among the Romans and later. It seems to follow from this that the incorrect rule, having arisen in the Occident, made its way to the Orient and there found its refutation.

Dirichlet.

[1]The present piece is an exact reprint of a manuscript written in G. Lejeune Dirichlet's hand, found in the estate among the correspondence between Lejeune Dirichlet and Alexander von Humboldt, and probably composed for the historical second volume of *Kosmos*. F.

Remarks on the Most Suitable Way to Combine Observations for the Determination of Unknown Elements[1]

In the applications of mathematics, especially in practical astronomy, a frequently recurring task is to determine, by experience, the elements or constants contained in the analytical expression of a phenomenon. If the observations that must be taken as aids in this were capable of absolute accuracy, one would be dealing with a purely mathematical question. It would suffice to compare with the analytical law as many observations as that law involves unknown elements, in order to obtain the equations needed for the determination of the unknowns. But because errors of observation are unavoidable, this procedure, which presupposes no more observations than there are elements to be found, is to be regarded only as a first approximation; and the attempt lies very near at hand to replace the lack of accuracy by the number of observations, and to combine them with one another in such a way that the errors with which they are affected compensate one another as much as possible. The arithmetic mean, as the simplest form of such a combination, has always been in use among observers; but only in more recent times has this procedure become the object of a more thorough investigation. Lagrange, in 1779,[2] showed in a memoir entitled “Sur l’utilité de la méthode de prendre le milieu” how, under certain assumptions about the law of the errors to which the individual observations are subject, one can determine the probability that the error of the arithmetic mean falls within assigned bounds, and thereby laid the first foundation for answering the question which combination of observations deserves preference over all others.

For my purpose it is not necessary to specify more closely here the numerous and important investigations to which this subject later gave occasion; and it would be all the more superfluous since our esteemed colleague Mr. Encke, to whom we owe interesting memoirs on this theory, has in one of them characterized with great clarity the various points of view under which the combination of observations has been considered. Thus, referring as regards the historical matter to that memoir, I wish to discuss somewhat more closely only the point to which the considerations I wish to present to the Academy are attached. My remarks concern the way in which Laplace, in the “Théorie analytique des probabilités”, grounds the method of least squares. At the end of the chapter devoted to this investigation,[3] he himself expresses the matter approximately as follows:

> If the elements sought, as one may always suppose, are already known approximately, then their corrections are the actual unknowns, and the conditional equations given by the observations pass, with respect to these corrections, into linear form. If the final equations are likewise to be linear, then the conditional equations can be combined only by multiplying them successively by factors which, for the equations that are not to be used, are equal to zero, and then uniting them by addition into one final equation. A second system of factors gives a second final equation, and so on, until their number is equal to the number of elements. It is now clear that the factors are to be arranged so that the mean error of each element, that is, the integral of each possible error multiplied by its probability, becomes a minimum. For a small number of observations, the factors to be chosen depend on the law of the errors to

[1]The following note seems to be the fuller treatment of the work printed in vol. I, p. 281, of this edition of G. Lejeune Dirichlet’s works. F.

[2]From Lagrange’s collected memoirs it appears that this memoir did not appear in 1779, but rather in 1770–1773 in the *Miscellanea Taurinensia*, tome V. F.

[3]*Théorie analytique des probabilités*, book II, chap. IV, no. 24, p. 348, third edition; complete works, vol. VII, p. 353. F.

> which the observations are subject. If, however, as is usually the case in astronomy, the observations are very numerous, then this dependence ceases, and the theory shows that the most advantageous factors coincide with those that correspond to the method of least squares. This method is therefore to be preferred to all others when there are numerous conditional equations, provided that one wishes to use only linear final equations.

To the restriction just stated, which in a certain sense seems to be imposed by practice, there is added in the analytical determination of the most advantageous system of factors still another: namely, that the factors should depend only on the coefficients of the elements, and not at all on the constant terms contained in the conditional equations, which are affected by the observational errors. If one considers that these coefficients are wholly given by the circumstances under which the observations are made, for example by the times at which they take place, then one sees at once, if for simplicity one restricts oneself to one element, that the analytical treatment given by LAPLACE corresponds in fact to the following problem:

> Observations are to be made at previously determined times. Which, then, among all systems of factors specified in advance by magnitude and order, is the one from whose application one may expect the most advantageous determination of the unknown element?

If one did not wish[4] to accept this second restriction, but instead took into account the actual outcome of the observations in determining the factors, one would easily find that, however numerous the observations may be, the factors to be chosen always remain dependent on the error law, and therefore cannot be represented when that law is unknown. But one would be very much mistaken if one wished to conclude from this that, when the error law is not known, no systems of factors can be given that promise in general results just as advantageous as those furnished by the method of least squares, to which, as is well known, there correspond factors proportional to the coefficients of the element in the individual conditional equations. I shall try to show this in the simplest case, where the quantity to be determined is itself immediately the object of observation and the result of the method of least squares passes into the arithmetic mean.

Suppose that there is a large odd number $(2n + 1)$ of observations, and let $B_1, B_2, \ldots, B_{2n+1}$ denote, arranged in order of magnitude, the values given by observation for the unknown u to be determined. Then one may choose for u that B which occupies the middle place, or is furnished with the index $n+1$. This long-known procedure is evidently to be regarded as a special case of the linear treatment of the conditional equations; in that form it appears when all factors are set equal to zero except the one which multiplies the $(n+1)$st equation, and which remains arbitrary and may, for example, be taken equal to 1.

But it is equally clear that this system of factors is not included among those which LAPLACE compares with one another in order, by this comparison, to find the one that gives reason to expect the most advantageous result. For in the method just indicated the factors are, as regards their magnitude, independent of the values $B_1, B_2, \ldots$; but in applying the method, attention is nevertheless paid to these values, which become known only after the observation has been made, insofar as the conditional equation multiplied by unity - that is, the equation used alone for determining the unknown u - is the one in which B lies, by magnitude, in the middle among all the others.

[4]Here begins the work printed in vol. I, p. 281, of this edition of G. Lejeune Dirichlet's works. Not verbatim. F.

Laplace mentions this procedure (*Introd.*, p. 56) and adds that, by applying it as the number of observations increases, one approaches the true value of u indefinitely, but that the arithmetic mean is nevertheless to be preferred as the most advantageous method. This assertion did not seem evident to me, since the way in which Laplace grounds the arithmetic mean affords no comparison with this procedure, and therefore leaves it wholly undecided which of the two methods deserves preference. I have therefore tried to compare the two procedures with respect to the error that they leave to be feared. With respect to the arithmetic mean, Laplace determined the bounds within which the error affecting it lies with a given probability. For the other procedure, the same method offers few difficulties and leads to a very simple result. Comparing this result with the one corresponding to the arithmetic mean, one finds that, for the same probability, the error bounds of the results of the two methods stand in a constant ratio, that is, a ratio independent of this probability, without it being possible in general to decide for which method the error bounds turn out narrower. The ratio is that of two constants, both of which depend on the unknown error law of the observations; the constant corresponding to the arithmetic mean is expressed by an integral extending over the whole range of the error curve, whereas the constant belonging to the other procedure is determined merely by the ordinate at the origin of the coordinates.

Note on a Theorem of Indeterminate Analysis

I read in the summary of the meetings of the Académie des Sciences, inserted in the *Annales de Chimie et de Physique*: "M. Lamé presents a memoir on the impossibility of the equation $x^5 + y^5 = 2^n z^5$, in integers." This recalls to me a theorem of the same kind on which I came upon more than a year ago, and whose statement follows.

The equation

$$x^3 \pm y^3 = 2^n z^3$$

is impossible in positive integers.

To prove this theorem it will be enough to prove the impossibility of the three equations

$$x^3 \pm y^3 = z^3, \qquad x^3 \pm y^3 = 2z^3, \qquad x^3 \pm y^3 = 4z^3. \tag{a}$$

For if one wished to suppose $x^3 \pm y^3 = 2^n z^3$, with $n = 0$ or $n > 3$, one need only put $n = 3k + r$, where r is one of the numbers $0, 1, 2$; this changes the preceding equation into

$$x^3 \pm y^3 = 2^r (2^k z)^3,$$

which falls under one of the equations (a) assumed impossible.

The impossibility of the first two equations (a) has long since been proved, as may be seen in the second volume of EULER's *Algebra* or in M. LEGENDRE's theory of numbers. I therefore have only to deal with the third.

Suppose, if possible, that $x^3 + y^3 = 4z^3$. Since we may suppose that x and y are relatively prime, they will both be odd. Put $x = p + q$ and $\pm y = p - q$; then p and q are integers, relatively prime, and moreover one is even and the other odd. Substitution gives

$$p(p^2 + 3q^2) = 2z^3,$$

which shows that p must be even, for otherwise the first member would be odd; hence $p/2$ is an integer and

$$\frac{p}{2}(p^2 + 3q^2) = z^3.$$

There are now two cases, according as p is or is not divisible by 3.

First case. If p is not divisible by 3, then $p/2$ and $p^2 + 3q^2$ are relatively prime; since their product is a cube, each of these numbers must be a cube. To make $p^2 + 3q^2$ a cube in the most general way, one need only put, as is known,

$$p = m^3 - 9mn^2, \qquad q = 3m^2 n - 3n^3.$$

From what has been said about p and q, it is easily seen that m must be prime to n, and moreover even and not divisible by 3. It follows that the three factors in

$$\frac{p}{2} = \frac{m}{2}(m + 3n)(m - 3n)$$

are relatively prime. Since their product $p/2$ is to be a cube, each of them must be a cube; consequently

$$\frac{m}{2} = \alpha^3, \qquad m + 3n = \beta^3, \qquad m - 3n = \gamma^3.$$

This gives $\beta^3 + \gamma^3 = 4\alpha^3$, an equation of the same form as $x^3 + y^3 = 4z^3$, but expressed in much smaller numbers.

Second case. If p is divisible by 3, put $p = 3r$; then r, like p, is even and relatively prime to q. Thus

$$\frac{9r}{2}(q^2 + 3r^2) = z^3.$$

It is easy to see that the two factors of the first member are relatively prime; hence each must be a cube. To make $q^2 + 3r^2$ equal to a cube, put, as in the first case,

$$q = m^3 - 9mn^2, \qquad r = 3m^2n - 3n^3.$$

Then n must be prime to m, and even; hence $9r/2$ is an integer and

$$\frac{9r}{2} = 27\frac{n}{2}(m+n)(m-n).$$

The factors $n/2$, $m+n$, and $m-n$ are relatively prime. Since their product multiplied by the cube 27 is to be a cube, each of them must be a cube. Thus

$$\frac{n}{2} = \alpha^3, \qquad m+n = \beta^3, \qquad m-n = \gamma^3,$$

and therefore $\beta^3 - \gamma^3 = 4\alpha^3$, an equation similar to the proposed equation $x^3 \pm y^3 = 4z^3$, but expressed in much smaller numbers.

Thus, by the ordinary infinite descent, the proposed equation is not possible.

Proof of Bernoulli's Theorem

Let l denote the sum of the terms in the expansion of $(p+q)^n$ (where $p+q=1$), from the term containing q^h up to the term containing $q^{h'}$. It is easy to see that

$$l = \frac{1}{2\pi}\int_{-\pi}^{+\pi}[p+qe^{z\sqrt{-1}}]^n e^{-(h+h')\frac{z}{2}\sqrt{-1}}\frac{\sin(h'-h+1)\frac{z}{2}}{\sin\frac{z}{2}}\,dz. \tag{1}$$

This expression can be put into the following other form, by setting $h=b-c$, $h'=b+c$:

$$l = \frac{1}{\pi}\int_0^{\pi}[p^2+q^2+2pq\cos z]^{\frac{n}{2}}\cos(n\psi-bz)\frac{\sin(c+\frac{1}{2})z}{\sin\frac{z}{2}}\,dz, \tag{2}$$

where

$$\psi = \operatorname{arctg}\frac{q\sin z}{p+q\cos z}.$$

From this one obtains

$$\psi = qz - Qz^3 + \cdots, \tag{3}$$

writing, for shortness,

$$Q = \frac{1}{3}q^3 - \frac{1}{2}q^2 + \frac{1}{6}q.$$

Furthermore,

$$\begin{cases} \log(p^2+q^2+2pq\cos z)^{\frac{n}{2}} = -\frac{n}{2}pqz^2 + n\left(\frac{pq}{24}-\frac{p^2q^2}{4}\right)z^4+\cdots, \\ (p^2+q^2+2pq\cos z)^{\frac{n}{2}} = e^{-\frac{n}{2}pqz^2}\left(1+n\left(\frac{pq}{24}-\frac{p^2q^2}{4}\right)z^4+\cdots\right), \end{cases} \tag{4}$$

$$\frac{1}{\sin\frac{z}{2}} = \frac{2}{z}\left(1+\frac{z^2}{24}+\cdots\right), \tag{5}$$

and therefore

$$l = \frac{2}{\pi}\int dz\, e^{-\frac{n}{2}pqz^2}\frac{\sin(c+\frac{1}{2})z}{z}\left(1+\frac{z^2}{24}+nkz^4+\cdots\right)\cos((nq-b)z-nQz^3+\cdots). \tag{6}$$

Let $nq=b$ and $c+\frac{1}{2}=\gamma\sqrt{n}$, $z=\varphi/\sqrt{n}$. Then

$$l = \frac{2}{\pi}\int d\varphi\, e^{-\frac{pq}{2}\varphi^2}\frac{\sin\gamma\varphi}{\varphi}\left(1+\frac{\varphi^2}{24n}+k\frac{\varphi^4}{n}+\cdots\right)\left(1-n^2\frac{Q\varphi^6}{n^3}+\cdots\right), \tag{7}$$

$$l = \frac{2}{\pi}\int_0^{\infty} d\varphi\, e^{-\frac{pq}{2}\varphi^2}\frac{\sin\gamma\varphi}{\varphi} = \frac{2}{\sqrt{\pi}}\int_0^{\frac{\gamma}{\sqrt{2pq}}} e^{-u^2}\,du. \tag{8}$$

The probability that, in n simple events, the event B (whose probability is q) lies between

$$qn \pm \gamma\sqrt{n}$$

is therefore

$$\frac{2}{\sqrt{\pi}}\int_0^{\frac{\gamma}{\sqrt{2pq}}} e^{-u^2}\,du.$$

Equivalently, if $\gamma/\sqrt{2pq}=\beta$, then

$$\frac{2}{\sqrt{\pi}}\int_0^{\beta} e^{-u^2}\,du$$

is the probability that Q/n lies between

$$q \pm \frac{\beta\sqrt{2pq}}{\sqrt{n}}.$$

Remarks on the Preceding Memoir

1) Formula (1) can be verified as follows. Rewrite the value of l as

$$l = \sum_{\alpha=0}^{n} \binom{n}{\alpha} p^{n-\alpha} q^{\alpha} \frac{1}{2\pi} \int_{-\pi}^{+\pi} e^{\alpha z i} e^{-(h+h')\frac{zi}{2}} \frac{\sin(h'-h+1)\frac{z}{2}}{\sin\frac{z}{2}}\, dz.$$

Set

$$J_\alpha = \frac{1}{2\pi} \int_{-\pi}^{+\pi} e^{\alpha z i-(h+h')\frac{zi}{2}} \frac{\sin(h'-h+1)\frac{z}{2}}{\sin\frac{z}{2}}\, dz.$$

With $e^{zi} = u$ this becomes

$$J_\alpha = \frac{1}{2\pi i} \int u^{\alpha-1-h'} (1+u+u^2+\cdots+u^{h'-h})\, du,$$

where the integration is over the circumference of the unit circle surrounding $u = 0$. Hence

$$J_\alpha = 0 \quad (\alpha < h \text{ or } \alpha > h'), \qquad J_\alpha = 1 \quad (h \le \alpha \le h').$$

Thus indeed

$$l = \binom{n}{h} p^{n-h} q^h + \binom{n}{h+1} p^{n-h-1} q^{h+1} + \cdots + \binom{n}{h'} p^{n-h'} q^{h'}.$$

2) Substituting $h = b-c$, $h' = b+c$ in (1) gives formula (2), after writing P for the modulus of $p + qe^{zi}$ and ψ for its argument.

3) For (6), expansion of the two factors gives

$$\begin{aligned} l = \frac{2}{\pi} \int_0^{\pi} dz\, e^{-\frac{n}{2}pqz^2} \frac{\sin(c+\frac{1}{2})z}{z} \left[1 + n\left(\frac{pq}{24} - \frac{p^2q^2}{4}\right) z^4 + \cdots\right] \\ \times \left[1 + \frac{z^2}{24} + \frac{7z^4}{5760} + \cdots\right] \cos((nq-b)z - nQz^3 + \cdots), \end{aligned}$$

so that in order to obtain the formula stated in the text one must put

$$nk = n\left(\frac{pq}{24} - \frac{p^2q^2}{4}\right) + \frac{7}{5760}.$$

4) In (7), $Q^2/2$ must be put in place of Q; that is, the final factor must read

$$\left(1 - \frac{n^2Q^2\varphi^6}{2n^3} + \cdots\right).$$

5) In (8) it is first assumed that n is very large. The second equality follows from the known formula

$$\int_0^\infty e^{-x^2} \cos 2\alpha x\, dx = \frac{\sqrt{\pi}}{2} e^{-\alpha^2},$$

whence

$$\frac{d}{d\alpha} \int_0^\infty e^{-x^2} \frac{\sin 2\alpha x}{x}\, dx = \sqrt{\pi} e^{-\alpha^2}.$$

Since the integral vanishes for $\alpha = 0$, one obtains

$$\int_0^\infty e^{-x^2} \frac{\sin 2\alpha x}{x}\, dx = \sqrt{\pi} \int_0^\alpha e^{-u^2}\, du.$$

Putting $x = \varphi\sqrt{pq/2}$ and $\alpha = \gamma/\sqrt{2pq}$ gives (8).

6) In the third-to-last line on p. 355 the letter Q occurs in a sense different from that used in equations (3), (6), and (7): it is the number of occurrences of the event B among the n trials. Thus l is the probability that

$$h = b - c \leq Q \leq b + c = h',$$

or, after $b = nq$ and $c + 1/2 = \gamma\sqrt{n}$ have been chosen, that Q lies between the limits

$$qn \pm (\gamma\sqrt{n} - \tfrac{1}{2}),$$

or, rounded,

$$qn \pm \gamma\sqrt{n}.$$

For this reason, in the penultimate line of the text $\sqrt{n}$ belongs not in the numerator but in the denominator: l is the probability that the ratio Q/n lies between

$$q \pm \frac{\gamma}{\sqrt{n}} = q \pm \frac{\beta\sqrt{2pq}}{\sqrt{n}}.$$

Prize Question

of the Physical-Mathematical Class of the Royal Prussian Academy of Sciences, for the Jubilee of the Accession of King Frederick II, for the year 1844

Announced on the anniversary of Frederick II's accession, 31 May 1840.

The theorem which science owes to ABEL, remarkable alike for its generality and its simplicity, seems to contain the germ of a complete theory of all integrals whose element is an algebraic function of the variable. In the simplest forms of this function Abel's theorem passes into the long-known fundamental equations of the trigonometric and elliptic functions; and, from the extent and importance that the theory of these two kinds of transcendents has attained through the repeated efforts of mathematicians, one may already infer, with great probability, the future significance of the general theory which Abel prepared by his discovery. What has so far been accomplished on the foundation laid by him, chiefly by LEGENDRE, JACOBI, and RICHELOT, may be regarded as a first important beginning of an extensive discipline, which will undoubtedly furnish analysts with material for the most comprehensive investigations for a long time to come. For these investigations the analogy that the subject presents with transcendents of a similar but simpler nature, already so widely explored, appears to offer a powerful aid; and one may promise oneself all the greater success from using it, since the complete transformation that the theory of elliptic functions has undergone in recent times has itself made that theory more similar to the earlier-developed doctrine of circular functions.

Although the enlargement and enrichment of integral calculus just mentioned, like all more significant analytical discoveries, did not arise from a single thought, but from the cooperation of several mutually supporting thoughts, one of them nevertheless appears to deserve the greatest importance, because it has been more effective than any other in producing this transformation and intimately permeates all parts of the new theory. Earlier workers on this subject regarded the elliptic integral as a function of its amplitude; the new mode of consideration proceeds essentially from the opposite point of view and treats the amplitude, or rather certain trigonometric combinations of it, as functions of the integral, just as one had earlier reached the most important properties of the transcendents depending on the circle by regarding sine and cosine as functions of the arc, and not the arc as a function of them. The numerous and brilliant results that have followed from this new treatment make it highly desirable that the same mode of consideration be applied to the more complicated transcendents introduced into science by ABEL and whose fundamental properties he established. JACOBI has already taken an important step in this direction by showing that the inverse functions corresponding to Abelian integrals contain two or more variables and possess the remarkable property of being fourfold or multiply periodic. This result sheds an entirely new light on the nature of these transcendents, but at the same time reveals the full extent of the difficulties that stand in the way of a complete representation of these inverse functions, and that must be overcome if the theory of Abelian transcendents is to be brought to the same stage of development as that which the theory of elliptic functions has already reached.

Convinced of the advantages that analysis must gain from the further development of this theory, the Royal Academy, being prompted by the memorial celebration of Frederick the Second's accession to the throne to open an extraordinary prize competition, believes that it makes a choice worthy of this celebration in submitting this subject to mathematicians for treatment. It therefore asks for:

> A detailed investigation of Abelian integrals, and especially of the functions of two or more variables which are to be regarded as their inverse functions.

The Academy abstains from any closer determination of the scope to be given to the treatment of the subject, since only the work itself can decide whether Abelian integrals can already be investigated with success in their full generality, or whether one must first restrict oneself to special classes of them, perhaps even to the class that follows immediately upon the elliptic functions.

On the Quadrature of the Circle

It is a highly remarkable phenomenon that, despite the great advances which mathematics has made in recent centuries - advances that may be said to exceed all expectations - several questions that must have presented themselves already in the earliest times, and that are in fact almost as old as science itself, have still not been decided in spite of all efforts. Among these problems is the quadrature of the circle, which had already attained a high degree of fame among the Greeks because of its difficulties. The ancients sought to solve all problems by pure geometrical construction; and for the solution of the problem in question here, understood in the same sense as the Greek mathematicians understood it, it would therefore be necessary to show by what geometrical construction, from the radius of a given circle, one can derive the side of a square having the same area as that circle. The question then arises whether such a derivation is possible at all. A complete answer requires either the specification of the construction by which this derivation can be performed, or the proof that the side of the square cannot be determined geometrically from the radius. Attempts have often been made to apply to this question the fruitful methods invented by the moderns, which have led to so many results that the ancients, with their limited aids, would have had to regard as unattainable; but so far this has not fully succeeded.

Before I pass to the presentation of the single satisfactory result that has arisen from these efforts, it is necessary to say in a few words under what form the question presents itself when translated into the language usual among modern mathematicians. By the quadrature of the circle one understands the derivation, from the radius of the circle, of the side of the square having the same area as a given circle; and this derivation must be performed by geometrical construction. By this name, as is well known, one denotes arbitrary but finite combinations of the following simple constructions or operations: drawing a straight line through two given points, and describing a circle whose centre and radius are given.

In analytic geometry it is shown in detail that, if one line can be derived from another in the manner just specified, then the ratio of the two lines can be represented by an expression in which, besides the four fundamental operations of arithmetic, only one or more extractions of square roots occur; and conversely, if such an expression is given, one can always find by geometrical construction a line that stands to a given line in the ratio determined by the expression. Thus, if the question is to be decided by calculation, it becomes: can the ratio of the diameter to the side of the square equal to the circle be represented by a finite combination of the operations just named? Instead of that ratio one may also consider the ratio of the diameter to the circumference, since each is very easily derived from the other; in other words, the quadrature and rectification of the circle are essentially one and the same problem. In fact the question is usually posed in the latter form, and to decide it one must investigate whether the ratio of the diameter to the circumference, denoted by π, can be represented by an expression of the kind defined above. Everything that has so far been attempted to answer this question, and especially some investigations of LAMBERT to be discussed presently, has made the impossibility of such a representation, and therefore of the quadrature of the circle, very probable. Yet it would still be very desirable to find a rigorous proof of the impossibility.

For apart from the fact that truths without practical applications - and the theorem just described as highly probable, the impossibility of the quadrature of the circle, must undoubtedly be counted among them - interest us only insofar as we have recognized them as truths by strict reasoning, it has also not rarely happened that what appeared, after repeated fruitless efforts, most probably impossible was finally recognized as possible and actually carried out. A notable example is provided by a problem with which the Greek mathematicians were already occupied, but which has been solved in full generality only in the most recent time: the division of the circle. Already in EUCLID's Elements one finds the geometrical construction for dividing the circle into 3, 5, and 15 parts, from which still others can be derived by repeated bisection of arcs. All later efforts to accomplish the division in other cases by geometrical construction remained fruitless, and it seemed at last to be generally believed that the cases just mentioned were in fact the only ones in which such a division was possible, until GAUSS showed the contrary in a manner as surprising as it was brilliant.

Draft for an Academic Address

Having discharged the office of public professor for more than twenty years, and having experienced throughout this long time the truth of the proverb that by teaching we learn, it does not seem inappropriate to me, when the occasion is given, to say a few things about the true way of cultivating analytical science. If I wished to pursue this subject with the fullness it deserves, neither time nor strength would suffice. It is enough to warn by a few examples of the danger one is in if, in matters which one has scarcely looked at, much less penetrated, one presumes to establish a fixed order. It is agreed by all that many geometers, generally following the chronological origin of the individual parts of doctrine, have established a certain order of topics in which one must ascend from each part to the next higher one. They have thought that this order must be observed so strictly that one not rarely falls into assertions such as these: in treating the calculus of partial differentials one must suppose that the integration of ordinary differential equations has already been completely perfected; and similarly, if a question belonging to higher analysis is proposed, the solution of algebraic equations must be ready to hand. How greatly nature itself mocks this artificial order of topics and difficulties, and how many aids would be lacking if one wished to preserve such an order in analytical investigations, I now undertake to show by a few examples.

What else is EULER's celebrated invention concerning the addition of integrals, which we now call elliptic, than a fundamental theorem concerning these integrals, drawn from a higher chapter? It is well known that this great man derived this distinguished truth from the consideration of a differential equation whose variables are separated. And who will doubt that the addition theorem is the true foundation of the entire doctrine of elliptic functions, if he considers not only that nearly all of the discoveries of the illustrious LEGENDRE in this chapter rest upon Euler's theorem, but also that on the same foundation rest the excellent truths by which the young ABEL and JACOBI, deserving their first laurels and contending without envy, enlarged this most fertile part of doctrine, and moved the admiration of that Nestor of geometers? In commemorating these brilliant discoveries there is renewed for me the grief which I share with all who care for the growth of analysis: inexorable fate took from the republic of letters two such great minds by premature death, one scarcely having entered the arena, the other diverted by so many and such great investigations that he was not allowed to put the final hand to them.

Another example of aid sought from parts of doctrine that seem to surpass the proposed question in difficulty is provided by the celebrated JACOBI's investigations on shortest lines on the surface of an ellipsoid. The problem is easily reduced to a problem of mechanics by means of the known principle that a point constrained by no accelerating force on a given surface moves with its initial velocity along a shortest line. Since the dynamical equations on which the motion on surfaces of the second degree depends do not admit integration by known methods, that illustrious man ascended to a higher region and, using a theorem of the illustrious HAMILTON, which he had previously reduced to a simpler form, reduced the question to a single equation containing partial differentials and was thereby able to solve it with excellent success. By this ingenious invention he founded an entirely new chapter in the theory of surfaces of the second order, in the cultivation of which the acute men JOACHIMSTHAL, CHASLES, LIOUVILLE, and ROBERTS found ample material for exercising their powers and enriching geometry with distinguished truths.

I now come to an argument than which one can find nothing more admirable, if one wishes to show by example how often the parts of analytical science are bound together by links of art, though at first sight hardly any connection is seen among them. I speak of the geometrical investigations concerning the section of the circle, a question known to have been discussed already in the time of EUCLID. The solution of the problem in the simplest cases - the only ones that seem able to dispense with the calculus of the moderns - having been completed by the Alexandrians, the general question awaited the efforts of moderns. Among them the great FRANÇOIS VIÈTE first succeeded in reducing the general division of the circle to an algebraic equation. After the interval of a century, the equation defining the division of the circle was reduced to the utmost degree of simplicity by the work of the illustrious COTES and MOIVRE, who showed that the problem depends on the simplest of all equations, in which the power of the unknown is equal to unity. After another century the sagacious VANDERMONDE undertook most ingenious investigations on the Cotesian equation. We now see that they rest on a very profound principle, though they seem scarcely to have come to the knowledge of contemporary geometers, perhaps because the author, already advanced in age when he first gave his attention to analysis, set them out somewhat obscurely. I shall briefly recount the very singular origin of these investigations, which the illustrious LAGRANGE rescued from oblivion in the notes added to his treatise on numerical solution of equations, as my revered teacher LACROIX, a man highly meritorious in the diffusion of analytical doctrine, once told it to me. VANDERMONDE, Dutch by nation, after practicing medicine for many years in his country and then living in Paris at an advanced age, frequented a certain establishment where many gathered for games. Seeing that the famous geometer CLAIRAUT was treated with singular reverence by everyone who came there, he, entirely untrained in analysis, asked someone whether anything remained to be perfected in analytical doctrine. He received the reply that, among other things - and we may add, many things - the general solution of algebraic equations was still lacking. Stirred by this reply, the acute man mastered the elements of algebra in a short time, devoted himself with the greatest zeal to this subject, and soon completed the investigations mentioned above.

At last appeared the illustrious GAUSS's *Disquisitiones arithmeticae*, a work by which its author inaugurated a century and restored to Germany the glory which, after LEIBNIZ's time, had belonged to foreigners, at a time when the cultivation of analysis among us was languishing and almost extinct. In the last section of the work, to pass over other things, the equation belonging to the section of the circle is solved generally - or rather, to define the matter more clearly, the equation is reduced to pure equations of lower degrees. By this reduction the section of the circle is carried out geometrically in infinitely many cases, the simplest after the previously known cases being the division into seventeen parts, a discovery that soon came to be on the lips even of beginners, while the other chapters of the work, no less worthy of admiration, scarcely became known to a few foreign geometers. The author himself is witness to this, for more than twenty years after the book was published he publicly declared that he had still found no one who had read it through. But I return to the matter.

The solution is built on a very singular principle taken from higher arithmetic. By its aid the roots of the equation are arranged in a cycle in such a way that each depends on the preceding in the same manner; once this is done the reduction to lower equations is accomplished without difficulty. Yet in solving this algebraic problem the great man did not rest content: looking more deeply into the proposed subject, with his characteristic sagacity he soon perceived that the solution was most closely connected with the most hidden properties of numbers. Algebra, returning with great interest the help it had borrowed from higher arithmetic, drew from this rich source most beautiful arithmetical theorems, which scarcely seem accessible by any other path. Following in the footsteps of this great man, the distinguished geometers ABEL, JACOBI, CAUCHY, GALOIS, EISENSTEIN, KUMMER, and LEBESGUE formed from this subject a very extensive discipline, which one can scarcely say should be referred more to algebra or to higher arithmetic. The boundaries of the two are so confounded that they cannot be discerned. To the same subject belong the observations of the celebrated POINSOT on primitive roots derived from the section of the circle. How this geometer could assert that by these observations he had presented the first application of algebra to the knowledge of numerical properties, I could hardly understand, unless I suppose that this acute man once bore some resemblance to the good Homer portrayed by HORACE in the *Ars poetica*.

I could bring other examples to show that great aids for analytical questions can be sought from regions apparently most remote, if I wished to speak of the use of integral calculus and of series with an infinite number of terms in higher arithmetic; but after the brilliant discoveries just recalled I fear it would not be fitting to discuss my own.

If I may comprehend in a few words all that I have said above and express it under the form of a trope, those very great studies and contests, to which through so many centuries men endowed with supreme genius and singular sagacity have devoted their labor, do not seem to me very different from the composition of a certain singular book, whose chapters, pages, and lines all of us, each according to his powers, try to compose. Since in that book so many passages are left to posterity to perfect and so many pages to fill in, no certain order seems to be established, nor will anyone presume so to join individual sentences together that Horace's rule, "let not the middle disagree with the beginning, nor the end with the middle," can rightly be applied.

To this book, then, there seem rightly to belong the words of a man of outstanding acuteness of mind and much meditation on the human mind:

> The last thing one finds in making a work is to know what must be put first.

On a Proof from the Calculus of Probability Due to Lejeune Dirichlet

After a communication by J. F. Encke

If the errors of observation are arranged according to their absolute magnitudes, without regard to their signs, then for m observations the error belonging to the index $\frac{1}{2}(m+1)$, when m is odd, or, when m is even, the arithmetic mean of the errors with indices $\frac{1}{2}m$ and $\frac{1}{2}m+1$, will give an approximate value for the probable error r.

If, already for power sums, the larger number of observational errors makes the accuracy with respect to probable limits increase so much, then with this method of counting this must happen all the more. Since GAUSS, in the *Zeitschrift für Astronomie*, vol. I, p. 195, gave the formula needed for this without proof, the following elegant proof, which I owe to the communication of my esteemed colleague Professor DIRICHLET, will be all the more valuable here, since the theorem itself has not yet been proved elsewhere.

We seek the probability that, among $(2n+1)$ observations, the distribution of errors is such that one error lies between t and $t+dt$, n errors lie between 0 and t, and n errors lie between $t+dt$ and ∞. Let the probability that an error is smaller than t be, quite generally,

$$\int_0^t \psi(\Delta) d\Delta = u.$$

The probability of an error $> t+dt$ then becomes

$$1 - \psi(t)dt - \int_0^t \psi(\Delta) d\Delta = 1 - u - \psi(t)dt,$$

since the probability of an error between t and $t+dt$ is $\psi(t)dt$.

Thus the compound probability of one arrangement of the errors, with n errors $< t$, one error between t and $t+dt$, and n errors $> t+dt$, is

$$u^n(1-u)^n\psi(t)dt,$$

if terms of the second order are neglected, since the result is of the first order. But such cases or arrangements can occur in as many ways as there are permutations of $2n+1$ elements when among them there are n equal elements of one kind, with probability u, and n equal elements of another kind, with probability $1-u$. Consequently the probability of all possible arrangements of this kind is

$$\frac{1\cdot 2\cdot 3\cdots(2n+1)}{(1\cdot 2\cdot 3\cdots n)^2} u^n(1-u)^n\psi(t)dt = U.$$

If the size dt of the interval between t and $t+dt$ is held constant, there is a value of t for which U is a maximum. Differentiation gives, for its determination,

$$\frac{n\psi(t)}{u} - \frac{n\psi(t)}{1-u} + \psi'(t) = 0.$$

Here $\psi'(t)$ has the same meaning as $\varphi'(\Delta)$ above; namely du, or the increment of $\int_0^t \psi(\Delta)d\Delta$ under an infinitely small change in the upper limit t, is equal to $\psi(t)dt$. The last equation can be put in the form

$$\frac{1}{u} - \frac{1}{1-u} + \frac{\psi'(t)}{n\psi(t)} = 0.$$

The last term becomes smaller the larger n is, or the more observations are given. For a sufficiently large number it may be neglected. Thus the value of t for which the maximum occurs approaches, as n increases, the value that follows from

$$\frac{1}{u} - \frac{1}{1-u} = 0,$$

that is,

$$u = \int_0^t \psi(\Delta) d\Delta = \frac{1}{2},$$

or, by the definition given above, the value r.

If one takes the integral of U between assigned limits, one obtains the probability that the middle error is contained within those limits. For the limits $r-\delta$ and $r+\delta$ this probability is therefore

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \int_{r-\delta}^{r+\delta} u^n (1-u)^n \psi(t) dt.$$

Since $\psi(t)dt = du$, if with respect to t the limits are written

$$\int_0^{r-\delta} \psi(t) dt = u', \qquad \int_0^{r+\delta} \psi(t) dt = u'',$$

the probability that the middle value lies between $r-\delta$ and $r+\delta$ becomes

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \int_{u'}^{u''} u^n (1-u)^n \, du.$$

The larger the number of observations, the narrower the limits between which t will lie with equal probability. If the observations are sufficiently numerous, then in expanding u' and u'' by Taylor's theorem it is permissible to retain only the first power of δ. Hence

$$u' = \int_0^r \psi(t) dt - \delta\psi(r) \cdots = \frac{1}{2} - \delta\psi(r),$$

and similarly

$$u'' = \frac{1}{2} + \delta\psi(r).$$

Both this form and the combination of u and $1-u$ in the integral show that a more convenient form is obtained by introducing another variable for u, most simply by

$$u = \frac{1}{2} + \frac{s}{2\sqrt{n}} = \frac{1}{2}\left(1 + \frac{s}{\sqrt{n}}\right),$$

so that

$$1 - u = \frac{1}{2} - \frac{s}{2\sqrt{n}} = \frac{1}{2}\left(1 - \frac{s}{\sqrt{n}}\right).$$

The corresponding limits in s are found from

$$\delta\psi(r) = \frac{s}{2\sqrt{n}}.$$

The integral therefore becomes

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \cdot \frac{1}{2^{2n+1}\sqrt{n}} \int_{-2\delta\sqrt{n}\psi(r)}^{+2\delta\sqrt{n}\psi(r)} \left(1 - \frac{s^2}{n}\right)^n ds,$$

or, since only even powers of s occur in the differential,

$$\frac{1\cdot 2\cdot 3\cdots(2n+1)}{(1\cdot 2\cdot 3\cdots n)^2}\cdot\frac{1}{2^{2n}\sqrt{n}}\int_0^{2\delta\sqrt{n}\psi(r)}\left(1-\frac{s^2}{n}\right)^n ds.$$

Let now $\delta\sqrt{n}$ be a finite quantity, $=\varrho$, so that the limit δ decreases in just the same measure as $\sqrt{n}$ increases. Then s remains finite within the assumed limit, however much n increases. For large n one may also put, by the development of logarithms in EULER's *Introductio*,

$$\left(1-\frac{s^2}{n}\right)^n = e^{-s^2}$$

and

$$\frac{1\cdot 2\cdot 3\cdots 2n}{(1\cdot 2\cdot 3\cdots n)^2}=\frac{2^{2n}}{\sqrt{n\pi}},$$

according to EULER, *Calc. Diff.*, part II, chap. VI, §§160–162, as the limiting value to which it continually approaches as n increases. Hence the expression becomes

$$\frac{2n+1}{n\sqrt{\pi}}\int_0^{2\delta\sqrt{n}\psi(r)} e^{-s^2}\,ds,$$

for which one may safely write

$$\frac{2}{\sqrt{\pi}}\int_0^{2\delta\sqrt{n}\psi(r)} e^{-s^2}\,ds.$$

This is the expression for the probability that, in numerous observations arranged by magnitude, the middle error r lies between $r-\delta$ and $r+\delta$.

This probability will therefore be $1/2$, or the probable limits are given by

$$2\delta\sqrt{n}\,\psi(r)=\varrho,$$

that is,

$$\delta=\frac{\varrho}{2\sqrt{n}}\cdot\frac{1}{\psi(r)}.$$

For the law of errors assumed above,

$$\psi(\Delta)=2\varphi(\Delta)=\frac{2h}{\sqrt{\pi}}e^{-hh\Delta\Delta},$$

the probable limits of r are therefore

$$r\pm\frac{\varrho e^{hhrr}\sqrt{\pi}}{4\sqrt{n}\,h}.$$

If, instead of $2n+1$, the number of observations is denoted by m, and if the equation

$$hr=\varrho$$

is used, this becomes

$$r\left\{1\pm\frac{e^{\varrho\varrho}\sqrt{\pi}}{\sqrt{8m}}\right\}.$$

The numerical value of $e^{\varrho\varrho}$ is 1.2554176, and the expression becomes

$$r\left\{1\pm\frac{0.786716}{\sqrt{m}}\right\}.$$

CORRESPONDENCE BETWEEN LEJEUNE DIRICHLET AND GAUSS[1]

I. Lejeune Dirichlet to Gauss

Most revered Court Councillor!

I have the honour of most respectfully presenting to Your Excellency my first mathematical attempt, as a token of my most heartfelt veneration and my deepest admiration. With the kind recommendation of Baron von HUMBOLDT, I venture to flatter myself with the hope that you will receive and judge this first work of a young German with kindly indulgence. If Your Excellency should not find it wholly unworthy of your attention, I would venture most respectfully to ask permission to write to you from time to time and to ask you for some hints for the guidance of my further scientific efforts. I would regard this permission as the highest happiness, since, with the preference with which I pursue the study of indeterminate analysis, I wish for nothing so earnestly as to see the author of the immortal *Disquisitiones arithmeticae* take some interest in my efforts. Since until now I have chiefly occupied myself with higher arithmetic, I have followed my inclination entirely, without duly considering how little, with my limited powers, I may hope to accomplish anything substantial in this difficult part of mathematics. Although I become more convinced every day of the difficulties connected with investigations of this kind, this occupation has nevertheless become too much a habit for me, and I would almost say too much a passion, for me to decide so easily to abandon the investigations once begun.

Toward the end of this year I shall return to my native country, Prussia, where I hope to be appointed through the kind offices of Baron von HUMBOLDT. Although I know how properly to value the advocacy of this great scholar, I cannot help admitting that it still leaves something to be desired. For it depends largely on the judgment that will be passed on my work in Berlin what sphere of activity will be assigned to me in my fatherland, and therefore, in view of the slight value of my attempt, it is of the highest importance to me that it be judged with indulgence, or at least that justice be done to it. But since even among the most distinguished mathematicians – as I have had occasion here to convince myself – only very few are familiar with indeterminate analysis, there is reason to fear that my treatise will receive a less favourable reception than might perhaps fall to it if, with otherwise equal intrinsic value, it had as its subject a problem from a better-known part of science, for example astronomy or the integral calculus. This circumstance will excuse me in some measure for daring to trouble Your Excellency with the most submissive request that you might, on some occasion, communicate your judgment of my work to one of the Berlin scholars with whom you are in correspondence. I believe I may entertain the hope that Your Excellency will not refuse this favour to a young German whose fortune would be so greatly advanced by your kind advocacy.

Since I intend to remain in Paris for some time, I take the liberty of offering you my services during my stay in this city, while at the same time begging Your Excellency, should you wish to honour me with a commission or a few lines, kindly to address your letter to my father, Post Commissioner in Düren near Aachen.

I remain in deepest veneration,

[1] Of Gauss's replies to the letters of Dirichlet printed below, only the two letters which are printed in Gauss, Werke, vol. II, pp. 514–518, were available. According to a communication from E. Schering, no other replies have been found in Gauss's estate. F.

Your Excellency's
most obedient
G. LEJEUNE DIRICHLET

Paris, 28 May 1826.

II. Gauss to Lejeune Dirichlet

To Monsieur
Monsieur LEJEUNE DIRICHLET in Paris.

I would already have expressed to you earlier my thanks for the treatise you so kindly sent me, and for the great pleasure you thereby gave me, had I not wished first to learn something of the success of what I have tried to do in Berlin with regard to your wishes, and I may add my own. I am extremely pleased now to see from a letter received from the Secretary of the Academy in Berlin that we may hope people will soon be inclined to provide you with an appropriate position in your fatherland.

It is for me all the more pleasing a phenomenon that you devote yourself with great inclination to that part of mathematics which has always been my favourite study, the rarer such an inclination is. I heartily wish you an external position in which, as much as possible, you remain master of your time and of the choice of your work. I myself, immediately after the appearance of my *Disquisitiones*, was very much hindered by employments of another kind, and later by my external circumstances, from following my inclination to the extent I would have wished. Instead of a second part of that work, which I had earlier intended, I shall in all probability have to restrict myself to providing, from time to time, a memoir on a single subject. The three papers of this kind which have so far appeared in the sixteenth volume of the local Commentationes and in the first and fourth volumes of the *Commentationes recentiores* contain, however (apart from part of the second), none of the subjects which I already had in view in 1801 for a continuation, but new ones; and likewise my later works of this kind also concern a new subject, namely the theory of biquadratic residues, which I intend to give in about three papers. The first of these will shortly be printed for the sixth volume of the *Comment. rec.*, and the main materials for the remainder, as well as for the analogous theory of cubic residues, although little of it has yet been put properly on paper, may be regarded as essentially settled.

Kindly commend me to Herr von HUMBOLDT, if he is still in Paris, and excuse me for not writing to him himself now, since I am afraid that my letter might not reach him, as I hear he intended to leave Paris.

With sincere esteem,
your devoted
C. F. GAUSS

Göttingen, 13 September 1826.

III. Lejeune Dirichlet to Gauss

Most highly revered Court Councillor!

I would already have made use of the permission so kindly granted to me by Your Excellency to write to you occasionally, in order to thank you for the goodwill with which you received me, had I not wished at the same time to send you a work with which I began to occupy myself soon after my arrival in Breslau. This work was prompted by the announcement in your notices of your paper on biquadratic residues. The elegance of the criteria communicated in that announcement for deciding whether ± 2 is a biquadratic residue or non-residue of a prime number $= 8n + 1$, the second of which seemed to me especially remarkable, aroused in me the wish to find proofs of them myself. As soon as I had become somewhat familiar here with my professional duties, I began to occupy myself seriously with this subject. My efforts remained fruitless for some time, until I succeeded in deriving your second criterion, whose proof, according to the course chosen by you, seems to require many auxiliary investigations, from the first. But after this first happy success my work again came to a serious halt, and for a long time I could find no means suitable for establishing the first criterion. Finally, toward the beginning of winter, I came upon the proof contained in the accompanying paper, which is so simple that one can hardly understand how it does not at once present itself, once one is familiar with the theorem to be proved. The continuation of my investigations, of which my paper contains only one part, has led me to a great number of results which one certainly would not have suspected in advance to stand in any relation to this subject, and in this work several remarkable examples of the often wonderful connection among arithmetical truths have presented themselves to me, in which you find the principal ground of the attraction which investigations of indeterminate analysis hold for us.

Since I could assume that the readers of my paper would already be somewhat familiar with indeterminate analysis, I expressed myself very briefly in the preliminaries. This circumstance has occasioned a restriction harmful to the generality of the investigation. By analogy I allowed myself to be led into the error that in the theory of biquadratic residues it is sufficient to determine the behaviour of any one prime number with respect to another prime number of the form $4n+1$, whereas the complete answer to all the questions belonging here requires that one also determine the behaviour of composite numbers, at least of those that are products of two primes, with respect to primes of the form $4n + 1$. Thus, for example, the question whether 21 is a biquadratic residue or non-residue of a prime number $84n + 1$, 25, 37 may very easily be answered by means of the theorems set up in the paper; but the case is different for primes p of the form $84n + 5$, 17, 41, which require new criteria. For the special case just mentioned I find the following criterion.

> Put $5p = \varphi^2 + \psi^2$ (where I assume that ψ is divisible by 6). Then 21 will be a biquadratic residue of $p = 84n + 5$, 17, 41 if one of the numbers φ, ψ is divisible by 7; in the contrary case 21 is a biquadratic non-residue of p.

I am now busily occupied in filling the gap in my work produced by the error indicated above.

During my stay in Göttingen you had the kindness to communicate to me some of the remarkable results that are the fruit of your recent geometrical researches. The wish, already arising then, to become acquainted with your investigations has become very lively in me since, through the abstract of your paper contained in the notices, I have obtained a clearer and more complete idea of the results only lightly indicated in conversation. On this occasion I have again felt very strongly how desirable it would be for learned societies, when publishing their Commentaries, to follow the example given by the London Society, which has a volume appear every half-year. Even your paper on biquadratic residues has still not reached me, although I have repeatedly urged the bookseller to procure it quickly, and the volume of your Commentaries which is to contain it was already announced in the previous catalogue.

May I ask Your Excellency to accept the assurance of the heartfelt veneration and gratitude with which I remain

Your Excellency's
obedient
LEJEUNE DIRICHLET

Breslau, 8 April 1828.

IV. Gauss to Lejeune Dirichlet

For your kind letter and the agreeable sending of your two papers, I offer you, my highly esteemed friend, my most binding thanks. I see with pleasure the increasing interest that is now beginning to be taken in investigations of higher arithmetic. The happy way in which you derive the second criterion relating to the biquadratic residuacity of the number 2 from the first pleased me very much.

Presumably the sixth volume of our Commentationes has now found its way to Breslau, and my *Commentatio prima* on biquadratic residues is therefore probably known to you. Should an opportunity present itself, I would also gladly send you a separate offprint of it. I could have chosen among several kinds of proof for the theorem occurring there; but it will not escape you why I preferred the one carried out there, chiefly because the classification of 2 for those moduli for which it is a quadratic non-residue (under B or D) must be regarded as an essential integral part of the theorem, to which most of the other methods of proof do not seem applicable.

The whole investigation, the material of which I have possessed completely for 23 years, but the proofs of the main theorems (to which the theorem in the first Commentation is not yet to be counted) for about 14 years – although I wish and hope still to simplify something in the latter, the proofs – I have calculated at about three papers. I have already now begun drafting the second, and hope to complete it in not too long a time, unless the surveying business recently imposed on me again causes some delay.

The final theorem $b \equiv \frac{1}{2}rr' \pmod p$ I had already made known three years ago in the local learned notices, and I had called attention to the remarkable knot that remained to be solved there. But so far I have not heard that anyone attempted it. A few days ago I really succeeded with one half, and this discovery gave me all the more pleasure because it is not at all founded on induction – for I admit that I would not have expected precisely this connection, but rather a priori from the combination of other, very entangled and interesting investigations, already 28 years old but not yet made known at all, of which a slight indication is given in the concluding note to the *Disquis. Arithm.* p. 668.

This is namely a sufficient criterion for the case in which p is of the form $8n+5$.

Let the number of classes which the binary forms in each of the two kinds form for determinant $-p$ be equal to k. The beginning of a table I constructed up to determinant -3000 stands in the *Disquis. Arithm.* p. 520. It is also still to be noted that, for a p of the assumed form, one always has $k = 2m+1$ when m denotes the number of decompositions of p into three positive squares (I say positive in order to exclude 0), as LEGENDRE found by induction and as was first proved in the *Disquisitiones Arithm.* from the theory of ternary forms. For example:

p	5	13	29	37	53	61	101	109	149	157
k	1	1	3	1	3	3	7	3	7	3
m	0	0	1	0	1	1	3	1	3	1
decompositions			$4+9+16$		$1+16+36$	$9+16+36$	$1+36+64$ $4+16+81$ $16+36+49$	$9+36+64$	$1+4+144$ $4+64+81$ $36+64+49$	$4+9+144$

Assuming this, that value of b which is congruent to $\frac{1}{2}rr' \pmod p$ is always

$$b \equiv 2k + a - 1 \equiv 4m + a + 1 \pmod 8,$$

by which the sign of b is completely determined. Here are 22 examples, by which I double the extent of the table given at the end of the paper:

p	k	a	b	p	k	a	b
5	1	$+1$	$+2$	181	5	$+9$	$+10$
13	1	-3	-2	197	5	$+1$	-14
29	3	$+5$	$+2$	229	5	-15	$+2$
37	1	$+1$	-6	269	11	$+13$	$+10$
53	3	-7	-2	277	3	$+9$	$+14$
61	3	$+5$	-6	293	9	$+17$	$+2$
101	7	$+1$	-10	317	5	-11	$+14$
109	3	-3	$+10$	349	7	$+5$	$+18$
149	7	-7	-10	373	5	-7	$+18$
157	3	-11	-6	389	11	$+17$	-10
173	7	$+13$	$+2$	397	3	-19	-6

One can therefore also express the rule as follows, always assuming $p \equiv 5 \pmod 8$. One has

$$b \equiv a + 1 \pmod 8 \quad \text{if } m \text{ is even},$$

and

$$b \equiv a + 5 \pmod 8 \quad \text{if } m \text{ is odd}.$$

I do not yet venture a conjecture whether an even simpler criterion is possible, by which one could distinguish in advance the case of even m from that of odd m, that is, without knowing the value of m itself, since, as I have noted above, this rapprochement is still quite new. In the case $p \equiv 1 \pmod 8$, the above congruence $b \equiv 2k + a - 1 \pmod 8$ does remain correct, but it no longer decides the sign of b, since it is simultaneously satisfied by the positive and the negative value of b. Here k is namely always even, $= 2m$ (if the meaning of m is stated as above) or $= 2m + 2$ if by m one understands the number of decompositions of p into three positive unequal squares, and $b \equiv 0 \pmod 4$, or $b \equiv -b \pmod 8$. I suppose that the case $p \equiv 1 \pmod 8$, or $b \equiv 0 \pmod 4$, is *altioris indaginis* and perhaps again

$$b \equiv 4 \pmod 8 \quad \text{is easier than} \quad b \equiv 0 \pmod 8,$$

$$b \equiv 8 \pmod{16} \quad \text{is easier than} \quad b \equiv 0 \pmod{16}, \quad \text{etc.}$$

With distinguished esteem I remain
your friendly devoted
C. F. Gauss

Göttingen, 30 May 1828.

V. Lejeune Dirichlet to Gauss

Receive, highly honoured patron, my most heartfelt thanks for the permission so kindly granted to me to spend some time near you. Unfortunately I am not able to make use of it during these holidays and to carry out a long-cherished wish. A few days ago I had a rather violent attack of dysentery, and now, when I am beginning to recover, my family are laid low with measles. I need hardly tell you how much I regret seeing my visit to Göttingen postponed once again. Occupied for years with the study of your works, nothing could be more enjoyable and more advancing for me than to owe to your kindness explanations of the origin and development of the splendid methods which appear in your writings in admirable perfection. Indeed, I will confess to you that in my hopes I go so far as to expect, from your goodwill, oral communication of part of your theoretical investigations on magnetism and electromagnetism, and I solemnly vow in advance, should you deem me worthy of such a favour, to take Herr KRONE's discretion as my model; he, in the enviable possession of this treasure, would reveal nothing except that these investigations are intimately connected with the beautiful method developed in your paper on the attraction of ellipsoids. Have you learned from the *Comptes rendus* of the Paris Academy of the discussion which has recently arisen over the problem just mentioned? POISSON seems to regard his solution as the only direct one, while in all others, in the case of an exterior point, the difficulty is merely eluded. I must frankly confess that statements of this kind have no clear sense for me. To reduce a double integral to a simple one by introducing new variables for which one integration can be performed does not seem to me more direct than any other means, for example differentiation under the integral sign. In my opinion everything depends on the route by which the reduction can be carried out most quickly and clearly; and in this respect your treatment, which gives the attraction for interior and exterior points at one stroke, has decisively the advantage over all known ones. If such distinctions were admitted, then, to be consistent, one would have to admit for almost all definite integrals that in their discovery the difficulty was merely eluded; yet by this, it seems to me, actually nothing at all would be said.

Permit me, highly honoured patron, to send you, enclosed, a short article which appeared in the last issue of Crelle's Journal. I was led to the method indicated there, which appears to me very fruitful, through the effort to prove the theorem that every arithmetic progression whose first term and difference have no common factor contains infinitely many primes. I would almost conjecture that my method has some analogy with the investigations indicated in the final note of the *Disq. arith.*, especially because you say of your investigations that they shed light on several parts of analysis, while my method stands in such intimate connection with the remarkable trigonometric series which represent discontinuous functions and whose nature, at the time of the appearance of the *Disq. arith.*, was still wholly unexplained. You may perhaps still remember that more than ten years ago, when I was still in Breslau, you communicated to me a criterion for deciding the question raised at the end of your first paper on biquadratic residues (for the case $p = 8n + 5$), with the remark that you had derived it from the investigations announced at the end of the *Disq. arith.* This criterion now follows very easily from the combination of the theorem that for determinant $-p$ the number of forms in each genus is equal to the excess of the number of quadratic residues $< \frac{p}{4}$ over the number of quadratic non-residues $< \frac{p}{4}$, with the theory of biquadratic residues, this theory itself taken only to the extent in which I have given it in Crelle's Journal. In a detailed working out of my investigations on the use of series in higher arithmetic, with which I intend shortly to occupy myself, may I mention this criterion as having been communicated to me by you in 1828?

Allow me to ask you to hand the enclosed letter to WEBER, and kindly accept the assurance of my deepest admiration and heartfelt veneration.

Yours gratefully devoted,
DIRICHLET

Berlin, 9 September 1838.

VI. Lejeune Dirichlet to Gauss

Most revered patron!

The low spirits caused in me by the illness from which my wife has recently suffered will, I hope, excuse me for having delayed so long in writing to you, however much I wished to offer you my heartfelt thanks for the exceedingly kind and benevolent reception you granted me. Another reason for postponing my intention lay in the fact that several of the books I wanted to consult, in order to give you the desired information about what has been written so far on the shortest surface, were just not in the library, and I therefore had to await their return.

I have found nothing at all on the subject named, but I have found investigations on other problems which, although different in object, coincide in essence with the determination of the shortest surface. These problems concern the theory of heat in the case of permanent temperatures, to which, according as one takes two or three dimensions into account, the equation

$$(1)\quad \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0,\qquad \text{or}\qquad (2)\quad \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=0$$

relates. If the given curve, only slightly removed from a plane, through which the shortest surface is to be laid, projects as a rectangle or a circle, then the solution of the problem is easily derived from the methods given by FOURIER in his *Théorie de la Chaleur*. The much more difficult case in which the curve projects as an ellipse also seems to be contained implicitly in a beautiful paper by LAMÉ (Journal de Liouville, May 1839), since there a function u is determined which, inside an ellipsoid given by

$$\left(\frac{x}{\alpha}\right)^2+\left(\frac{y}{\beta}\right)^2+\left(\frac{z}{\gamma}\right)^2=1,$$

satisfies differential equation (2) and on the surface reduces to an arbitrary function of the two coordinates determining each point of it. If one sets $\gamma=\infty$ and lets the given arbitrary function become independent of z, one obtains the equation of the shortest surface whose boundary curve projects as an ellipse.

Since my return I have occupied myself somewhat with quadratic forms for complex numbers and have convinced myself that almost all the results and methods of section 5 of the *Disq. arith.* remain applicable to this case *mutatis mutandis*. In a similar way to the case of real numbers, the number of forms can also be determined; the remarkable result arises that this number, as it is connected for real positive determinants with the division of the circle, stands here in an analogous relation to the division of the lemniscate. These investigations have led me to a theorem which seems not unimportant for the theory of indeterminate equations of higher degree and which offers a remarkable generalization of the theorem according to which the equation $t^2-Du^2=1$ is always solvable.

Let $a,b,\dots,g,h$ be integers, and suppose the equation

$$s^n+as^{n-1}+bs^{n-2}+\cdots+gs+h=0$$

has no rational factor and has among its roots $\alpha,\beta,\gamma,\dots$ at least one real root. Then integers

$$\lambda,\mu,\nu,\dots,\varrho\qquad(\text{not with }\lambda=1,\ \mu=0,\ \nu=0,\dots)$$

can always be found, and in infinitely many different ways, of such a kind that, if for abbreviation one puts

$$\lambda+\mu\alpha+\nu\alpha^2+\cdots+\varrho\alpha^{n-1}=F(\alpha),$$

the product

$$F(\alpha)F(\beta)F(\gamma)\cdots = 1$$

holds. The proof of this theorem is in the highest degree simple and can be dispatched with the aid of geometrical considerations in two or three pages.

I cannot close this letter without repeating the wish already expressed to you in Göttingen, that you might be pleased to favour me with a separate offprint of your beautiful investigations on the general theory of attractions. I would most reluctantly have to wait until the completion of the whole next issue of the *Resultate* etc. to become acquainted with them. In expectation of your kind granting of my request, I remain

your gratefully devoted
LEJEUNE DIRICHLET

Berlin, 3 January 1840.

May I ask you to hand the enclosed letter to WEBER, or, if he should be in Leipzig, to send it there by post.

VII. Lejeune Dirichlet to Gauss

Highly honoured patron!

If I communicate to you so late the result of the inquiries undertaken at your commission concerning the Russian books, this delay, I hope, will find some excuse in the good success of my efforts. Like you, Privy Councillor von VARNHAGEN found great difficulties in obtaining Russian books. The connections formerly made by him with great effort for this purpose now largely no longer exist, since for years he has greatly neglected his Russian studies and consequently lacked opportunity to cultivate the earlier connections. In that earlier period, however, when he pursued Russian with great zeal, he acquired a not inconsiderable stock of books, which he places at your disposal for whatever use you wish, with great readiness. At my request he drew up a list of the most important of these books, which I enclose. Please kindly tell me which of these works you wish to read. I shall then immediately ask H. von V. for what you desire and send it to you without loss of time.

I returned from my holiday journey just in time to see poor EISENSTEIN at least once more. When I visited him the day after my arrival, I found him changed to a frightening degree and so wholly occupied with his suffering that even the attempt to tell him of you and so to distract him for a moment remained without success. With such boundless suffering, the death that occurred that same night had to appear to me as a benefit for our poor friend.

As I hear from WEBER, you desire information on the origin of the mathematical library which is to be sold here in March, unless it is earlier acquired as a whole – for which there seems to be some likelihood. This library was formerly in the possession of a Privy Councillor HEILIGENSTEIN, who stayed in Paris at the beginning of the century, and then later passed into the possession of his son, who is said to have been an astronomer. With this library, for simultaneous auctioning, some other things have been joined, among which are, for example, the two manuscripts by you – of which one is that of the *Determinatio attract.*; I do not immediately recall the content of the other – which come from DIRKSEN's estate.

You recently seemed to wish that my proof for the convergence of trigonometric series be extended to the case in which the maxima and minima of the function to be expanded are no longer countable, and at the same time you expressed the supposition that the proof of the theorem under this extended hypothesis would probably offer no essential difficulties. On closer consideration I find your supposition completely confirmed, provided one disregards certain quite singular cases. The whole foundation of the proof is formed by the theorems according to which

$$\lim \int_0^c f(\beta)\frac{\sin k\beta}{\sin\beta}\,d\beta = \frac{\pi}{2}f(0), \qquad \text{and} \qquad \lim \int_b^c f(\beta)\frac{\sin k\beta}{\sin\beta}\,d\beta = 0,$$

where $0 < b < c \leq \frac{\pi}{2}$ is assumed and the symbol lim refers to the positive quantity k becoming infinite.

The extension of these theorems to the case just mentioned, in which $f(\beta)$ presents an infinite number of maxima and minima, is quite easy, at least for the second theorem and also for the first, provided only that infinitely many maxima and minima do not lie in the immediate neighbourhood of $\beta = 0$. Namely, one need only separate from the integrals those parts within which such an accumulation of maxima and minima occurs, and one can arrange these, by bringing the bounds of these partial integrals sufficiently close to each other, so that they remain arbitrarily small, however large k becomes. From this simple remark, and by the way in which convergence is concluded from the theorems above, it already follows that the trigonometric series for representing an arbitrary function $\varphi(x)$ always converges even when the maxima and minima of $\varphi(x)$ are no longer countable, provided only that one assigns to x no value in the immediate neighbourhood of which an infinite accumulation of maxima and minima of $\varphi(x)$ occurs. To prove convergence for the exceptional values just mentioned, it must be shown that the first of the above theorems remains valid when $f(\beta)$, from $\beta = 0$ to $\beta = \delta$, where δ is arbitrarily small, passes infinitely often from increasing to decreasing and conversely. If then $f(\beta)$ is so constituted that the difference $f(\beta) - f(0)$ can be represented as a product whose first factor $\varphi(\beta)$ remains finite for infinitely small β, while the second, which I suppose positive and shall call $\psi(\beta)$, has the property that the integral

$$\int_0^\delta \psi(\beta)\frac{d\beta}{\beta},$$

as is the case for example for every positive power β^p, itself becomes infinitely small when δ is infinitely small, then one may decompose

$$\int_0^\delta f(\beta)\frac{\sin k\beta}{\sin\beta}\,d\beta$$

only into the two components

$$f(0)\int_0^\delta \frac{\sin k\beta}{\sin\beta}\,d\beta + \int_0^\delta \varphi(\beta)\sin k\beta\frac{\psi(\beta)}{\sin\beta}\,d\beta.$$

Of these, for fixed δ, the first passes as k increases into $\frac{\pi}{2}f(0)$, while the second, for a suitably small chosen δ, always remains smaller than an arbitrarily small quantity. From this, and since

$$\int_\delta^c f(\beta)\frac{\sin k\beta}{\sin\beta}\,d\beta$$

has zero as its limit, the validity of the first theorem follows, at least under the earlier hypothesis on $f(\beta) - f(0)$.

Forgive, revered patron, this long, almost unletterlike mathematical exposition, which I would not have dared to write had I foreseen, just as I began it, its great extent.

Please accept, highly honoured patron, the expression of my deep admiration and heartfelt veneration.

yours gratefully
devoted
DIRICHLET

Berlin, 20 February 1853.

G. Lejeune Dirichlet

Werke, Band II, XXXVII

Correspondence between G. Lejeune Dirichlet and Leopold Kronecker

communicated by Ernst Schering

Complete English reading translation with mathematical notation retained in type

Introductory note by Ernst Schering

The following correspondence between G. Lejeune Dirichlet and Leopold Kronecker was communicated by Ernst Schering. Schering explains that, apart from the first letter, the letters belong to the period of Dirichlet's stay in Göttingen, and that this local connection gives them a special interest. Since the letters also have considerable historical importance for mathematical science, he asked Kronecker's permission to print them, together with some letters addressed by Kronecker to Dirichlet to which Dirichlet's replies refer. Schering knew that these Kronecker letters had been returned to their author by Dirichlet's son after Dirichlet's death. Kronecker granted permission and also allowed the letters addressed by him to Dirichlet, with Dirichlet's replies, to be handed over to the University Library at Göttingen.

I. Dirichlet to Kronecker

I have, dear friend, made myself guilty of such great negligence toward you that I can almost despair of obtaining your forgiveness. Being a young man, you will hardly find credible what I can put forward in my excuse: you can scarcely form an idea how much, in the middle age into which I entered some time ago, one lacks the necessary flexibility for quickly finding one's way into matters with which one is not already somewhat familiar. When your first letter arrived I was so overloaded with thirteen lectures a week and with other business that the little time I could devote to studying your interesting communication did not suffice to make your investigations clear to me.

I hoped that during the holidays I would be able to penetrate better into the nature of your work. My preliminary efforts had at least convinced me that in it you had taken up a much-treated subject from a new and, as it seemed to me, very happily chosen side. In this hope I was unfortunately disappointed, for immediately after the beginning of the holidays I was visited by an attack of influenza that made me incapable of any exertion for more than three weeks. Now, after having in recent days come nearer, I believe, to your work, I hasten to tell you that it seems to me very desirable that you publish a short note on your investigations, since your continuing ill health unfortunately prevents you for the present from thinking of a full redaction of them. If that note is so arranged that every mathematician not foreign to the subject can understand it completely, and if it does not exceed the limits set for articles admitted into the Academy's Monthly Report, I shall gladly present it to the Academy and move that it be admitted into the Report.

The rest verbally. I hope that your presence here will not fall, or at least not wholly, in the period from 12 to 22 May, during which I accompany my wife, who is to visit Carlsbad. With my wife's and my own best compliments to you and to your wife, I call myself, your faithfully devoted Dirichlet.

Berlin, 3 May 1853.

Schering's notes. The two letters from Kronecker to which Dirichlet refers in this letter were not found in Dirichlet's papers. Their scientific content is reproduced in the note communicated by Dirichlet to the Berlin Academy in June 1853 and printed in the corresponding Monthly Report. Kronecker further informed Schering that at the end of May 1853, on a journey to Paris for consultation with a physician, he passed through Berlin and personally handed Dirichlet the manuscript of that note.

II. Kronecker to Dirichlet

Most honoured Professor, I have heard with great joy from many sides that you are very pleased with your new home and have every reason to be satisfied with the change, although that change has brought us here so grave a loss. I hardly need tell you how deeply I am involved in that loss and how keenly I feel it, for you know yourself how infinitely much I owe you, and you know also that the main purpose of my move to Berlin was to be nearer to you.

I had intended to visit you this month, so that for at least a few days I might have the long-missed pleasure of personal intercourse with you. But Borchardt told me that you are already going to Paris in a few days, probably to remain there four to six weeks; I shall therefore postpone the execution of that plan until after your return, provided that you still maintain the permission you earlier gave me orally to visit you. I shall learn in due time, if not directly from you then indirectly through Borchardt or other acquaintances here, when you are back in Göttingen.

For that time I could of course reserve all other communications; but who knows what may intrude between intention and execution. Since I still believe myself assured of your old interest, I shall add a few words about my mathematical life during the past winter. I worked quite steadily during the whole time since my return from the seaside, and the results obtained have for the most part rewarded my efforts. I did not solve the principal question to which I came about a year ago, but I gained an insight into it that is very valuable to me. Moreover, without that solution I have attained the original purpose for whose sake I believed that I had to solve that question, so completely that nothing remains to be desired in that respect.

I have found a method for deriving all the properties of solvable equations of prime degree; in simplicity and rigour it should satisfy every just demand. The method requires no higher standpoint of mathematical conception than the problem itself, which is disposed of by it. I shall devote the summer to a careful working out of these matters. The work will become fairly extensive, not because of obscurity but because very many individual results come out, and because I must not spare space in precisely designating certain quite new points of view, especially those missing in the Galois and Abel fragments, without which the required rigour cannot be observed. Some pleasing results–or more exactly, pleasing ways of viewing things–which I obtained while occupied with these methods, I shall soon publish in a small note, because in the complete papers now to be drafted these matters would come quite last and thus would not reach publication for about a year.

In that note I shall also give the most general root of an Abelian integral equation represented as a complex number. I have now succeeded in putting it into so simple a form that the necessary preliminary number-theoretic remarks are only quite insignificant. While in this way I brought the investigation of the solvability of equations of prime degree completely to an end, I have also occupied myself in many ways with more general work: on the one hand by undertaking the investigation of solvability for degrees that are not prime, and on the other by no longer restricting the investigation of equations to solvability, but by extending it to the discovery of all properties that an equation can have at all.

In this latter direction I have admittedly only reached the point of seeing the whole great new field before me, that is, of knowing clearly what is to be sought; but in an investigation of such generality I do not think that insignificant. The complete solution of the precise problem I have set myself here will, if possible at all, only fall to my weak powers after a considerable time. Yet it is certainly worthwhile to expend time and effort on a single question that comprises the entire theory of algebraic equations in one unknown. I add only that I have solved this problem for the first degrees, including the seventh; that the difficulties begin only with degree seven; and that already in this special case I obtained results which, by their novelty and by their remarkable connection with complex number theory, seem very interesting to me.

The naturalness of these abstract algebraic investigations appeared to me in the noteworthy circumstance that, just as all solvable and Abelian equations occur in theories of transcendental analytic functions, so also all the "affects"–to use Jacobi's expression–which I had found a priori to be possible actually occur in certain known equations of the seventh degree. As for the solvability of equations whose degree is an arbitrary composite number, I have also made substantial progress, though without having brought the matter to complete closure. My work on equations of prime degree will supply the simple points of view by which the algebraic and the number-theoretic are properly separated in this whole domain, and by which the clarity and security hitherto everywhere lacking are obtained.

You will surely approve that I now interrupt the general investigations just mentioned in order to begin the redaction of the material–considerable at least in extent, if not in content–which has accumulated through nearly three years' work. I also want to finish at once a few very small papers to send to Liouville. You would greatly oblige me if, in Paris, you would occasionally use your weighty word to recommend their prompt acceptance in the Journal. In the first I shall explain the small point about Bernoulli numbers that I found more than ten years ago; in the second I shall give, in a form suitable for lectures on cyclotomy, an extremely simple way of determining the sign of

$$\sum e^{\frac{2k^2\pi i}{p}}.$$

In the third I shall give a very simple proof of the irreducibility of

$$1 + x + x^2 + \cdots + x^{p-1},$$

and the fourth will contain an essential shortening of a method used by Kummer in his paper on complex numbers. These, most esteemed Professor, are my good intentions for the next weeks. I postpone the working out of many other things I did this winter and earlier because I do not yet consider them ripe; in my work on solvable equations I have learned that I must warm matters for a long time before they attain the maturity needed for publication.

I have surely put your patience as a reader to too hard a test, and therefore I shall finally close these lines, wishing you a very happy journey and recommending my wife and myself most warmly to you and to your wife. With most sincere reverence and attachment, your grateful pupil, Leopold Kronecker.

Berlin, 3 March 1856. I add Kummer's best greetings.

III. Kronecker to Dirichlet

Most honoured Professor, when I wrote to you on 3 March I had already considered the possibility that my intended trip to Göttingen might be prevented by something, but there was nothing then to fear directly. Since then my old ailment has again tormented me; my little son, ill throughout the preceding winter, needed a change of air as soon as possible; and so for several weeks we have been here in Kösen, from where I shall go to a sea-bath at the end of July in order to bring my own health as much as possible back on track. Since in this way I am again thrown back upon preserving myself in your memory in writing, you receive these few lines merely as an accompanying letter to the offprint of a little article, whose occasion I already described to you on 3 March. I hope that occasion will sufficiently excuse, in your eyes, the existence of the enclosed trifle, should you glance into it.

In that case I mention only that the numbers denoted by k on page 209 and their properties cost me more trouble than they look, and that the derivations and proofs of the results communicated about Abelian equations offered various difficulties precisely in the detailed working-out that was required. Moreover those numbers k are interesting in more than one respect, and there is no doubt that for certain enumerations of forms they play the same role as the numbers denoted by you by a and b in the theory of quadratic forms.

If, keeping the meanings assigned to the letters n and m in the little article, one considers in general forms of degree n in several variables, then those corresponding to the simplest complex numbers can be defined as follows: the forms must be decomposable into linear factors, and the cyclic functions of these linear factors themselves must already be rational functions of the variables, that is, have rational coefficients. Such forms of degree n with determinant m are nothing other than the norm forms, and their associates, of complex numbers built from the periods $\omega(\varrho)$; and it is in the number of such forms that the k's must appear.

In this way one has a simple explanation of the forms that are plainly the nearest at hand, and none of the accidental peculiarities occurs which Eisenstein had to call upon in explaining the cyclotomic forms of the third degree in three variables. As for my work, I have advanced only little in my main occupation, because I had to occupy myself with several small elaborations. But in hopeful prospects, in what I can prove, I have often run ahead and gained views of the nature of equations which, if they stand up, should afford an entirely new and substantial interest. Yet because the field on which I am now working is wholly new and because its complexity is scarcely to be mastered, I believe that I shall need years of work before I arrive at strict results. Therefore, honoured Professor, do not expect something from me in this respect very soon, especially since I shall devote a good part of my time to the most careful elementary treatment of my older results. And

I have nothing to miss, provided that my health improves with the years rather than becomes worse.

One thing more. In connection with a small proposition about integral equations, which will probably be printed in the first number of the Journal appearing under Borchardt's name, I returned several times to your note on complex units in the Monthly Report of the Academy. Allow me to say that your presentation there, as I believe, can lead to misunderstandings, and that in my opinion it would be better partly to begin the matter from the end. Suppose–as you show can always be found–one has a system of units

$$\Phi_1(\alpha), \quad \Phi_2(\alpha), \quad \ldots, \quad \Phi_{h-1}(\alpha),$$

for which, if α' is conjugate to α,

$$\Phi_k(\alpha)\Phi_k(\alpha') = a_k, \qquad \log a_k = A_k,$$

and if the determinant $\sum \pm A_1 B_2 C_3 \cdots$ does not vanish, then there are no non-zero values of $m_1, m_2, \ldots, m_{h-1}$ for which all the equations

$$\begin{aligned} A_1 m_1 + A_2 m_2 + \cdots + A_{h-1} m_{h-1} &= 0, \\ B_1 m_1 + B_2 m_2 + \cdots + B_{h-1} m_{h-1} &= 0, \\ &\vdots \end{aligned}$$

would be satisfied. Thus it cannot be the case that

$$\Phi_1(\alpha)^{m_1}\Phi_2(\alpha)^{m_2}\cdots\Phi_{h-1}(\alpha)^{m_{h-1}} = 1$$

for any integral values of $m_1, m_2, \ldots, m_{h-1}$; for otherwise, by the irreducibility of the equation on which everything rests, it would follow that

$$\begin{aligned} \Phi_1(\alpha')^{m_1}\Phi_2(\alpha')^{m_2}\cdots &= 1, \\ \Phi_1(\beta)^{m_1}\Phi_2(\beta)^{m_2}\cdots &= 1, \\ \Phi_1(\beta')^{m_1}\Phi_2(\beta')^{m_2}\cdots &= 1, \quad \text{etc.} \end{aligned}$$

Multiplying the equations two by two gives, for all h equations,

$$a_1^{m_1}a_2^{m_2}\cdots = 1, \qquad b_1^{m_1}b_2^{m_2}\cdots = 1, \qquad \text{etc.},$$

which is precisely impossible. Hence $\Phi_1(\alpha), \Phi_2(\alpha), \ldots$ form a system of independent units.

I think such a reversal of the order of your conclusions would guard against certain misinterpretations. The words on page 106 and the following passage could lead one to believe, first, that for a system of independent units no single one of the above h equations could hold, whereas it is only impossible that all of them hold; and, secondly, that the equation $a_1^{m_1}a_2^{m_2}\cdots = 1$ implies the equation $b_1^{m_1}b_2^{m_2}\cdots = 1$, whereas only from the existence of

$$\Phi_1(\alpha)^{m_1}\Phi_2(\alpha)^{m_2}\cdots = 1$$

does the existence of all the others follow. The equation

$$\Phi_1(\alpha)^{m_1}\Phi_1(\alpha')^{m_1}\Phi_2(\alpha)^{m_2}\Phi_2(\alpha')^{m_2}\cdots = 1$$

gives no conclusion about the persistence of the same equation when α is changed into β.

I mention only the cases in which, for a system of independent units $\Phi_1(\alpha), \ldots, \Phi_{h-1}(\alpha)$, one or more, though of course not all, of the equations

$$a_1^{m_1} a_2^{m_2} \cdots = 1, \qquad b_1^{m_1} b_2^{m_2} \cdots = 1,$$

are in fact satisfied. If, for example, there exists a unit $\Psi(\alpha)$ whose analytic modulus is 1 but which is not a root of unity, then by your fundamental theorem one must have

$$\omega\Psi(\alpha) = \Phi_1(\alpha)^{m_1} \Phi_2(\alpha)^{m_2} \cdots \Phi_{h-1}(\alpha)^{m_{h-1}},$$

where $\Phi_1(\alpha), \ldots$ are the fundamental units and ω denotes a root of unity. If one changes $\sqrt{-1}$ into $-\sqrt{-1}$, this becomes

$$\omega^{-1}\Psi(\alpha') = \Phi_1(\alpha')^{m_1} \Phi_2(\alpha')^{m_2} \cdots \Phi_{h-1}(\alpha')^{m_{h-1}},$$

and, multiplying and taking account of $\Psi(\alpha)\Psi(\alpha') = 1$, one obtains

$$1 = a_1^{m_1} a_2^{m_2} \cdots a_{h-1}^{m_{h-1}},$$

where the m's are non-zero.

There would be many more things to mention here, but they are not worth the writing. I leave this subject only to ask you whether you consider the observation interesting that one can find cyclometric solutions of

$$x^2 + Dy^2 = 4p$$

when $p \equiv 1 \pmod{D}$. This is easily seen as a generalisation of Jacobi's Ψ's, from which only the solutions of

$$x^2 + 3y^2 = 4p, \qquad x^2 + 7y^2 = 4p$$

result. The whole observation has merely fallen to me from more general considerations, which otherwise have not yet shown themselves sufficiently fruitful.

For greater simplicity take $D = \lambda$, a prime of the form $4n + 3$. If $\alpha^\lambda = 1$, $x^p = 1$, and $p = k\lambda + 1$, then in Jacobi's notation

$$(\alpha, x) = x + \alpha x^g + \alpha^2 x^{g^2} + \cdots .$$

The product $\Pi(\alpha^a, x)$, extending over all quadratic residues a of λ, is easily seen to be of the form $\frac{1}{2}u + \frac{1}{2}v\sqrt{-\lambda}$, where u and v are integers. One has

$$\frac{u + v\sqrt{-\lambda}}{u - v\sqrt{-\lambda}} = \left(\frac{U + V\sqrt{-\lambda}}{U - V\sqrt{-\lambda}}\right)^{\frac{\sum b - \sum a}{\lambda}},$$

where U and V are the numbers in the solution of $U^2 + \lambda V^2 = 4p$. Thus for given p and λ one can set up an equation of degree

$$\frac{\sum b - \sum a}{\lambda},$$

which, if $U^2 + \lambda V^2 = 4p$ is solvable in integers, has a single rational root from which U and V follow immediately.

I have, to my alarm, again become far too expansive, and I therefore beg your pardon. I also ask you to commend me and my wife most warmly to your wife, and to preserve for me always your very precious goodwill. With most intimate attachment and reverence, your grateful pupil, Leopold Kronecker.

Kösen, 26 June 1856.

IV. Kronecker to Dirichlet

Most honoured Professor, I have just written a few lines to Dr. Dedekind, and my old attachment drives me also to send you some words, even at the risk that your interest in me may meanwhile have cooled somewhat. I do not really see a prospect of seeing you personally soon, for you always travel westward and, for many reasons, I cannot come to Göttingen.

My mathematical work has gone strangely this past winter. Both in my algebraic elaborations and in the further investigations on the affects of equations I was constantly interrupted by things apparently little connected with them, though I had no reason to be dissatisfied with these interruptions. I found many very interesting matters. Since I feel so independent that not the least trace of ambition troubles me, and since my joy lies solely in the recognition of what is true, it matters little to me on what exactly I spend my time, provided only that I use it well.

The two principal episodes of my work concerned the complex multiplication of elliptic functions and the theory of the most general ideal numbers. Regarding the first, I already communicated my first results to you here in December. In the first three months of this year I found many similar results, proved everything rigorously, and really completed the investigation in question. The little mathematical region explored in this connection was of the greatest interest to me in many respects: the elegance of the results, the simplicity of the methods of proof, the connection between analysis and number theory, and the further prospects opened by the matter–all this combined to spur my zeal in the investigation and afterwards to reward it.

I shall mention briefly the most essential results. First, the moduli k for which multiplication by $a + b\sqrt{-D}$ occurs are roots of a solvable equation of degree H, whose coefficients are complex numbers of the form $a + b\sqrt{-D}$, where H denotes the number of quadratic forms for determinant $-D$. Secondly, the equation has the character of the equations treated by Abel: all its roots are rational functions of one of them, and if $k, \theta(k), \theta_1(k)$ are such roots, then

$$\theta\theta_1(k) = \theta_1\theta(k).$$

The number of cycles or periods arising here according to Abel equals Gauss's exponent of irregularity for $-D$. If this number is 1, so that the equation is, as I call it, "Abelian," then the determinant is regular. I do not consider it hopeless that I shall still be able to determine whether that equation is Abelian or not, and hence whether the determinant is regular or irregular. For the moment I possess only this one criterion, but it is remarkable enough.

Thirdly, from the H moduli k one can express transcendently the coefficients of a system of non-equivalent quadratic forms of determinant $-D$. This result is all the more important since it could probably become a means of passing from the theory of elliptic functions to that of quadratic forms. Fourthly, in setting up the Abelian equations with complex multiplication I was led to transformations of theta-products and forms which in several cases give very short proofs and new forms of already known theorems.

Fifthly, I believe I have found a theorem on the number of forms for a real determinant in the complex theory of numbers. The formula is as follows. Let $F(m)$ be the number of forms for determinant $-m$. Then

$$F(n) + 2F(n-1) + 2F(n-4) + 2F(n-9) + 2F(n-16) + \cdots = \Phi(n),$$

where $\Phi(n)$ denotes the sum of those divisors of n which are greater than a certain bound indicated in the original computation, and where $n \equiv 3 \pmod 4$. If n is prime, then $\Phi(n) = n$, and a noteworthy induction property appears for the determinants

$$n, \quad n-1, \quad n-4, \quad n-9, \quad n-16, \ldots.$$

For the reduced forms belonging to these determinants there are congruence relations modulo n, so that their coefficients, in the appropriate order, are congruent modulo n.

Sixth and seventh, the same circle of ideas gives relations among modular equations and form numbers, and in several cases the necessary computations amount essentially to reading off the terms. Eighth, the moduli k for which modular equations have equal roots are of the kind already described. Kronecker breaks off here, remarking that enough has been said and that these details must await a more orderly exposition. He adds that he is not yet ready to write on the composition of forms, and that he hopes above all to discuss the matter personally with Dirichlet.

With this he closes the mathematical part of the letter, asking Dirichlet to preserve his valued goodwill, sending greetings to Dirichlet's wife, and signing as his devoted pupil, Leopold Kronecker. Berlin, 17 May 1857, Köthener Strasse 12.

V. Dirichlet to Kronecker

Dr. Kronecker, Berlin, Köthener Strasse No. 12. Göttingen, 4 July 1857.

Since you know from experience, dear friend, how deeply rooted in me is the bad habit of answering even the letters of my dearest friends very late, you will hardly be surprised that the news brought here by Borchardt and Ernst, namely that you were absent from Berlin, prevented me from carrying out my firm resolve to write to you during the Pentecost holidays. But since, as I learn from Dedekind, you have returned and will soon leave Berlin again, I cannot delay longer in thanking you most warmly for your very interesting letter.

For the great pleasure caused by the communication of your beautiful discoveries, I can find no more suitable expression than to cry to you with full conviction, *macte virtute*. At the same time I cannot conceal that a little egoism is mixed with this pleasure, for in all modesty I cannot deny myself the testimony that I first introduced you into the lower regions of one of the sciences on whose heights you now stride as a master. I deliberately speak only of one of these sciences, for as regards your algebraic greatness I must declare myself wholly innocent.

My interest in the remarkable relations you have found among quadratic forms, algebra, and elliptic functions is all the livelier because I myself have sometimes traced similar relations. A point of view connected with this is indicated in my paper abbreviated "R. s. d. a.", paragraph 7; yet I later noticed that what is said there is much obscured by the condition assumed there, namely that in the double summations over x and y the trinomial

$$ax^2 + 2bxy + cy^2$$

should be relatively prime to D. I have followed the subject somewhat further from the point of view, much simplified by removal of that condition, but if I interpret correctly your unfortunately very brief communications, the remarks I made have hardly anything in common with your beautiful results. Your results seem essentially to rest on specialising the modulus, while my consideration presupposes the indeterminacy of k and of the corresponding q.

I am therefore all the more eager to learn the foundation and proper starting point of your investigations. I believe I may count on their early publication, since you yourself describe the work as essentially finished and of moderate extent. As for myself, I am now engaged in working out the hydrodynamic paper whose subject we touched upon briefly at Christmas. The main result can be formulated approximately as follows.

If an incompressible homogeneous fluid mass, whose parts attract one another according to Newton's law and which at its surface is under a constant pressure, or a pressure depending only on time, initially has the shape of an ellipsoid; and if the initial motion is such that it can be decomposed into two simpler motions–a rotation about an axis through the centre with the same angular velocity for all mass elements, and a second motion in which all mass elements move, with respect to three mutually perpendicular planes through the centre, so that their velocities resolved perpendicular to each of these planes are proportional to their distances from them–then at any later time the surface will have the form of an ellipsoid concentric with the original one, whose axes, however, vary in length and direction with time. For the motion of the individual elements the same holds as was assumed at the first instant: at every time the instantaneous motion is composed of two simple motions of the kind just defined, but with the axis of rotation and the three corresponding planes generally variable with time.

It is easy to see that the assumed initial state of motion involves eight arbitrary quantities, and it is remarkable that for such an initial state the general character of the resulting motion can be specified. I say the general character, because for complete knowledge of the motion nine functions of time must be determined, defined by as many equations–one finite equation and eight differential equations of the second order–for which one can give generally seven first integrals. A complete solution can be carried out only in special cases; one of the simplest is when the ellipsoid has two equal axes and the motion begins without initial velocity.

From my equations follows the solution of a problem that seems to present some interest, even though the result is purely negative. The known results found by Maclaurin and Jacobi lead naturally to the question in what cases a fluid homogeneous ellipsoid, whose elements attract one another according to Newton's law, can rotate like a rigid body about its centre of gravity. The answer is that this is possible only when the motion takes place about a fixed axis coinciding with one of the principal axes of the ellipsoid; this is the case studied by Maclaurin and Jacobi.

The rest of the letter conveys family greetings and an apology from Dirichlet's wife for having been unable to visit Kronecker's wife during her short and hurried stay the previous winter. Dirichlet closes: *Vale et mihi fave*. Göttingen, 4 July 1857.

VI. Dirichlet to Kronecker

I hasten, dear friend, to thank you for your friendly letter and at the same time to send you, according to your wish, the earlier one from May. It goes without saying that you must later return the latter to me, since it is my property; but there is no haste, for I have at every moment a copy made by Dedekind. I take formal note, as the jurists say, of your friendly promise to visit me, though I hope perhaps to see you even earlier in Berlin. As to your commission concerning Riemann, whose present whereabouts I do not know with certainty–he was for a long time in Harzburg–I discharge it by writing today to Dedekind, so that through him your request may reach Riemann.

Please recommend me and my wife most warmly to your wife, and greet our friends Ehlers, Kummer, Weierstrass, and Borchardt, if he has returned, from your faithfully devoted Dirichlet. Göttingen, 4 October 1857.

VII. Kronecker to Dirichlet

Most honoured Professor, first of all my heartiest thanks for the extremely prompt fulfilment of my recent request. Meanwhile I have been struck by a very severe loss: my mother, in the fifty-ninth year of her life, died so suddenly, though of a chronic illness, that when I hurried to Liegnitz six weeks ago at the first news, I no longer found her alive. I remained there several weeks with my poor father and was myself greatly shaken in mind and body by the blow. Afterwards there were several minor illnesses in my little family here. You can imagine that my work advanced only slowly, and I must ask your indulgence.

I take the liberty of enclosing the small note I have drafted on my "elliptic-function work" and ask that you look through it. Since spring I have altered the treatment in two respects. First, I have dropped for the time being the assertion about the irreducibility of the equations on which the modulus values depend, for my proof of it was false. I am still convinced that those equations are irreducible, but because I cannot yet prove it, I had to abandon for the time being the conclusion about the irregularity exponent founded on it. As long as that irreducibility is not established, the conclusion could be made only in an alternative form, and that is neither beautiful nor simple enough to be included in the enclosed note.

Secondly, in spring I had the mistaken opinion that I was, as it were, finished with the mathematical field under investigation, whereas now a fullness of new and rich material offers itself. Nevertheless I shall this time, contrary to my habit and inclination, proceed to publication before I have worked the matter through completely. In two parts of my work I shall give the solvability of those equations and their application to the theory of quadratic forms of negative determinant; the further applications, especially to the composition of forms, I shall probably postpone.

I hope you will approve this moderation in your poor pupil, for I finally see that I can no longer strive after my ideal: to produce, even in lower regions, a work as complete in itself as yours all are. It is my cross that I am always led into fields requiring better intellectual powers than nature has given me and more time than I can count on with my actually very unreliable body. In my exposition I shall presuppose nothing from the theory of quadratic forms, in order to show clearly that the elliptic functions make all this for themselves. But before I do the same for composition, I must absolutely speak with you, the master of all these regions.

I hope you will come here in the Christmas holidays; if not, I shall later come to Göttingen. Kummer, who has progressed very beautifully with his proof of the reciprocity law, asks me to convey his warmest greetings. Please commend me and my wife to your wife, who I hope has arrived there safely and who has not yet entirely overcome her influenza. I ask you yourself, honoured Professor, to preserve for me the goodwill that I value above all. Berlin, 2 December 1857. Your faithful and grateful Leopold Kronecker. Please give Riemann and Dedekind my warm greetings.

VIII. Dirichlet to Kronecker

To Dr. Kronecker, from Berlin, presently in Ilsenburg. Göttingen, 23 July 1858.

Since you, dear friend, recently expressed at the Harzburg station the wish to learn the starting point of my investigations on the connection between quadratic forms of negative determinant and elliptic functions, I shall communicate my fundamental formula for the special case in which the determinant is a negative prime p of the form $4n+3$, in a few words and as well as can be done without entering into lengthy discussions for which I lack leisure. I already indicated this connection in the second part of my paper "R. s. d. a. etc."; but the matter is complicated there by the fact that numbers having a common divisor with the determinant are excluded from the representation. If this assumption is removed, the formula for the special case just mentioned, using Jacobi's notation, is

$$\frac{k}{\sqrt{p}}\frac{K}{\pi}\sum\left(\frac{s}{p}\right)\sin\operatorname{am}\left(s\frac{4K}{\pi}\right)=\sum q^{\frac{1}{4}(ax^2+2bxy+cy^2)}+\text{etc.}$$

The sum on the left extends from $s=1$ to $s=p-1$, and the double sums on the right refer to all systems x, y for which the particular quadratic form (a, b, c), of the first kind or properly primitive, is odd. The condition just stated could easily be removed if one also adjoined forms of the second kind. Each double sum can then, as already noted in the cited place, be decomposed into a finite number of products of two simple sums of the form

$$\sum_{x=-\infty}^{\infty} q^{(\delta x+\varepsilon)^2},$$

where δ and ε are rational numbers. Every such simple sum can, as follows easily from Jacobi's properties of ϑ, though to my knowledge this has not previously been observed, be brought into the form

$$\sqrt{\frac{K}{a}}\,\varphi(k),$$

where $\varphi(k)$ is an algebraic function of the modulus k belonging to q. Since, on the other hand, every $\sin\operatorname{am}$ occurring on the left is likewise an algebraic function of k, the equation expresses a relation among algebraic functions of the still variable k. One can therefore say that the whole theory of forms of negative determinant and of the numbers represented by them comes back to relations among roots of algebraic equations and must be derivable from those relations–a very noteworthy rapprochement, since no trace of such a connection has so far appeared for forms of positive determinant.

Since our recent conversation on the ride from Ilsenburg to Harzburg I have succeeded in narrowing substantially the estimate of the sum

$$\sum_{s=1}^{n}\left[\frac{n}{s}\right],$$

where the brackets denote, after Gauss, the greatest integer. Previously I could give it only with an error of order $\sqrt{n}$. The discovery of the method used for this, which will probably also apply to the following cases, gives me great pleasure, but it comes at an inconvenient time because it draws me away from completing the hydrodynamic paper, which must finally be finished.

Please recommend me to your wife and to the other ladies in Ilsenburg, as well as to Dr. Benzler. Göttingen, 31 July 1858. Your faithful Dirichlet.

Schering's note. In July Dirichlet visited Kronecker, who spent the summer in Ilsenburg, for several days. On Dirichlet's return journey to Göttingen Kronecker accompanied him by carriage from Ilsenburg to Harzburg.

IX. Dirichlet to Kronecker

Göttingen, 31 July 1858. Since I wrote to you, dear friend, yesterday week, I have received from my wife in Prussia a letter from which it seems to follow that she did not see Borchardt, who had expressed the intention of visiting her in Charlottenburg. I must therefore fear that the condition of our friend has again worsened, and I ask you to tell me in two words what you have learned about him from Berlin. A letter from Heine received yesterday does not mention with a syllable that you had summoned him to the Harz; it will probably now not take place at all, since your departure for Heligoland, as you recently told me, must be imminent. With my best recommendations to your wife, your most devoted Dirichlet.

X. Kronecker to Dirichlet

Ilsenburg, Sunday 1 August 1858. Most honoured Professor, I have just received your letter of yesterday and hasten to give you news of our friend Borchardt, with whom I was in Braunschweig a week ago today. How that happened without my being able to fulfil the arrangement made with you I must explain in a few words. The fairly persistent fine weather on the Monday after your departure from here impelled me to carry out my plan of making a several-day excursion to the Brocken and the Lower Harz. Weather and other disturbances prolonged my absence until Saturday a week ago. In the meantime I had written neither to Borchardt nor to Heine, since according to Kummer's earlier communication on Borchardt's health I regarded his departure as still postponed for some time.

To my astonishment, on my return here Saturday I found Borchardt's notice that he would arrive in Braunschweig at noon on Sunday. It was therefore quite impossible to inform you in time, especially since I knew that you would again lecture on Monday and that even a longer waiting by Borchardt in Braunschweig would not have helped. I therefore travelled over alone on Sunday and was with Borchardt from one to eight o'clock. He naturally regretted very much being deprived–perhaps also of Heine's visit–by circumstances whose prevention was perhaps partly within my power. You now have all the data, honoured Professor, to pronounce the verdict on me yourself.

So much for my excuses. As for Borchardt, I found him admittedly very weak and impaired; nevertheless I hope that he will have completed happily the journey to the Isle of Wight, which he made in short daily stages in the company of an experienced servant. He expects to meet on the Isle of Wight his cousin Helborn, now married Sussmann, with her husband, so that he will feel all the less abandoned there. Borchardt had, as I heard from Berlin a few days ago, had another small attack a few weeks earlier, and from that time dates the real improvement, whereas until then the illness had run a more creeping course but had also never really yielded. Borchardt plans to stay about five weeks in England and, if he then feels well, to go for a time to Sorrento. The two sea coasts lie far apart, but railways and steamships run quickly.

I shall learn Borchardt's address on the Isle of Wight only in Heligoland, where I intend to travel the day after tomorrow with my son Ernst, and where, to judge from a Berlin letter received yesterday, Weierstrass and Kummer may perhaps follow me.

After all these personal matters I come now to thank you most warmly for your friendly letter of 23 July and to tell you how flattered I feel by this prompt communication, and still more how greatly the content interested me. That you have advanced so far in determining

$$\sum \left[\frac{n}{s}\right]$$

is splendid indeed, and no consideration should lessen your joy in it. Certainly no hesitation will keep you from the further pursuit of these investigations, for the force of their attraction is too great. Your comprehensive mind will surely have no difficulty working out finished investigations alongside work in quite different fields. That is precisely the advantage of our great men: they not only possess an abundance of ideas, but the abundance does not hinder them from exploiting those ideas.

Your beautiful communications about elliptic functions I shall use properly as soon as I return to Berlin. I shall try to obtain the transformation you indicated,

$$\sum q^{\frac{1}{4}(ax^2+2bxy+cy^2)}$$

into products of two simple sums

$$\sum q^{(\delta x+\varepsilon)^2},$$

and especially to determine the rational numbers δ and ε. These probably depend directly on the coefficients of the forms, at least in such a way that–while the left side of your principal equation evidently contains algebraic functions depending on the modular equation of order p–the right side contains algebraic functions arising from modular equations of other orders. Such relations are of the highest algebraic interest to me. Known examples already exist, but those resulting from your equation are certainly quite different from the known ones. The number-theoretic consequences of your equation must likewise be very different from those that I drew from the theory of elliptic functions. I believe that your equation will yield interesting results also when one gives the modulus k one of the special numerical values I have investigated.

All my ideas and intentions in this direction are naturally still very indefinite, for Ilsenburg is not a good climate for mathematical studies. But from Berlin I hope to show my gratitude for your valuable communications. My dear wife asks to be warmly recommended to you, and I myself ask you to preserve for me your very precious friendship. Your faithfully devoted Kronecker.

Best greetings to your son Ernst and my best recommendations to your mother, as well as to Weber, Listing, and Riemann. I ask you to say nothing to anyone else for the moment about the ideas I have communicated to you concerning the algebra of the future, especially concerning the use of Jacobi's linear equations for $\sqrt{n}$. I should first like to see whether my speculations succeed.

G. Lejeune Dirichlet
Werke, Band II, XXXVIII

Extract from the correspondence between Alexander von Humboldt and G. Lejeune Dirichlet

English translation; tables and symbols reset in type

It follows from several letters from A. von Humboldt to G. Lejeune Dirichlet that Dirichlet supplied contributions to the *Kosmos*. Thus a letter of Humboldt, which must have been written shortly before the printing of the second volume, that is around 1847, says:

> I write these lines to you amid many idle and futile agitations; but it would be a lack of feeling and gratitude on my part not to tell you, my dear and respected friend, how deeply I am touched by the sacrifice of time you made in order to answer my request. You have satisfied me, and perfectly satisfied me, in what alone could enter into the composition of my historical survey. You did it with that measure and precision in the expression of ideas which so highly characterizes everything that comes from your pen. I do not name you; that would make you "the responsible minister", but I put the passages in quotation marks, which indicates, as I said on p. XIV of the preface, that the sentence is borrowed from one of my friends. Since you are less Indophile than I am, despite duties of kinship, you nurture the malicious hope that one day some fragment of Diophantus will be discovered which will give us what so far is found only among the Indians. I hope this malice will not succeed for you; and if I knew as much mathematics as I do not know, I too would place in the scale of probabilities the particular character of analysis, the local flavor of the Indian algebraists, metrical poets and colorists through symbols. Do not let these phrases make you fear that I have added to the sacramental phrase. I have carefully avoided doing so.

This letter seems to refer to the following passage in the *Kosmos* (vol. II, p. 262):

> In the algebraic works of the Indians there is found the general solution of indeterminate equations of the first degree, and a more developed treatment of those of the second degree than in the writings of the Alexandrians that have come down to us; hence there can be no doubt that, if the works of the Indians had become known to Europeans two centuries earlier, and not only in our own day, they would have had a promoting influence on the development of modern analysis.

A. von Humboldt to G. Lejeune Dirichlet

Here, my dear friend and colleague, is my long-announced question concerning the naming of the days of the week by the use of periodic series. The ancients–in the *Somnium Scipionis*, and in a famous passage of Dio Cassius, Ideler vol. I, p. 179–reckoned the planets as follows:

Saturn ♄, Jupiter ♃, Mars ♂, Sun ☉, Venus ♀, Mercury ☿, Moon ☾.

The Jews, however, let the planetary names of the seven-day week follow as Sabbath or Saturday, Sunday, Monday, Tuesday, Wednesday, Thursday, and Friday, corresponding respectively to Saturn, the Sun, the Moon, Mars, Mercury, Jupiter, and Venus.

There are two widespread solutions of the problem of how the latter sequence arises from the former. According to Dio Cassius' explanation, if one begins to count the hours of day and night from the first hour of the day, dedicating the first to Saturn, the second to Jupiter, and the third to Mars, then the last of the twenty-four hours will be Mars, and the first hour of the day following the Sabbath, Sunday, will fall on the Sun; the third day begins with an hour governed by the Moon. According to Letronne the problem is solved by placing, in the zodiac, twelve signs under thirty-six decans, three decans belonging to each sign. The thirty-six decans are named after the planets in the order of the first table. If one looks at Clerac's *Musee de Sculpture*, p. 832, one finds that the planet which is first in the sign names the weekday:

Leo	♄	♃	♂
Virgo	☉	♀	☿
Libra	☾	♄	♃
Scorpio	♂	☉	♀.

The weekdays are therefore ♄, ☉, ☾, ♂, ...

My question is: how can I show briefly and arithmetically that the path of the periodic series 7 and 24 leads to the same result as the same periodic series distributed by threes in 12?

To this question I add another, one that interests me less. The foolish law of planetary distances of Titius–often wrongly called Bode's–is

$$4, \quad 4+3, \quad 4+6, \quad 4+12, \quad 4+24, \quad 4+48.$$

If Saturn's distance from the Sun is 100, this gives Mercury, Venus, Earth, Mars, the small planets, Jupiter, and Saturn the relative distances

$$\frac{4}{100}, \frac{7}{100}, \frac{10}{100}, \frac{16}{100}, \frac{28}{100}, \frac{52}{100}, \frac{100}{100}.$$

Gauss has already said that it is a mad series which begins with 4, since Mercury should not be 4 but $4+\frac{1}{4}=5\frac{1}{4}$. Maedler calls this empirical law a geometrical progression, Bode calls it a harmonic one. It is surely neither. Titius seems to have had dimly in mind the idea of doubling from Mercury onward, not from the Sun onward: $3, 6, 12, 24, \ldots$; and by groping he found for Mercury the constant quantity 4, which, added deformingly to each member of the geometric progression, approximately gives the actual distances. Is that correct? Is not a harmonic progression one in which the first term is to the third as the difference between the first and the second is to the difference between the second and the third,

$$a:b:c:d, \qquad a:c=b-a:c-b?$$

Where is anything of that sort to be found in Titius' sequence $4, 7, 10, 16, \ldots$, as Bode wants? $4:10=3:3$. Nonsense.

Do not smile too much at my meagre knowledge. On account of the obscure idea of doubling, I recall the actual distances of the planets from the Sun: Mercury 8 million geographical miles, Venus 15, Earth 20.6, Mars 31.5, small planets 56, Jupiter 107.5, Saturn 197, Uranus 396, Neptune 620.8.

Berlin, Sunday night, 2 o'clock.

With intimate friendship,
A. v. Humboldt.

G. Lejeune Dirichlet to A. von Humboldt

According to two of the three hypotheses proposed for the origin of the planetary weekday names, they arose as follows. In the periodic series formed from the planetary sequence

$$\text{Saturn } a,\ \text{Jupiter } b,\ \text{Mars } c,\ \text{Sun } d,\ \text{Venus } e,\ \text{Mercury } f,\ \text{Moon } g,$$

namely

$$(1.)\quad abcdefg,\ abc\ldots,$$

one always took one member and then skipped two, giving the new periodic series

$$(2.)\quad adgcfbe,\ ad\ldots.$$

This is so whether, according to the first explanation of Dio Cassius, this skipping of two terms arose from the application of the fourth, or whether, as Letronne thinks, one inscribed every three following terms of the first sequence in a sign of the zodiac and used only the first as the weekday name. According to the third hypothesis, Dio Cassius' second explanation, successive weekdays are named after the planets governing the first hour of the day, so that twenty-three terms are always skipped. But in a periodic series such as (1) it is plainly immaterial whether one skips a certain number of terms or that number increased by any multiple of the number of terms in the period, here 7. Hence skipping $23 = 3 \cdot 7 + 2$ must give exactly the same result as skipping 2, namely the series (2).

Concerning Bode's assertion that the Titius series is harmonic, I know nothing else to say than that the term harmonic progression is known to me in no sense other than the one given by Your Excellency, and that nothing else is found in Kluegel either.

Your Excellency's most devoted,
Dirichlet.

Wednesday, 26 May 1851.

The remarks relating to the names of the weekdays were used by Humboldt in the *Kosmos*, where he discusses the question. In the third volume of the *Kosmos*, on pages 474 ff., there is an almost literal printing of the two preceding letters. In the following letter A. von Humboldt informs Dirichlet that he has used his communications.

A. von Humboldt to G. Lejeune Dirichlet

My dear friend! I am in great haste to return a proof to Stuttgart in order to finish this unfortunate third, entirely astronomical, volume. Kindly cast a hurried glance over pages 10 and 11[1] to see whether I have properly grasped your explanation of the agreement of the two methods, that of the planetary hours and that of the 36 decans. If you order changes, give them to me in German and send everything back with the note in your handwriting.

[1]The indication of the page number probably refers to the manuscript of the *Kosmos*.

G. Lejeune Dirichlet
Werke, Band II, XXXIX

Remark on Dirichlet's Works, vol. I, p. 348, line 7

By R. Dedekind; English translation

If the prime $p \equiv 5 \pmod 8$, then the two numbers h and k satisfying

$$h^2 - pk^2 = -4$$

are not always odd, as is asserted there; they can also both be even. The latter case first occurs for $p = 37$, where $h = -12$ and $k = 2$. The method of solving Pell's equation by means of cyclotomy arose from the investigations on the arithmetic progression and on the class number of binary quadratic forms of positive determinant; and that assertion would have found its correction of itself in the memoir *Recherches sur diverses applications de l'analyse infinitesimale a la theorie des nombres*, if Dirichlet had there returned to the consideration of the individual cases. The lectures he gave in Göttingen in the winter 1856–1857 likewise broke off before this point was settled; only while preparing them for publication did I notice the error just mentioned.

Let T, U, T', U', T'', U'' denote the least natural numbers satisfying

$$T^2 - pU^2 = 1, \qquad T'^2 - pU'^2 = 4, \qquad T''^2 - pU''^2 = -4.$$

Putting the fundamental unit in the form

$$\frac{T'' + U''\sqrt{p}}{2} = \varepsilon,$$

one sees at once that

$$\frac{T' + U'\sqrt{p}}{2} = \varepsilon^2, \qquad T' \equiv U' \equiv T'' \equiv U'' \pmod 2.$$

If one further sets

$$T + U\sqrt{p} = \varepsilon^{2m},$$

Dirichlet shows (vol. I, pp. 470–471) that $m = 1$ or $m = 3$, according as the numbers T', U', and hence also T'', U'', are even or odd. If n and n' denote the class numbers, there denoted by h and h', of the forms of the first and second kind belonging to determinant p, then at the same time

$$mn = 3n'.$$

By §11 of the same memoir (vol. I, p. 495), where one has to put $P = p$, $Y_1 = q = pk$, and $Z_1 = -h$, the class number n is determined by

$$(T + U\sqrt{p})^n = \left(\frac{-h + k\sqrt{p}}{2}\right)^6.$$

Since the numbers $-h$ and k are always positive, it follows that

$$\frac{-h + k\sqrt{p}}{2} = \varepsilon^{n'}.$$

Consequently h, k will certainly be even when T'', U'', hence also T', U', are even, that is, when $m = 1$, and therefore $n = 3n'$. In the *Disquisitiones Arithmeticae*, Art. 256 VI, a passage cited by Dirichlet at the end of §8 for comparison, Gauss notes that this case $n = 3n'$ occurs for

$$37, 101, 197, 269, 349, 373, 389, 557$$

and for no other prime below 600. For example,

$$-h = 12, \quad k = 2, \quad n' = 1, \quad n = 3 \quad (p = 37),$$

$$-h = 20, \quad k = 2, \quad n' = 1, \quad n = 3 \quad (p = 101).$$

The preceding equation for the class number n immediately yields the general criterion that h, k are even always and only when n is divisible by 3. This certainly happens in the case $m = 1$, because then $n = 3n'$; but it can also happen in the other case $m = 3$, $n = n'$, although T', U', T'', U'' are odd. The smallest prime p for which this phenomenon occurs is 229; the extremely laborious computation of the functions Y, Z gives

$$-h = 3420, \qquad k = 226 \qquad (p = 229).$$

The same result is obtained much more quickly by setting up the reduced forms, from which

$$T'' = 15, \qquad U'' = 1, \qquad n = n' = 3$$

follows.

R. Dedekind.

G. Lejeune Dirichlet

Werke, Band II, XXXX

Remark by Heinrich Weber

English translation

Kirchhoff told the undersigned, in a conversation held in Heidelberg at the beginning of the 1870s, that Dirichlet–as Kirchhoff knew from Dirichlet's own mouth–had possessed the solution of the problem of determining the distribution of static electricity on the surface of a conducting cube, and that this too, like so much else, has been lost to us. Nothing more precise about it has been ascertained.

Strassburg, March 1897.

H. Weber.

G. Lejeune Dirichlet

Werke, Band II, XXXXI

List of translations of Dirichlet's papers not included in the Collected Works

English translation of headings and bibliographic notes; French titles preserved

I. *Demonstration de cette proposition: Toute progression arithmetique, dont le premier terme et la raison sont des entiers sans diviseur commun, contient une infinite de nombres premiers.*
Liouville, *Journal de Mathematiques*, Series I, vol. IV, pp. 393–422 (translated by Terquem).
Vol. I, pp. 313–342 of this edition of G. Lejeune Dirichlet's Works.

II. *Recherches sur la theorie des nombres complexes.*
Liouville, *Journal de Mathematiques*, Series I, vol. IX, pp. 245–269 (translated by Faye).
Vol. I, pp. 509–532 of this edition.

III. *Note sur la stabilite de l'equilibre.*
Liouville, *Journal de Mathematiques*, Series I, vol. XII, pp. 474–478 (translated by Kopp).
Vol. II, pp. 3–8 of this edition.

IV. *Sur la reduction des formes quadratiques a trois indeterminees entieres.*
Liouville, *Journal de Mathematiques*, Series II, vol. IV, pp. 209–232 (translator not indicated).
Vol. II, pp. 27–48 of this edition.

V. *Sur la determination des valeurs moyennes dans la theorie des nombres.*
Liouville, *Journal de Mathematiques*, Series II, vol. I, pp. 353–370 (translated by Houel).
Vol. II, pp. 49–66 of this edition.

VI. *Sur une nouvelle formule pour la determination de la densite d'une couche spherique infiniment mince, quand la valeur du potentiel de cette couche est donnee en chaque point de la surface.*
Liouville, *Journal de Mathematiques*, Series II, vol. II, pp. 57–80 (translated by Houel).
Vol. II, pp. 67–88 of this edition.

VII. *Sur la possibilite de la decomposition des nombres en trois carres.*
Liouville, *Journal de Mathematiques*, Series II, vol. IV, pp. 233–240 (translated by Houel).
Vol. II, pp. 89–96 of this edition.

VIII. *Sur un probleme relatif a la division.*
Liouville, *Journal de Mathematiques*, Series II, vol. I, pp. 371–376 (translated by Houel).
Vol. II, pp. 97–104 of this edition.

IX. *De la composition des formes binaires du second degre.*

Liouville, *Journal de Mathematiques*, Series II, vol. IV, pp. 389–398 (translated by Lebesgue).
Vol. II, pp. 105–114 of this edition.

X. *Sur la premiere demonstration donnee par Gauss de la loi de reciprocite dans la theorie des residus quadratiques.*
Liouville, *Journal de Mathematiques*, Series II, vol. IV, pp. 401–420 (translated by Houel).
Vol. II, pp. 121–138 of this edition.

XI. *Eloge de Charles-Gustave-Jacob Jacobi.*
Liouville, *Journal de Mathematiques*, Series II, vol. II, pp. 217–243 (translated by Houel).
Vol. II, pp. 225–252 of this edition.

XII. *Sur le caractere biquadratique du nombre 2, extrait d'une lettre adressee a M. Stern.*
Liouville, *Journal de Mathematiques*, Series II, vol. IV, pp. 367–368 (translated by Houel).
Vol. II, pp. 259–262 of this edition.

G. Lejeune Dirichlet
Werke, Band II, Errata

Errata list for volume I

English translation

Page 26, line 7 from the bottom, should read:

$$(dd' \pm aee')^2 - a(d'e \pm de')^2 = l^2,$$

page 27, line 6 from the bottom, should read:

$$D'^2 - aE'^2 = k,$$

and on page 289, in formula (2), it should read:

$$\frac{n(n-1)}{2(2n-1)} \quad \text{instead of} \quad \frac{n(n-1)}{2(2n-2)}.$$